

Lexical Functions: The Naked Truth

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Textbooks in Language Sciences ??

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Writing conventions

Abbreviations

a	: a given arc in a network / graph
$\mathcal{A}$	: a set of arcs in a network / graph
ACC	: accusative case
Adj	: Adjective
Adv	: Adverb
$A_{(poss)}$	: possessive adjective (<i>my, your, ...</i>)
ART	: article or other determiner – e.g. <i>to have</i> ART <i>dream</i>
Claus	: Clausative
circumst	: circumstantial surface-syntactic relation
DAT	: dative case
determ	: determinative surface-syntactic relation
dir-obj	: direct-objectival surface-syntactic relation
DSynt-	: deep-syntactic
DSyntA	: deep-syntactic actant
DSyntS	: deep-syntactic structure
f	: A given lexical function
$f(L)$	: Application of the lexical function f to its argument (lexical unit) L
fr-LN	: French Lexical Network
$\mathcal{G}$	: a given graph
$\mathcal{G}_{LS}$	: a given Lexical System graph
GEN	: genitive case
GER	: gerund
INSTR	: instrumental case
L	: a given lexical unit, i.e. a lexeme or an idiom

Note. In the body of the book, we allow ourselves to use the symbol “L” in the sense of ‘referent of lexical unit L’ – e.g., [...] *prototypically used to do L* in the description of S_{instr} (Ch. 4, [17], p. xcvi).

L _{GENER}	: generic term (of a semantic field)
$\mathcal{L}$	: a given natural language
LF	: lexical function
LN	: lexical network
LOC	: locative case
LDOCE	: <i>Longman Dictionary of Contemporary English</i> (online)
modif	: modificative surface-syntactic relation
<i>n</i>	: a given node in a network / graph
$\mathcal{N}$	: a set of nodes in a network / graph
N	: Noun
NOM	: nominative case
obl-obj	: oblique-objectival surface-syntactic relation
PL	: plural
PoS	: part of speech
possess	: possessive surface-syntactic relation
prepos	: prepositional surface-syntactic relation
PREPOS	: prepositional case
S	: substantive [= ‘noun’; in the name of a nominal lexical function]
SAE	: Standard Average European [language type; roughly, Germanic, Romance and Slavic languages]
Sem-	: semantic
SemA	: semantic actant
SemS	: semantic structure
SG	: singular
SSyntS	: surface-syntactic structure
subj	: subjectival surface-syntactic relation
subj-adnom	: subjectival-adnominal surface-syntactic relation
V	: Verb

Symbols

*A : A is ungrammatical

$A \approx A$	: A and B are approximately equal; in the present volume, the symbol “ $\approx$ ” mostly stands for <i>approximately</i>
$A \equiv B$	: A and B are equivalent
$A \cong B$	: A and B are approximately equivalent
$s_1 \oplus s_2$	: linguistic union of (components of) linguistic signs s_1 and s_2
$E C$	: constraint C on the expression E – e.g., $\text{Loc}_{in}(\textit{weather}) = \textit{by}$ [ART ~ with a modifier] (i.e., one can say <i>by a / the weather</i> only if the noun <i>weather</i> has a modifier)
$A \Rightarrow B$	: A implies B
$A \Leftrightarrow B$	: linguistic entity A of the level n of representation corresponds (= is semantically equivalent) to linguistic entity B of level $n + 1$, and vice versa; for instance: ‘Joey $\leftarrow$ 1–sleep’ [SemS] $\Leftrightarrow$ JOEY $\leftarrow$ 1–SLEEP _(V) [DSyntS].
$f_{\supset}, f_{\subset}, f_{\cap}$	: lexical function that is semantically richer ($\supset$) or poorer ($\subset$) than f , or has semantic intersection ($\cap$) with f
$L_o \leftarrow \dots \rightarrow_o L'$	: in a syntactic representation, indicates that the nodes L and L' are coreferential – i.e., they have the same referent
$L_1 \vdash \text{LexRel} \rightarrow L_2$	: L_2 is the target of the lexical relation LexRel originating from L_1
$L_1 \text{--Dep} \rightarrow L_2$	: L_2 semantically / syntactically depends on L_1 by semantic / syntactic dependency Dep
$\lceil L_1 L_2 \dots L_n \rceil$	: idiom – e.g., $\lceil \textit{bite the bullet} \rceil$
$\sim$	: the lexical unit under description in a given formula – e.g., $\text{Real}_1(\textit{bicycle}) = \textit{ride}$ [ART ~]
	: [in a Russian wordform] boundary between a morphological stem and a suffix – e.g., <i>sobak a</i> ‘dog-SG.NOM’
—	: graphical connector for segments of one wordform that is traditionally spelled as several words – e.g. <i>at_last</i>
σ	: a given meaning
σ^f	: meaning of the lexical function f
Ω	: external participant of the situation denoted by L (i.e., a participant that is not expressed as one of L’s semantic actants)
$\emptyset$	: zero sign

In the list of an LF application values [*Val*]:

$Val_i; Val_2$	there is a notable semantic difference between Val_i and Val_2
$Val_1 < Val_2$	Val_2 is “more” than Val_1
$\boxdot Val$	Val is a fused value element (for a syntagmatic LF)

Special font formats

- SMALL CAPITALS** : lexical unit supplied, when needed, with its lexicographic number – e.g., the lexeme STEP_(V) 1.1 and the idiom ‘BLOW THE WHISTLE’
- monospace font** : lexical function’s name – e.g., Syn, A₁, ...
- italicized sans serif font* : important notion, when first introduced – e.g., *idiom*
- small bold sans serif font** : usage label – e.g., **archaic, child language, colloquial, ...**

Russian examples are in international scientific transliteration; it is as follows:

Aa – Aa	Ěě – Ěě	Ja ja – Яя	Oo – Oo	Tt – Tt	Žž – Жж
Bb – Бб	Èè – Ээ	Ju ju – Юю	Pp – Пп	Uu – Уу	’ – Ьь
Cc – Цц	Ff – Фф	Kk – Кк	Rr – Рр	Vv – Вв	” – Ъъ
Čč – Чч	Gg – Гг	Ll – Лл	Ss – Сс	Xx – Хх	
Dd – Дд	Ii – Ии	Mm – Мм	Šš – Шш	Yy – Ыы	
Ee – Ее	Jj – Ъъ	Nn – Нн	Šč, šč – Щщ	Zz – Зз	

Simple standard lexical functions: alphabetical list

- Numbers to the right of lexical function [LF] names indicate their order of presentation in Chapter 4 – paradigmatic LFs, cf. “P” – and Chapter 5 – syntagmatic LFs, cf. “S.”
- Boxing of the numbers indicates the most productive and important LFs.
- A “systemic” presentation of standard LFs is given in Ch. 3, Sec. 3.8 – see, in particular, Table 3.5, p. lxvii.

A ₀	9	P	p. lxxxvi	Gener	4	P	p. lxxix	Pred	12	P	p. xc
A _i	21	P	p. xcix	Germ	5.1.0.1	S	p. cxii	Prepar	5.4.3.4	S	p. cxliii
Able _i	22	P	p. c	Imper	27	P	p. cvi	Propt _i	5.3.0.5	S	p. cxxviii
Adv ₀	10	P	p. lxxxviii	Imperf	29	P	p. cviii	Prox	5.4.4.4	S	p. cxlvii
Adv _i	24	P	p. ci	Incep	5.4.4.1	S	p. cxlvi	Qual _i	23	P	p. c
Anti; Non	3	P	p. lxxv	Instr	5.3.0.4	S	p. cxxvii	Real _i	5.4.3.1	S	p. cxxxix
Bon	??	S	p. ??	Involv	5.4.7.3	S	p. clvi	Redun	5.2.0.2	S	p. cxv
Cap	15	P	p. xciv	Labor _{ij}	5.4.2.3	S	p. cxxxvi	Result _i	30	P	p. cviii
Caus	5.4.5.1	S	p. cxlix	Labreal _{ij}	5.4.3.3	S	p. cxlii	S ₀	7	P	p. lxxxv
Claus ₀	11	P	p. lxxxix	Liqu	5.4.5.2	S	p. cl	S _i	13	P	p. xcii
Cont	5.4.4.3	S	p. cxlvii	Loc _{ab}	5.3.0.3	S	p. cxxv	S _{instr}	17	P	p. xcvi
Contr	6	P	p. lxxxiv	Loc _{ad}	5.3.0.2	S	p. cxxv	S _{loc}	16	P	p. xcvi
Conv _{ijk}	2	P	p. lxxiii	Loc _{in}	5.3.0.1	S	p. cxxv	S _{med}	18	P	p. xcvi
Copul	5.4.1.1	S	p. cxxx	Magn	5.2.0.3	S	p. cxvi	S _{mod}	20	P	p. xcvi
Culm	5.1.0.2	S	p. cxii	Manif	5.4.7.2	S	p. clv	S _{res}	19	P	p. xcvi
Epit	5.2.0.1	S	p. cxiv	Minus	5.2.0.7	S	p. cxxiv	Sing	25	P	p. cv
Equip	14	P	p. xciii	Mult	26	P	p. cvi	Son	5.4.7.1	S	p. cliv
Excess	5.4.6.3	S	p. clii	Obstr	5.4.6.1	S	p. clii	Stop	??	S	p. ??
Fact _i	5.4.3.2	S	p. cxlii	Oper _i	5.4.2.1	S	p. cxxxii	Sympt _{ijk}	5.4.7.4	S	p. clvii
Figur	5	P	p. lxxxiii	Perf	28	P	p. cvii	Syn	1	P	p. lxxii
Fin	5.4.4.2	S	p. cxlvi	Perm	??	S	p. ??	V ₀	8	P	p. lxxxv
Func _i	5.4.2.2	S	p. cxxxv	Plus	5.2.0.6	S	p. cxxii	Ver	??	S	p. ??

1 Introduction

1.1 Our object of study: lexical functions

This book is about *lexical functions*. Therefore, our very first task is to explain – at least, approximately, in simple terms and with the help of most trivial examples – what lexical functions are. Let's do it!

Consider the phrases in (1) and (2):¹

- (1) a. **pay** tribute [to someone]
b. ***give** tribute [to someone]
- (2) a. **give** recognition [to someone]
b. ***pay** recognition [to someone]

The verbs *pay* and *give*, used in (1a) and (2a) with the nouns *tribute* and *recognition*, respectively, are not interchangeable in spite of the following facts:

- (i) The two nouns are semantically close.
- (ii) The two verbs are, so to speak, pure verbalizers of their nominal objects: they express the very general, poor meaning, approximately 'do', and add no semantic content to the nouns.

These verbs are empty of meaning in the given contexts to such an extent that *pay tribute* is semantically reducible to a hypothetical verb **to tribute* [someone], and *give recognition*, to the verb *to recognize* [someone (during a ceremony)]. For this reason, verbs such as *pay* and *give* are called *support*, or *light*, verbs (see Ch. 5, Sec. 5.4.2, p. cxxxi^{ff}).

¹As is customary, we use the asterisk * in front of an expression to indicate that this expression is ungrammatical. Let it be emphasized that for an expression to be ungrammatical does not imply that it cannot be found in speech. For instance, at the time of writing, the one-billion-word Corpus of Contemporary American English (Davies 2008) contained 1,676 occurrences of *pay tribute* and 13 occurrences of the ungrammatical phrase **give tribute* (for all inflected forms of these phrases). Additionally, the acceptability of word combinations can fluctuate significantly due to geographical variation – we are grateful to Alastair Butler for the suggestion to use (i) and (ii) as illustrations.

The choice of the appropriate support verb by the Speaker is, as one sees in (1) and (2), contingent upon a specific nominal lexeme – the direct object of this verb: the meaning ‘do’ is expressed by the verb *pay* if you ‘do’ tribute, but by the verb *give* if you ‘do’ recognition. In other words, the meaning ‘do’ is expressed **not freely**, that is, not by any convenient verb the Speaker is able to think of. It is expressed in a **constrained way**: by the verb that the noun imposes. To borrow from the language of mathematics, the meaning ‘do’ is expressed **as function of** the direct object of the corresponding verb. This is where the noun *function* in the term *lexical function* comes from. The reason for the adjective *lexical* is obvious: such functions apply to lexical units and return lexical entities (on lexical entities, see Ch. 2, Table 2.6, p. xxiv).

What is illustrated in (1) and (2) is a **syntagmatic** lexical function called Oper_1 (see Ch. 5, [5.4.2.1], p. cxxxii), which describes a particular pattern of constrained lexical unit combinations – that is, a particular type of a two-component phrase where one lexical unit, freely selected by the Speaker, determines the selection of the other lexical unit. In this specific case, borrowing again from mathematical formalization, we write:

$$\begin{aligned}\text{Oper}_1(\textit{tribute}) &= \textit{pay} [\textit{tribute}] \\ \text{Oper}_1(\textit{recognition}) &= \textit{give} [\textit{recognition}]\end{aligned}$$

Constrained lexical combinations described by syntagmatic lexical functions are termed *collocations* (see Ch. 3, Sec. 3.3.2, p. xlix). They are extremely varied – support-verb constructions (presented above), realization-verb constructions (*pay a debt*, *satisfy a desire*, *ring a bell*), (de-)intensifying modifier constructions (*generous / meager salary*, *remember clearly / vaguely*), etc. – and their description requires a wide range of syntagmatic lexical functions.

Lexical functions are not limited to constrained combinations of **lexical units** within **phrases**: they also deal with constrained combinations of **morphemes** within **wordforms**. For instance, in English, the derivational means used to produce from a verb *V* the noun that means ‘person who does *V*’ is contingent upon *V*, as illustrated in (3):

- | | | | |
|-----|---|--|--------------------------------|
| (3) | <i>teach</i> | $\Rightarrow$ <i>teacher</i> | : -er suffixation |
| | <i>represent</i> [a social group] | $\Rightarrow$ <i>representative</i> _(N) | : -(at)ive suffixation |
| | <i>chair</i> _(V) [a meeting] | $\Rightarrow$ <i>chairperson</i> | : compounding |
| | <i>guide</i> _(V) | $\Rightarrow$ <i>guide</i> _(N) | : V $\Rightarrow$ N conversion |
| | <i>steal</i> _(V) | $\Rightarrow$ <i>thief</i> | : suppletive form |

In parallel with what has been said about the meaning ‘do’ for our first set of illustrations, we say that the meaning ‘person who does [V]’ is expressed as function of the verb V in order to return the corresponding noun.

The constrained combinations of morphemes illustrated in (3) are also representable by means of a lexical function; in this case, a **paradigmatic** lexical function called S_1 (see Ch. 4, [13], p. xcii). Accordingly, we write:

$S_1(\text{teach}) = \text{teacher}$
 $S_1(\text{represent}) = \text{representative}_{(N)}$
 $S_1(\text{chair}_{(V)}) = \text{chairperson}$
 $S_1(\text{guide}_{(V)}) = \text{guide}_{(N)}$
 $S_1(\text{steal}_{(V)}) = \text{thief}$

Notation. As one sees, we indicate in subscript the part of speech of a lexical unit if it has a homonym of a different part of speech.

Summing up, lexical functions are a formal tool designed to describe specific patterns of constrained combinations of linguistic signs – of lexical units as well as of morphemes.² As such, lexical functions account for an important part of natural languages’ phraseology:³ lexical as well as morphological phraseology (Mel’čuk 2021a).

The set of lexical functions is organized around a rigorously structured system of **simple standard** lexical functions, whose number is under 70 (see Ch. 3, Table 3.5, p. lxxvii). Their defining property is that they are language-universal (see Ch. 3, Sec. 3.2.1, p. xlii).

Caution! To avoid misunderstanding, let us stress that lexical functions are by no means semantic units – a kind of “elementary” meanings. They do not exist on the semantic level of linguistic representation. They are special – namely, deep – lexical units, appearing in the deep-syntactic structures of utterances and participating in deep-syntactic paraphrasing (see Ch. 3, Sec. 3.2.2, p. xliiiff).⁴ One can say that a lexical function is, metaphorically speaking, a lexical unit “of the second order.” Like any lexical unit, it possesses a signified, a signifier and

²The morphemes in question are mostly derivational morphemes, that is, morphemes whose meanings are fairly close to lexical meanings.

³On the notions of phraseme and phraseology, see Ch. 2, Sec. 2.1, p. xixff.

⁴Recall the equivalence mentioned above: $\text{give}_{\text{oper}_1(\text{recognition})} \text{recognition} \equiv \text{recognize}$.

1 Introduction

a syntactics – a set of combinatorial properties;⁵ the remarkable fact is that a lexical function’s signifier is a set of alternating lexical expressions.

Lexical Functions: The Naked Truth presents the system of lexical functions as fully as possible, but with the following two restrictions.

- 1) Our approach is exclusively synchronic, which means that we avoid the diachronic perspective on word formation or collocations (cf. Blanco 2018), as well as microdiachronic study of lexical dynamics, such as the transformation of idioms into collocations (cf. Pausé & Polguère 2020).
- 2) Our presentation is restricted to theoretical and descriptive considerations and we do not touch on domains of practical applications of lexical functions such as:

- pedagogical applications, that is, language teaching and learning;
- psycholinguistics, that is, Mental Lexicon issues such as language acquisition, access to and loss of lexical knowledge;
- Natural Language Processing – e.g., extraction / interpretation of phraseological expressions, training of “Language Models” for Artificial Intelligence, etc.

However, one essential application of lexical functions is systematically taken into consideration throughout the book: it is the lexicographic modeling of systems of lexical units, performed according to theoretical and descriptive principles of Explanatory Combinatorial Lexicology (Mel’čuk et al. 1995, Mel’čuk 2013a: Ch. 11). This includes dictionary models of the lexicon, such as Explanatory Combinatorial Dictionaries (Mel’čuk & Zholkovsky 1984, Mel’čuk & Žolkovskij 2016, Mel’čuk et al. 1984, 1988, 1992, 1999), lexical databases (Polguère 2000b, Alonso Ramos et al. 2010, Barrios Rodríguez & Boguslavsky 2019), as well as lexical network models, such as Lexical Systems (Polguère 2014).

1.2 Our aims

Lexical functions – which are hypothesized as being language-universal – operate at the core of the structuring and functioning of natural languages: organization of lexical knowledge, lexical access / selection, paraphrasing, etc. It is thus

⁵On the syntactics – third component of a linguistic sign, see Mel’čuk (2013b).

important for the notion of lexical function to be understood by as many people as possible, inside and outside the field of linguistic research. However, the hard fact is that lexical functions are familiar to only a limited circle of “insiders” and, when used, they are often characterized in a simplistic way, which makes these characterizations mostly ineffective. This may be the result of a certain lack of pedagogy and attractiveness of texts presenting lexical functions. Yet the core cause of this state of affairs is the fact that the notion of lexical function is inherently difficult to assimilate, for at least two reasons.

First, the mastering of lexical functions requires the mastering of a rather rich set of underlying linguistic notions, which are relevant to all structural modules of natural languages. They pertain to semantics, syntax and morphology, that is, to the lexicon as well as to the grammar of natural languages. For instance, it is impossible to master the above-mentioned syntagmatic lexical function Oper_1 without mastering Meaning-Text use of (dependency) syntax, the distinction between semantic and deep-syntactic actants and, of course, the principles of lexicographic definition. In other words, to really “understand” lexical functions, one needs to have a bird’s-eye view of the organization of natural languages and not focus solely on specific linguistic modules one happens to have a specific interest in. Lexical functions are fully embedded in the Meaning-Text approach to language studies (Mel’čuk 1981, 2016b, Kahane 2003, Milićević 2006) and have to be acquired as such. Figure 1.1 visualizes the architecture of a Meaning-Text model of natural languages: levels of utterance representation and linguistic modules that establish correspondences ($\Leftrightarrow$) between those levels. Though correspondences are bidirectional, this figure has to be read from bottom to top – according to the Speaker’s perspective, i.e., in the perspective of speech production.

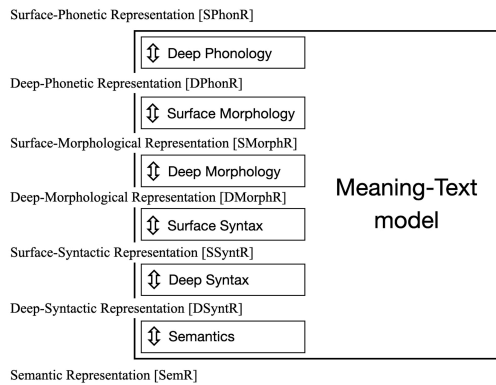


Figure 1.1: Architecture of a Meaning-Text model

Second, we have to acknowledge that the formal language of lexical functions is far from being streamlined and self-evident. It tries to reconcile two mutually incompatible imperatives: on the one hand, formal precision; on the other hand, legibility by humans and the possibility of easily applying the researcher's linguistic intuition when reasoning on lexical function formulas. Consequently, the language of lexical functions is sometimes making use of unusual formal conventions – for instance, the use of actantial subscripts (to the names of lexical functions) with different interpretations depending on the context, which is not acceptable in a genuine formal language (Ch. 3, Secs. 3.2.3 and 3.4.3).⁶

Faced with these difficulties, we decided to deal with the situation frontally, without cutting corners. However, because our ambition is, first of all, to be as much pedagogical as possible, we made a systematic effort at providing readers with all background and, even, insider knowledge needed to achieve an in-depth understanding of lexical functions. We hope therefore that the book can prove useful to a vast range of readers: linguists, lexicographers, researchers and developers in Natural Language Processing, language teachers, translators, etc. – in a nutshell, to everyone who deals with or is interested in Language.

1.3 How do we describe lexical functions?

As we just said, this book is intended as a fresh, more pedagogical presentation of the system of lexical functions, which has already been the topic of many publications – e.g., Alonso Ramos (1993), Mel'čuk (1996), Mel'čuk (2007), Mel'čuk & Milićević (2014: 210–223), Mel'čuk (2015b: Ch. 14), etc. So how are we proceeding to achieve this pedagogical goal?

1.3.0.0.1 Chapter organization To start with, the chapter organization follows a progression that aims at a gradual acquisition of lexical functions:

- Chapter 2 introduces some basic lexicological, semantic and syntactic notions the knowledge of which is a prerequisite for the assimilation of lexical functions. The presentation of these preliminary notions encompasses important terminology and formal conventions used throughout the remainder of the book.

⁶For an alternative (so-called *algebraic*) notation of lexical functions, see Kahane & Polguère (2001). Note that what is gained with this notation in terms of formal rigor and calculability is counterbalanced by a clear loss in terms of possible recourse to linguistic intuition when a lexicographer, a linguist, etc., wants to reason using lexical functions.

- Chapter 3 is an in-depth discussion of the notion of lexical function. It is intended to familiarize readers with all conceptual and formal aspects of lexical functions on which the description of individual lexical functions in the following chapters is based.
- Chapter 4 presents, in a logical order, all paradigmatic simple standard lexical functions.
- Chapter 5 does the same for syntagmatic simple standard lexical functions.
- Chapter 6 concludes the book with the discussion of the central role played by lexical functions in the structuring of natural language lexicons modeled as lexical networks.

While readers are expected to study Chapters 2 and 3 in a linear way, Chapters 4 and 5 are designed for random, or “opportunistic,” access to information on individual lexical functions. A profusion of cross-references helps establish connections between conceptually related lexical functions, thus allowing readers to browse a **system** of lexical functions, rather than following a one-way path through a **list**.

1.3.0.0.2 Detailed description template of lexical functions The presentation of individual lexical functions follows a well-specified template that is explained in Ch. 3, Sec. 3.7, p. lxivff. We have paid special attention to providing information required for **active** learning, where readers can not only understand what a given lexical function is, but also develop the ability to use it in various contexts (lexicographic work, language teaching, etc.).

As the book contains updates to the system of standard lexical functions that could be confusing with respect to information and practices found in earlier publications, **Historical remarks** have been added at the relevant points to help the readers in this respect.

1.3.0.0.3 Extensive linguistic exemplification Though standard lexical functions are postulated to be language-universal, it would have been a colossal task to systematically illustrate each given lexical function with examples borrowed from a broad range of typologically different language families. However, to take into consideration the cross-linguistic relevance of lexical functions, our illustrations in the presentation of lexical functions are systematically done in English, French and Russian – the two latter being the mother tongues of the authors.

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1.3.0.0.4 Facilitation of multiple access to information We did not hesitate to introduce some redundancies in the volume in order to facilitate the different types of usage by readers: in-depth learning of the nature and use of lexical functions vs. punctual consultation of the description of individual lexical functions. In the same spirit, indexes (of notions and of lexical functions) are extensive, with some form of redundancy at times, in order to meet the multiple needs of individual readers.

Note that the full list of standard lexical functions is available in two forms: as an alphabetical list (“, p. ix); and as a “logical” list (Ch. 3, Table 3.5, p. lxvii), where lexical functions are enumerated in the order they appear in Chs. 4 and 5.

1.3.0.0.5 Alternative approaches As explained in Section 1.1, lexical functions serve to describe both collocations, embodying syntagmatic lexical relations, and derivations, embodying paradigmatic lexical relations. There exists a wealth of literature on these topics, which are generally treated as two non-related lexical phenomena; but the systematic presentation of alternative approaches to lexical functions is beyond the scope of the present book. We establish connections between the lexical function approach to the description of lexical relations and other well-known approaches when it is especially relevant to do so. In particular, current approaches to the description of collocations in lexicographic models are examined at the end of Chapter 5, on syntagmatic lexical functions (Sec. 5.5, p. clxvff.); semantic lexical relations for the structuring of so-called *-Net* models (cf. WordNet) are contrasted with the use of paradigmatic lexical functions in Chapter 6, on lexical networks (Sec. 6.1, p. clxviii ff.).

Acknowledgments

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2 Preliminary linguistic notions

The description of lexical functions requires a preliminary set of linguistic notions that we need to make use of throughout the present volume. This chapter offers the reader a concise presentation of those notions, which are grouped into three classes:

- Lexical notions (2.1)
- Semantic notions (2.2)
- Syntactic notions (2.3)

What follows is a series of definitions supplied with explanations and illustrations. Two types of definitions are used: analytical and enumerative definitions.

Analytical definitions constitute the basic type; they follow the Aristotelian definition pattern “by next kind and specific differences” (*per genus proximum et differentia specifica*) – e.g., Definitions 2.1 and 2.3 below, for *lexeme* and *idiom*. These definitions are of almost the same structure as lexicographic definitions described in Mel’čuk & Polguère (2018).¹ The first component of the definition is the *central* (or *generic*) *component*, and it corresponds to the Aristotelian “next kind.” It is followed by a series of *peripheral components*, which correspond to “specific differences”; each peripheral component is introduced by a bullet (•).

Marginally, *enumerative definitions* are also used. They are structured as disjunctive lists of previously defined notions – e.g., Definition 2.4 below (p. xxii), for *lexical unit* (= lexeme or idiom).

2.1 Lexical notions

Lexical functions, as their name suggests, are primarily about the lexicon and the units that constitute it, i.e., words and phraseological expressions. Therefore, we begin with linguistic notions related to words, on the understanding that the

¹For examples of actual lexicographic definitions, see the definition of DEBT in Ch. 5 (Figure 5.8, p. cxI) and the definition of ANGER_N in the Appendix (p. cxcvii).

2 Preliminary linguistic notions

word *word* is both too ambiguous and vague to belong to linguistic (in particular, lexicological) terminology. Let us start with the notion of *lexeme*.

Table 2.1: Notion definition: Lexeme

A **lexeme** is a set of wordforms and analytic-form phrases such that

- their semantic, syntactic and morphological differences can be fully described in terms of inflectional significations.

Example. The lexeme AMAZE = {*amaze, amazes, amazed, amazing, have amazed, will amaze, am amazed, being amazed, will have amazed, ...*}.

2.1.0.0.1 Notation A lexeme name is printed in SMALL CAPITALS, perhaps accompanied by its part of speech and *lexicographic number* (in case of polysemy / homonymy) – e.g., PICTURE_(N) 1.1 ‘image’, PICTURE_(N) 1.2 ‘movie’, PICTURE_(N) II ‘mental representation’.² Lexicographic numbers used in this book are not borrowed from a particular lexicographic description, unless specified otherwise. They are, so to speak, conventional and serve only to distinguish wordsenses when needed.

Lexemes are one of the two types of lexical units, the second one being idioms (see below). Lexemes are the “primary” type, as will be shown shortly. The above definition relies on the morphological notions of *wordform* and *inflection*, which have to be taken for granted here. A lexeme can be conceived of as a generalization over a set of wordforms and analytic-form phrases that all carry the same lexical meaning in their signified, while being related by shared signifiers.

The next notion to be defined is that of (lexemic) *phaseme*. It is a core preliminary notion for two reasons:

1. Idioms (e.g., ‘PULL THE PLUG’) – the second type of lexical units that we need to define – are a particular class of phrasemes.
2. Lexical functions are especially relevant for the description of collocations (e.g., *wide awake* or *pay attention*), which are also a particular case of phrasemes (Ch. 3, Sec. 3.3.2, p. xlix).

²On lexicographic numbers used in Explanatory Combinatorial Lexicology, see Mel’čuk (2013a: Ch. 13, Sec. 3.1, 321ff.).

2.1.0.0.2 Terminological remark In the following definition, the term *Speaker*, written with a capital *S*, refers to the producer of the utterance under consideration (as opposed to a *speaker* of a language). The corresponding receiver of the utterance is referred to as the *Addressee*, with a capital *A*.

Table 2.2: Notion definition: (Lexemic) phraseme

A (lexemic) **phraseme** is a complex lexemic expression such that

- the selection of at least one of its lexemes by the Speaker is not free – i.e., it is constrained by another lexeme of this expression.

Example. The phraseme *wide awake*, where the Speaker's choice of *wide* to express 'fully' is constrained by the adjective *awake*; *wide awake* can be contrasted with the non-idiomaticity of **wide dressed* 'fully dressed', **wide ready* 'fully ready', **wide known* 'fully known' (cf. idiomatic phrase *widely known*), etc.

The presence of phrasemes is overwhelming in natural languages; the set of all phrasemes of a given natural language is called its *phraseology*. In this volume, we concentrate on lexemic phrasemes (in particular, idioms and collocations) and disregard other families of phrasemes, such as syntactic and morphemic phrasemes (Mel'čuk 2015b: Ch. 16, 2021a, 2023). Therefore, the modifier *lexemic* will be omitted.

Equipped with the definition of phraseme, we can define the second type of lexical unit (along with lexemes): *idiom*.

Table 2.3: Notion definition: Idiom

An **idiom** is a phraseme such that

- it is semantically non-compositional.

Example. The idiom 'GREEN THUMB' 'great ability to take care of plants'.

2.1.0.0.3 Notation An idiom name is enclosed between two top corners: 'L₁ L₂ ... L_n'.

In the above definition, the phrase *is semantically non-compositional* refers to the fact that the meaning of an idiom is not a regular linguistic union of the meanings of its lexical components,³ so that a lexicographic definition is required to describe the idiom's meaning. Three types of idioms are distinguished as regards their meaning: strong, semi- and weak idioms.

³On the operation of linguistic union, see Ch. 3, Sec. 3.4, p. liii.

2 Preliminary linguistic notions

- A *strong idiom* has a meaning that does not contain the meaning of any of its components – e.g., ‘SHOOT THE BREEZE’ ‘talk leisurely’.
- A *semi-idiom* has a meaning that contains the meanings of some of its lexical components – e.g., ‘MORNING SICKNESS’ ‘nauseous feeling experienced by a pregnant woman typically in the **morning**’.
- A *weak idiom* has a meaning that contains the meanings of all its lexical components plus additional semantic material – e.g., ‘BACON AND EGGS’ ‘dish made of fried **eggs and** fried **bacon**’.

2.1.0.0.4 Warning Semantic non-compositionality is often mistaken for opacity / transparency, which is relevant to the perspective of the Addressee, i.e., to understanding of speech. Non-compositionality, however, is about the Speaker, who has to produce the idiom out of the semantic content he seeks to express. Idioms, especially semi- and weak idioms, are often easily interpretable by Addressees who do not know them. Even strong idioms can be based on metaphors that are quite transparent, at least in the contexts of use; such is the case of ‘HAVE ONE’S HANDS FULL’ ‘be very busy’, ‘IN A NUTSHELL’ ‘concisely’, etc. The semantic (non-)compositionality of an expression should therefore be estimated **strictly** by contrasting its meaning as a whole with that of the lexical (and grammatical) material it is formally made up from, without recourse to hypothetical understanding processes.

(Non-)compositionality is relevant in the perspective of speech production, while transparency is strictly about speech understanding. Since this book takes the Meaning-Text approach, it is exclusively (non-)compositionality that we consider in order to characterize idioms.

Now that both notions of lexeme and idiom have been defined and clarified, we can offer an enumerative definition of *lexical unit*:

Table 2.4: Notion definition: Lexical unit

A lexical unit is a lexeme or an idiom.
--

2.1.0.0.5 Remark There is another, marginal type of lexical unit, which we do not mention in the above definition in order to simplify the presentation: *compounding elements*, also called *combining forms* (Prčić 2005, 2008) – e.g., ANTHROPO- (in *anthropomorphic*, ...), NEURO- (*neuroscience*, ...), or -PHILE (*francophile*, ...),

-PHOBIA (*arachnophobia*, ...). They are defective lexemes, in the sense that they cannot be used independently: they function only in compounding.

A lexical unit is the unit of the linguistic description of a language's lexicon. To put it differently, it is a unit of lexicographic description. That is, a lexical unit is described by a *lexical entry*, of which it is the *headword*.

In conformity with the principles of the description of linguistic signs (Mel'čuk 2013a: Ch. 11, Sec. 2), the lexical entry for a given lexical unit L accounts for three major components of L:

1. L's *signified* – described in the *semantic zone* of the lexical entry, which is centred around L's lexicographic definition; other possible elements of this zone are, for instance, *lexical connotations* (Iordanskaja & Mel'čuk 2009).
2. L's *signifier* – described in the *phonological/graphematic zone* of the entry.
3. L's *restricted cooccurrence* (*cooccurrence*, for short) – described in the *cooccurrence*, or *lexical function zone* of the entry.

2.1.0.0.6 Restricted cooccurrence L's restricted cooccurrence is its capacity to combine in utterances with other linguistic units, according to the following two facets:

- L as a particular syntactic unit belonging to a part of speech, characterized by a set of syntactic features and having a government pattern (Sec. 2.3, paragraph "Active syntactic valence," p. xxxii); this is L's *syntactic cooccurrence*;
- L as a particular lexical unit; this is L's *lexical cooccurrence*.

The expression *lexical cooccurrence of L* covers two different linguistic phenomena, both described by lexical functions:

- derivation, which involves the cooccurrence of lexical unit L with particular word formation means (Ch. 3, Sec. 3.3.1, p. xlvii^{ff});
- collocation, which involves the cooccurrence of lexical unit L with particular lexical units (Ch. 3, Sec. 3.3.2, p. xlix^{ff}).

2 Preliminary linguistic notions

Due to the existence of *polysemy*, lexical units are grouped within *vocables*:

Table 2.5: Notion definition: Vocable

A **vocable** is a set of lexical units such that

- all of them share a common signifier;
- their signifieds share important enough common semantic components.

Examples. The vocable $\text{PICTURE}_{(N)}$ contains the lexemes $\text{PICTURE}_{(N)} \text{ i.1}$ ‘image’, $\text{PICTURE}_{(N)} \text{ i.2}$ ‘movie’ and $\text{PICTURE}_{(N)} \text{ ii}$ ‘mental image’; the vocable ‘COME A CROPPER’ contains the idioms ‘COME A CROPPER’ i ‘have a bad fall’ and ‘COME A CROPPER’ ii ‘fail badly – ‘as if’ ‘coming a cropper’ i’.

Finally, we give the definition of the notion of *lexical entity*. This notion covers a much larger spectrum of linguistic elements than the notion of lexical unit; it is necessary to encompass all possible “values” of lexical functions.⁴ It includes:

- lexical units – lexemes and idioms;
- compositional phrasemes – collocations and linguistic clichés, such as *What time is it?* and *Have a nice day* (Mel’čuk 2020);

Table 2.6: Notion definition: Lexical entity

A **lexical entity** is a lexeme or one of the two other linguistic entities semantically similar to a lexeme: an idiom – i.e., non-compositional phraseme – or a compositional phraseme (a collocation or a linguistic cliché).

All concrete signs that constitute these entities – such as the wordforms and analytic-form phrases – are also considered as lexical entities.

FURTHER READING. On morphological notions (wordform, inflection, etc.), see Mel’čuk (2006); on phraseology and types of phrasemes, see Mel’čuk (2023).

2.2 Semantic notions

This section introduces several notions that pertain to the domain of linguistic meanings. We start with *semantemes*, which are main semantic entities.

⁴On values of lexical functions, see Ch. 3, Sec. 3.1.1, p. xxxvi.

Table 2.7: Notion definition: Semanteme

A **semanteme** of language $\mathcal{L}$ is a meaning such that

- it is expressed by a lexical unit of $\mathcal{L}$.

Examples. In English, the meaning ‘not deep’ (as of a body of water) is a configuration of two semantemes equivalent to a single semanteme ‘shallow’; by contrast, in French, the corresponding configuration ‘peu profond’ has no equivalent semanteme – French does not have an adjective corresponding to SHALLOW_(Adj).

Semantemes come in two main families as regards their combinatorial characteristics.

A) A semanteme of the first family, called *semantic name*, is “self-contained”: it does not have open slots expected to be filled with other semantemes – in order to make it informationally complete. A semantic name denotes an *entity* (object, substance, being, ...) and, in terms of part of speech, the lexical unit expressing it is necessarily nominal – e.g., STONE_(N), SAND_(N), TIGER, ALEXANDER, VESUVIUS, MONTREAL, etc. The majority of semantic names are proper names.

B) A semanteme of the second family is “binding” rather than being self-contained: it is *predicative*, i.e., it controls open slots to be filled in by other semantemes. These latter are called *semantic actants* [*SemAs*] of the semanteme in question, and the slots themselves are called *actant slots*; they are represented by variables ‘X’, ‘Y’, ‘Z’, ... Predicative semantemes are further subdivided:

- A predicative semanteme that denotes a *fact* (an event, an action, a state, a characteristic, ...) is a (*semantic*) *predicate* – e.g., ‘X collides with Y’ = ‘collision of X with Y’, ‘war between X and Y for Z’, ‘X kisses Y on Z’, ‘happy X’, ‘tall X’, ‘X under Y’, etc. As can be seen, the lexical expressions of semantic predicates can be of any part of speech.
- A predicative semanteme that denotes an entity is a (*semantic*) *quasi-predicate* – e.g., ‘X’s bed’, ‘X’s blood’, ‘X, mother of Y’, etc. Like semantic names, quasi-predicates are expressed by nouns.

A predicative semanteme denotes an extralinguistic *situation* (or an entity implicated in a situation, for quasi-predicates), which is characterized by a set of obligatory *participants* – as one can see from the examples above. A participant of an extralinguistic situation corresponds to a semantic actant of the semanteme that denotes this situation.

2.2.0.0.1 Semantic valence and “external” participant ‘ Ω ’ The cooccurrence of any given lexical unit whose corresponding semanteme is predicative – e.g., the lexeme $\text{KISS}_{(V)}$ [‘X kisses Y on Z’] – is largely determined by the set of actant slots it controls, called its *semantic valence*. However, speaking of a particular lexical unit L (mainly, about its definition and its lexical functions), it can be necessary to refer to an *external participant* of the situation denoted by L, i.e., a participant that is not part of L’s active semantic valence: it is not expressed as one of L’s SemAs. This participant is designated with the symbol ‘ Ω ’ – see, for instance, the presentation of Real_{Ω} in Ch. 5 ([5.4.3.1], Remark 2, p. cxli) illustrated with collocations such as [Tony_{Ω}] *climbs/ scales the mountain*.⁵

Table 2.8 recapitulates the above typology of semantemes, based on two parameters of classification: 1) predicativity and 2) type of denotation – fact vs. entity. This gives four classes of semantemes, of which one is not present in natural languages: non-predicative semantemes denoting facts are logically impossible since, by definition, a fact is denoted by a predicative semanteme.

Table 2.8: Classification of semantemes

	Predicative (binding)	Non-predicative (self-contained)
Denotes facts	semantic predicate: ‘X snores’, ‘doubt of X about Y’, ‘strange X’, ...	
Denotes entities	semantic quasi-predicate: ‘nose of X’, ‘[X is a] neighbor of Y’, ‘hat of X’, ...	semantic name: ‘sand’, ‘giraffe’, ‘Montreal’, ...

The relation between a predicative semanteme ‘ σ ’ and its SemA ‘ σ_{actant} ’ is a *semantic dependency*, which is represented – as all linguistic dependencies are – by an arrow oriented from ‘ σ ’ to ‘ σ_{actant} ’: ‘ σ ’ $\text{--sem--}\rightarrow$ ‘ σ_{actant} ’. Since a predicative semanteme can control more than one actant slot – e.g. ‘X kisses Y on Z’, the different SemAs must be distinguished. This is done by semantic actant numbers 1, 2, 3, ..., as illustrated below with sentence (1).

- (1) Judy kissed Ben on the cheek.

⁵Note that MOUNTAIN – the base of the collocation – is a semantic name: it does not control any semantic actant slot.

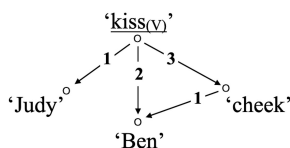


Figure 2.1: Semantic structure of the sentence *Judy kissed Ben on the cheek*.

The configuration of semantemes seen in Figure 2.1 is the *semantic structure* [SemS] of sentence (1) (and of all its paraphrases – see Ch. 3, Sec. 3.2.2.3, p. xlv). It calls for the following three comments.

Note. The above SemS does not encode inflectional meanings, that is, *grammemes* involved: the indicative and the past tense of the verb $\text{kiss}_{(V)}$, as well as the singular of the noun CHEEK . For simplicity's sake, we follow the same policy in all SemSs in this volume.

2.2.0.0.2 1) Communicatively dominant semanteme The sentence *Judy kissed Ben on the cheek* is about an act of kissing: therefore, the semanteme 'kiss' is *communicatively dominant*, which is indicated by underscoring. By contrast, the noun phrase *Ben's cheek that was kissed by Judy* is about the cheek (of Ben) and possesses an identical semantic structure, except for the semanteme 'cheek', which is communicatively dominant (i.e., underscored in the semantic structure of this phrase). Communicative dominance is an essential component of the *communicative structure* of utterances, as theorized in Polguère (1990) and Mel'čuk (2001). For the sake of simplicity, other components of communicative structures – such as the Rheme vs. Theme opposition – are not encoded in utterance representations in this volume.

2.2.0.0.3 2) Semantic actants' numbering Numbering of SemAs of a predicative meaning depends on its internal structure, that is, on its semantic decomposition. Kissing someone is to act upon someone, and the Agent of an action is taken to be SemA 1, the Patient undergoing an action is SemA 2, and the Target of the kiss (part of the Patient's body) is SemA 3.

2.2.0.0.4 3) Semantic structure as a network Figure 2.1 shows that the SemA 3 of 'kiss' is a quasi-predicate that controls one actant slot: 'cheek of X'. This is due to the fact that 'cheek' is 'part of the body of X [such that...]'; all names of body parts necessarily express quasi-predicative semantemes. In this case, the SemA 1

of the quasi-predicate ‘cheek’ is ‘Ben’, and ‘Ben’ is itself the SemA 2 of ‘kiss’. This configuration of semantic dependencies forms a *semantic network*.

FURTHER READING. On semantemes, see Mel’čuk & Polguère (2008), Mel’čuk (2012: Part II, Sec. 2.2, 193–205); on communicative dominance, see Mel’čuk (2001: Part I, Sec. 2.3.1.1); on SemA numbering, see Mel’čuk (2015b: Ch. 12, Sec. 3.4.2.3) and Mel’čuk (2012: Ch. 4, Sec. 3); on Meaning-Text semantic networks, see Polguère (1997).

2.3 Syntactic notions

From the viewpoint of syntactic structures, two levels of representation are distinguished, which can be approximately characterized as follows.

The *deep-syntactic structure* [DSyntS] is the most direct lexical and syntactic expression of the corresponding SemS. It contains only the semantically full lexical units of the sentence to be produced, connected by labeled *deep-syntactic dependencies*, called *deep-syntactic relations*. The resulting structure is a *dependency tree*.

The *surface-syntactic structure* [SSyntS] is geared to the actual sentence in the sense that it serves to determine word order and morphological dependencies between lexemes, if any. It contains all and only the lexemes of the sentence to be produced, connected by labeled surface-syntactic dependencies, called *surface-syntactic relations*. The resulting structure is, of course, also a dependency tree.

Let us illustrate the *correspondences* between the SemS, the DSyntS and the SSyntS of a sentence, using sentence (2) as example.

(2) Judy shoots the breeze with Ben for hours.

2.3.0.0.1 Notation $A \Leftrightarrow B$ means ‘linguistic entity A of the level n of representation corresponds (= is semantically equivalent) to linguistic entity B of level $n + 1$; for instance:

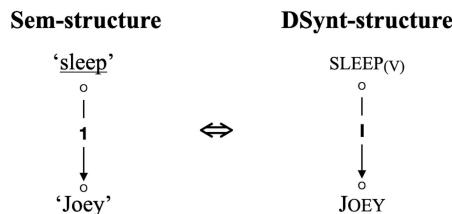


Figure 2.2: Example of a SemS $\Leftrightarrow$ DSyntS correspondence

In what follows we focus on the two particular correspondences between utterance structures: $\text{SemS} \Leftrightarrow \text{DSyntS}$ and $\text{DSyntS} \Leftrightarrow \text{SSyntS}$.

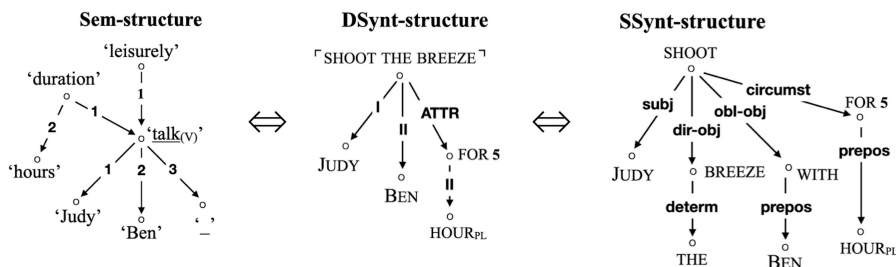


Figure 2.3: $\text{SemS} \Leftrightarrow \text{DSyntS} \Leftrightarrow \text{SSyntS}$ correspondences for the sentence *Judy shoots the breeze with Ben for hours*.

The SemS has been introduced and explained in Sec. 2.2. Here we need to explain the DSyntS and the SSyntS.

2.3.0.0.2 Deep-syntactic structure [DSyntS] of sentence (2) The nodes of this tree are labelled with *deep lexical units*, i.e., semantically full lexical units, which express semantemes of the SemS. The idiom "SHOOT THE BREEZE" expresses the semantic configuration 'leisurely-1→talk_(V)', and the meaningful preposition FOR 5, the semanteme 'duration'.⁶ The other nodes are self-explanatory.

The arcs (branches) of this tree are labeled with DSynt-relations of two types:

- I and II are *actantial DSynt-relations*; the dependent members of these relations – JUDY and BEN – are DSynt-actants of their governor, the idiom "SHOOT THE BREEZE" ('X "shoots the breeze" with Y'). The notion of DSynt-actant will be defined shortly (Table 2.10).
- ATTR is a non-actantial DSynt-relation: it is not foreseen by the meaning of its governor and represents a dependency inversion – called *head-switching* – with respect to the corresponding semantic relation, as shown in Figure 2.4.

⁶The lexicographic number of FOR 5 comes from the online *Longman Dictionary of Contemporary English* (LDOCE 2025), henceforth LDOCE.

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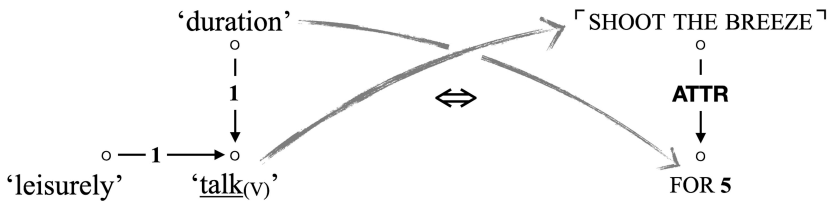


Figure 2.4: SemS ↔ DSyntS dependency inversion (= head-switching) involving the DSynt-relation **ATTR**

At the DSynt-level several other DSynt-relations are used – listed in Table 2.9. These relations are language-universal. In the present volume only actantial and **ATTR**(ibutive) DSynt-relations are relevant.

Table 2.9: Language-universal DSynt-relations

I	: 1 st deep-syntactic relation / 1 st deep-syntactic actant
II	: 2 nd deep-syntactic relation / 2 nd deep-syntactic actant [Joey←I—loves—II→food]
...	
APPEND	: appenditive; it links an “extra-structural” element – such as a parenthetical or an interjection – to its governor – the syntactic head of the clause [Sorry,←APPEND—[I]—am late]
ADDRESS	: addressative; it links a direct address to its governor –, which is the syntactic head of the clause [Yes,—ADDRESS→Sir]
ATTR	: attributive; it links a restrictive modifier / circumstantial to its governor [inquisitive←ATTR—student; arrived—ATTR→yesterday]
ATTR_{qual}	: qualificative attributive; it links a qualificative modifier / circumstantial to its governor [these students,—ATTR _{qual} →hungry for knowledge]
COORD	: coordinative; it links the dependent member of a syntactic coordination to its governor [Ben—COORD→and Judy]
PSEUDO-COORD	: pseudo-coordinative; it links the dependent member of elaboration or clarification to its governor [She came last year,—PSEUDO-COORD→in the spring]

A notion playing a key role in the presentation of lexical functions is that of *deep-syntactic actant* [DSyntA], mentioned above. It is defined as follows:

Table 2.10: Notion definition: Deep-syntactic actant [DSyntA]

A **deep-syntactic actant** I, II, ... of lexical unit L is a DSynt-dependent of L such that

- it expresses, in a prototypical case, a SemA of L and it is governed by L via an actantial DSynt-relation I, II, ...

There is also a frequent enough possibility of a DSyntA of L coming, or being “raised,” to L from one of its DSyntAs. Such *syntactic raising* is illustrated in Figure 2.5:

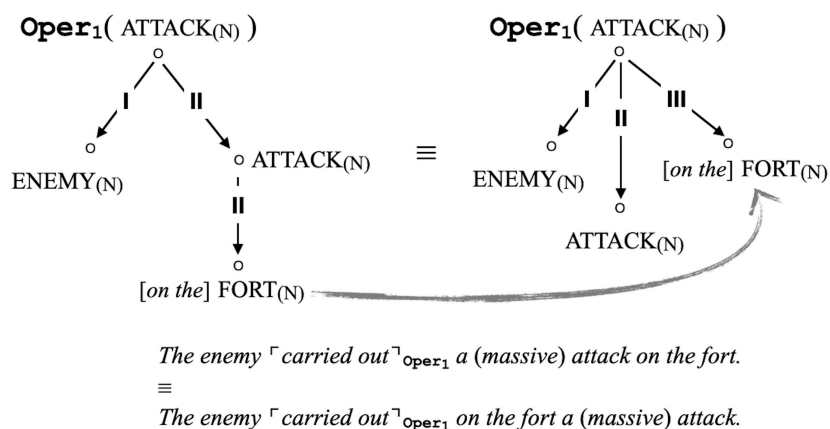


Figure 2.5: Syntactic raising of a DSyntA

This figure presents one of the support verbs mentioned in Ch. 1 (sec. 1.1), namely the lexical function Oper_1 (see Ch. 5, [5.4.2.1], p. cxxxii). $\text{Oper}_1(\text{attack}(\text{N}))$ is expressed, in the SSynt-structure (see below), by the lexical units ‘CARRY OUT’ or CONDUCT(V).

The *actant transfer* illustrated in Figure 2.5 is common with verbal syntagmatic lexical functions (support and realization verbs) – see Ch. 5, “Transfer of L’s actants to support verbs,” p. cxxxiv.

Remark. The second sentence (with raising) in Figure 2.5 is dubious: speakers seem to accept it only with a special prosody, namely, pauses before and after *on the fort*: *The enemy carried out – on the fort – a massive attack*. Other languages, such as French and Russian, allow for this construction without problem:

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- (3) a. **Fr.** L'ennemi a mené contre le fort une attaque massive.
 b. **Rus.** Vrag provël na fort massirovannuju ataku.

2.3.0.0.3 Surface-syntactic structure [SSyntS] of sentence (2) The nodes of the SSynt-tree for (2) are labelled with all lexemes of the sentence, including lexical components of the idiom (SHOOT, BREEZE and THE) and a governed preposition (WITH).

The arcs of this tree are labeled with SSynt-relations, which are language-specific – i.e., non-universal; there are several dozens of SSynt-relations in a language. Table 2.11 lists SSynt-relations featured in Figure 2.3 (p. xxix) and in the whole volume.

Table 2.11: Some SSynt-relations of English

circumstantial	: <i>talked</i> ←circumst→ <i>for</i> [<i>hours</i>]
determinative	: <i>the</i> ←determ→ <i>breeze</i> ; <i>my</i> ←determ→ <i>daughter</i>
direct-objectival	: <i>shoot</i> ←dir-obj→ <i>breeze</i> ; <i>Marry</i> ←dir-obj→ <i>me</i> ; <i>want</i> ←dir-obj→ <i>to</i> [<i>go</i>]; <i>declare</i> ←dir-obj→ <i>that</i> [<i>it's OK</i>]
modificative	: <i>big</i> ←modif→ <i>deal</i>
oblique-objectival	: <i>shoot</i> ←[<i>breeze</i>]←obl-obj→ <i>with</i> [<i>Ben</i>]; <i>depend</i> ←obl-obj→ <i>on</i> [<i>you</i>]
possessive	: <i>tomorrow's</i> ←possess→ <i>event</i>
prepositional	: <i>with</i> ←prepos→ <i>Ben</i> ; <i>for</i> ←prepos→ <i>hours</i>
subjectival	: <i>Judy</i> ←subj→ <i>shoots</i> [<i>the breeze</i>]; <i>it</i> ←subj→ <i>is</i> [<i>late</i>]
subjectival-adnominal	: <i>speech</i> ←subj-adnom→ <i>of</i> [<i>the President</i>]

2.3.0.0.4 Active syntactic valence SemS ⇔ DSyntS ⇔ SSyntS correspondences are essentially based on the actantial structures – i.e., sets of actants – of lexical units. More precisely, these correspondences are controlled by the *active syntactic valence* of each lexical unit L they involve: how L's SemAs are expressed by means of DSyntAs, and how the latter are expressed at the surface-syntactic and morphological levels.⁷

Consequently, to implement SemS ⇔ DSyntS ⇔ SSyntS correspondences, the information about active syntactic valence of lexical units is needed. It is specified in their lexical entries by means of a table, called *government pattern*. As an illustration, Figure 2.6 gives the (simplified) government pattern of the verb KISS_(V), appearing earlier in sentence (1) – Sec. 2.2, p. xxvi.

⁷Unlike the situation with semantic valence (cf. Sec. 2.2, p. xxvi above), the adjective *active* is required – since active syntactic valence contrasts with passive syntactic valence: see Ch. 3, Sec. 3.5.1, p. lviii.

'X' ↔ I	'Y' ↔ II	'Z' ↔ III
1. –subj→N	1. –dir-obj→N	1. –obl-obj→on N

Figure 2.6: Government pattern of the verb $\text{KISS}_{(V)}$ (simplified)

Another example of government pattern, for the noun $\text{ANGER}_{(N)}$, is found in the Appendix, p. cxcvii.

The government pattern of a given lexical unit *L* describes *L*'s restricted **syntactic** cooccurrence in *L*'s lexical entry. By contrast, lexical functions – the topic of the present volume – describe restricted **lexical** cooccurrence of lexical units.

FURTHER READING. On the notion of DSyntA, see Mel'čuk (2015b: Ch. 12); on the correspondence between the SemS and the DSyntS, see Polguère (1990) and Mel'čuk (2013a: Ch. 10); on SSynt-relations, see Mel'čuk (2015a, 2016a, 2021b: Ch. 2); on government patterns, see Milićević (2009) and Mel'čuk (2015b: Ch. 13).

3 Notion of lexical function [LF]

This chapter presents the notion of lexical function – hereafter, LF – in eight steps:

- What is an LF? (3.1)
- Simple standard LFs (3.2)
- Two major classes of simple standard LFs (3.3)
- Simple standard LF combinations (3.4)
- Parts of speech of simple standard LFs (3.5)
- Regularities in standard LFs (3.6)
- Description template for simple standard LFs (3.7)
- System of simple standard LFs (3.8)

3.1 What is an LF?

3.1.1 Definition of the notion of LF

A *lexical function* [LF] is similar to a *function* in the mathematical sense of the term. Recall that a **mathematical** function f is a relation between two sets A and B such that:

- f applies to each individual element a_i of A , called f 's *argument* – this application is denoted as $f(a_i)$;
- $f(a_i)$ returns, in constant proportion, an element b_i of B , called $f(a_i)$'s *value*:
 $f(a_i) = b_i$;
- by the application $f(a_i)$, each a_i is associated to one and only one b_i .

3 Notion of lexical function [LF]

For instance, arithmetical function f_1 specified by the formula $f_1(x) = x + 3$ is a relation that associates ($\mapsto$) to any positive integer x another positive integer that is superior by three units: $0 \mapsto 3$, $1 \mapsto 4$, $2 \mapsto 5$, ...

In the case of **lexical** functions, the arguments and the values are, of course, not numbers:

- the argument of an LF f is a particular lexical unit L of the language under description;
- the *value* of its application to L – i.e., $f(L)$ – is a set of (quasi-)synonymous lexical entities that represent alternative choices for the expression of a given meaning.¹

The lexical entities that constitute the value of an LF application can be lexemes, idioms, collocations, compounding elements or derivational affixes. Example (1) is an illustration with the LF A_1 : adjective that typically characterizes the DSyntA $\uparrow$ of its argument L (see Ch. 4, [21], p. xcix); in this case, L is $\text{DOUBT}_{(V)}$ and its DSyntA $\uparrow$ the person who doubts.

$$(1) \quad A_1(\text{doubt}_{(V)}) = \text{doubtful}, \text{ in } \text{doubt}_{(N)}$$

The equality (1) tells us that the LF A_1 establishes a relation between the lexical unit $\text{DOUBT}_{(V)}$ and the set of near-synonymous lexical entities: the **lexeme** **DOUBTFUL** and the **collocation** *in doubt*.²

3.1.1.0.1 Terminological remark We use the term *value* (of an LF) either to refer to a set of alternative lexical entities or, as a shortcut, to one of these lexical entities. For instance, we can say that the lexeme **DOUBTFUL** is a value of $A_1(\text{doubt}_{(V)})$, where a *value* is an abbreviation for *an element of the value*. Similarly, we write formulas such as “ $L \equiv \text{Syn}(L)$ ” (cf. Sec. 3.3.1, p. xlix) to state that the lexical unit L is equivalent to one of its synonyms (and not, of course, to the set of its synonyms).

Let us now propose a definition of lexical functions:

This definition entails that LFs model such selection of lexical expressions for a given meaning ‘ σ^f ’ that is contingent upon the lexical expression of another meaning with which ‘ σ^f ’ is combined. Let us explain this.

¹The notions of lexical units and lexical entities have been defined in Ch. 2, Sec. 2.1 (p. xixff).

²The phrase *in doubt* is a deep-syntactic adjective (on deep parts of speech, see Sec. 3.5.1, p. lviii).

Table 3.1: Notion definition: Lexical function [LF]

A **lexical function** f is a function such that

- it expresses a particular meaning ‘ σ^f ’;
- it applies to a lexical unit L as its argument – this application being denoted as $f(L)$;
- it returns as value a set of (quasi-)synonymous lexical entities each of which expresses ‘ σ^f ’ as a function of L .

Linguistic meanings can belong to two major classes from the viewpoint of their lexical expression: to an open class of “free” meanings or to a closed class of “constrained” meanings.

- A *free meaning* is directly expressible by a lexical means with no regard for other lexical choices made in the production of the same sentence – e.g., ‘not expected’ can be freely expressed as *unexpected*. No matter what exactly is not expected – a birth, a thunderstorm or silly behavior, it can be characterized as *unexpected*. This is, so to speak, the default situation.
- A *constrained meaning*, by contrast, is lexically expressed depending on another lexical choice. Thus, the meaning ‘intense’ in ‘intense X ’ has to be lexically expressed as a function of the lexical expression of ‘ X ’: ‘**intense** desire’ can be expressed as *burning desire*, and ‘**intense** despair’ can be expressed as *deep despair* (while **deep desire* and **burning despair* are both ill-formed).

It is for constrained meanings that LFs come into play. Each LF f is associated to a constrained meaning – written ‘ σ^f ’: an application $f(L)$ computes the lexical expression of ‘ σ^f ’ as a function of the lexical unit L .

The expression “meaning ‘ σ^f ’ associated to an FL” is an approximation. In fact, the semantic content ‘ σ^f ’ is not necessarily a genuine meaning. Two cases can be distinguished.³

1) ‘ σ^f ’ is a genuine meaning – e.g., ‘the one who does [L]’ for the LF S_1 (Ch. 4, [13], p. xcii), as in $S_1(\textit{build}) = \textit{builder}$. Alternatively, the LF can correspond to a disjunction of genuine meanings – e.g., Magn (Ch. 5, [5.2.0.3], p. cxvi) corresponds to ‘intense’ (*mad* [love]), to ‘of big magnitude’ (*heavy* [weight]), to ‘to a high

³Kahane & Polguère (2001) proposes a formal modeling of the meaning expressed by each LF, as well as of *combinations* of LFs presented in Sec. 3.4 below.

3 Notion of lexical function [LF]

degree' ([*stubborn*] 'as a mule'), etc. Note that there 'σ^f' can be a zero meaning: support verbs, such as *oper_i* (Ch. 5, [5.4.2.1], p. cxxxii).

2) 'σ^f' is, strictly speaking, not a meaning, but an indication of meaning identity of L and f(L):

- with preservation of L's part of speech; see, for instance, the LF Syn (Ch. 4, [1], p. lxxii): Syn(*buttocks*) = *butt*;
- with change of the part of speech; see, for instance, the LF S_θ (Ch. 4, [7], p. lxxxv): S_θ(*pursue*) = *pursuit*.

To conclude this introductory presentation of the notion of LF, let us stress that the meaning 'σ^f' of an LF f determines its *domain of application*, i.e. the set of its possible arguments: the meanings of the latter must be compatible with 'σ^f'. Thus, the above-mentioned LF A₁ takes as arguments only lexical units with predicative meanings (Mel'čuk & Polguère 2008, Polguère 2012c). Two cases are possible.

- application of an LF to a lexical unit that does not belong to its domain – i.e., to an illegitimate argument; for instance: *A₁(*poppy*), since the noun POPPY is not predicative;
- application of an LF to a legitimate argument that returns an empty value (no lexical entity) in the given language; for instance: A₁(*swim_(V)*) = –, since there is no special adjective in English to characterize somebody who is swimming.⁴

3.1.1.0.2 LFs and lexical networks A natural language lexicon is essentially a huge set of lexical entities connected by a set of relations: in other words, it is a *lexical network*. LFs participate in the structuring of the lexicon in a fundamental way: each LF application f(L) is equivalent to an elementary subnetwork. This is illustrated in Figure 3.1 with two LF applications involving the lexeme PROMISE_(N) as argument.

⁴The expression *swimming* is an inflectional form (present participle) of the verb SWIM_(V), whose use is accounted for by the grammar.

“Traditional” lexicographic representation:

$\text{Syn}_n(\text{promise}_{(N)})$	=	<i>commitment, engagement; oath, pledge_(N), vow_(N)</i>
$\text{Real}_1(\text{promise}_{(N)})$	=	<i>“carry out”, fulfill, keep_(V) [ART ~]; honor_(V) [ART ~]</i>

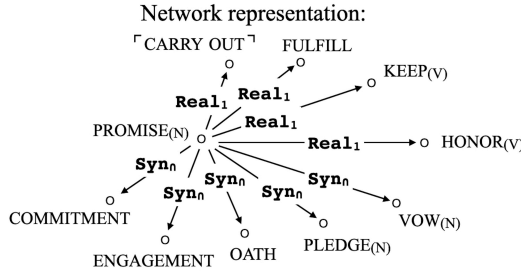


Figure 3.1: Lexical network representing a set of LF applications

The important issue of the structuring of lexical networks by LFs is considered in Ch. 6.

3.1.2 Notation for LF applications

Throughout this book, we use a set of notations (writing conventions and symbols) to present examples of LF applications. The general format of presentation for an LF application is as follows:

$$(2) \quad S_0(\textit{leave}_{(V)}) = \textit{departure}$$

The argument and the elements of the value of the LF are written in *italics* in the above formula – though, technically, they are lexemes and should be written using the standard convention for lexical units: *LEAVE_(V)* and *DEPARTURE*. However, this violation of conventions allows for easier legibility; additionally, values may not be lexical units (e.g., they can be collocations) and it is easier to use a uniform presentation for values, disregarding their lexical status. Note that lists of values in our examples are not necessarily complete; we often limit ourselves to only one value element.

As illustrated in example (3), three symbols are used to separate the elements of an LF value: “,” for no or minimal semantic difference, “<” for the difference of degree, and “;” for a significant semantic difference.

$$(3) \quad \text{Magn}(\textit{salary}) = \textit{big, generous, hefty, large, substantial} < \textit{enormous, huge, massive; exorbitant}$$

[The value *exorbitant* expresses an additional semanteme: ‘too much’.]

3 Notion of lexical function [LF]

The $\boxtimes$ symbol used in (4) below indicates a fused value element – see “Synthetic (= fused) values of syntagmatic LFs,” p. lii.

$$(4) \quad \text{Magn}(\text{amaze}) = \text{utterly}, \boxtimes \text{stun}_{(V)} [\equiv \text{utterly amaze}]$$

Finally, two indications can follow a given value element L' .

1) L' 's government pattern, written between square brackets, specifies the syntactic configuration that L' can govern. This configuration is presented in simplified linear form; for instance in (5) below, where “ $\sim$ ” stands for the LF's argument ($\text{kiss}_{(N)}$) and “ART” stands for a determiner:

$$(5) \quad \text{Oper}_1(\text{kiss}_{(N)}) = \text{give} [\text{ART} \sim \text{to } N_Y]$$

On government patterns, see Ch. 2, Sec. 2.3, p. xxxii.

2) Semantic or grammatical constraints associated with L' are introduced by a vertical bar “ $|$ ”; for instance:

$$(6) \quad \text{Oper}_1(\text{visit}_{(N)}) = \text{pay} [\text{ART} \sim \text{to } N_Y] \mid N_Y \text{ denotes a person}$$

3.2 Simple standard LFs

The LFs as defined in Section 3.1 constitute a *system* built around a core of about seventy LFs, called *simple standard LFs*; they are referred to simply as *LFs* in the subsequent discussion. Let us present the fundamental properties of these LFs.

The set of simple LFs has been empirically identified over the last 60 years in the context of the study of typologically varied languages (first publication on LFs: Žolkovskij & Mel'čuk 1965). The lexicographic projects of Russian and French Explanatory Combinatorial Dictionaries have allowed the researchers to sharpen the LF system and to consolidate the very notion of LF, under the elaboration of lexical entries, where LFs are systematically used (Mel'čuk & Zholkovsky 1984, Mel'čuk & Žolkovskij 2016, Mel'čuk et al. 1984, 1988, 1992, 1999, Mel'čuk & Polguère 2007). At the same time, Meaning-Text research on *syntactic paraphrasing* (Milićević 2007, Mel'čuk 2013a: Ch. 9) and dependency syntax (Mel'čuk & Pertsov 1987, Mel'čuk 1988, 2009, 2015a, 2016a, 2021b) have heavily contributed to the maturation of the notion.

Two topics are discussed below: the two fundamental properties of LFs (3.2.1); and LFs in deep syntax (3.2.2).

3.2.1 Two fundamental properties of simple standard LFs

Simple standard LFs have two fundamental properties: a vast domain of application and a vast range of possible values. These properties follow from the character of the constrained meanings (Sec. 3.1.1, p. xxxvii) they express: these meanings play such an important role in linguistic communication that they tend to induce these properties in all languages.

Property 1: Vast domain of application

A given LF f takes, in a given natural language, a large number of arguments: it has a vast domain of application. To put it differently, the meaning ' σ^f ' must be sufficiently vague (i.e., poor) to be compatible with a large set of lexical meanings.

Consequently, an f involves a significant subset of a language's lexicon, and this subset is semantically rather heterogenous. For instance, one could be tempted to postulate a simple standard LF with meaning 'young of L ' to describe *lexical relations* of the following type:

- (7) $cat \vdash_{\text{young of}} \rightarrow kitten$
 $dog \vdash_{\text{young of}} \rightarrow puppy$
 $pig \vdash_{\text{young of}} \rightarrow piglet$

This, however, would not be correct since this LF's meaning specifies a very constrained *semantic class* of potential arguments: that of animate beings. Such a lexical relation can be described by an LF, but by a *non-standard* LF (see further below the paragraph "Non-standard LFs").

3.2.1.0.1 Notation A lexical relation LexRel – of any type – between lexical units L_1 and L_2 is denoted as $L_1 \vdash_{\text{LexRel}} \rightarrow L_2$.

Property 2: Vast range of possible values

The application $f(L)$ returns a large number of different values for different arguments: f is, therefore, not a trivial relation. An LF that would express the meaning 'of white color' would have a trivial value: $\text{WHITE}_{(\text{Adj})}$ for any lexical unit L . Such trivial cases must not be considered when identifying standard LFs.

The two above properties entail another significant property of standard LFs. Given their frequency in the language and their regularity, one can expect the expression of ' σ^f ' by morphological means. This can happen, in a given language, either massively or sporadically; for instance:

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- to implement the LF S_1 (Ch. 4, [13], p. xcii), English uses the very productive derivational suffix -ER (*driv+er*, *advis+er*, ...);
- by contrast, one finds in French the unproductive nominal derivational suffix -AILLON that expresses the complex LF AntiBon (Ch. 5, [??], p. ??): Fr. *avoc+aillon* ‘bad lawyer’ (from *avocat* ‘lawyer’), *écriv+aillon* ‘bad writer’ (from *écrivain* ‘writer’), and a few other nouns.

3.2.1.0.1 Universality of LFs LFs that have both above properties are linguistically *universal*; by this, we mean two things.

Firstly, each LF is supposed to be found in all or most world languages. We say *most* because there are languages where a given LF is trivial: it is always expressed there by the same linguistic means – a given lexical unit or a given affix.

Secondly, the system of LFs proposed in the present volume is presumed complete. It is believed (until proven wrong) that one cannot find in a language a lexical relation with the above properties of simple standard lexical functions that would not be included in the proposed system.

Given LFs’ linguistic universality, it seems natural that their names come from Latin, so that they are not linked to any specific contemporary language.⁵ To keep LF formulas relatively short, Latin sources are truncated to a maximum of four or five characters – e.g., Syn, from Lat. *synonymum*. In a few cases, pseudo-Latin terms are used, preceded by an asterisk – e.g., Conv_{ijk}, from Lat. **conversivum*.

3.2.1.0.2 Non-standard LFs Any natural language possesses an open set of constrained meanings ‘ σ^f ’ such that:

- Each one of them is much richer than meanings of standard LFs.
- Because of this, a ‘ σ^f ’ can apply only to a semantically limited set of arguments; in point of fact, its domain of application is a clearly identifiable semantic field or a handful of lexical units. Quite frequently, its domain is even reduced to one lexical unit.
- There is not much variability in its expression – most of the time, just one lexical unit.

⁵The LF Equip (Ch. 4, [14], p. xciii), from Fr. *équipage* ‘crew’, is an exception.

Such meanings are still expressed as a function of their argument and therefore they can be described by LFs, but by LFs of a completely different nature than standard LFs: these are *non-standard LFs*.

Three examples are given in (8): *young* of, mentioned earlier in (7), which only applies to the semantic class of “higher-order” animals; *overprotective* of their children, which applies only to *PARENT_(N)*, *MOTHER_(N)* and *FATHER_(N)* (and a few of their synonyms); of 366 *days*, which applies only to the lexical unit *YEAR*:

- (8) young of(*cat*_(N)) = *kitten*_(N)
 overprotective of her child(*mother*_(N)) = *helicopter*_(N) [*~*]
 of 366 days(*year*) = *leap*_(N) [*~*]

Non-standard LFs are quite different from standard LFs in the following aspects:

- They are not universal, but language-specific.
- Their number in a given language is unlimited, and they cannot be theoretically foreseen.
- They are so diverse that it is difficult to normalize them, and they have to be specified by the lexicographic definition of their value.
- Prototypically, they are not deep lexical units: they do not appear in deep-syntactic structures and do not participate in universal paraphrasing, which are two essential characteristics of standard LFs – see Sec. 3.2.2 below; their only job is to ensure correct lexical selection in the transition from the semantic to the deep-syntactic structure.

FURTHER READING. On standard *vs.* non-standard LFs, see Polguère (2007) and Mel'čuk (2023: Appendix).

3.2.2 Standard LFs in deep syntax

3.2.2.1 LFs as deep lexical units

An application $f(L)$ specifies a set of alternative lexical units: it is, so to speak, a “meta-unit.” It appears in the deep-syntactic structure [DSyntS] of a sentence, to be replaced by a specific lexical unit selected for the surface-syntactic structure [SSyntS]. As such, an application $f(L)$ is a deep lexical unit. This is illustrated by Figure 3.2.

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3.2.2.1.1 Reminder For the semantic and syntactic notions used in the present section – as well as for the representation of semantic and syntactic structures, see Ch. 2, Secs. 2.2 and 2.3.

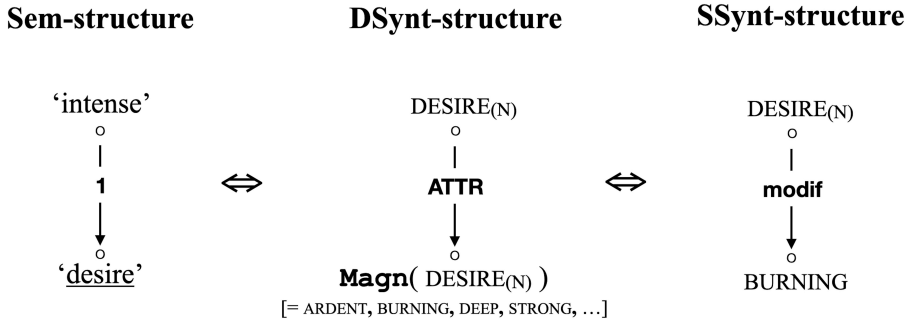


Figure 3.2: Magn in the DSyntS of the phrase *burning desire*

Figure 3.2 shows as well that, in linguistic representations, LFs appear only in DSyntSs.

Being a lexical element in DSyntS, an LF has its own combinatorial properties, in particular, a deep part of speech – see Sec. 3.5, p. lviii^{ff}.

LFs fulfill two tasks in deep syntax: they ensure correct lexical selection (3.2.2.2) and DSynt-paraphrasing (3.2.2.3).

3.2.2.2 LFs and lexical selection

Lexical selection is an important part of what is known as *lexicalization* in speech production – e.g., in psycholinguistics research (Kempen & Huijbers 1983) and in computer text generation (Stede 1994).

Under lexical selection, LFs ensure correct choices in case a lexical unit has to be selected contingent upon a previously selected lexical unit.

Thus, Figure 3.2 above represents the following steps in lexicalization:

1. At Sem-level, the semanteme ‘intense’ is identified as a constrained meaning (Sec. 3.1.1, p. xxxvii), to be lexically expressed as function of the expression of the semanteme ‘desire’.
2. At DSynt-level, the semanteme ‘intense’ is expressed by the LF Magn. This LF is applied to the lexeme DESIRE_(N): the application Magn(DESIRE_(N)) represents a constrained lexical choice, more precisely, a set of usable syntactic modifiers.

3. At the SSynt-level, the lexical selection is completed: the lexeme BURNING is introduced as a modifier of $\text{DESIRE}_{(N)}$; this is done on the basis of the lexical entry of $\text{DESIRE}_{(N)}$, where the value of the application $\text{Magn}(\text{DESIRE}_{(N)})$ is specified.

3.2.2.3 LFs and paraphrasing

A vital feature of natural languages is their ability to express, in a general case, a given meaning by a large set of quasi-synonymous utterances, i.e., *paraphrases*, as shown below.

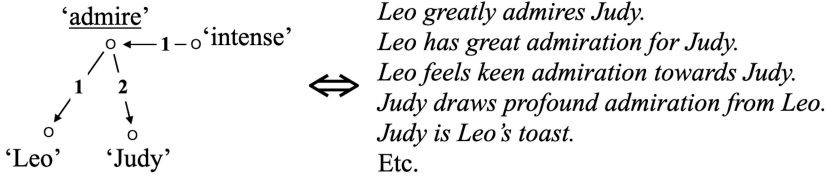


Figure 3.3: Illustration of the paraphrasing potential of a natural language

This is but a modest illustration of what a natural language can do with regard to paraphrasing: a natural language is a paraphrasing device (Mel'čuk 2012: Ch. 2). Paraphrasing as such is not often used in actual speech production – only in specific situations as editing a text, rephrasing for better explanation, etc. However, the paraphrasing power of a language offers, at any given stage of speech production, multiple alternatives that enable the Speaker to speak fluently. Speaking a second language, a person generally doesn't master the same paraphrasing wealth and therefore can be slowed down or outright blocked.

The system of standard LFs plays a core role in the use of language, because it is in terms of LFs that *deep-syntactic paraphrasing rules* are formulated. Since LFs are language-universal (see above, p. xlii), the resulting system of paraphrasing rules is also universal. Such a paraphrasing system is described in Mel'čuk (1974: 141–176), Milićević (2007: Ch. 6) and Mel'čuk (2013a: Ch. 9). Here, we limit ourselves to citing two paraphrasing rules: R_{Mult} and R_{Oper_1} .

$$R_{\text{Mult}}: L_{(N)\text{PL}} \equiv \text{Mult}(L_{(N)})_{\text{SG}}$$

This rule describes the universal semantic equivalence between a plural noun $L_{(N)\text{PL}}$ and the corresponding collective noun $\text{Mult}(L)$ – Ch. 4, [26], p. cvi; for instance:

3 Notion of lexical function [LF]

- (9) *workers* $\cong$ *workforce*
 lies $\cong$ *bunch of lies*

3.2.2.3.1 Note This equivalence holds for a specific meaning of the nominal plural: ‘collective of [Ns]’. $\text{Mult}(L)$ itself is in the singular, as stated in the rule $\mathbf{R}_{\text{Mult}}$.

$$\boxed{\mathbf{R}_{\text{oper}_1}: L_{(V)} \equiv \text{oper}_1(S_0(L_{(V)})) \dashv\!\!\dashv\!\!\rightarrow S_0(L_{(V)})}$$

This rule describes the universal semantic equivalence between a verb $L_{(V)}$ and the verbal collocation consisting of (i) the S_0 nominalization of $L_{(V)}$ (Ch. 4, [7], p. lxxxv) and (ii) an oper_1 support verb of this nominalization (Ch. 5, [5.4.2.1], p. cxxxii). For instance:

- (10) *Leo admires Judy.* $\equiv$ *Leo has admiration for Judy.*

3.2.3 Actantial subscripts in simple standard LFs

Actantial subscripts to LF names have frequently appeared in the preceding text. An explanation of their interpretation is supplied here for the names of simple standard LFs. The corresponding explanation concerning the names of LFs in combinations is provided in Sec. 3.4.3, p. lvi.

In general case of an application $f_i(L)$ where f_i is a simple LF, the subscript “ $_i$ ” refers to the $\text{DSyntA } i$ of L . Here are two illustrations, with LFs S_2 (Ch. 4, [13], p. xcii) and Func_3 (Ch. 5, [5.4.2.2], p. cxxxv).

- (11) a. $S_2(\text{eat} [\text{'X eats Y'}]) = \text{food}$
 b. $\text{Func}_3(\text{ultimatum} [\text{'X's ultimatum to Y to do Z before time T'}]) = \text{call}_{(V)} [\text{for } N_Z]$
 Ex.: *The ultimatum calls_{Func3(ultimatum)} for the release_Z of prisoners.*

However, there are at least three special cases of LFs where the interpretation of actantial subscripts is not straightforward:

- $\text{Adv}_{(i)}$, which stands for absolute adverbials (such as *in the presence of*) – see Ch. 4, [24], Comment 2, p. ciii;
- Sympt_{ijk} , which describes somatic phrasemes of psychological or physiological states – see Ch. 5, [5.4.7.4], p. clvii;
- Caus , Liqu and Perm , which are verbs of causing – see Ch. 5, Sec. 5.4.5 (p. cxlviii^{ff.}).

3.3 Two major classes of simple standard LFs

LFs are divided in two classes, based on the two axes of functioning of natural languages (Saussure 1972[1916]: Part 2, Ch. V):

- the *paradigmatic axis*, on which the Speaker performs the **selection** of linguistic units to be used;
- the *syntagmatic axis*, on which the Speaker performs the **combination** of selected linguistic units into higher-level units.

These two axes and the corresponding linguistic operations (selection vs. combination) determine the two classes of LFs: paradigmatic LFs (3.3.1) and syntagmatic LFs (3.3.2).⁶

3.3.1 Paradigmatic LFs

The application of a *paradigmatic LF* to a lexical unit L returns as value a set of lexical entities of which each element L' stands in a particular paradigmatic relation to L – i.e., L' is a possible substitute for L in the text. More precisely, L' is a *semantic derivative* of L.

Table 3.2: Notion definition: Semantic derivative

A **semantic derivative** of lexical unit L is a lexical entity L' whose meaning 'L'' includes 'L' such that

- the semantic difference ' σ ' = 'L' – L' is found in a large enough number of lexical pairs of the language – i.e., ' σ ' is regular;
- the language has dedicated means for expressing ' σ ' – e.g., morphological means.

L is called the *base of the derivative* L'.

As anticipated by the above definition, there is a particular case of semantic derivative: *morphological derivative*.

⁶In the process of text production the operations of selection and combination intermingle: a particular selection of a linguistic unit specifies the conditions for eventual combinations, and a particular combination can require an alternative selection.

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Table 3.3: Notion definition: Morphological derivative

A **morphological derivative** of lexical unit L is a semantic derivative L' of L such that

- the semantic difference ' σ ' = ' $L' - L$ ' is expressed by a morphological means.

A typical case of derivatives is that of *Agent noun*, or more precisely, the name for the first semantic actant [SemA 1] 'X' of L. Here are a few illustrations:

- (12)
- | | | |
|-------------------------------|--------------|--|
| <i>support</i> _(V) | ⊢Agent noun→ | <i>supporter</i> 'X who supports' |
| <i>agression</i> | ⊢Agent noun→ | <i>agressor</i> 'X who commits an agression' |
| <i>congress</i> | ⊢Agent noun→ | <i>congressman / congresswoman</i> 'X who is a member of a country's congress' |
| <i>monstrous</i> | ⊢Agent noun→ | <i>monstrosity</i> 'X that is monstrous' [What a monstrosity!] |
| <i>account</i> _(N) | ⊢Agent noun→ | <i>account holder</i> 'X who has a bank account' |
| <i>steal</i> _(V) | ⊢Agent noun→ | <i>thief</i> 'X who steals' |
| <i>annoying</i> | ⊢Agent noun→ | <i>vulgar</i> 'pain in the ass' 'X that is annoying' |
| <i>heal</i> | ⊢Agent noun→ | <i>remedy</i> _(N) 'X that heals' |

Among these semantic derivatives, the first four are also morphological derivatives.

Semantic derivation is described by paradigmatic LFs in the following way:

- a given paradigmatic LF f specifies a particular family of semantic derivatives that express the meaning ' σ^f ';
- it takes as arguments the bases of derivatives;
- it returns as values the derivatives themselves.

For instance, the semantic derivation of Agent nouns is described by the paradigmatic LF S_1 with the meaning 'somebody who... / something that...' (Ch. 4, [13], p. xcii). Thus, for the above illustrations (12), it is encoded as in (13):

- (13)
- | | | |
|-------------------------------|---|-----------------------------------|
| $S_1(\textit{support}_{(V)})$ | = | <i>supporter</i> |
| $S_1(\textit{agression})$ | = | <i>agressor</i> |
| $S_1(\textit{congress})$ | = | <i>congressman, congresswoman</i> |
| etc. | | |

3.3.1.0.1 Remark on the LF Syn The LF Syn 'synonym' (Ch. 4, [1], p. lxxii) occupies a special place among paradigmatic LFs. It models a particular case of the

most important relation underlying the functioning of natural languages: *synonymy in the broad sense*, i.e., semantic equivalence between linguistic expressions. Consequently, Syn underlies the simplest paraphrasing rule:

$$\boxed{R_{\text{Syn}}: L \equiv \text{Syn}(L)}$$

Note that, in fact, all paraphrasing rules – e.g., rules R_{Mult} and R_{Oper_1} (Sec. 3.2.2.3, p. xlv) – establish the synonymy relations between linguistic expressions. Even more importantly, the Meaning-Text approach is built on the notion of *Meaning* (written with a capital *M*), which is the invariant of a set of paraphrases, i.e., synonymous expressions (Mel’čuk 1974: 10–11, 2012: 44–45, Mel’čuk & Milićević 2014: 102–103).

3.3.2 Syntagmatic LFs

The application of a *syntagmatic LF* to a lexical unit L_1 returns as value a set of lexical entities of which each element L_2 stands in a particular syntagmatic (i.e., cooccurrence) relation with L_1 . More precisely, L_2 forms a *collocation* with L_1 .

Table 3.4: Notion definition: Collocation

A **collocation** is a compositional phraseme consisting, prototypically, of two lexical units L_1 and L_2 such that

- L_1 is selected by the Speaker freely, i.e., independently of any other lexical choice;
- L_2 is selected by the Speaker to express a particular meaning ‘ σ^f ’ in combination with L_1 as a function of L_1 .

L_1 is called the *base of the collocation* and L_2 , its *collocate*.

Examples. In the collocation *deep feeling*, *feeling* is the base and *deep*, the collocate; in *pay a visit*, *visit* is the base and *pay*, the collocate.

3.3.2.0.1 Remark on the lexical nature of collocates The above definition would necessitate some fine-tuning. In practice, instances of collocations can be found where the collocate L_2 is not a lexical unit but is itself a collocation. For instance, the lexeme ALCOHOL is the base (L_1) of the collocation *under the effect of alcohol*, whose collocate (L_2) is not a lexical unit – that is, neither a lexeme nor an idiom, but is itself a collocation with EFFECT_(N) as the base. For a description of the simple collocation *under the effect* in terms of the LF A_2 , see Ch. 4, [21], p. xcix. For

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a description of the “double shot” collocation *under the effect of alcohol* in terms of the LF Propt₁, see Ch. 5, [5.3.0.5], p. cxxix.

An emblematic case of collocation is that of intensification; here are three examples where collocates are bolded:

- (14) a. **nagging** doubt
b. Thanks **a million**!
c. drink ‘as a **sponge**’

Syntagmatic LFs take as arguments lexical units that are bases of collocations and return as values lists of corresponding collocates. For instance, intensification collocations are accounted for by the syntagmatic LF Magn (Ch. 5, [5.2.0.3], p. cxvi), so that collocations given in (14) are described as (15):

- (15) Magn(*doubt*_(N)) = *strong < gnawing, nagging; lingering*
Magn(*Thanks!*) = *a bunch* | postposed, *a million* | postposed
Magn(*drink*_(V)) = ‘as a *sponge*’, *heavily*, ‘like a *fish*’

| FURTHER READING. For more on the notion of collocation, see Hausmann (1979), Wanner (1996), Mel’čuk (1998) and Mel’čuk (2023: Ch. 6).

3.3.2.0.2 Semantic target of a syntagmatic LF For a full (= semantically loaded) syntagmatic LF such as Magn,⁷ the meaning of f(L) – i.e., the meaning of L’s collocate that is returned as value – necessarily applies to a component of L’s lexicographic definition (Iordanskaja & Polguère 2005). Let us call such a component the (semantic) *target* of the LF.

In the most cases, the LF target in L’s definition is its central component and need not be explicitly indicated in the encoding of the LF. If, however, the target is not the central component, it must be specified as a subscript to the LF name; for instance, for Magn:

- (16) Magn(*earthquake*) = *powerful, strong < major, massive*
Magn_{·damage·}(*earthquake*) = *catastrophic, devastating; deadly*

Note. The approximate definition of EARTHQUAKE is ‘sudden shaking of the earth surface, able to cause serious **damage** around location Y’.

Follows an illustration for the verbal syntagmatic LF Real₁ (see Ch. 5, [5.4.3.1], cxxxix):

⁷By contrast, support verb LFs (see Ch. 5, Sec. 5.4.2, p. cxxxi^{ff}) are examples of semantically empty syntagmatic LFs.

- (17) $\text{Real}_{1, \text{'hostile action'}}(\text{anger}_{(N)}) : \text{vent } [A_{(\text{poss})}X \sim \text{on } N_Y]$
 $\text{Real}_{1, \text{'loss of self-control'}}(\text{anger}_{(N)}) : \text{'be beside oneself'} [with _]$

The lexical entry for $\text{ANGER}_{(N)}$ in the Appendix (p. cxcvii ff.) presents more examples of the same kind.

3.3.3 Paradigmatic and syntagmatic LFs in contact

Both classes of LFs – paradigmatic and syntagmatic ones – are designed to fulfill the same role, namely, to specify lexical expressions of constrained meanings (Sec. 3.1.1, p. xxxvii). Therefore, it has to be expected for them to be frequently intertwined; and in fact, they are. Their interactions are of two kinds: with respect to the values they return (3.3.3.1) and with respect to their use in linguistic description (3.3.3.2).

3.3.3.1 Quid pro quo in LF values

The paradigmatic LFs express derivational meanings and are supposed to return semantic derivatives of their arguments; the syntagmatic LFs express collocational meanings and are supposed to return collocates of their arguments.

However, there is no airtight partition between the two LF classes with respect to the lexical nature of their values.

This happens because both types of LFs do essentially the same job: they supply an appropriate expression means for a meaning to be added to the meaning of the base (of a derivative or of a collocation). The fundamental difference between the two types of LFs in this respect is as follows: for paradigmatic LFs, these means tend to be affixal – at least, in *Standard Average European* [SAE]⁸ languages; for the syntagmatic LFs, the expressive means are normally separate lexical units. As the result, the values of the paradigmatic LFs are rather “synthetic,” and those of the syntagmatic LFs, rather “analytical.” But, as is always the case with natural languages, this distinction is not implemented 100% systematically: there are analytical values of paradigmatic LFs and synthetic values of syntagmatic LFs.

3.3.3.1.1 Analytical values of paradigmatic LFs A paradigmatic LF f applied to a lexical unit L can return (instead of a “whole-made” semantic derivative) a collocation of L , which is, so to speak, an analytical semantic derivative of L . Such is the case of two previously mentioned examples: the analytical value *account holder* for $S_1(\text{account}_{(N)})$ and *in doubt* for $A_1(\text{doubt}_{(N)})$; see (18):

⁸Roughly, SAE languages are Germanic, Romance and Slavic languages (Whorf 1941).

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- (18) $S_1(\text{account}_{(N)}) = \sim \text{holder}$
 $A_1(\text{doubt}_{(N)}) = \text{doubtful, in } \sim$

This phenomenon is quite typical of some paradigmatic LFs, in particular of the collective Mult (Ch. 4, [26], p. cvi):

- (19) $\text{Mult}(\text{fish}_{(N)}) = \text{school of } \sim$
 $\text{Mult}(\text{attack}_{(N)}) = \text{spate of } \sim s$
 $\text{Mult}(\text{activity}) = \text{flurry of } \sim$

3.3.3.1.2 Synthetic (= fused) values of syntagmatic LFs A syntagmatic LF f applied to L can return – instead of a collocate of L – a semantic derivative of L: a lexical unit that synthetically expresses the linguistic union of the meanings ‘L’ and ‘ σ^f ’. For instance, in the case of the attenuation LF AntiMagn (antonym of Magn):

- (20) $\text{AntiMagn}(\text{rain}_{(N)}) = \text{light, } \boxtimes \text{drizzle}_{(N)} [\cong \text{light rain}]$

The noun DRIZZLE_(N) is a semantic derivative of RAIN_(N), namely, a richer synonym Syn_▷ rather than a genuine collocate, such as LIGHT_(Adj).

The phenomenon under consideration is called *fusion*, and the corresponding value is a *fused value*. Fusion is indicated, as shown above, by the symbol $\boxtimes$.

3.3.3.2 Interaction of paradigmatic and syntagmatic LFs

3.3.3.2.1 1) Cohabitation of LFs in paraphrasing rules Syntactic paraphrasing rules (cf. Sec. 3.2.2.3), which are crucial for the functioning of language, actively exploit both families of LFs, often making them collaborate within the same rule. This latter case is illustrated by rule $R_{0\text{per}_1}$ (Sec. 3.2.2.3, p. xlvi), which simultaneously uses the paradigmatic S_0 and the syntagmatic 0per_1 .

3.3.3.2.2 2) Semantic connections between paradigmatic and syntagmatic LFs A paradigmatic LF and a syntagmatic LF can show interesting semantic connections; here are two examples.

- The paradigmatic LF S_1 applied to a verb L means ‘somebody / something that does L’ – L is, so to speak, what S_1 is supposed to do; and the syntagmatic LF Fact_0 applied to a noun L’ means ‘[L’] does what it is supposed to do’. From this it follows:

$$(21) \quad S_1(L_{('action')}) = L' \quad \text{implies} \quad \text{Fact}_\theta(L') = L_{('action')}$$

Examples

$$\begin{array}{llll} S_1(ski_{(V)}) & = skier & \text{implies} \text{Fact}_\theta(skier) & = ski_{(V)} \\ S_1(sell_{(V)}) & = seller & \text{implies} \text{Fact}_\theta(seller) & = sell_{(V)} \end{array}$$

- A modifying syntagmatic LF – Magn, Ver or Bon – in case of fusion adds to L a meaning such that the resulting L' can be considered a richer synonym $\text{Syn}_\supset$ of L, $\text{Syn}_\supset$ being a typical paradigmatic LF:

$$(22) \quad \text{Magn}(L) = \boxtimes L' \quad \text{implies} \quad \text{Syn}_\supset(L) = L'$$

Examples

$$\begin{array}{llll} \text{Magn}(tired) & = \boxtimes exhausted & \text{implies} \text{Syn}_\supset(tired) & = exhausted \\ \text{Bon}(musician) & = \boxtimes virtuoso_{(N)} & \text{implies} \text{Syn}_\supset(musician) & = virtuoso_{(N)} \end{array}$$

3.4 Simple standard LF combinations

Simple LFs are combinable with each other; let us explain when such combinations are necessary.

A given constrained meaning ' σ ' can be representable as the *linguistic union* of two constrained meanings ' σ_1^f ' and ' σ_2^f ', which correspond to two standard LFs f_1 and f_2 . This situation is described by the following expression, where the symbol $\oplus$ stands for the operation of linguistic union: ' $\sigma = \sigma_1^f \oplus \sigma_2^f$ '.⁹

In such a case, ' σ ' has to be associated with an LF constructed by the combination of f_1 and f_2 . There are two types of such combinations: complex LFs (3.4.1) and configurational LFs (3.4.2).

3.4.1 Complex LFs

An LF f_1 can take another LF f_2 as its argument to form a specific combination of LFs, called a *complex LF* and denoted as $f_1 f_2$. This complex LF applies to L and returns L' expressing ' $\sigma^{f_1 \rightarrow f_2}$ '.

Here is an illustration where ' σ^{f_1} ' is 'begin' – expressed by the LF Incep – and ' σ^{f_2} ' is 'utilize' – expressed by the LF Real₁:

$$(23) \quad \text{IncepReal}_1(\text{road}) = \text{take} [\text{ART } _]$$

$$[\text{take a road} \cong \text{begin}_{\text{Incep}} \text{ using}_{\text{Real}_1} \text{ a road}]$$

⁹The operation of linguistic union is defined as follows in Mel'čuk (2006: Ch. 7, Sec. 2, def. 7.3, p. 386): "[...] an operation applicable to pairs of linguistic units (of language $\mathcal{L}$) – in particular, to signs – which unites them, according to their syntactics and/or the general rules of (the grammar of) $\mathcal{L}$."

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The name of a complex LF $f_1 f_2$ is an abbreviation for $f_1(f_2)$. This means that f_1 applies to f_2 to produce a complex LF $f_1(f_2)$, which then applies as a whole to L – i.e., $[f_1(f_2)](L)$.

This situation is quite different from a *composition* of LFs (in the mathematical sense). In a composition of LFs f_1 and f_2 , there is no actual production of an LF $f_1(f_2)$: the LF f_2 applies to L , then f_1 applies to the application $f_2(L)$, which is written as $f_1(f_2(L))$. The contrast between a complex LF $A_2\text{Real}_1$ and the corresponding composition of the same LFs is shown in (24):

- (24) $\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'}) = \text{use} [\text{ART } \sim]$
 $A_2(\text{use}_{(V)}) = \text{used}$
 $A_2\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'}) = \text{occupied}$
 $A_2(\text{Real}_1(\text{toilet}_{(N)} \text{ 'bathroom'})) = \text{used}$

As one can see, in this case, the composition of A_2 and Real_1 fails to return the idiomatic adjective expressing the meaning '[toilet] that is being utilized': namely, *occupied*, which will actually appear on a locked toilet door. (For an interesting case of LF composition with the LF *Son*, see Ch. 5, [5.4.7.1], p. cliv.)

3.4.1.0.1 Remark While complex LFs are overwhelmingly present in the lexicon, composition of LFs is very seldom relevant in lexicological modeling. It is mentioned here only because complex LFs are frequently mistaken for cases of LF composition.

As regards the two major classes of LFs – paradigmatic vs. syntagmatic (Sec. 3.3), there are four possible cases of complex LFs:

1. A paradigmatic LF applies to a paradigmatic LF
 $S_{\text{loc}}\text{Mult}(\text{senator}) = \text{senate } 2 \text{ [building]}^{10}$
2. A syntagmatic LF applies to a paradigmatic LF
 $\text{MagnMult}(\text{gunshot}) = \text{hail of } \sim s$
3. A paradigmatic LF applies to a syntagmatic LF
 $S_{\text{loc}}\text{Real}_1(\text{「racing car」}) = \text{「racing track」}$
4. A syntagmatic LF applies to a syntagmatic LF
 $\text{MagnSon}(\text{engine}) = \text{roar}_{(V)}$
 $\text{IncepSon}(\text{phone}_{(N)}) = \text{「go off」}$

¹⁰The connection between *SENATOR* and *SENATE 2* can also be established in two steps: $\text{Mult}(\text{senator}) = \text{senate } 1$ and $S_{\text{loc}}(\text{senate } 1) = \text{senate } 2$.

A complex LF $f_1 f_2$ encodes a deep-syntactic structure [DSyntS] (Ch. 2, Sec. 2.3, p. xxviii) that is determined by the nature of f_1 and f_2 . Thus, for the case of MagnSon vs. IncepSon, we have:

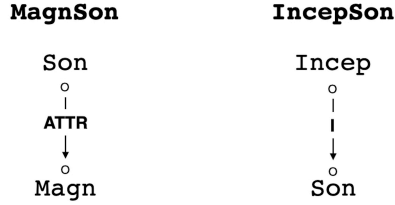


Figure 3.4: DSyntSs of complex LFs MagnSon and IncepSon

3.4.1.0.2 Part of speech of a complex LF The head (= top node) of the DSyntS of a complex LF gives its part of speech to the value elements of this LF. For instance, the value of $S_{loc}Real_1$ – whose DSyntS is $S_{loc} \dashv \dashv \rightarrow Real_1$ – is a noun (as S_{loc}), and the value of MagnSon – whose DSyntS is $Magn \leftarrow ATTR \dashv Son$ – is a verb (as Son). For details on the part of speech of LFs, see Sec. 3.5, p. lvii ff.

A complex LF can consist of more than two simple LFs. For instance, Figure 3.5 shows the DSyntSs of such complex LFs exemplified in (25):

- (25) a. $AntiBonReal_1(chair_{(N)}) = slouch_{(V)}, slump_{(V)} [in ART _]$
 b. $CausPlusFact_0(screw_{(N)}) = tighten [ART _]$
- Note.** CausPlusFact₀ is an abbreviation of CausPlusMagnFact₀, see Ch. 5, [5.2.0.6], Remark, on p. cxxiv.

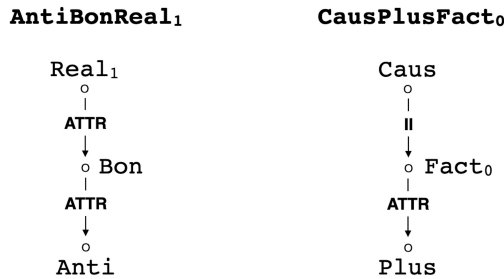


Figure 3.5: DSyntSs of complex LFs AntiBonReal₁ and CausPlusFact₀

3.4.2 Configurational LFs

The combination of two LFs f_1 and f_2 can be such that both LFs apply to L independently of each other and return a single value expressing jointly both LFs. This is represented by the following formula: $f_1+f_2(L)$.

An f_1+f_2 configuration is called a *configurational LF*. Compare the following pair of LFs:

- (26) $A_1(\text{sweat}_{(V)}) = \text{sweaty}$ 'who is sweating'
 $\text{Magn}+A_1(\text{sweat}_{(V)}) = \text{drenched in sweat}_{(N)}$ 'who is sweating a lot'

3.4.2.0.1 Order of components In a configurational LF f_1+f_2 , the order of components is relevant: f_2 – the rightmost LF – is dominant, both syntactically and informationally.

In the configuration f_1+f_2 where f_1 and f_2 are of different parts of speech, the component f_2 imposes its part of speech on the value of the configurational LF, i.e., it dominates f_1 syntactically. For instance, in the example below, the value is a verb, in accordance to the Oper_1 component of the configurational LF:

- (27) $\text{Magn}+\text{Oper}_1(\text{desire}_{(N)}) = \text{simmer}$ [with ~]

If f_1 and f_2 are of the same part of speech – e.g., both are modifying adjectives – the meaning of f_2 constitutes the informationally dominant component in the meaning of the value. For instance, the contrast between $\text{Bon}+\text{Magn}$ and $\text{Magn}+\text{Bon}$ is illustrated in (28):

- (28) $\text{Bon}+\text{Magn}(\text{meal}) = \text{hearty}$ 'copious, and also good'
 $\text{Magn}+\text{Bon}(\text{meal}) = \text{sumptuous}$ 'good, and also copious'

3.4.3 Actantial subscripts in LF combinations

Actantial subscripts in complex LFs

Two cases are to be distinguished for the actantial subscript to the name of a simple LF f_i used as a component of a complex LF (except for the rightmost one in the formula, which is interpretable in the standard way).

1) If f_i in a complex formula $f_i g$ is not a verb of causing (Ch. 5, Sec. 5.4.5, p. cxlviii^{ff}), its argument is the next simple LF g , and its actantial subscript " i " refers to the $\text{DSyntA } i$ of g . For instance, in the complex LF $A_1\text{Real}_2$, illustrated in (29), " 1 " refers to the $\text{DSyntA } 1$ of Real_2 – that is, to the sick person taking the medication.

(29) $A_1 \text{Real}_2(\text{medication} [\text{prescribed by } X \text{ to } Y \text{ against } Z]) = \text{medicated, on } \sim$

The actantial subscripts of the LF $A_1 \text{Real}_2$ used in (29) are interpretable as follows:

- The lexical unit **MEDICATION** controls three DSyntAs: DSyntA Ⅰ is the doctor *X*; DSyntA Ⅱ, the sick person *Y*; DSyntA Ⅲ, the illness *Z*.
- $\text{Real}_2(\text{medication}) - Y \text{ takes a medication} -$ controls, by definition of the LF Real_1 (Ch. 5, [5.4.3.1], p. cxxxix), two DSyntAs:
 - DSyntA Ⅰ, the sick person *Y* – cf. “₂” in Real_2 (since *Y* is DSyntA Ⅱ of **MEDICATION**);
 - DSyntA Ⅱ, the noun **MEDICATION**.
- The subscript “₁” of A_1 in the formula $A_1 \text{Real}_2$ refers to the DSyntA Ⅰ of its argument Real_2 , that is, to the sick person *Y* (and not to the DSyntA Ⅰ of **MEDICATION**, that is, to the doctor *X*).

2) If f_i is a verb of causing, its actantial subscript “_i” refers to the corresponding DSyntA of *L*, just as it does when this LF is used alone. For instance, in the complex LF $\text{Caus}_2 \text{Oper}_2$ illustrated in (30), the subscript “₂” of Caus_2 refers to the DSyntA Ⅱ of $\text{trust}_{(N)}$, that is, to the person who is trusted:

(30) $\text{Caus}_2 \text{Oper}_2(\text{trust}_{(N)} [\text{of } X \text{ in } Y]) = \text{win}_{(V)} [\text{ART } \sim]$
 Ex.: *Joey_Y won_{Caus₂Oper₂(trust_(N)) the trust of Chandler_X}*.

Actantial subscripts in configurational LFs

There is nothing particular to be said about actantial subscripts in configurational LFs: the components of these LFs are either simple LFs, or complex LFs, and the explanations about actantial subscripts in these two types of LFs have already been provided.

3.5 Parts of speech of simple standard LFs

Since an LF is a deep lexical unit (cf. 3.2.2.1, p. xliii), it belongs to a part of speech. More specifically, as we will see, it has only a deep-syntactic part of speech.

3.5.1 Notion of part of speech [PoS]

A *part of speech* [PoS] is a major class of lexical units that feature, at the syntactic level, the same *passive syntactic valence*. By passive syntactic valence of a lexical unit, we mean its ability to participate in particular syntactic constructions as the dependent member (Mel'čuk 2015b: 109):¹¹

- a verb – in a finite form – is typically the syntactic head of a clause and need not to be dependent;
- a noun is typically a syntactic actant;
- an adjective is typically a modifier of a noun;
- etc.

Two types of PoS are distinguished: deep PoS and surface PoS. There are five *deep parts of speech*:

- *Verb* [V],
- *Noun* [N],
- *Adjective* [Adj],
- *Adverb* [Adv],
- *Clausative* [Claus].

V, N, Adj and Adv do not call for further remarks. Claus, on the other hand, needs an explanation: a clausative is a lexical expression that is syntactically equivalent to an independent clause and is not a verb.¹² Clausatives form a very heterogenous lexical class, which includes emotional interjections (Wow!, Yuck!) and onomatopoeic interjections (MEOW!, BANG!), as well as interjection-like lexemes (WELL) and fixed idiomatic phrases (‘BLESS YOU’ [to someone who just sneezed]).

Deep PoS are linguistically universal in the sense that they are necessary and sufficient for describing deep-syntactic structures of sentences in any language.

In contrast to deep PoS, *surface PoS* are not linguistically universal: each language features its own set of surface PoS. Thus, English has prepositions, while

¹¹Passive syntactic valence is to be contrasted with active syntactic valence, introduced in Ch. 2, Sec. 2.3, p. xxxii.

¹²Tesnière (1959: Ch. 45) calls such expressions *mots phrases* or *phrasillons*.

Hungarian has only postpositions; English has determiners, while Russian has not. A given surface PoS is a “particular case” of a deep PoS: prepositions and conjunctions are subtypes of deep adverbs (Adv);¹³ determiners are subtypes of deep adjectives (Adj); etc. On deep vs. surface PoS, see Mel’čuk (2013a: Ch. 7, Sec. 2.1.3, 49–51).

In what follows, only deep PoS are considered, so that the adjective *deep* can be omitted.

To conclude, note that the PoS of a lexical unit L is a core element of L’s cooccurrence; as such, it can be expected to be the first piece of information given in L’s lexical entry – see e.g., the lexical entry of $\text{ANGER}_{(N)}$ in the Appendix (p. cxvii).

3.5.2 LF’s inherent PoS

Most simple standard LFs possess an inherent PoS: e.g., $S_i(L)$ is always a noun, and $\text{Oper}_i(L)$ is always a verb. However, several paradigmatic LFs borrow their PoS from their argument. Such LFs are:

- the three “meta”-LFs, which underlie the whole LF system: Syn , Conv_{ijk} and Anti (with Contr) – Ch. 4, Sec. 4.1;
- the “semantic pronoun”: Gener – Ch. 4, [4], p. lxxix;
- two LFs whose meanings are close to those of grammemes: the singulative Sing and the collective Mult – Ch. 4, Sec. 4.8.

For the PoS of complex LFs, see Sec. 3.4.1, “Part of speech of a complex LF” (p. lv).

3.5.2.0.1 Part of speech tables for LF applications An LF f calls for two syntactic characteristics as regards PoS: (i) the PoS of its possible arguments L and (ii) the corresponding PoS of its value – a deep lexical unit $f(L)$.

This correspondence is shown in a *PoS table* for each LF. Take as example the PoS table of Magn , intensification modifier:

Magn: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Magn(L)	Adv	Adj	Adv	Adv	Adj / Adv

¹³More precisely, prepositions and conjunctions form a subclass of adverbs that have an obligatory $\text{DSyntA} \#$ (on DSyntA , see Ch. 2, Sec. 2.3, p. xxxi).

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The table header indicates that Magn is a modifier that functions as an adjective or an adverb. The body of the table specifies the correspondence between the PoS of L and that of the deep lexical unit Magn(L) – i.e., the PoS of Magn itself, for short.

When a PoS correspondence is theoretically possible, but not attested to our knowledge, the symbol [?] is added. The PoS table of Gener illustrates this situation.

Gener: borrows L's PoS					
L	V	N	Adj	Adv	Claus
Gener(L)	V	N	Adj	Adv	Claus[?]

In the header of this table, the statement “borrows L’s PoS” indicates that Gener is an LF without inherent PoS.

3.6 Regularities in standard LFs

In spite of the basically idiosyncratic character of LFs, in numerous cases a given LF has the same values for quite a few different arguments: you *take a bath*, *a shower*, *a rest*, ...; you *are in despair*, *love*, *a rage*, ...; you *follow the rules*, *prescriptions*, *recommendations*, *instructions*, ... The reason is, of course, semantic proximity of these LF’s arguments: semantically related lexical units tend to feature the same values for a given LF.

In other words, there exist *regularities* among LFs of any natural language. No doubt that such regularities play an important role in the acquisition of LFs by speakers, as the former allow the latter to make predictions about LFs. These regularities should be exploited in such domains as language teaching, translation, etc. It is therefore important to extract them, and formulate *generalizations*, which should be stored in models of the lexicon in a way that supports the *inheritance* of regularities (Mel’čuk & Wanner 1996).

The elements of the values of an LF *f* shared by several argument lexical units can be generalized by extracting them from individual lexical entries of these units and storing them together: either on the argument side (for a group of lexical units) or on the LF side (attaching them to the LF itself). In practice, both methods can be combined in different proportions. We propose below four types of generalizations: types of regularities with associated means of accounting for them.

3.6.0.0.1 1) “Local” generalizations at vocable’s level If several lexical units of a given vocable or of several semantically-related vocables share LF values, these values can be given under one of these units, with a cross-reference in the lexical entries of the others. For instance, we have for the verb $\text{ESCAPE}_{(V)}$:

- (31) S_1 : *escaper*
 $S_1\text{Perf}$: *escapee, fugitive_{(N)}, runaway_{(N)}*
 $S_1\text{Able}_1$: ‘*flight risk*’
 S_2 : ‘*place of detention*’
 A_1 : *fugitive_{(Adj)}, runaway_{(Adj)}*
 Able_2 : *escapable*

As the same LF relations hold for the corresponding noun $\text{ESCAPE}_{(N)}$, which is the S_0 (nominalization) of the verb, we can simply indicate the following cross-reference in its lexical entry:

- (32) $S_1, S_1\text{Perf}, S_1\text{Able}_1, S_2, A_1, \text{Able}_2$: $\uparrow \text{ESCAPE}_{(V)}$

Note that (32), which concerns two specific lexical units ($\text{ESCAPE}_{(V)}$ and $\text{ESCAPE}_{(N)}$), is in point of fact a consequence of a series of more general statements concerning the entire system of LFs itself, namely:

- (33) $S_i(L_{(V)}) = S_i(S_0(L_{(V)}))$
 ...
 $A_i(L_{(V)}) = A_i(S_0(L_{(V)}))$
 ...

3.6.0.0.2 2) Generalizations associated with a generic term There are semantic classes of lexical units such that:

1. they all belong to the same semantic field;
2. they possess the same generic term L_{GENER} ¹⁴ – for instance, CAR , $\text{TRUCK}_{(N)}$, VAN , $\text{BUS}_{(N)}$, $\text{MOTORBIKE}_{(N)}$, ... form a semantic class united under the same L_{GENER} VEHICLE in the sense of ‘motor vehicle’;
3. the meaning of L_{GENER} is sufficiently rich so that it belongs to the same semantic field as these lexical units – cf., VEHICLE belongs to the same semantic class as CAR , $\text{TRUCK}_{(N)}$, etc.

¹⁴On generic terms, see the LF Gener, Ch. 4, [4], p. lxxix.

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Elements of such semantic classes may share the same values for some LFs. (This is only a tendency, with numerous exceptions, yet in some cases this tendency is quite clear.) If such is the case, generalizations can be made by storing all common values of LFs supplied with necessary constraints under a special *public zone* of L_{GENER} 's lexical entry. For instance, if the following is stored in the public zone of the lexical entry for **VEHICLE**, it does not need to be repeated in the lexical entry for all names of vehicles:

VEHICLE, noun	
...	
Public	
...	
S_1	: <i>driver; rider</i> for two-wheel vehicle
S_2	: <i>passenger</i>
...	
Real_1 [i.e., 'use ~ as driver']	: <i>drive</i> _(V) [ART ~]; <i>ride</i> _(V) 1 [ART ~] for two-wheel vehicle
Real_2 [i.e., 'use ~ as passenger']	: <i>ride</i> _(V) 2 [(in / on) ART ~], <i>go</i> [by ~]
...	

Figure 3.6: Public zone in the lexical entry of **VEHICLE**

A rather straightforward particular case of the above type generalization is the use of LFs with proper names. If the argument of an LF f happens to be a proper name $N_{(\text{prop})}$, the value of f must be computed substituting for $N_{(\text{prop})}$ the lexical unit L that denotes the entity named by $N_{(\text{prop})}$. In other words, the values of LFs for **RHINE**, **GANGES**, **VOLGA**, **THAMES**, **MISSISSIPPI**, etc. are those of the LFs for **RIVER** (or **WATERCOURSE**):

- (34) a. The Rhine <This river> **flows** from Switzerland to the North Sea.
b. The Ganges <This river> **empties** into the Bay of Bengal.
c. The Oka <This river> is a **tributary** of the Volga.
d. The Thames <This river> was **swollen** and **flooding** the riverbanks.

3.6.0.0.3 3) “Joker” values associated with given LFs The level of generality of LF values might be such that it becomes feasible to associate with a given LF some general default elements of its value: we refer to such default elements as *joker values*. Thus, a generally valid expression in English of the LF S_0 for a verb $L_{(V)}$ is the phrase *the fact of V-ing* – or just simply *V-ing*; e.g.:

- (35) joker $S_0(\text{chew}_{(V)})$: *the fact of chewing, the chewing*
joker $S_0(\text{ponder})$: *the fact of pondering, the pondering*

For the support verbs $0per_1(L)$, the joker value elements are DO, MAKE, GIVE, HAVE, TAKE, ‘CARRY OUT’ and maybe some more. Of course, for each of those, semantic restrictions on L have to be specified. For $Func_0(L)$ – another type of support verbs, joker value elements are, among others, ‘TAKE PLACE’, HAPPEN (for L denoting a social event), etc. For intensifiers $Magn(L)$, when they have nominal arguments, such general elements are: $BIG_{(Adj)}$, $HIGH_{(Adj)}$ (for L denoting non-linear physical parameters such as $PRESSURE_{(N)}$, $SPEED_{(N)}$, $TEMPERATURE$), $UTMOST_{(Adj)}$ / $UTTER_{(Adj)}$ (for feelings), etc. An obvious $Magn$ joker value element for adjectival lexemes as arguments is VERY.

The generalizations considered here are directly associated with LFs and their description presupposes the existence of special lexical entries dedicated to LFs.

3.6.0.0.4 4) Lexical entries for collocative lexical units *Collocative lexical units* are “designed” to be selected by the Speaker as values of syntagmatic LFs: they have virtually no autonomy and are essentially used as collocates within collocations. To ensure a sufficiently systematic description of collocative lexical units shared by many collocation bases, one has to compile for them special lexical entries, where semantic and syntactic conditions on their use are specified. Thus, the support verb *deliver* is an element of the value of the LF $0per_1$ for many nouns of public activities (*deliver a speech / a lecture / a performance*). It receives a lexical entry of its own in the vocable DELIVER; accordingly, it needs not be lexicographically described in all lexical entries for nouns it combines with but could be selected and used following indications found in its own entry. For an example of a dictionary of collocations with entries for collocates, see Bosque (2010).

To round up the topic, still another perspective of possible generalizations for LFs is opened by accounting for the links between the meaning of a lexical unit L and the LFs applicable to it; for instance:

- L that is predicative expects actantial paradigmatic derivatives – cf. LFs S_i , A_i , ..., and syntagmatic verbal collocates such as support verbs ($0per_i$, ...) and possibly realization verbs ($Real_i$, ...).
- L that denotes an artefact expects realization verbs such as $Real_1$, which expresses ‘make use of L’.
- L that denotes something producing sounds expects values for the LF Son.
- L that denotes an inner state of a human expects values for the LFs $Manif$ ([L] manifests itself in (part of the body of) X) and $Sympt_{ijk}$.

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- L that denotes a place expects values for the LF $\text{Loc}_{\text{in/ad/ab}}$.

The identification and proper modeling of such generalizations is an open research question.

FURTHER READING. On semantic regularities in standard LF applications, see Apresjan (2009); on lexical inheritance of LF regularities, see Mel'čuk & Wanner (1996), Barrios (2009); on accounting for regularities in lexical entries for collocates, see Reuther (1996); on the lexicographic description of collocates, see Alonso Ramos & Mantha (1996), Alonso Ramos (2003); on extracting regularities – polysemy and lexicographic definitions – in lexical networks with the LF Gener, see Polguère (2022).

3.7 Description template for simple standard LFs

In Chs. 4 and 5, we follow a three-part template (**A–C**) for the presentation of each individual LF. This template is illustrated below with the example of the LF S_θ (Ch. 4, [7], p. lxxxv).

3.7.0.0.1 A Identification number between square brackets and name of the LF together with its Latin source also between square brackets (cf. the remark on the naming of LFs on p. xlii), followed by a rough characterization.

[7] S_θ [Lat. *substantivum*]: nominalization

3.7.0.0.2 B Semantic-syntactic characterization of the LF, followed by its PoS table.

S_θ applies to a non-nominal lexical unit L and returns a noun having the same meaning as L.

S_θ : nominal				
L	V	Adj	Adv	Claus
$S_\theta(L)$	N	N	N	N

If a series of LFs have an identical PoS table, it is extracted and mentioned just once before the presentation of these LFs – see, e.g., the shared PoS table for Syn, Conv_{ijk} and Anti (Ch. 4, Sec. 4.1, p. lxxii).

3.7.0.0.3 C Illustration of the LF in English, French and Russian (sample).

English

$S_0(\text{present}_{(V)}) = \text{presentation}$

$S_0(\text{leave}_{(V)}) = \text{departure}$

French

$S_0(\text{présenter}) = \text{présentation}$

$S_0(\text{partir}) = \text{départ}$

Russian

$S_0(\text{predstavljat'}) = \text{predstavlenie}$

$S_0(\text{uezžat'}) = \text{ot'ezd}$

As mentioned at the beginning of this chapter (Sec. 3.1.2, p. xxxix), our illustrations do not provide exhaustive lists for LF values; in most cases, only one value element is indicated.

3.7.0.0.4 Active syntactic valence of L The use of an LF f that carries actantial subscripts (Conv_{12} , S_1 , Oper_2 , etc.) presupposes a preliminary identification of L's active syntactic valence (Ch. 2, Sec. 2.3, p. xxxii). Thus, to understand (36) below, one must first establish that $\text{FEAR}_{(N)}$ is a lexeme that (i) denotes a type of feeling and (ii) controls two SemAs expressible as DSyntAs: $X's_{[1 \Leftrightarrow I]} \text{fear of } Y_{[2 \Leftrightarrow II]}$.

- (36) $\text{Able}_1(\text{fear}_{(N)}) = \text{fearful}$
 $\text{Able}_2(\text{fear}_{(N)}) = \text{scary}$

Note. LFs being deep lexical units, subscripts $_1$ and $_2$ refer to $\text{FEAR}_{(N)}$'s DSyntAs I and II (on Able_i , see Ch. 4, [22], p. c); these DSyntAs correspond respectively to SemAs 1 and 2 – cf. Ch. 2, Sec. 2.3, p. xxxi.

It would be tedious to supply each $f(L)$ illustration with the actantial structure of L; readers are expected to establish L's actantial structure themselves. This is, by the way, an excellent training in Explanatory Combinatorial Lexicology.

3.8 System of simple standard LFs

The next two chapters introduce LFs in an enumerative form and constitute a repository of paradigmatic LFs (Ch. 4) and syntagmatic LFs (Ch. 5). Though LFs are listed linearly in these chapters, it should be stressed that, in reality, the set of LFs constitutes a *system* where all elements are interconnected.¹⁵ Table 3.5

¹⁵On the systemic organization of LFs and its computational modeling, see Jousse (2010) and Fonseca et al. (2016).

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(p. lxvii) lists all simple standard LFs following a logic that takes into account their systemic organization. Three strategies are used to account for this organization.

3.8.0.0.1 1) Identification of central LFs Not all LFs are of equal importance and some can be considered as being truly at the heart of the system of simple standard LFs. At least two features determine whether a given LF is central:

- Its massive presence in the lexical network of the language (cf. Sec. 3.1.1, “LFs and lexical networks,” p. xxxviii) – some LFs (e.g., *Syn*, *Magn*, *Oper_i*, ...) apply to many different arguments and return many elements of their values; others (*Figur*, *S_{mod}*, *Obstr*, ...) apply much more marginally.
- Its implication in paraphrasing and, more specifically, in the universal system of syntactic paraphrasing rules (Sec. 3.2.2.3, p. xlv), which demonstrates its importance in the semantics ↔ syntax correspondence.

The mastering of the system of LFs has to develop starting from the most central LFs. They are signaled in Table 3.5 below by their boxed number.

3.8.0.0.2 2) Subclassification of paradigmatic and syntagmatic LFs The ordering of LFs in Table 3.5 – cf. identification numbers – is logically motivated, starting with LFs whose meaning is closer to that of their argument. Thus, paradigmatic LFs (semantic derivatives) are listed before syntagmatic LFs (collocates). Subclassifications are established within each of the two main classes of LFs; they are as follows:

Paradigmatic LFs (Chaper 4, p. lxxiff.)

- Three pillars of the LF system: *Syn*, *Conv_{ijk}*, *Anti* (4.1)
- Three LFs related to *Syn* and *Anti*: *Gener*, *Figur*, *Contr* (4.2)
- Structural (= syntactically motivated) derivatives (4.3)
- Predicative derivatives (4.4)
- Nominal derivatives (4.5)
 - Nominal actantial derivatives (4.5.1)
 - Nominal circumstantial derivatives (4.5.2)
- Adjectival actantial derivatives (4.6)
- Adverbial actantial derivatives (4.7)
- Singulative, collective and other inflection-like derivatives (4.8)

Syntagmatic LFs (Chaper 5, p. cxiff.)

- Nominal collocates (5.1)
- Adjectival / adverbial collocates (5.2)
- Prepositional collocates (5.3)
- Verbal collocates, distributed into seven subclasses corresponding to their increasing “semantic load” (5.4):
 - Copula verb (5.4.1)
 - Support [= light] verbs (5.4.2)
 - Realization verbs (5.4.3)
 - Phasic verbs (5.4.4)
 - Verbs of causing (5.4.5)
 - Verbs of mode of functioning (5.4.6)
 - Other verbal collocates (5.4.7)

Table 3.5: System of standard simple lexical functions [LFs]

Paradigmatic LFs [Ch. 4]				Syntagmatic LFs [Ch. 5]			
[1]	Syn	19	S _{res}	5.1.0.1	Germ	[5.4.3.1]	Real _i
[2]	Conv _{ijk}	20	S _{mod}	5.1.0.2	Culm	[5.4.3.2]	Fact _i
[3]	Anti; Non	[21]	A _i	5.2.0.1	Epit	[5.4.3.3]	Labreal _{ij}
[4]	Gener	[22]	Able _i	[5.2.0.2]	Redun	5.4.3.4	Prepar
5	Figur	23	Qual _i	[5.2.0.3]	Magn	[5.4.4.1]	Incep
6	Contr	24	Adv _i	[??]	Ver	[5.4.4.2]	Fin
[7]	S _θ	[25]	Sing	[??]	Bon	[5.4.4.3]	Cont
[8]	V _θ	[26]	Mult	5.2.0.6	Plus	5.4.4.4	Prox
[9]	A _θ	27	Imper	5.2.0.7	Minus	[5.4.5.1]	Caus
[10]	Adv _θ	28	Perf	5.3.0.1	Loc _{in}	[5.4.5.2]	Liqu
[11]	Claus _θ	29	Imperf	5.3.0.2	Loc _{ad}	[??]	Perm
12	Pred	30	Result _i	5.3.0.3	Loc _{ab}	5.4.6.1	Obstr
[13]	S _i			5.3.0.4	Instr	??	Stop
14	Equip			5.3.0.5	Propt _i	5.4.6.3	Excess
15	Cap			5.4.1.1	Copul	5.4.7.1	Son
16	S _{loc}			[5.4.2.1]	Oper _i	5.4.7.2	Manif
17	S _{instr}			[5.4.2.2]	Func _i	5.4.7.3	Involv
18	S _{med}			[5.4.2.3]	Labor _{ij}	5.4.7.4	Sympt _{ijk}
Tip. In the electronic version of the present volume, LF identification numbers are clickable hyperlinks leading to the description of corresponding LFs.							

Table 3.5 lists LFs in a revised order in comparison to the one used in previous Meaning-Text publications. For instance, Conv_{ijk} is introduced after Anti in

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Mel'čuk (1996: 48–50), a reference text on LFs. More importantly, LFs' classification has been revised in some instances – e.g., Germ and Culm, now listed among syntagmatic LFs, were formerly classified as paradigmatic LFs.

3.8.0.0.3 3) Cross-referencing between LFs' descriptions A profusion of cross-references in Chs. 4 and 5 make explicit the interconnections between individual (or families of) LFs. Their aim – together with the classification we propose – is to help the reader assimilate the set of simple standard LFs as a system, rather than a simple list of theoretical entities.

Before we proceed with the detailed presentation of paradigmatic and syntagmatic LFs, let us answer a legitimate question: how trustworthy are LF-based descriptions with respect to formalization? In particular, can it be expected that any given lexical relation $L_1 \vdash_{\text{LexRel}} L_2$ will be modeled with the same LF application $f(L_1) = L_2$ by different individuals?

For seasoned lexicographers, with extensive experience of the use of LFs in the construction of lexical resources, this is a non-issue. We do know that the system of LFs is a reliable formal language for collaborative work on modeling paradigmatic and syntagmatic relations. However, it can be expected from our readers – faced with the theoretical and formal complexity of LFs – to question the actual applicability of the approach.

As with any formal descriptive tool, disagreements between researchers, though marginal, do exist in the context of encoding LF relations. There can be three reasons for this.

- 1) Lexicographers being human, they make mistakes, and those mistakes simply have to be fixed.
- 2) More interestingly, some lexical relations can give rise to alternative encodings that are formally equivalent – cf. example (22) of Section 3.3.3.2 (p. liii), fused values for Magn that entail a corresponding $\text{Syn}_{\supset}$ relations.
- 3) Finally, tricky cases do exist. A disagreement on the semantic content of L_1 and/or L_2 involved in $L_1 \vdash_{\text{LexRel}} L_2$ can entail different choices of a corresponding LF. The triplet of adjectival / adverbial LFs Magn, Ver and Bon is a notorious source of problems in this respect – see discussion in Chapter 5, Section 5.2, p. cxx.

In spite of such unavoidable situations, the LF system proves to be reliable in many contexts of application, and in lexicography in particular. An argument

for the robustness and non-aleatory nature of LF modeling can be found in the fact that it is possible to implement the automatic diagnosis of LF relations. In particular, LF-based descriptions can be used to train *machine learning models* in Natural Language Processing in order to perform the automatic identification of LF relations. For a synthesis of work done on the identification of syntagmatic LF relations, see Wanner (2022); on the extraction and LF encoding of collocations using generative Artificial Intelligence, with an almost perfect inter-evaluators agreement on the experiment results, see Fisas & Wanner (forthcoming).

4 Paradigmatic lexical functions

Paradigmatic LFs fall into eight major classes; they are presented here in the same order in which they appear in the LF zone of a lexical entry in an Explanatory Combinatorial Dictionary (Mel'čuk & Zholkovsky 1984, Mel'čuk & Žolkovskij 2016, Mel'čuk et al. 1984, 1988, 1992, 1999):

- Three pillars of the LF system: Syn, Conv_{ijk}, Anti (4.1)
- Three LFs related to Syn and Anti: Gener, Figur, Contr (4.2)
- Structural (= syntactically motivated) derivatives (4.3)
- Predicative derivatives (4.4)
- Nominal derivatives (4.5)
 - Nominal actantial derivatives (4.5.1)
 - Nominal circumstantial derivatives (4.5.2)
- Adjectival actantial derivatives (4.6)
- Adverbial actantial derivatives (4.7)
- Singulative, collective and other inflection-like derivatives (4.8)

The presentation order of paradigmatic LFs corresponds to the importance of their semantic contribution: from Syn, whose semantic import is null, to LFs that carry relatively rich semantic load.

4.1 Three pillars of the LF system: Syn, Conv_{ijk}, Anti

The lexical function Syn (synonym) is the most important in the system of LFs and in the structuring of the lexical stock of languages, given its intimate relation with paraphrasing – cf. the remark on rule **R_{Syn}**, Ch. 3, Sec. 3.3.1 (p. xlix).

Syn expresses the identity of meanings, and the identity of meanings underlies the LF Conv_{ijk} (conversive), which expresses the identity of meanings with

4 Paradigmatic lexical functions

the permutation of the DSyntAs; Anti (antonym) expresses the contrariness of meanings.

We call Syn, Conv_{ijk} and Anti pillars of the LF system because these LFs are, so to speak, “meta-functions”: they are extensively used to express the semantic connections between LFs themselves.

These three LFs do not have their own inherent part of speech [PoS]: they are neither nominal, nor verbal, etc., but borrow the PoS of their argument. This is reflected in their PoS table, where L’s PoS and that of f(L) are always identical. All three LFs possess the same PoS table, namely:

Syn, Conv _{ijk} , Anti: borrow L’s PoS					
L	V	N	Adj	Adv	Claus
Syn / Conv _{ijk} / Anti(L)	V	N	Adj	Adv	Claus

[1] Syn [Lat. *synonymum*]: synonym

Syn(L) has the same meaning and PoS as L and is, therefore, a possible substitute for L in the syntactic structure of the utterance with preservation of its meaning. L’, a given value element of Syn(L), is a special case of L’s semantic derivatives (Ch. 3, Sec. 3.3.1, p. xlvii): one where the semantic difference between ‘L’ and ‘L’ (i.e., ‘ σ^{Syn} ’) is zero – see Ch. 3, “Remark on the LF Syn,” p. xlviii.

As illustrated below, synonyms of L can belong to different registers and have different stylistic properties; what counts for synonymy is just identity of meanings.

English

Syn(urinate)	=	informal <i>pee</i> _(V) , colloquial <i>piss</i> _(V)
Syn(car)	=	<i>auto</i> _(N) , <i>automobile</i> _(N)
Syn _⊃ (murder _(N))	=	<i>assassination</i>
Syn _⊂ (assassination)	=	<i>murder</i> _(N)
Syn _∩ (murder _(N))	=	<i>killing</i> ; <i>massacre</i> _(N) ; <i>slaughter</i> _(N)
Syn _⊂ (colloquial ‘like a bat out of hell’)	=	<i>fast</i> _(Adv)
Syn _⊃ (fast _(Adv))	=	colloquial ‘like a bat out of hell’
Syn(Ouch!)	=	Ow!

4.1.0.0.1 Notation The three set-theoretical subscripts $\supset$, $\subset$ and $\cap$ are used in the following sense:

- Syn_⊃ means ‘richer (= narrower) synonym’ – the meaning of Syn_⊃(L) includes that of L;

4.1 Three pillars of the LF system: *Syn*, *Conv_{ijk}*, *Anti*

- $Syn_{\subset}$ means ‘less rich (= poorer, broader) synonym’ – the meaning of $Syn_{\subset}(L)$ is included into that of L ;
- $Syn_{\cap}$ means ‘intersecting synonym’ – the meanings of $Syn_{\cap}(L)$ and L have an intersection, while each possesses a distinctive component.

These subscripts can accompany the names of other LFs as well – always with the same meaning.

French

$Syn(uriner)$	=	colloquial <i>pisser</i>
$Syn(voiture)$	=	<i>auto, automobile, colloquial bagnole</i>
$Syn(\text{‘doigt de pied’})$	=	<i>orteil</i>
$Syn_{\sup}(meurtre)$	=	<i>assassinat</i>
$Syn_{\subset}(assassinat)$	=	<i>meurtre</i>
$Syn_{\cap}(meurtre)$	=	<i>‘mise à mort’; carnage, massacre</i>
$Syn_{\subset}(\text{‘à fond de train’})$	=	<i>vite</i>
$Syn_{\sup}(vite)$	=	<i>‘à fond de train’</i>
$Syn(Aïe !)$	=	<i>Ouille !</i>

Russian

$Syn(močit' sja)$	=	colloquial <i>pisat'</i>
$Syn(mašina)$	=	<i>avtomobil', dated avto, slang tačka</i>
$Syn_{\sup}(prestuplenie)$	=	<i>ubijstvo</i>
$Syn_{\subset}(ubijstvo)$	=	<i>prestuplenie</i>
$Syn_{\cap}(ubijstvo)$	=	<i>formal umerščvlenie; bojnja, reznja</i>
$Syn_{\subset}(\text{‘kak strela’})$	=	<i>bystro</i>
$Syn_{\sup}(bystro)$	=	<i>‘kak strela’</i>
$Syn(Aj!)$	=	<i>Oj!</i>

As can easily be seen in the above illustrations, the LF *Syn* is reversible:

if $Syn(L_1) = L_2$, then $Syn(L_2) = L_1$.

Of course, in case poorer / richer synonyms are involved, the reversibility is as follows:

if $Syn_{\subset}(L_1) = L_2$, then $Syn_{\sup}(L_2) = L_1$.

The intersecting synonyms are reversible:

if $Syn_{\cap}(L_1) = L_2$, then $Syn_{\cap}(L_2) = L_1$.

The same remark applies to *Syn*’s “fellow” LFs *Conv_{ijk}* and *Anti*.

[2] **Conv_{ijk}** [Lat. **conversivum*]: **conversive**

$Conv_{ijk}(L)$ has the same meaning and PoS as L and is, therefore, a possible substitute for L in the deep-syntactic structure of the utterance, with preservation

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of meaning, but with the necessary permutation of L's DSyntAs. This relation of paraphrase between L and its converse is accounted for by the $R_{\text{Conv}_{ijk}}$ paraphrasing rule:

$$\boxed{R_{\text{Conv}_{ijk}}: L \cong \text{Conv}_{ijk}(L)}$$

The pair of sentences (1a–b) illustrates the semantic quasi-equivalence of two converse lexical units:

- (1) a. Lightning **precedes** thunder.
 $\cong$
 b. Thunder **follows**_{Conv₂₁(precede)} lightning.

The equivalence between sentences (1a) and (1b) is approximate because they have different communicative structures: in (1a), the *Theme* is *lightning*, and in (1b), *thunder*. As (1a–b) shows, converse substitutions allow the Speaker to express relevant communicative information.

English

Conv ₂₁ (<i>include</i> [X is a set])	=	<i>belong</i>
Conv ₃₂₁₄ (<i>lend</i>)	=	<i>borrow</i>
[N _X lends N _Y to N _Z for period N _W]		
Conv ₃₂₁₄ ∩(<i>buy</i> _(V))	=	<i>sell</i> _(V)
[N _X buys N _Y from N _Z for N _W]		
Conv ₃₂₁₄ ∩(<i>sell</i> _(V))	=	<i>buy</i> _(V)
Conv ₁₄₃₂ ⊃(<i>buy</i> _(V))	=	<i>pay</i> _(V)
Conv ₂₁ ([<i>be</i>] <i>parent</i> _(N))	=	[<i>be</i>] <i>child</i>
Conv ₂₁₃ ([<i>be</i>] <i>teacher</i>)	=	[<i>be</i>] <i>pupil</i> [<i>Professor Wanner's pupil in mathematics</i>]

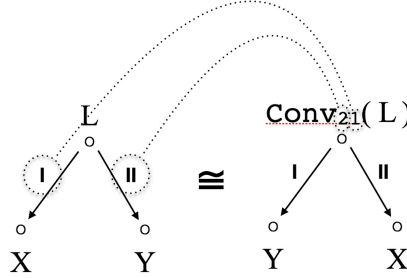
4.1.0.0.1 Warning The LF Conv_{ijk}, as in fact the majority of LFs, is based, in quite an essential way, on the notion of DSyntA, presented in Ch. 2, Sec. 2.3 (p. xxviii ff.).

The symbol Conv is supplied with a string of actantial subscripts; this string encodes the relation between the active syntactic valence of L and that of Conv_{ijk}(L). Each number in the string refers to a DSyntA of L, and this number's position in the string refers to the corresponding DSyntA of Conv_{ijk}(L). Thus, the number 2 in the **first** position of the numerical subscript of Conv₂₁ means that DSyntA II of L is expressed by DSyntA I of Conv₂₁(L); the number 1 in the **second** position of the subscript indicates that L's DSyntA I corresponds to DSyntA II of Conv₂₁(L).

Figure 4.1 visualizes the correspondence between the DSyntAs of L and those of Conv₂₁(L) under the semantic quasi-equivalence of the two deep-syntactic

4.1 Three pillars of the LF system: *Syn*, *Conv_{ijk}*, *Anti*

structures: that constituted by L with its DSyntAs and that constituted by L's *Conv₂₁*-conversive with its DSyntAs.



For instance: $X \text{ follows } Y. \equiv Y \text{ precedes}_{\text{Conv}_{21}(\text{follow})} X.$

Figure 4.1: Correspondence between L's DSyntAs and those of *Conv₂₁*(L)

French

<i>Conv₂₁</i> (<i>inclure</i> [X is a set])	=	<i>appartenir</i> , 'faire partie'
<i>Conv₃₂₁₄</i> (<i>prêter</i>)	=	<i>emprunter</i>
[N _X prête N _Y à N _Z pour la période N _W]		
<i>Conv₃₂₁₄</i> (<i>acheter</i>)	=	<i>vendre</i>
[N _X achète N _Y à N _Z pour N _W]		
<i>Conv₃₂₁₄</i> (<i>vendre</i>)	=	<i>acheter</i>
<i>Conv₁₄₃₂</i> (<i>acheter</i>)	=	<i>payer</i>
<i>Conv₂₁</i> ([être] <i>parent</i>)	=	[être] <i>enfant</i>
<i>Conv₂₁₃</i> ([être] <i>maître</i>)	=	[être] <i>élève</i> [du professeur Ménard en math]
<i>Conv₂₁</i> (<i>meilleur</i>)	=	<i>pire</i>

Russian

<i>Conv₂₁</i> (<i>soderžat'</i> [X is a set])	=	<i>prinadležat'</i>
<i>Conv₃₂₁₄</i> (<i>odolžit'</i> I [N _{DAT}])	=	<i>odolžit'</i> II [u N _{GEN}]
[N _{X-NOM} odolžil N _{Y-ACC} N _{Z-DAT} na N _{W-ACC}]		
<i>Conv₃₂₁₄</i> (<i>kupit'</i>)	=	<i>prodat'</i>
[N _{X-NOM} kupil N _{Y-ACC} u N _{Z-GEN} za N _{W-ACC}]		
<i>Conv₃₂₁₄</i> (<i>prodat'</i>)	=	<i>kupit'</i>
<i>Conv₁₄₃₂</i> (<i>kupit'</i>)	=	<i>zaplatit'</i>
<i>Conv₂₁</i> ([byt'] <i>roditel'm</i>)	=	[byt'] <i>rebënkom</i>
<i>Conv₂₁₃</i> ([byt'] <i>učitel'm</i>)	=	[byt'] <i>učenikom</i> [professora Menara po matematike]
<i>Conv₂₁</i> (<i>tjaželee</i>)	=	<i>legče</i>

[3] *Anti* [Lat. **antonymum*]: antonym

Anti(L) has a meaning "opposite" to that of L and is, therefore, a possible substitute for L in the syntactic structure of the utterance with preservation of the

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meaning, but under the condition that a negation is added to *Anti(L)* as illustrated in (2a–b).

- (2) a. Chris was **absent** from the meeting.
 ≡
 b. Chris was **not present**_{*Anti(absent)*} at the meeting.

However, such equivalence is valid only for contradictory antonyms (see type i of antonyms after the illustrations below); with other types of antonyms, the semantic equivalence is only approximate or does not obtain at all. The paraphrastic relation holding between a lexical unit *L* and *Anti(L)* + Negation is too complex to be summarized here with a single paraphrasing rule, and it is considered below separately for each of the four types of antonyms.

English

<i>Anti(respect_(N))</i>	=	<i>disrespect_(N)</i>
<i>Anti(small)</i>	=	<i>big_(Adj)</i>
<i>Anti(hot)</i>	=	<i>cold_(Adj)</i>
<i>Anti(with)</i>	=	<i>without</i>
<i>Anti(dress_(V))</i>	=	<i>undress_(V)</i>
<i>Anti_∩(build_(V))</i>	=	<i>destroy</i>
<i>Anti(Yummy!)</i>	=	<i>Yuck!</i>

French

<i>Anti(respect)</i>	=	<i>irrespect</i>
<i>Anti_⊃(mentionner)</i>	=	‘ <i>passer sous silence</i> ’
<i>Anti(petit)</i>	=	<i>grand</i>
<i>Anti(chaud)</i>	=	<i>froid</i>
<i>Anti(avec)</i>	=	<i>sans</i>
<i>Anti(habiller)</i>	=	<i>déshabiller</i>
<i>Anti_∩(construire)</i>	=	<i>détruire</i>
<i>Anti(Miam !)</i>	=	<i>Beurk !</i>

Russian

<i>Anti(uvaženie)</i>	=	<i>neuváženie</i>
<i>Anti_⊃(upomjanut’)</i>	=	‘ <i>obožiti molčaniem</i> ’
<i>Anti(malen’kij)</i>	=	<i>bol’šoj</i>
<i>Anti(gorjačij)</i>	=	<i>xolodnyj</i>
<i>Anti(s)</i>	=	<i>bez</i>
<i>Anti(odet’)</i>	=	<i>razdet’</i>
<i>Anti_∩(stroit’)</i>	=	<i>razrušat’</i>
<i>Anti(Ux_ty!)¹</i>	=	<i>Fu!</i>

¹The grey underscoring of the space between two words, as in *UX_TY*, shows that, linguistically speaking, this expression is a single wordform, which is traditionally spelled in two words (cf. *AT_ALL*, *AT_LAST* or *BY_FAR*).

Antonyms come in four types: contradictory, contrary, scalar and positioning antonyms.

Contradictory antonyms also known as *complementary*. A contradictory antonym is opposed to its “partner” by a negation of the highest level of its meaning, i.e. that applies to the central component of its lexicographic definition (Mel’čuk & Polguère 2018: Sec. 2.4.1): ‘Anti(L)’ = ‘non L’. Two mutually contradictory antonyms represent the two values of a binary category. For instance, PRESENT_(Adj) [*in a room*] is a contradictory antonym of ABSENT_(Adj) [*from a room*]:

‘[X] absent from Y’ = ‘[X] **not** present in Y’

Contradictory antonyms do not (or hardly) allow for gradation: **more* / **less present in a room*. In contrast, the remaining three types of antonym are gradable.

Contrary antonyms. Two mutually contrary antonyms are opposed by a negation that is deeply embedded in the lexicographic definition of one of them, i.e. that applies to a peripheral component. For instance, CLOSE_(V) [*a door*] is a contrary antonym of OPEN_(V) [*a door*]:

‘X opens Y’ = ‘X causes that Y becomes open_(Adj)’

‘X closes Y’ = ‘X causes that Y becomes **not** open_(Adj)’

These semantic decompositions show that ‘close_(V)’ ≠ ‘not open_(V)’ and, therefore, that the verbs CLOSE_(V) and OPEN_(V) are not contradictory antonyms.

Scalar antonyms. They are opposed by the components ‘more’ ~ ‘less’ in their lexicographic definitions: ‘more’ / ‘less’ indicates the position on a gradable scale, or on an axis. For instance, HEAVY_(Adj) and LIGHT_(Adj) are scalar antonyms:

‘X is heavy’ = ‘X’s weight is **higher** [= ‘more high’] than the normal weight of Xs’

‘X is light’ = ‘X’s weight is **lower** [= ‘less high’] than the normal weight of Xs’

The opposition between ‘more’ and ‘less’ is reducible to the opposition by negation (Mel’čuk 2015b: 190).

4.1.0.0.1 Note If two adjectives are scalar antonyms, their comparative degrees are conversives as well; thus, we have ‘X is heavier than Y’ = ‘Y is lighter than X’.

leftmivrgivn=*,font= *Positioning antonyms.* These antonyms denote two opposite spatial / temporal positions (in a literal or metaphorical sense). For instance, TOP_(N) [*the top of the shelves*] and BOT-TOM_(N) [*the bottom of the shelves*] are positioning antonyms:

‘top of X’	=	‘part of X that is the furthest [= ‘least close’] surface on which X is located’
‘bottom of X’	=	‘part of X that is the closest [= ‘most close’] to the face on which X is located’

As is seen, all types of antonyms are based on negation. It is the position of negation in the lexicographic definition of lexical units that are Anti(L) that determines the type of antonymy. In the enumeration of the four antonym types above, we started with the negation occupying the uppermost position in the definition and concluded with the negation that is the most deeply embedded.

A lexical unit can have antonyms of different types. Such is, for instance, the lexeme DESTROY, appearing in (3):

- (3) The enemy destroyed the bridge.
 ‘The enemy caused that the bridge ceased to exist’
 = ‘The enemy caused the bridge to **not**[exist]’

DESTROY has two antonyms: a contradictory one – SPARE_(V), cf. (4a), and a contrary one – BUILD_(V), cf. (4b).

- (4) a. The enemy spared the bridge.
 ‘The enemy **not**[caused that the bridge ceased to exist]’
 b. The enemy built the bridge.
 ‘The enemy caused that the bridge **not**[**not**[exist]]’
 = ‘The enemy caused that the bridge exists’
 (‘**not**[**not**[exist]]’ = ‘exist’)

For all four types of antonyms only one LF is used: Anti. Theoretically speaking, it is easy to distinguish the four Anti LFs by means of super- or subscripts – e.g. Anti^{contrad}, Anti^{contrary}, etc. This is, however, superfluous, since each antonym should have its own lexicographic description – in particular, its definition – in a lexicographic model, which allows the user of the model to see the meaning relation between L and Anti(L).

4.1.0.0.2 Note The cases where a lexical unit has antonyms of different types are infrequent enough.

Sometimes the negation underlying the antonymy has to be encoded as such in LF formulas; then the LF Non ‘not’ is used. Thus, Non – rather than Anti – combines with semantically empty LFs, such as those that represent support verbs (*Oper_i*, *Func_i* and *Labor_{ij}*, in Ch. 5, Sec. 5.4.2, p.cxxxi*ff*). And, from a logical viewpoint, a semantically empty verb cannot have an antonym: an antonym of L is supposed to have an “opposite” meaning with respect to that of L, while a light verb has no meaning. Therefore, we have:

- (5) *Oper₁*(*consent_(N)*) = *give, grant* [ART ~]
NonOper₁(*consent_(N)*) = *refuse* [ART ~]

4.1.0.0.3 Historical remark The previous presentations of LFs – for instance, Mel’čuk et al. (1995: 125–152) and Mel’čuk et al. (1999: 75–81) – did not introduce the LF Non explicitly. In these publications the LF Non (as Non or non) was simply smuggled into complex LF formulas.

4.2 Three LFs related to *Syn* and *Anti*: *Gener*, *Figur*, *Contr*

[4] ***Gener*** [*Lat. genus*]: generic term

Gener(L) – a generic term for L – stands for the name of one of the semantic classes to which L’s meaning may belong. It applies to L of any PoS and, like *Syn*, it borrows L’s PoS:

Gener: borrows L’s PoS					
L	V	N	Adj	Adv	Claus
Gener(L)	V	N	Adj	Adv	Claus[?]

English

- Gener*(*love_(N)*) = *feeling* [of ~]; *emotion* [of ~]
Gener(*gas_(N)*) = [*gaseous*] *substance*
Gener(*see_(V)*) = *perceive* [*visually*]

French

- Gener*(*amour*) = *sentiment* [*d’~*]; *émotion*
Gener(*gaz*) = *substance* [*gazeuse*]
Gener(*voir*) = *percevoir* [*visuellement*]

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Russian

Gener(<i>ljub ov'</i>)	=	čuvstvo [-vi]; <i>emocija</i>
Gener(<i>gaz</i>)	=	[gazoobraznoe] <i>veščestvo</i>
Gener(<i>videt'</i>)	=	<i>vosprinimat'</i> [zritel'no]

4.2.0.0.1 Identifying Gener(L) For a lexical unit to be an element of the value of Gener(L) it is necessary and sufficient to appear in the following constructions:

- for nouns: L, L', L'', ... *and other* Gener(L)

(6) *melons, raspberries, mangoes, ... and other fruits*_{Gener(melon)}

- for verbs, adjectives and adverbs: L, L', L'', ... *and* Gener(L) *in other ways*

(7) a. *see, smell, ... and perceive*_{Gener(see)} *in other ways*
b. *fried, boiled, ... and cooked*_{Gener(fried)} *in other ways*

These examples show that Gener is a powerful classifier: it identifies numerically important semantic classes of lexical units. For instance:

- the noun DOG is Gener of BASSET, BULLDOG, COLLIE, DACHSHUND, DALMATIAN, 'GERMAN SHEPHERD', HUSKY, LABRADOR, POODLE, SETTER, ...
- the noun FEELING is Gener of ADORATION, DESPAIR_(N), DREAD_(N), EXCITEMENT, FEAR_(N), GRATITUDE, LOVE_(N), REMORSE, ...

The compatibility of a Gener with the above constructions allows for the distinction between Gener and Syn_C. Thus, DREAM_(N) is a Syn_C of NIGHTMARE, rather than a Gener, because English does not have an important enough class of lexical units denoting different types of dreams.

4.2.0.0.2 Gener-based paraphrasing The lexical stock and the lexical cooccurrence of the language allowing, a given Gener(L) can be used in the two following equivalences grouped under paraphrasing rule **R_{Gener}**.²

²These equivalences are described by two distinct paraphrasing rules (Rules 4 and 5) in Mel'čuk (2013a: p. 152).

R_{Gener}:	a) $L_{(N)} \equiv \text{Gener}(L_{(N)}) - r \rightarrow L_{(N)}$ $\text{anger}_{(N)} \equiv \text{feeling}_{\text{Gener}(\text{anger}_{(N)})} \text{ of anger}$ $\text{submission} \equiv \text{attitude}_{\text{Gener}(\text{submission})} \text{ of submission}$ $\text{smell}_{(N)} \equiv \text{sense}_{\text{Gener}(\text{smell}_{(N)})} \text{ of smell}$ b) $L \equiv \text{Gener}(L) - r \rightarrow A_0 / \text{Adv}_0 / \text{Adv}_1(L)$ For $L_{(N)}$: $\text{anger}_{(N)} \equiv \text{angry}_{A_0(\text{anger}_{(N)})} \text{ feeling}_{\text{Gener}(\text{anger}_{(N)})}$ For $L_{(V)}$: $\text{see} \equiv \text{perceive}_{\text{Gener}(\text{see})} \text{ visually}_{\text{Adv}_1(\text{see})}$ For $L_{(\text{Adj})}$: $\text{visual} \equiv \text{visually}_{\text{Adv}_0(\text{visual})} \text{ perceptible}_{\text{Gener}(\text{visual})}$ See below for the description of the LFs A_0 (Sec. 4.3, [9], p. lxxxvi), Adv_0 (Sec. 4.3, [10], p. lxxxvi) and Adv_1 (Sec. 4.7, [24], p. ci).
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4.2.0.0.3 Note In subrules **a)** and **b)** above, the actantial deep-syntactic relation **R** on the right-hand side of the equivalence remains unspecified, as it varies according to the semantic valence and government pattern of each particular $\text{Gener}(L)$; for instance:

- (8) a. $\text{COLOR}_{(N)}$, ‘color of X that is Y’: $\text{the color}_{\text{Gener}(\text{purple}_{(N)})} - \text{II} \rightarrow \text{purple}$
b. FEELING (‘feeling of X towards Y that is Z’): $\text{the feeling}_{\text{Gener}(\text{love}_{(N)})} - \text{III} \rightarrow [\text{of}] \text{ love}$
c. $\text{NUMBER}_{(N)}$: $\text{the number}_{\text{Gener}(\text{nine})} - \text{ATTR} \rightarrow \text{nine}$

4.2.0.0.4 Multiplicity of the values of $\text{Gener}(L)$ A lexical unit L can have multiple elements of the value for $\text{Gener}(L)$. This happens in two cases.

The first case is trivial. As with any other LF, there can be multiple elements of the value of $\text{Gener}(L)$ that are (quasi-)synonyms; for instance:

- (9) $\text{Gener}(\text{fear}_{(N)}) = \text{feeling} [\text{of } \sim]; \text{emotion} [\text{of } \sim]$
 $[\text{Syn}_{\supset}(\text{feeling}) = \text{emotion}]$
 $\text{Gener}(\text{bin}_{(N)}) = \text{container}; \text{receptacle}$
 $[\text{Syn}_{\cap}(\text{container}) = \text{receptacle}]$

The second case is more remarkable as it is conditioned by the special character of L ’s meaning. Two subcases are distinguished:

- L denotes more than one kind of things and, therefore, L belongs to more than one semantic class. This phenomenon corresponds to L ’s semantic ambivalence (Milićević & Polguère 2010, Smirnova & Tolochin 2018), that is, to the presence of a disjunction in L ’s definition.³

³See also the related notions of *facet* / *subsense* (Cruse 1995) and *logical polysemy* (Pustejovsky 1995).

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- L's definition contains a classifying component – called *semantic-class marking component* (Mel'čuk & Polguère 2018: Sec. 2.4.1.2) – that underlies a value of $\text{Gener}(L)$. If, at the same time, the central component of L's definition underlies also a value of $\text{Gener}(L)$, which is a prototypical case, L obtains more than one type of $\text{Gener}(L)$.

Let us take an example: the definition of the lexeme $\text{SNOW}_{(N)}$, which illustrates both situations.

<i>snow on Y</i> :	frozen water that is falling or has fallen from the sky on geographical place Y
	• in the form of white flakes
	or this falling
	• which is a weather phenomenon

Figure 4.2: Lexicographic definition of $\text{SNOW}_{(N)}$

In the above definition, the presence of a disjunction in the central component ('or') and the semantic-class marking component ('which is a weather phenomenon') combine to produce two types of $\text{Gener}(\text{snow}_{(N)})$:

- (10) $\text{Gener}(\text{snow}_{(N)}) =$ (i) *substance*;
(ii) *precipitation*, 'weather condition'

Example (10) presupposes that we are dealing with one single sense of $\text{SNOW}_{(N)}$, that is, with one lexeme, and not with a case of polysemy, that is, with two lexemes – one denoting a substance and one denoting the event of its falling. This can be demonstrated using the "Unifying cooccurrence" criterion presented in Mel'čuk (2013a: Ch. 11, Sec. 3.2.2). Indeed, a given occurrence of $\text{snow}_{(N)}$ in an utterance can be used to refer to both substance and event simultaneously:

- (11) The snow that **started** [snow as precipitation] this morning
is beautifully **fluffy** [snow as substance].

For this reason, both types of denotation of $\text{SNOW}_{(N)}$ (substance and meteorological event) have to be considered as belonging to one lexical unit, with a disjunctive central component in its definition.

The classifying power of *Gener* makes this LF particularly important in the structuring of natural language lexical networks and in the description of lexical regularities – see Ch. 3, Sec. 3.6, paragraph "2) Generalizations associated with a generic term" (p. lxi).

FURTHER READING. For more on generic terms and lexicographic definitions, see Goddard (2017).

[5] **Figur** [*Lat. figuraliter* ‘figuratively’]: **figurative term**

Figur(L) is a metaphorical expression having the same meaning as the noun L plus an “artistic representation” of L; it is either a lexical unit or a collocation whose base is L and the collocater, a noun that takes L as its DSyntA 1. The relation of paraphrase between L and Figur(L) is accounted for by the R_{Figur} paraphrasing rule:

$$R_{\text{Figur}}: L \cong \text{Figur}(L)$$

The equivalence established by this rule is approximate because the use of Figur(L) necessarily adds to the meaning of L a rhetorical content, as illustrated by (12a–b):

- (12) a. He never experienced **passion**.
 $\cong$
 b. He never experienced the **flame**_{Figur(*passion*)} of **passion**.

Figur: nominal	
L	N
Figur(L)	N

Being a figurative **substitute** explains the status of Figur as a paradigmatic (rather than syntagmatic) LF. It is used for stylistic effects, and for this reason it is close to the syntagmatic LF Epit – Ch. 5, [5.2.0.1], p. cxiv. See also the remark on the syntactic similarity of Figur to two other syntagmatic LFs: Germ and Culm – Ch. 5, [5.1.0.1-5.1.0.2], p. cxiii.

English

Figur([bubonic] <i>plague</i> _(N))	=	‘ <i>Black Death</i> ’
Figur(<i>power</i> _(N))	=	<i>reins of</i> ~
Figur(<i>passion</i>)	=	<i>flame of</i> ~
Figur(<i>fog</i> _(N))	=	<i>curtain of</i> ~

French

Figur(<i>pétrole</i>)	=	‘ <i>or noir</i> ’
Figur(<i>pouvoir</i>)	=	<i>rênes du</i> ~
Figur(<i>passion</i>)	=	<i>flamme de la</i> ~
Figur(<i>brouillard</i>)	=	<i>rideau de</i> ~

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Russian

Figur(<i>vlast'</i>)	=	「 <i>brazdy pravlenija</i> 」
Figur(<i>strast'</i>)	=	<i>plamja</i> <i>..i</i>
Figur(<i>tuman</i>)	=	<i>pelena</i> <i>..a</i>

[6] Contr [Lat. *contrarium*]: contrastive term

Contr(L) is a lexical entity that expresses a meaning that is in contrastive opposition with that of L, while not being an Anti of the latter.

The contrastive opposition of L and the value of Contr(L) is not just conceptual, but truly lexical – i.e., language-dependent. In particular, it has to be manifested in phrasemes of the language, such as the bolded idiom in (13) below for ICE_(N) and its Contr FIRE_(N).

- (13) They are 「**like ice and fire**」, but friends nonetheless.

Contr: borrows L's PoS					
L	V	N	Adj	Adv	Claus
Contr(L)	V	N	Adj	Adv	Claus

English

Contr(<i>sea</i>)	=	<i>land</i> _(N)
Contr(<i>ice</i> _(N))	=	<i>fire</i> _(N)
Contr(<i>black</i> _(Adj))	=	<i>white</i> _(Adj)
Contr(<i>here</i>)	=	<i>there</i>

French

Contr(<i>manger</i> _(V))	=	<i>boire</i> _(V)
Contr(<i>terre</i>)	=	<i>mer</i>
Contr(<i>eau</i>)	=	<i>feu</i>
Contr(<i>nous</i>)	=	<i>eux</i>
Contr(<i>noir</i> _(Adj))	=	<i>blanc</i> _(Adj)
Contr(<i>ici</i>)	=	<i>là</i>

Russian

Contr(<i>more</i>)	=	<i>suša</i>
Contr(<i>lěd</i>)	=	<i>plamen'</i>
Contr(<i>my</i>)	=	<i>oni</i>
Contr(<i>čěrnij</i>)	=	<i>belyj</i>
Contr(<i>zdes'</i>)	=	<i>tam</i>

4.3 Structural (= syntactically motivated) derivatives

The following five LFs – S₀, V₀, A₀, Adv₀ and Claus₀ – represent the five possible derivations that modify L's deep part of speech, but do not entail semantic increments to L: nominalization, verbalization, adjectivization, adverbialization and

4.3 Structural (= syntactically motivated) derivatives

clausativization. The “₀” subscript in the name of these LFs indicates that they are structural (purely syntactic) derivatives.

[7] **S₀** [Lat. *substantivum*]: nominalization

S₀ applies to a non-nominal lexical unit L and returns a noun having the same meaning as L.

S ₀ : nominal				
L	V	Adj	Adv	Claus
S ₀ (L)	N	N	N	N

English

S ₀ (<i>present</i> _(V))	=	<i>presentation</i>
S ₀ (<i>leave</i> _(V))	=	<i>departure</i>
S ₀ (<i>fall</i> _(V))	=	<i>fall</i> _(N)
S ₀ (<i>close</i> _(Adj))	=	<i>proximity</i>
S ₀ (<i>Bang!</i>)	=	<i>gunshot</i>

French

S ₀ (<i>présenter</i>)	=	<i>présentation</i>
S ₀ (<i>partir</i>)	=	<i>départ</i>
S ₀ (<i>tomber</i>)	=	<i>chute</i>
S ₀ (<i>proche</i>)	=	<i>proximité</i>
S ₀ (<i>près</i>)	=	<i>proximité</i>
S ₀ (<i>Pan !</i>)	=	‘ <i>coup de feu</i> ’

Russian

S ₀ (<i>predstavljat'</i>)	=	<i>predstavlenie</i>
S ₀ (<i>uezžat'</i>)	=	<i>ot'ezd</i>
S ₀ (<i>padat'</i>)	=	<i>padenie</i>
S ₀ (<i>blizkij</i>)	=	<i>blizost'</i>
S ₀ (<i>rjadom</i>)	=	<i>blizost'</i>
S ₀ (<i>child language Pif-paf!</i>)	=	<i>vystrel</i>

[8] **V₀** [Lat. *verbum*]: verbalization

V₀ applies to a non-verbal lexical unit L and returns a verb having the same meaning as L.

V ₀ : verbal				
L	N	Adj	Adv	Claus
V ₀ (L)	V	V	V	V

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Obviously, only lexical units being genuine predicates (that is, denoting facts) can have a V_θ .

English	
$V_\theta(\textit{presentation})$	= <i>present</i> _(V)
$V_\theta(\textit{oath})$	= <i>swear</i> _(V)
$V_\theta(\textit{compliment}_{(N)})$	= <i>compliment</i> _(V)
$V_\theta(\textit{age}_{(N)})$	= <i>be</i> [NUM + N] _Y <i>old</i> N denotes a time unit – e.g., MONTH, YEAR, CENTURY [<i>She is 33 years old.</i>]
$V_\theta(\textit{stinking}_{(Adj)})$	= <i>stink</i> _(V)
$V_\theta(\textit{energetic})$	= colloquial <i>‘be full of beans’</i>
$V_\theta(\textit{impatient})$	= colloquial <i>‘champ at the bit’</i>
$V_\theta(\textit{Bang!})$	= <i>fire</i> _(V)
$V_\theta(\textit{Wow!})$	= <i>be impressed</i>
French	
$V_\theta(\textit{présentation})$	= <i>présenter</i>
$V_\theta(\textit{serment})$	= <i>jurer</i>
$V_\theta(\textit{erreur})$	= <i>se tromper</i>
$V_\theta(\textit{âge})$	= <i>avoir</i> [NUM + N] _Y N denotes a time unit – e.g., MOIS, AN, SIÈCLE [<i>Elle a 33 ans.</i>]
$V_\theta(\textit{colloquial puant})$	= colloquial <i>puer</i>
$V_\theta(\textit{énergique})$	= colloquial <i>‘avoir bouffé du lion’</i>
$V_\theta(\textit{impatient})$	= <i>s’impatier</i> ; <i>‘prendre le mors aux dents’</i> , <i>‘ronger son frein’</i>
$V_\theta(\textit{Pan !})$	= <i>‘faire feu’</i>
$V_\theta(\textit{Waouh !})$	= <i>être en admiration</i>
Russian	
$V_\theta(\textit{predstavlenie})$	= <i>predstavljat’</i>
$V_\theta(\textit{kljatva})$	= <i>pokljast’ sja</i>
$V_\theta(\textit{ošibka})$	= <i>ošibit’ sja</i>
$V_\theta(\textit{vozrast})$	= [N _{X-DAT}] <i>byt’</i> [NUM + N _Y] N _Y denotes a time unit – e.g., MESJAC, GOD, VEK [<i>Ej 33 goda.</i>]
$V_\theta(\textit{vonjučij})$	= <i>vonjat’</i>
$V_\theta(\textit{ščastlivyj})$	= <i>‘byt’ na sed’ mom nebe’</i>
$V_\theta(\textit{child language Pif-paf!})$	= <i>vystrelit’</i>
$V_\theta(\textit{Ux ty!})$	= <i>byt’ v vostorge</i>

[9] A_θ [Lat. *adjectivum*]: adjektivization

A_θ applies to a non-adjectival lexical unit L and returns an adjective having the same meaning as L.

A_θ : adjectival				
L	V	N	Adv	Claus
$A_\theta(L)$	Adj	Adj	Adj	Adj

4.3 Structural (= syntactically motivated) derivatives

English

$A_0(\text{compare})$	=	<i>comparative</i>
$A_0(\text{time}_{(N)})$	=	<i>temporal</i>
$A_0(\text{school}_{(N)})$	=	<i>scholar</i>
$A_0(\text{rotation})$	=	<i>gyratory</i>
$A_0(\text{rapidly})$	=	<i>rapid</i> _(Adj)
$A_0(\text{closely})$	=	<i>close</i> _(Adj)

French

$A_0(\text{comparer})$	=	<i>comparatif</i>
$A_0(\text{temps})$	=	<i>temporel</i>
$A_0(\text{port})$	=	<i>portuaire</i>
$A_0(\text{rotation})$	=	<i>giratoire</i>
$A_0(\text{rapidement})$	=	<i>rapide</i> _(Adj)
$A_0(\text{près})$	=	<i>proche</i>

Russian

$A_0(\text{sravnivat'})$	=	<i>sravnitel'nyj</i>
$A_0(\text{vremja})$	=	<i>vremennoj</i>
$A_0(\text{port})$	=	<i>portovyj</i>
$A_0(\text{vraščenie})$	=	<i>vraščatel'nyj</i>
$A_0(\text{bystro})$	=	<i>bystryj</i>
$A_0(\text{rjadom})$	=	<i>blizkij</i>

Contrary to $S_0(L)$ and $V_0(L)$, the LF application $A_0(L)$ returns an adjective that does not have exactly the same meaning as L . This is so for the following reason: $A_0(L)$ is used as adjectival modifier of a noun – for instance, *the temporal coordinate* means ‘the coordinate relative to time’;⁴ and the syntactic relation of modification (deep-syntactic relation **ATTR**) between the lexemes COORDINATE and TEMPORAL [= $A_0(\text{time})$] expresses the semanteme ‘relative [to]’, which connects ‘coordinate’ and ‘time’, as shown in Figure 4.3.

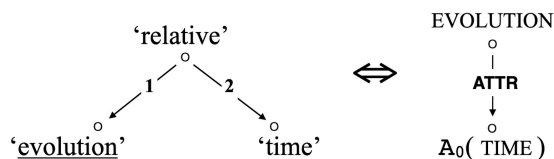


Figure 4.3: Semantic content carried by an A_0 syntactic modifier

The semanteme ‘relative [to]’ is always included in the meaning of A_0 . Given this semantic increment, it is not the case that $\text{TIME}_{(N)}$ is S_0 of TEMPORAL; the S_0

⁴On the lexicographic definition of lexical units that are A_0 s, see Mel’čuk & Polguère (2018: Sec. 4.1.1).

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of TEMPORAL is TEMPORALITY. The correspondences between these three lexemes are illustrated in paraphrases (14a–c).

- (14) a. We measure the dimension of existence relative to **time**.
b. We measure the **temporal** dimension of existence.
c. We measure the dimension of existence relative to its **temporality**.

The LF that associates TEMPORAL to TEMPORALITY is called A_1 – an adjective characterizing the first deep-syntactic actant of L (that is, of TEMPORALITY): $A_1(\text{TEMPORALITY}) = \text{TEMPORAL}$. The LF A_1 is presented below (Sec. 4.6, [21], p. xcix).

The lexical triplet $\text{TIME}_{(N)}$, TEMPORAL and TEMPORALITY forms a configuration of lexical links that is visualized in Figure 4.4. This type of configuration is highly recurrent in the lexical network of English and other languages.

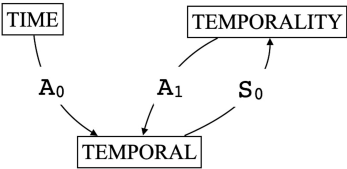


Figure 4.4: Typical configuration of lexical links for $A_0 \sim S_0 \sim A_1$

[10] **Adv_θ** [Lat. *adverbium*]: adverbialization

Adv_θ applies to a non-adverbial lexical unit L and returns an adverb having the same meaning as L.

Adv _θ : adverbial				
L	V	N	Adj	Claus
Adv _θ (L)	Adv	Adv	Adv	Adv

English

- Adv_θ(*see*_(V)) = visually [*horrible to see* = visually horrible]
Adv_θ(*vision*) = visually
Adv_θ(*alphabet*) = alphabetically
Adv_θ(*rapid*_(Adj)) = rapidly
Adv_θ(*close*_(Adj)) = closely

French

- Adv_θ(*voir*) = visuellement [*insupportable à voir* = visuellement insupportable]
Adv_θ(*vue*) = visuellement

4.3 Structural (= syntactically motivated) derivatives

$\text{Adv}_0(\textit{alphabet})$	=	<i>alphabétiquement</i>
$\text{Adv}_0(\textit{rapide}_{(\text{Adj})})$	=	<i>rapidement</i>
$\text{Adv}_0(\textit{proche})$	=	<i>près</i>

Russian

$\text{Adv}_0(\textit{videt}')$	=	「 <i>na vid</i> 」 [<i>protivno videt</i> = <i>protivnyj</i> 「 <i>na vid</i> 」]
$\text{Adv}_0(\textit{zrenie})$	=	<i>zritel'no</i>
$\text{Adv}_0(\textit{alfavit})$	=	<i>po alfavitu</i>
$\text{Adv}_0(\textit{bystryj})$	=	<i>bystro</i>
$\text{Adv}_0(\textit{blizkij})$	=	<i>blizko</i>

The Adv_0 LF should not be confounded with another adverbialization LF, namely Adv_1 , which is examined below (Sec. 4.7, [24], p. ci):

- (15) Eng. $\text{Adv}_1(\textit{see} [\text{N}_Y])$ = *at the sight* [*of N_Y*] ($\equiv$ *seeing* [*N_Y*])
 Fr. $\text{Adv}_1(\textit{voir} [\text{N}_Y])$ = *à la vue* [*de N_Y*]
 Rus. $\text{Adv}_1(\textit{videt}' [\text{N}_{Y-\text{ACC}}])$ = *pri vide* [*N_{Y-GEN}*]

[11] Claus_0 [_{Lat.} **clausativum*]: clausativization

Claus_0 applies to a non-clausative lexical unit L and returns a clausative having the same meaning as L.

$\text{Claus}_0(\text{L})$ is a clausative lexical unit – that is, a lexical unit syntactically equivalent to a sentence (cf. Ch. 3, Sec. 3.5.1, p. lviii). Claus_0 turns the meaning of its lexical argument L into the meaning of an utterance communicatively dominated by the semantic component ‘I signal [that L ...]’.⁵ In this respect, Claus_0 is not a pure structural derivation, just like A_0 (see above, p. lxxxvii); it adds a piece of communicative information (‘I signal [that L ...]’) to the derivative.

Claus_0 : clausative				
L	V	N	Adj	Adv
$\text{Claus}_0(\text{L})$	Claus	Claus	Claus	Claus

Claus_0 should not be confounded with the Imper LF (Sec. 4.8, [27], p. cvi), whose values are syntactically also clausatives, but which carry an additional meaning: ‘I signal that I want you to do [L]’.

English

$\text{Claus}_0(\textit{congratulations})$	=	<i>Congratulations!</i> , <i>Congrats!</i>
--	---	--

⁵*To signal* is a technical term (Mel’čuk 2001: 249–250): in contrast to *to communicate*, it denotes an enunciation mode that precludes negation and interrogation. A signaled expression does not state something about the world, but manifests an internal (mental, emotional or physiological) state of the Speaker.

4 Paradigmatic lexical functions

$\text{Claus}_\theta(\text{meow}_{(V)})$	=	<i>Meow!</i>
$\text{Claus}_\theta\text{Son}(\text{cat})$	=	<i>Meow!</i>

Note. For the LF Son, see Ch. 5, Sec. 5.4.7, [5.4.7.1], p. cliv.

$\text{Claus}_\theta(\text{explosion})$	=	<i>Boom!</i>
$\text{Claus}_\theta(\text{gunshot})$	=	<i>Bang!, Pow!</i>
$\text{Claus}_\theta(\text{informal yucky})$	=	<i>Yuck!</i>

French

$\text{Claus}_\theta(\text{féliciter})$	=	<i>(Mes) Félicitations !; Chapeau !</i>
$\text{Claus}_\theta(\text{miauler})$	=	<i>Miaou !</i>
$\text{Claus}_\theta\text{Son}(\text{chat})$	=	<i>Miaou !</i>
$\text{Claus}_\theta\text{Son}(\text{chute})$	=	<i>Badaboum !</i>
$\text{Claus}_\theta(\text{explosion})$	=	<i>Boum !</i>
$\text{Claus}_\theta(\text{「coup de feu」})$	=	<i>Pan !</i>
$\text{Claus}_\theta(\text{dégoûtant})$	=	<i>Beurk !</i>

Russian

$\text{Claus}_\theta(\text{pozdravlenije})$	=	<i>Pozdravljaju!</i>
$\text{Claus}_\theta(\text{mjaukat'})$	=	<i>Mjau!</i>
$\text{Claus}_\theta\text{Son}(\text{koška})$	=	<i>Mjau!</i>
$\text{Claus}_\theta\text{Son}(\text{padenie})$	=	<i>Bax!</i>
$\text{Claus}_\theta(\text{vystrel})$	=	<i>child language Pif-paf!</i>
$\text{Claus}_\theta(\text{protivnyj})$	=	<i>Fi!; Fu!</i>

4.3.0.0.1 Historical remark The Claus_θ LF was introduced in Mel'čuk & Polguère (2021) in order to better dovetail the set of structural derivative LFs with that of deep parts of speech. Claus_θ cancels the “neological” LF Enun, which was used in the *DiCo* lexicographic project (Polguère 2000b).

4.4 Predicative derivatives

[12] **Pred** [_{Lat.} **praedicātum* '[a] predicate']: predicativization

$\text{Pred}(L)$ is a verbal lexical unit that means ‘ $\text{be}_{(\text{copul})} L$ ’; it serves to turn a predicative noun, an adjective or an adverb L into a verb. Thus, Pred plays the role of the copula; it is the paradigmatic counterpart of the syntagmatic LF Copul (Ch. 5, [5.4.1.1], p. cxxx).

Note that for adjectives and adverbs, the LF Pred is semantically equivalent to V_θ (see above, [8], p. lxxxv):

$\text{Pred}(\text{unexpected}) = V_\theta(\text{unexpected}) = \text{「come as a bolt from the blue」}$.

For quasi-predicative nouns denoting Agents, Pred is semantically equivalent to Fact_θ (Ch. 5, [5.4.3.2], p. cxlii):

$\text{Pred}(\text{cook}_{(N)}) = \text{Fact}_\theta(\text{cook}_{(N)}) = \text{cook}_{(V)}$ ‘be a cook’.

Pred: verbal			
L	$N_{(\text{predic})}$	Adj	Adv
Pred(L)	V	V	V

English

Pred(*beggar*) = *beg* ‘be a beggar’ [*He begs for a living.*]
 Pred(*cook*_(N)) = *cook*_(V) ‘be a cook’ [*He cooks in a starred restaurant.*]
 Pred(*unexpected*) = ‘*come as a bolt from the blue*’
 Pred(*inseparable*) = ‘*be (as) thick as thieves*’, ‘*be joined at the hip*’

French

Pred(*mendiant*) = *mendier*
 Pred(*cuisinier*) = *cuisiner*
 Pred(*aisé* [financièrement]) = ‘*avoir les moyens*’
 Pred_⊃(*riche*_(Adj)) = **colloquial** ‘*péter dans la soie*’
 Pred(*inséparables*_(Adj)) = ‘*être comme les (cinq) doigts de la main*’,
 ‘*être comme les deux doigts de la main*’

Russian

Pred(*niščij*) = *niščenstvovat’*
 Pred(*direktor*) = *direktorstvovat’*
 Pred_⊃(*bogatyj*) = ‘*katat’ sja kak syr v masle*’
 Pred(*neožidannyj*) = **literary** ‘*pojavit’ sja kak čert iz tabakerki*’
 Pred(*nerazlučnyj*) = **colloquial** ‘*[N_{X-ACC}] vodoj ne razol’ eš’*’

4.4.0.0.1 Pred in configurational / complex LFs Pred is mostly combined with adjectival / adverbial LFs Magn, Ver and Bon (Ch. 5, Sec. 5.2, [5.2.0.3 / ?? / ??], p. cxviff.) to form configurational or complex LFs.

- (16) Eng. Magn+Pred(*angry*) = *fume*_(V)
 Eng. Magn+Pred(*hungry*) = *starve*
 Fr. Magn+Pred(*chauve*_(N) ‘bald person’) = **vulgar**
 ‘*avoir une perruque en peau de fesse*’
 lit. ‘have a wig made of buttock skin’
 Eng. PredMagn(*animosity*) =
 Magn+Func_θ(*animosity*) = ‘*run rampant*’
 Eng. PredVer(*dress*_(N)) = *fit*_(V); ‘*sit well*’ [with N]

Note that cases of the configurational **paradigmatic** LF Magn+Pred were mistakenly described as cases of the complex **syntagmatic** LF PredMagn in Mel’čuk (2015b: 216–217). For more on the use of Pred in complex LFs, see the description of the syntagmatic LF Plus (Ch. 5, Sec. 5.2, [5.2.0.6], p. cxxii).

4.5 Nominal derivatives

Nominal derivatives fall into two classes: actantial (4.5.1) vs. circumstantial (4.5.2). Note that adjectival (4.6) and adverbial (4.7) derivatives are only actantial.

4.5.1 Nominal actantial derivatives

This section presents, in the first place, the LFs $S_1, S_2, S_3, \dots$ – for short, S_i . S_i applies to a predicative or quasi-predicative lexical unit and returns the name for its DSyntA $\mathbf{i}, \mathbf{II}, \mathbf{III}, \dots$

Also included in this section are two more specific LFs intimately related to S_i : Equip and Cap.

[13] S_i : typical name for L’s DSyntA $\mathbf{i}$

$S_i(L)$ is a nominal lexical entity that is the name for L’s DSyntA $\mathbf{i}, \mathbf{II}, \mathbf{III}, \dots$

S_i : nominal					
L	V	N	Adj	Adv	Claus
$S_i(L)$	N	N	N	N	N

English

- $S_1(sell_{(V)})$ = *seller*
[N_X sells N_Y to N_Z for N_W]
- $S_2(sell_{(V)})$ = *merchandise*
- $S_3(sell_{(V)})$ = *buyer*
- $S_4(sell_{(V)})$ = *price*
- $S_1(chemistry)$ = *chemist*
[*chemistry done by N_X*]
- $S_1(mute_{(Adj)})$ = *mute_(N)*
[N_X is *mute*]
- $S_1(lazy)$ = *colloquial lazybones*
[N_X is *lazy*]

French

- $S_1(vendre)$ = *vendeur*
- $S_2(vendre)$ = *marchandise*
- $S_3(vendre)$ = *acheteur*
- $S_4(vendre)$ = *prix*
- $S_1(chimie)$ = *chimiste*
- $S_1(muet_{(Adj)})$ = *muet_(N)*

Russian

$S_1(\textit{prodavat'})$	=	<i>prodavec</i>
$S_2(\textit{prodavat'})$	=	<i>tovar</i>
$S_3(\textit{prodavat'})$	=	<i>pokupatel'</i>
$S_4(\textit{prodavat'})$	=	<i>cena</i>
$S_1(\textit{ximija})$	=	<i>ximik</i>
$S_1(\textit{nemoj}_{(Adj)})$	=	<i>nemoj}_{(N)}</i>

[14] Equip [_{Fr} *équipage* ‘crew’]: team that allows L’s referent to function

Equip(L) is a nominal lexical entity that designates an organized set of individuals (or, more generally, of living beings) involved in the functioning of L’s referent; in other words, Equip(L) designates the set of L’s DSyntAs *ι*. Consequently, L is a quasi-predicative noun denoting a functioning entity.

Equip: nominal	
L	N
Equip(L)	N

English

Equip(<i>ship</i> _(N))	=	<i>crew</i>
Equip(<i>theater</i>)	=	<i>company</i>
Equip(<i>newspaper</i>)	=	‘ <i>editorial staff</i> ’

French

Equip(<i>navire</i>)	=	<i>équipage</i>
Equip(<i>théâtre</i>)	=	<i>troupe</i>
Equip(<i>journal</i>)	=	<i>rédaction</i>

Russian

Equip(<i>sudno</i>)	=	<i>èkipaž</i>
Equip(<i>teatr</i>)	=	<i>truppa</i>
Equip(<i>gazeta</i>)	=	<i>redakcija</i>

According to its nature, Equip applies only to lexical units whose DSyntA *ι* denotes a participant – $S_1(L)$ – that makes the entity L function. In other words, Equip means ‘set of S_1 ’; compare the above English illustrations of Equip with (17):

- (17) $S_1(\textit{ship}_{(N)})$ = *sailor* [(*ship*’s) *crew* = set of sailors]
 $S_1(\textit{theater})$ = *actor* [(*theater*’s) *company* = set of actors]
 $S_1(\textit{newspaper})$ = *journalist* [‘*editorial staff*’ = set of journalists]

The meaning ‘set [of Xs]’ is represented by the LF Mult (Sec. 4.8, [26], p. cvi). It is thus possible to propose a paraphrasing rule **R_{Equip}** that states the following semantic equivalence:

4 Paradigmatic lexical functions

$\mathbf{R}_{\text{Equip}}: \text{Equip}(\mathbf{L}) \equiv \text{MultS}_1(\mathbf{L})$ $\text{crew [of a ship]} \equiv \text{set of sailors}_{\text{S}_1(\text{ship})} [\text{of a ship}]$

We can foresee a possible generalization of the LF Equip in order to cover cases corresponding to MultS_2 , MultS_3 , etc. With such a generalization, it would be necessary to distinguish Equip_1 , Equip_2 , Equip_3 , etc.

[15] **Cap** [_{Lat.} *caput* ‘head’]: person in charge of L’s referent

$\text{Cap}(\mathbf{L})$ is a nominal lexical entity that designates the individual in charge of a functioning entity $\mathbf{L}$ (in fact, in charge of the set of $\mathbf{L}$ ’s DSyntAs $\mathbf{I}$). As for Equip, $\mathbf{L}$ is a quasi-predicative noun.

Cap: nominal	
L	N
Cap(L)	N

$\text{Cap}(\mathbf{L})$ denotes a specific element of the set of $\text{S}_1(\mathbf{L})$. Two cases have to be distinguished:

1. if $\mathbf{L}$ denotes a group of individuals (e.g. a tribe, a pack of animals, etc.), $\text{Cap}(\mathbf{L})$ denotes the main member of this group;
2. if $\mathbf{L}$ denotes a functioning entity (e.g. a plane, a school, etc.) run by a group of individuals – that is, run by $\text{Equip}(\mathbf{L})$, $\text{Cap}(\mathbf{L})$ is the main member of $\text{Equip}(\mathbf{L})$.

English

Cap(<i>kingdom</i>)	=	<i>king</i>
Cap(<i>mafia</i>)	=	<i>Don, godfather</i>
Cap(<i>ship</i> _(N))	=	<i>captain</i> ₁
Cap(<i>plane</i> _(N))	=	<i>captain</i> ₂
Cap(<i>school</i> _(N))	=	<i>principal</i> _(N)
Cap(<i>tribe</i>)	=	<i>chief (of ART ~)</i>
Cap(<i>[wolf] pack</i> _(N))	=	<i>leader (of ART ~)</i>

French

Cap(<i>royaume</i>)	=	<i>roi</i>
Cap(<i>mafia</i>)	=	<i>parrain</i>
Cap(<i>navire</i>)	=	<i>capitaine</i> ₁
Cap(<i>avion</i>)	=	<i>capitaine</i> ₂ , <i>commandant (de bord)</i>
Cap(<i>gare</i>)	=	‘ <i>chef de gare</i> ’

Cap(*tribu*) = *chef* 1 (*de* (ART) ~)
 Cap(*meute*) = *chef* 2 (*de* (ART) ~)

Russian

Cap(*korolevstvo*) = *korol'*
 Cap(*mafija*) = *boss*
 Cap(*sudno*) = *kapitan*
 Cap(*samolët*) = 'komandir korablja'
 Cap(*plem|ja*) = *vožd'* (..eni)
 Cap(*sta|ja*) = *vožak* (..i)

4.5.2 Nominal circumstantial derivatives

As seen above, S_i applies to a predicative or quasi-predicative lexical unit L and returns the general name for $DSyntA_i$ of L . In contrast, the next five LFs – S_{loc} , S_{instr} , S_{med} , S_{res} and S_{mod} – return for L the name of L 's prototypical *circumstantials* (a.k.a. adverbials, or adjuncts): locative, instrumental, means, result and manner circumstantials.

The domain of application of a circumstantial LF is determined by its semantic content, namely:

- S_{loc} applies to an L that denotes either an entity or a fact (action, event, process, state, ...), since both entities and facts can have localizations;
- S_{instr} and S_{med} apply to an L that denotes an action – i.e. a particular type of fact, since only actions involve instruments and means;
- S_{res} and S_{mod} take an L that denotes a dynamic fact (action, event, process, ...): only dynamic facts have results and happen in a certain way.

Because of their particular semantic content, the circumstantial LFs do not apply to adjectives and adverbs.

All circumstantial LFs have the same PoS table :

$S_{loc/instr/med/res/mod}$: nominal			
L	V	N	Claus
$S_{loc/instr/med/res/mod}(L)$	N	N	N

The order of presentation of circumstantial LFs follows, roughly speaking, the hierarchy of circumstantials in natural languages: from more to less linguistically privileged circumstantials.

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4.5.2.0.1 Historical remark A different ordering of circumstantial LFs was used in previous publications, namely: S_{loc} , instead of heading the enumeration, was arbitrarily placed in the penultimate position.

[16] S_{loc} [Lat. *locus* ‘place’]: name of the prototypical location

$S_{loc}(L)$ is a nominal lexical entity that is the name of the prototypical location of the referent of L.

English

$S_{loc}(\textit{surgey})$	=	‘operation room’
$S_{loc}(\textit{smoke}_{(V)})$	=	‘smoking area’; ‘smoking room’
$S_{loc}(\textit{formal excrete})$	=	bathroom, restroom, toilet _(N) , washroom; privy; outhouse
$S_{loc}(\textit{orphan})$	=	orphanage
$S_{loc}(\textit{cattle})$	=	stable _(N)

French

$S_{loc}(\textit{opération [chirurgicale]})$	=	‘salle d’opération’; ‘bloc opératoire’
$S_{loc}(\textit{fumer})$	=	espace fumeur(s); fumoir
$S_{loc}(\textit{‘faire ses besoins’})$	=	vulgar <i>chiottes, toilettes</i> , W.-C.; formal ‘lieu d’aisance(s)’; latrine
$S_{loc}(\textit{orphelin})$	=	orphelinat
$S_{loc}(\textit{bétail})$	=	étable

Russian

$S_{loc}(\textit{operacija})$	=	<i>operacionnaja</i>
$S_{loc}(\textit{kurit’})$	=	‘mesto dlja kurenija’, colloquial <i>kurilka</i>
$S_{loc}(\textit{colloquial ‘spravljat’ nuždu’})$	=	<i>tualet, ubornaja</i> ; colloquial <i>sortir</i>
$S_{loc}(\textit{sirota})$	=	‘detskij dom’
$S_{loc}(\textit{skot})$	=	<i>xlev</i>

[17] S_{instr} [Lat. *instrumentum* ‘tool’]: name of the prototypical instrument

$S_{instr}(L)$ is a nominal lexical entity that is the name of the instrument, in the most general sense, which is prototypically used to do L.

English

$S_{instr}(\textit{drink}_{(V)})$	=	<i>glass; cup</i>
$S_{instr}(\textit{shower}_{(V)})$	=	<i>shower</i> _(N) [equipment]
$S_{instr}(\textit{wash}_{(V)} [\textit{oneself}])$	=	<i>washcloth</i>
$S_{instr}(\textit{fry}_{(V)})$	=	‘frying pan’, <i>skillet; fryer</i>
$S_{instr}(\textit{smell}_{(V)})$	=	<i>nose</i> _(N)
$S_{instr}(\textit{murder}_{(N)})$	=	‘murder weapon’
$S_{instr}(\textit{Snip!})$	=	<i>scissors</i>

French

S_{instr} (<i>boire</i>)	=	<i>verre; tasse</i>
S_{instr} (<i>se doucher</i>)	=	<i>douche</i> II [equipment]
S_{instr} (<i>se laver</i>)	=	「gant de toilette」
S_{instr} (<i>frire</i>)	=	[une] <i>poêle; friteuse</i>
S_{instr} (<i>sentir</i>)	=	<i>nez</i>
S_{instr} (<i>crime</i>)	=	「arme du crime」
S_{instr} (<i>Glouglou !</i>)	=	<i>verre</i>

Russian

S_{instr} (<i>pit'</i>)	=	<i>stakan; čaška; rjumka</i>
S_{instr} ([<i>prinimat'</i>] <i>duš</i> I)	=	<i>duš</i> II [equipment]
S_{instr} (<i>pričesyvat' sja</i>)	=	<i>grebešok, rasčeska</i>
S_{instr} (<i>žarit'</i>)	=	<i>skovoroda; fritjurnica</i>
S_{instr} (<i>njuxat'</i>)	=	<i>nos</i>
S_{instr} (<i>prestuplenie</i>)	=	「orudie prestuplenija」
S_{instr} (<i>Čik!</i>)	=	<i>nož; nožnicy</i>

4.5.2.0.1 Comment In the above examples, Eng. *Snip!*, Fr. *Glouglou !* and Rus. *Čik!* are clausative elements of the value of the LF $Claus_0SonS_0$ for the verbs Eng. *cut*, Fr. *boire* and Rus. *rezat'* – cf. Son, Ch. 5, Sec. 5.4.7, [5.4.7.1], p. cliv.

[18] S_{med} [Lat. *medius* ‘middle’; via such expressions as *by inter+med+iary of*]: name of the prototypical means

$S_{med}(L)$ is a nominal lexical entity that is the name of the means which is prototypically used to do L.

English

S_{med} (<i>shower</i> _(V))	=	<i>water</i> _(N) ; <i>soap</i> _(N)
S_{med} (<i>fry</i> _(V))	=	<i>oil</i> _(N)
S_{med} (<i>glue</i> _(V))	=	<i>glue</i> _(N)

French

S_{med} (<i>se doucher</i>)	=	<i>eau; savon</i>
S_{med} (<i>frire</i>)	=	<i>huile</i>
S_{med} (<i>coller</i>)	=	<i>colle</i>

Russian

S_{med} ([<i>prinimat'</i>] <i>duš</i> I)	=	<i>voda; mylo</i>
S_{med} (<i>žarit'</i>)	=	<i>maslo</i>
S_{med} (<i>kleit'</i>)	=	<i>klej</i>

S_{med} and S_{instr} are semantically distinguished by the fact that a means is “consummated” in the accomplishment of L, while an instrument remains with the user:

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- (18) $S_{\text{med}}(\text{nail}_{(V)}) = \text{nail}_{(N)}$
 $S_{\text{instr}}(\text{nail}_{(V)}) = \text{hammer}_{(N)}$

[19] S_{res} [Lat. *resultāre* ‘have as an outcome’]: name of the prototypical result

$S_{\text{res}}(L)$ is a nominal lexical entity that is the name of the prototypical expected result of L.

English

$S_{\text{res}}(\text{treat}_{(V)} [\text{medically}])$	=	(good) <i>health</i>
$S_{\text{res}}(\text{harm}_{(V)})$	=	<i>harm</i> _(N)
$S_{\text{res}}(\text{murder}_{(V)})$	=	<i>corpse, (dead) body</i>
$S_{\text{res}}(\text{reflexion})$	=	<i>fruits of ART ~</i>

French

$S_{\text{res}}(\text{soigner})$	=	(bonne) <i>santé</i>
$S_{\text{res}}(\text{nuire})$	=	<i>dommage</i>
$S_{\text{res}}(\text{assassiner})$	=	<i>mort</i> _(N) ; <i>corps</i>
$S_{\text{res}}(\text{réflexion})$	=	<i>fruit de ART ~</i>

Russian

$S_{\text{res}}(\text{lečit'})$	=	(dobroe) <i>zdorov'e</i>
$S_{\text{res}}(\text{vredit'})$	=	<i>vred</i>
$S_{\text{res}}(\text{ubit'})$	=	<i>ubityj; trup; (mërtvoe) telo</i>
$S_{\text{res}}(\text{razdum 'ja})$	=	<i>plod ~ij</i>

[20] S_{mod} [Lat. *modus* ‘manner’]: generic name of manner

$S_{\text{mod}}(L)$ is a nominal lexical entity that is the name of the manner in which L takes place.

4.5.2.0.1 Comment We do not use *prototypical(ly)* in the characterization of S_{mod} for the following reason. In contrast to the preceding four circumstantial LFs, $S_{\text{mod}}(L)$ is the name of the parameter “L’s manner” and not the name of a **particular** manner in which L takes place. For instance, the noun *gait* – see English illustrations below – is the value of $S_{\text{mod}}(\text{walk}_{(V)})$ since it means just ‘manner of walking’, while *limping* denotes a particular manner of walking.

English

$S_{\text{mod}}(\text{live}_{(V)})$	=	<i>lifestyle</i> ; ‘ <i>standard of living</i> ’
$S_{\text{mod}}(\text{walk}_{(V)})$	=	<i>gait</i>
$S_{\text{mod}}(\text{box}_{(V)})$	=	[<i>boxing</i>] <i>style, technique; footwork</i>

French

$S_{\text{mod}}(\text{vivre})$	=	‘ <i>mode de vie</i> ’; ‘ <i>train de vie</i> ’
--------------------------------	---	---

$S_{\text{mod}}(\text{marcher})$	=	<i>démarche</i>
$S_{\text{mod}}(\text{boxer})$	=	<i>style, technique</i> [<i>de boxe</i>]; <i>「jeu de jambes」</i>
Russian		
$S_{\text{mod}}(\text{žit'})$	=	<i>「obraz žizni」</i>
$S_{\text{mod}}(\text{xodit'})$	=	<i>poxodka</i>
$S_{\text{mod}}(\text{boksirovat'})$	=	<i>[boksërskij] stil', [boksërskaja] texnika; 「rabota nog」</i>

4.6 Adjectival actantial derivatives

The three next LFs account for adjectival derivatives relative to L's DSyntAs. They all have the same PoS table:

$A_i/\text{Able}_i/\text{Qual}_i$: adjectival				
L	V	N	Adj	Claus
$A_i/\text{Able}_i/\text{Qual}_i(\text{L})$	Adj	Adj	Adj	Adj

[21] A_i : adjectival characterizer of L's DSyntA i

$A_i(\text{L})$ is an adjectival lexical entity that denotes the characteristic of being the DSyntA i , II, III, ... of L.

$A_i(\text{L})$ is a “pure” semantic derivative, contrary to two other adjectival actantial derivatives carrying additional chunks of meaning: Able_i and Qual_i (see [22] and [23]).

For adjectival arguments, $A_1(\text{L}_{(\text{Adj})})$ returns $\text{L}_{(\text{Adj})}$ itself and therefore should not be used: an adjective is, by definition, its own A_1 . Of course, A_2 , A_3 , etc. can apply to adjectives, as illustrated below.

English

$A_1(\text{harm}_{(\text{V})})$	=	<i>harmful</i>
$A_1(\text{victory})$	=	<i>victorious</i>
$A_2(\text{victory})$	=	<i>defeated</i>
$A_2(\text{discussion})$	=	<i>under ~</i>
$A_2(\text{effect}_{(\text{N})})$	=	<i>under the ~</i> [<i>of N_X</i>]
$A_2(\text{informed})$	=	<i>known</i>
$A_2(\text{Yuck!})$	=	<i>disgusting, informal yucky</i>

French

$A_1(\text{nuire})$	=	<i>nuisible</i>
$A_1(\text{victoire})$	=	<i>victorieux</i>
$A_2(\text{victoire})$	=	<i>vaincu</i> _(Adj)
$A_2(\text{discussion})$	=	<i>en ~</i>
$A_2(\text{effet})$	=	<i>sous l'~</i> [<i>de N_X</i>]
$A_2(\text{jaloux}_{(\text{Adj})})$	=	<i>jalousé</i>

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A ₁ (<i>près</i>)	=	<i>proche</i> _(Adj)
A ₂ (<i>Beurk !</i>)	=	<i>dégoûtant</i>

Russian

A ₁ (<i>vredit'</i>)	=	<i>vrednyj</i>
A ₁ (<i>pobeda</i>)	=	<i>pobedonosnyj</i>
A ₂ (<i>pobeda</i>)	=	<i>pobežděnnij</i> _(Adj)
A ₂ (<i>vozdějstvi e</i>)	=	<i>pod ~em</i> [N _X -GEN]
A ₂ (<i>Fu!</i>)	=	<i>protivnyj</i>

[22] **Able_i** [Lat. *habilis* 'able, manageable']: adjectival characterizer of L's potential DSyntA i

Able_i(L) is an adjectival lexical entity that denotes the characteristic of being the potential DSyntA i, ii, iii, ... of L.

English

Able ₁ (<i>chat</i> _(V))	=	<i>chatty</i>
Able ₂ (<i>read</i> _(V))	=	<i>legible</i>
Able ₁ (<i>doubt</i> _(V))	=	<i>doubtful a</i> [Joey is doubtful of my success.]
Able ₂ (<i>doubt</i> _(V))	=	<i>doubtful b</i> [For Joey, my success is doubtful.]
Able ₁ (<i>emotion</i>)	=	<i>emotional</i>

French

Able ₁ (<i>bavarder</i>)	=	<i>bavard</i> _(Adj)
Able ₂ (<i>lire</i>)	=	<i>lisible</i>
Able ₁ (<i>peur</i>)	=	<i>peureux</i> _(Adj)
Able ₂ (<i>peur</i>)	=	Quebec <i>épeurant</i> < <i>effrayant</i>

Russian

Able ₁ (<i>boltat'</i>)	=	<i>boltilivnyj</i>
Able ₂ (<i>čitat'</i>)	=	<i>razborčivnyj</i> [počerk]
Able ₁ (<i>bojat' sja</i>)	=	<i>bojazlivnyj</i>
Able ₁ (<i>èmocija</i>)	=	<i>èmocional'nyj</i>
Able ₂ (<i>somnevat' sja</i>)	=	<i>somnitel'nyj</i>

Able_i is an actantial derived adjective enriched, with respect to A_i (see [21] above), by the semantic content of potentiality, corresponding to such meaning as 'be able to', 'be capable of', 'tend to', 'ought to', etc. The exact meaning of each particular Able_i is specified by its lexicographic definition in its own lexical entry, just like for all similar cases.

[23] **Qual_i** [Lat. *qualitas* 'quality']: adjectival characterizer of L's probable DSyntA i

Qual_i(L) is an adjectival lexical entity that denotes the characteristic of an external participant Ω of the situation denoted by L such that Ω can be expected to be

a DSyntA I, II, III, ... of L with a high probability.

Qual_i(L) is a derived actantial adjective whose relation to L is of a more conceptual or cultural nature than a semantic one. Consequently, while Able_i(L) is very often a morphological derivative of L, a Qual_i(L) never is.

English

Qual ₁ (<i>chat</i> _(V))	=	<i>voluble</i>
Qual ₁ (<i>deceive</i>)	=	<i>dishonest</i>
Qual ₂ (<i>deceive</i>)	=	<i>naive</i> < <i>gullible</i>
Qual ₁ (<i>fear</i> _(N))	=	<i>cowardly</i> _(Adj)
Qual ₂ (<i>fear</i> _(N))	=	<i>dangerous</i>

French

Qual ₁ (<i>bavarder</i>)	=	<i>voluble</i>
Qual ₁ (<i>tromper</i>)	=	<i>malhonnête</i>
Qual ₂ (<i>tromper</i>)	=	<i>naïf</i>
Qual ₁ (<i>peur</i>)	=	<i>lâche</i>
Qual ₂ (<i>peur</i>)	=	<i>dangereux</i>

Russian

Qual ₁ (<i>boltat'</i>)	=	<i>razgovorčivyyj</i>
Qual ₁ (<i>obmanyvat'</i>)	=	<i>nečestnyj</i>
Qual ₂ (<i>obmanyvat'</i>)	=	<i>naivnyj</i>
Qual ₁ (<i>ponimat'</i>)	=	<i>umnyj</i>
Qual ₂ (<i>ponimat'</i>)	=	<i>lëgkij, prostoj</i>

4.7 Adverbial actantial derivatives

[24] Adv_i: adverb that means ‘while being L’s DSyntA i’

Adv_i(L) is an adverbial lexical entity with the following three properties:

- it is an adverbial that syntactically depends on a verb;
- it is subject-oriented, that is, it semantically bears on the DSyntA I of this verb;
- it denotes the way L in which DSyntA I of the verb in question does what it does being at the same time DSyntA i of L (in L’s government pattern).

In prose, the above means that Adv₁(L) is equivalent to *L-ing* / *being L* – see (19a), and Adv_{2/3/...}(L) is equivalent to *being L-ed* / *undergoing L* – see (19b).

(19) a. Chris looked at Tony **suspiciously**_{Adv₁(suspect_(V))}. [= ... suspecting something.]

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- b. Chris was arrested **under suspicion**_{Adv₂(*suspect*_(V))} of murder. [= ... being suspected of murder.]

Adv _i : adverbial				
L	V	N	Adj	Claus
Adv _i (L)	Adv	Adv	Adv	Adv[?]

English

Adv ₁ (<i>admire</i>)	=	<i>admiringly</i>
Adv ₁ (<i>deceive</i>)	=	<i>deceivingly</i>
Adv ₁ (<i>silence</i> _(N))	=	<i>in ., silently</i>
Adv ₁ (<i>confrontation</i>)	=	<i>confrontationally</i> ; 'head to head'
Adv ₁ (<i>alone</i>)	=	<i>single-handedly</i>
Adv ₁ (<i>many</i>)	=	'in droves'
Adv ₂ (<i>see</i> _(V) / <i>look</i> _(V))	=	'under the eyes' [of N _X]
Adv ₂ (<i>try</i> _(V) [in court])	=	'in the dock'

French

Adv ₁ (<i>admirer</i>)	=	<i>admirativement</i>
Adv ₁ (<i>tromper</i>)	=	<i>trompeusement</i>
Adv ₁ (<i>boiter</i>)	=	<i>clopin-clopant</i>
Adv ₂ (<i>voir</i>)	=	'à la vue' [de N _X]
Adv ₂ (<i>regarder</i>)	=	'sous l'œil' [de N _X]
Adv ₂ (<i>dire</i>)	=	'aux dires' [de N _X]
Adv ₁ (<i>majorité</i>)	=	<i>majoritairement</i>
Adv ₂ (<i>procès</i>)	=	'sur le banc des accusés'

Russian

Adv ₁ (<i>vosxiščat' sja</i>)	=	<i>s vosxiščeniem; v vosxiščenii</i>
Adv ₁ (<i>obmanyvat'</i>)	=	<i>obmančivo</i>
Adv ₁ (<i>odin</i>)	=	'v odinočku'
Adv ₂ (<i>govorit'</i>)	=	<i>po slovam</i> [N _{X-GEN}]
Adv ₂ (<i>videt' / smotret'</i>)	=	'na glazax' [N _{X-GEN}]
Adv ₂ (<i>sudit'</i>)	=	'na skam'e podsudimyx'

Adv_i does not express any semantic addition with respect to L, and therefore it should not be confused with the semantically full syntagmatic prepositional LFs Instr, which implies utilization, and Propt_i, which implies causation (Ch. 5, Sec. 5.3, [5.3.0.4] and [5.3.0.5], p. cxxvii^{ff}).

4.7.0.0.1 Comment 1 For Adv₁(L), DSyntA₁ of L almost always⁶ coincides with DSyntA₁ of the governing verb; see (20), where CHRIS is DSyntA₁ of ANSWER_(V) and, at the same time, DSyntA₁ in the government pattern of HESITATE, and also of HESITATINGLY [= Adv₁(*hesitate*)]:

⁶See Comment 2 below.

(20) Chris answers **hesitatingly**_{Adv₁(hesitate)}.

The semantics $\Leftrightarrow$ syntax correspondence for sentence (20) looks as follows:

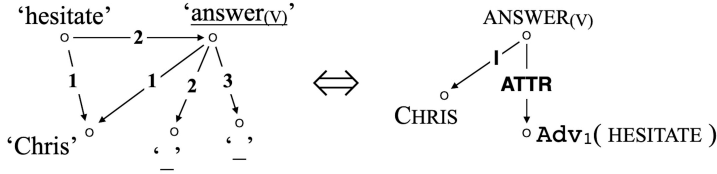
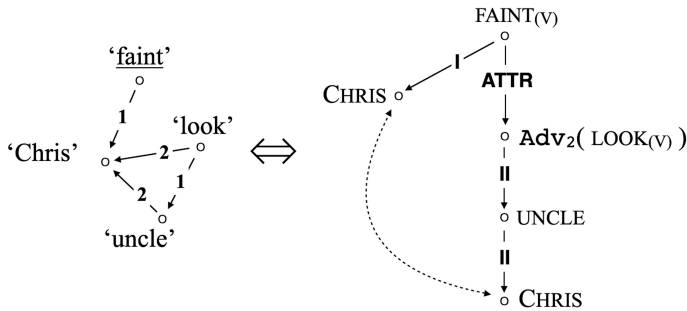


Figure 4.5: Sem $\Leftrightarrow$ DSynt correspondence for the sentence *Chris answers hesitatingly*.

For Adv_{2/3...}(L) things are more complicated since these adverbials involve a passive-like use of L; for instance, see (21), together with the corresponding Sem $\Leftrightarrow$ DSynt correspondence in Figure 4.6.

(21) Chris fainted **under the eyes**_{Adv₂(look_(V))} of his uncle.



The dashed double-headed arrow in the DSynt-structure indicates a coreference relation.

Figure 4.6: Sem $\Leftrightarrow$ DSynt correspondence for the sentence *Chris fainted under the eyes of his uncle*.

In the DSynt-tree of Figure 4.6, UNCLE appears as DSyntA || of Adv₂(look_(V)) because this adverbial is equivalent to a passive form of LOOK_(V): *under the eyes of the uncle* = *being looked at by the uncle*.

4.7.0.0.2 Comment 2 A specific subset of lexical entities that are Adv₁(L) constitutes an exception to the statement in Comment 1: DSyntA I of these adverbials

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does not coincide with DSyntA₁ of the governing verb. Thus, in (22), DSyntA₁ of Adv₁ – i.e. CHRIS – does not coincide with DSyntA₁ of VOTE_(V) – i.e. COMMITTEE:

(22) The committee voted **in the absence** of Chris.

Other Adv₁(L) of the same type are *in the presence* [of N_X], *with the departure* [of N_X], *with the illness* [of N_X], ...

All such adverbials are encoded by putting the 1 actantial subscript between parentheses: Adv₍₁₎(L). This notation indicates that the subscript does not “do its normal job,” but marks simultaneously two things:

- the non-coincidence between L’s DSyntA₁ and that of the governing verb;
- the obligatory character of the expression of L’s DSyntA₁ with Adv₍₁₎(L).

Consequently, the adverbial expression *in the absence* in (22) is described by the following LF formula:

Adv₍₁₎(*absent*) = *in the absence* [of N_X]

The Adv₁ ~ Adv₍₁₎ contrast is parallel to the contrast between English regular V_{ING} phrases with DSyntA₁ coinciding with DSyntA₁ of the governing verb (23a) and special V_{ING} phrases whose DSyntA₁ does not (23b):

- (23) a. **Having left the house**, I heaved a sigh of relief.
b. **My father having left the house**, I heaved a sigh of relief.

The V_{ING} construction in (23b) is known as *absolute construction*. Since Adv₍₁₎ are, so to speak, lexical embodiment of absolute construction, it is convenient to call them *absolute adverbials*.

4.7.0.0.3 Historical remark In several previous publications, Adv_i was erroneously classified as a syntagmatic LF – see, for instance, Mel’čuk (1982), Mel’čuk & Zholkovsky (1984), Mel’čuk et al. (1995, 1999). The reason is that an element of an Adv_i is very often a preposition governing a noun L, such as *in silence* in the English illustrations on p. cii.

4.8 Singulative, collective and other inflection-like derivatives

The LFs introduced in this section correspond to semantic derivatives manifesting a strong analogy with the expression of such inflectional meanings as (noun)

4.8 Singulative, collective and other inflection-like derivatives

singular vs. plural, etc. These LF's come in two groups: (i) those that have to do with quantity as regards L – Sing and Mult – and (ii) those that characterize the accomplishment of event L – Imper, Perf, Imperf and Result_i.

[25] **Sing** [_{Lat.} *singulus* 'singular, unique']: **singulative**

Sing(L) is a lexical entity that denotes a unit / quantum of L's referent.

Sing: verbal or nominal		
L	V	N
Sing(L)	V	N

English

S₀Sing(*paint*_(V)) = 'brush stroke'

Note. This is not a "pure" Sing for a verb, but an S₀Sing, as it returns a nominal value. See remark on verb singulatives after the illustrations.

Sing(*rice*) = grain of ~
Sing(*rain*_(N)) = raindrop
Sing(*parquet*) = ~ slat
Sing(*body*) = 'body part'
Sing(*fur*_(N)) = hair **3** [cf. *hair* **12** under Mult, p. cvi]
Sing(*vocabulary*) = word

French

S₀Sing(*peindre*) = 'coup de pinceau'
Sing(*riz*) = grain de ~
Sing(*pluie*) = goutte de ~
Sing(*parquet*) = lame de ~
Sing(*corps*) = 'partie du corps'
Sing(*pelage*) = poil
Sing(*vocabulaire*) = mot

Russian

S₀Sing(*pisat'* [*kartinu*]) = mazok
Sing(*ris*) = 'risovoe zërnyško', risinka
Sing(*dožd'*) = kaplja ~ja
Sing(*parket*) = parketina
Sing(*telo*) = 'čast' tela'
Sing(*mex*) = volos, volosok
Sing(*slovar'*) = slovo

In a *Standard Average European* [SAE] language, the singulative typically applies to nouns, less frequently to verbs. Here are examples of verbal singulatives:

- (24) Eng. Sing(*beat*_(V)) = hit_(V)
Fr. Sing(*battre*) = frapper
Rus. Sing(*bit'*) = udarit'

4 Paradigmatic lexical functions

Russian, with its inflectional category of aspect, has an important set of verbs whose perfective form expresses Sing:

- (25) Sing(*prygat'*) 'jump-IMPERF' = *prygnut'* 'jump-PERF = do one jump'
 Sing(*udarjat'*) 'hit-IMPERF' = *udarit'* 'hit-PERF = give one blow'
 Sing(*celovat'*) 'kiss-IMPERF' = *pocelovat'* 'kiss-PERF = give one kiss'

[26] **Mult** [Lat. *multum* 'multitude']: collective (for nouns) / multiplicative (for verbs)

Mult(L) is a lexical entity that denotes a set of Ls forming a whole.

Mult: verbal or nominal		
L	V	N
Mult(L)	V	N

English

- Mult(*dog*) = *pack*_(N) 1.1 of ~s
 Mult(*lie*_(N)) = *pack*_(N) 1.4 of ~s
 Mult(*hair*₁) = *hair*₂; 'head of hair'

French

- Mult(*chien*) = *meute de* ~s
 Mult(*mensonge*) = *tas de* ~s
 Mult(*cheveu*₁) = *cheveux*₂, *chevelure*

Russian

- Mult(*sobak*|*a*) = *svora* ~
 Mult(*l*|*ož'*) = *kuča* ~*ži*
 Mult(*volos*₁) = *volosy*₂; *ševeljura*

In parallel with singulative verbs, SAE languages show multiplicative verbs:⁷

- (26) Eng. AntiMagn+Mult(*jump*_(V)) = *hop*_(V)
 Fr. AntiMagn+Mult(*taper*) = *tapoter*
 Rus. AntiMagn+Mult(*xlopat'*) = *poxlopyvat'*

[27] **Imper** [Lat. *imperāre* '[to] command, order']: imperative

Imper(L) is an utterance by which the Speaker incites the Addressee to do L.

Imper: verbal or clausative	
L	V
Imper(L)	V / Claus

⁷On the AntiMagn LF used in (26), see Ch. 5, p. cxvii.

4.8 Singulative, collective and other inflection-like derivatives

English

Imper(「go away」)	=	colloquial 「Beat it!」
ImperPred(<i>quiet</i> _(Adj))	=	<i>Quiet!</i> , <i>Shush!</i> , <i>Shh!</i>
Imper(「give up」)	=	「Let it go!」
Imper(<i>Bang!</i>)	=	<i>Fire!</i>
ImperPrepar ₁ Real ₁ (<i>meal</i>)	=	<i>Breakfast / Lunch / Dinner / Supper is ready!</i>

French

Imper(<i>partir</i>)	=	offensive <i>Dégage !</i> ; colloquial <i>Zou !</i>
Imper(<i>se taire</i>)	=	<i>Chut !</i> ; offensive 「Ta gueule !」
Imper(「tenir le coup」)	=	<i>Courage !</i>
Imper(<i>venir</i>)	=	<i>Ici !</i>
Imper ₃ (「laisser le passage」)	=	「Chaud devant !」
Imper(<i>Pan !</i>)	=	<i>Feu !</i>
Imper(<i>abandonner</i>)	=	「Laisse tomber !」
ImperPrepar ₁ Real ₁ (<i>repas</i>)	=	<i>À table !</i>

Russian

Imper(<i>ujti</i>)	=	offensive <i>Von (otsjuda)!</i> , colloquial offensive <i>Pošël von (otsjuda)!</i>
Imper(<i>ne šumet'</i>)	=	<i>Tixo!</i> , <i>Tss!</i> , <i>Tšš!</i>
Imper(<i>vzjat'</i>)	=	<i>Na(te)!</i>
Imper(child language <i>Pif-paf!</i>)	=	<i>Ogon'!</i>
ImperPrepar ₁ Real ₁ (<i>eda</i>)	=	(<i>Prošu vsex</i>) <i>za stol!</i> , (<i>Prošu vsex</i>) <i>k stolu!</i>

[28] Perf [Lat. *perfectum*]: perfective

Perf(L) is a verbal or clausative lexical unit that denotes the fact of reaching the “goal / limit” inherent to the meaning of L – cf. the comment that follows the illustrations for the LF Result_i below ([30], p. cix).

4.8.0.0.1 Note Perf and the following LF Imperf correspond to aspectual meanings grammaticalized in Slavic languages, where they are inflectional.

Perf: verbal or clausative	
L	V
Perf(L)	V / Claus

English

Perf(<i>end</i> _(V))	=	「be over」
Perf(<i>undress</i>)	=	<i>have undressed</i>
Claus ₉ Perf(<i>dive</i> _(V))	=	<i>Splash!</i>

4 Paradigmatic lexical functions

French

Perf(<i>revenir</i>)	=	‘être de retour’
Perf(<i>se déshabiller</i>)	=	<i>s’être déshabillé</i>
Claus ₀ Perf(<i>plonger</i>)	=	<i>Plouf!</i>

Russian

Perf(<i>razdevat’ sja</i>)	=	<i>razdet’ sja</i>
Perf(<i>odevat’ sja</i>)	=	<i>odet’ sja</i>
Claus ₀ Perf(<i>padat’</i> [in a liquid])	=	<i>Bux!, Pljux!</i>

Values of Perf(L) and Imperf(L) are mostly trivial, in the sense that they are usually grammatical forms of a verb L. Perf is mainly used in complex LF formulas such as S₁Perf, A₁Perf or PerfOper_i.

- (27) S₁Perf(‘give birth’) = ‘new mother’
 A₂Perf(*cook*_(V)) = *cooked*
 PerfOper₁(*divorce*_(N)) = *get* [ART ~]

[29] Imperf [Lat. *imperfectum*]: imperfective

Imperf(L) is a verbal or clausative lexical unit that denotes fact L as ongoing process and, eventually (as in Slavic languages), as the iteration of fact L.

Imperf: verbal or clausative	
L	V
Imperf(L)	V / Claus

English

Imperf(‘be over’)	=	<i>end</i> _(V) , <i>finish</i> _(V)
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French

Imperf(<i>se réveiller</i>)	=	colloquial <i>émerger</i>
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Russian

Imperf(<i>vernut’ sja</i>)	=	<i>vozvraščat’ sja</i>
Imperf(<i>odet’ sja</i>)	=	<i>odevat’ sja</i>

[30] Result_i [Lat. **resultāre* ‘[to] result’]: resultative

Result_i(L) is a verbal or clausative lexical unit that denotes a situation necessarily resulting from reaching the “goal / limit” inherent to the meaning of L. DSyntA_i of Result_i(L) corresponds to DSyntA_i of L.

Let it be emphasized that the situation denoted by Result_i(L) is the necessary result of reaching L’s “goal / limit.” This is in contrast with some cases of Real_i(L) that are considered in Ch. 5 (Sec. 5.4.3, [5.4.3.1], property 3, p. cxli), where the situation referred to by Real_i(L) can be achieved in different ways.

4.8 Singulative, collective and other inflection-like derivatives

Result: verbal or clausative	
L	V
Result _i (L)	V / Claus

English

Result_i(*reach*_(V) [N_Y]) = *be* [Loc_{in} N_Y]

Note. For the syntagmatic LF Loc_{in} (locative preposition meaning ‘being located in’), see Ch. 5, [5.3.0.1], p. cxxv.

Result_i(**colloquial** ‘*drift off*’) = *sleep*_(V); *Zzz!*
 Result_i(*agonize*) = *die*_(V)
 Result_i(*die*_(V)) = **colloquial** ‘*push up daisies*’
 Result_i(*acquire*) = *own*_(V)
 Result₂(**colloquial** ‘*shut up*’₂ [N_Y]) = **colloquial** [N_Y] ‘*shut up*’₁

French

Result_i(*atteindre* [N_Y]) = *être* [Loc_{in} N_Y]
 Result_i(*s’endormir*) = *dormir*; *Zzz!*
 Result_i(*agoniser*) = *mourir*
 Result_i(*mourir*) = **colloquial** ‘*manger les pissenlits par la racine*’
 Result_i(*acquérir*) = *posséder*
 Result₂(**colloquial** ‘*clouer le bec*’ [à N_Y]) = [N_Y] *se taire*

Russian

Result_i(*dostič’* [N_{Y-ACC}]) = *byt’* [Loc_{in} N_Y]
 Result_i(*zasnut’*) = *spat’*; *Xrrr...!*
 Result_i(*agonizirovat’*) = *umeret’*
 Result_i(*umeret’*) = **colloquial** ‘*byt’ na tom svete*’
 Result_i(*priobresti*) = *vladet’*
 Result₂(**colloquial** ‘*zatknut’ rot*’ [N_{Y-DAT}]) = [N_{Y-NOM}] *zamoľčat’*, **colloquial** *zatknut’ sja*;
*Ni_gugu*_(Interj)

Result_i must be distinguished from Perf, with which it could be confounded. For instance, (28a) has a different meaning from (28b):

- (28) a. *have reached*_{Perf(reach_(V))} *Montreal*
 b. *be*_{Result_i(reach_(V))} *in Montreal*

Perf(L) denotes the same fact as L but in a different state: ‘have reached’ is still ‘reach’, but ‘completed’. On the contrary, Result_i(L) denotes a different fact: the situation which is necessarily entailed by reaching L’s “goal / limit” and in which L’s DSyntA *i* finds itself.

5 Syntagmatic lexical functions

Syntagmatic LFs are grouped in four major classes according to their part of speech [PoS] – i.e., the PoS of the collocate they encode. They are presented below in the same order as in the LF zone of a lexical entry (where they appear after paradigmatic LFs):

- Nominal collocates (5.1)
- Adjectival / adverbial collocates (5.2)
- Prepositional collocates (5.3)
- Verbal collocates, distributed into seven subclasses corresponding to their increasing semantic load – i.e., the richness of their meaning ‘ σ^f ’ (5.4)
 - Copula verb (5.4.1)
 - Support [= light] verbs (5.4.2)
 - Realization verbs (5.4.3)
 - Phasic verbs (5.4.4)
 - Verbs of causing (5.4.5)
 - Verbs of mode of functioning (5.4.6)
 - Other verbal collocates (5.4.7)

We conclude this chapter with a summary of alternative approaches to the description of collocations, with a focus on lexicographic models – dictionaries and lexical databases (5.5).

5.1 Nominal collocates

The two nominal collocates – Germ and Culm – are syntactic governors of L.

5 Syntagmatic lexical functions

5.1.0.1 [31] Germ [Lat. *germen* ‘embryo’]: origin

Germ(L) is a nominal lexical unit that denotes L’s initial state, that is, something that will become L.

Germ: nominal	
L	N
Germ(L)	N

English

Germ(<i>flower</i> _(N))	=	[~] <i>bud</i>
Germ([<i>a</i>] <i>painting</i>)	=	<i>sketch</i> [of ART ~]
Germ(<i>idea</i>)	=	<i>germ</i> [of (ART) ~]

French

Germ(<i>fleur</i>)	=	<i>bouton</i> [de (ART) ~]
Germ([<i>une</i>] <i>peinture</i>)	=	<i>esquisse</i> [de ART ~]
Germ(<i>idée</i>)	=	<i>embryon, germe</i> [de (ART) ~]

Russian

Germ(<i>cvet</i> <i>ok</i>)	=	<i>buton</i> [~ <i>ka</i>]
Germ(<i>kartin</i> <i>a</i>)	=	<i>nabrosok</i> [~ <i>y</i>]
Germ(<i>ide</i> <i>ja</i>)	=	<i>zarodyš</i> [~ <i>i</i>]

5.1.0.2 [32] Culm [Lat. *culmen* ‘highest point’]: culmination

Culm(L) is a nominal lexical unit that denotes L’s culminating / central point.

Culm: nominal	
L	N
Culm(L)	N

English

Culm(<i>anger</i> _(N))	=	<i>paroxysm</i> [of (ART) ~]
Culm(<i>crisis</i>)	=	<i>apogee, peak</i> [of ART ~]
Culm(<i>problem</i>)	=	<i>crux, heart</i> [of ART ~]
Culm(<i>night</i>)	=	<i>dead</i> [of (the) ~]
Culm(<i>earthquake</i>)	=	specialized focus, specialized hypocenter [of ART ~]; (specialized) epicenter [of ART ~]
Culm(<i>cyclone</i>)	=	<i>center, eye</i> [of ART ~]

Note. The phrases *center / eye* [of a cyclone] – the values of Culm(*cyclone*) – denote the central part of a cyclone where the intensity of the wind is minimal, but which is nonetheless seen as the most dangerous place to be in.

French

Culm(<i>colère</i>)	=	<i>paroxysme</i> [de ART ~]
Culm(<i>joie</i>)	=	<i>comble</i> [de ART ~]
Culm(<i>problème</i>)	=	<i>cœur, noyau</i> [de ART ~]
Culm(<i>nuit</i>)	=	<i>cœur</i> [de ART ~]
Culm(‘ <i>tremblement de terre</i> ’)	=	<i>foyer, specialized hypocentre</i> [de ART ~]; (specialized) <i>épïcèntre</i> [de ART ~]
Culm(<i>cyclone</i>)	=	<i>œil</i> [de ART ~]

Russian

Culm(<i>ciklon</i>)	=	<i>glaz, oko</i> [~a]
Culm(<i>zemetrijaseni</i> e)	=	<i>èpicèntèr, očag</i> [~ja]; specialized <i>gipocèntèr</i> [~ja]
Culm(<i>problem</i> a)	=	<i>sut’</i> [~y]

Culm often appears in combination with the syntagmatic LF Loc_{in} (Sec. 5.3, [5.3.0.1], p. cxxv), forming a complex LF:

English

Loc _{in} Culm(<i>night</i>)	=	<i>in the dead</i> [of (the) ~]
Loc _{in} Culm(<i>storm</i>)	=	<i>at the heart</i> [of ART ~]

French

Loc _{in} Culm(<i>nuit</i>)	=	‘ <i>en pleine</i> ’ [~], ‘ <i>au plus noir</i> ’ [de la ~]
Loc _{in} Culm(<i>saison</i> [touristique])	=	‘ <i>au plus gros</i> ’ [de la ~]

Russian

Loc _{in} Culm(<i>sezon</i>)	=	<i>v (samyj) razgar</i> [~a]
Loc _{in} Culm(<i>slav</i> a)	=	<i>na veršine</i> [~y]

5.1.0.2.1 Note The expressions of Loc_{in}Culm for English and Russian in the above examples are themselves collocations – on collocates being collocations, see Ch. 3, Sec. 3.3.2 (p. xlix).

5.1.0.2.2 Denotation of Culm The LF Culm covers the former pair of LFs Culm and Centr (Lat. *centrum* ‘center’) – the latter LF having been dropped. Note that the values of Culm(L) do not denote constitutive parts of L. The top of a tree is its extreme top part rather than its “culmination,” and the same can be said for the bow of a ship or the nose of a plane. The pairs *top* ~ *tree*, *bow* ~ *ship*, *nose* ~ *plane*, ... must be described as cases of *meronymy* (i.e., part / whole) relations. Such relations are encoded in the French Lexical Network [fr-LN] (Lux-Pogodalla & Polguère 2011) and other lexical network models by means of a formal tool called Mero (initially proposed in Jousse 2010: 125–127).

Both syntagmatic LFs Germ and Culm are syntactically quite similar to the paradigmatic LF Figur; cf. *flame of passion* = Figur(*passion*) – see Ch. 4, [5],

p. lxxxiii. However, Germ and Culm, on the one hand, and Figur, on the other, belong to two different LF families: Figur(L) is used by the Speaker as a substitute for L, whereas the syntagmatic Germ(L) and Culm(L) are used as collocates of L in order to designate something else than L.

5.2 Adjectival / adverbial collocates

The first five LFs in this section – Epit, Redun, Magn, Ver and Bon – are syntactic modifiers. Therefore, they are considered as deep adjectives when they modify a noun and as deep adverbs when they modify a verb, an adjective or another adverb; it allows us to speak of them as *adjectivals* / *adverbials*.

5.2.0.0.1 **Note** This treatment corresponds to the fact that all modifiers – be they modifiers of nouns or otherwise – are represented in deep-syntactic structure as dependents of the same DSynt-relation, namely **ATTR**.

5.2.0.1 [33] **Epit** [_{Lat.} *epitheton*]: clichéd modifier used for stylistic effects

Epit(L) is an adjectival / adverbial lexical unit used by the Speaker as a modifier of L in order to “fit in” by using a widely accepted cliché. For instance, in the collocation *chaotic jumble*, the adjective *chaotic* is semantically redundant – therefore, unnecessary (since *jumble* itself includes the semanteme ‘chaotic’); but it is used by the Speaker to abide by conventions of the corresponding genre, in this case, the journalese. Note that values of Epit(L) are not necessarily pleonastic in the strict sense and can encapsulate meanings such as that of (Anti)Magn, (Anti)Ver or (Anti)Bon. In such cases, they may be “ideologically” motivated in the context of a given genre; see, for instance, the Russian journalistic collocation of the Soviet era *svetloe buduščee* lit. ‘full.of.light future’ below.

Epit: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Epit(L)	Adv	Adj	Adv	Adv	Adj / Adv[?]

English		
Epit(<i>chat</i> _(V))	=	<i>casually</i>
Epit(<i>mane</i> _(N))	=	<i>unruly</i>
Magn+Epit(<i>chasm</i>)	=	<i>deep < bottomless, yawning</i>
AntiBon+Epit(<i>crime</i>)	=	<i>heinous</i>
Epit(<i>winner</i> [lottery announcement])	=	<i>lucky</i>

Bon+Epit(<i>speaker</i> [academic conference])	=	<i>distinguished</i>
French		
Epit(<i>insinuer</i>)	=	「 <i>en douce</i> 」
Bon+Epit(<i>deviser</i>)	=	<i>gaiement</i>
Magn+Epit(<i>abîme</i>)	=	<i>profond</i> < 「 <i>sans fond</i> 」
Bon+Epit(<i>chevalier</i> [romance of chivalry])	=	<i>archaic preux</i> anteposed
Bon+Epit(<i>princesse</i> [children's tale])	=	<i>belle, jolie</i> anteposed
Epit(<i>gagnant</i> [lottery announcement])	=	<i>heureux</i> anteposed
Russian		
Epit(<i>idti</i>)	=	<i>peškom</i>
Epit(<i>pocelovat' sja</i>)	=	「 <i>drug s drugom</i> 」
Epit(<i>zabit'</i>)	=	<i>nasmert'</i>
Magn+Epit(<i>bezdna</i>)	=	<i>bezdonnaja, zijajučaja</i>
Ver+Epit(「 <i>na vysote</i> 」)	=	<i>dolžnaja</i> [<i>na dolžnoj vysote</i>]
Bon+Epit(<i>buduščee</i> [Soviet era journalism])	=	<i>svetloe</i>
Magn+Epit(<i>sveršenija</i> [Soviet era journalism])	=	<i>velikie</i>
Bon+Epit(<i>dévica</i> [folklore])	=	<i>archaic krasna</i>
Bon+Epit(<i>archaic mólodec</i> [folklore])	=	<i>dobryj</i>

5.2.0.2 [34] Redun [Lat. *redundāre* 'be in excess']: pleonastic modifier used for word sense identification

Redun(L) is an adjectival / adverbial lexical unit used to help the Addressee identify the meaning of a lexical unit L used by the Speaker if L belongs to a polysemous vocable. Values of Redun(L) are lexical units that express explicitly a semantic component of L.

Redun is overwhelmingly used in terminologies (Polguère 2023), in the construction of an important subclass of so-called *complex terms* such as *chemical reaction*, *computer disk*, *money transfer*, ...

Redun: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Redun(L)	Adv	Adj	Adv	Adv	Adj / Adv[?]

English

Redun(<i>desire</i> _(V))	=	<i>sexually</i>
Redun(<i>desire</i> _(N))	=	<i>sexual</i>
Redun(<i>call</i> _(N))	=	<i>phone</i> _(N) [-]
Redun(<i>operation</i>)	=	<i>surgical</i>

French

Redun(<i>désirer</i>)	=	<i>sexuellement</i>
Redun(<i>appel</i>)	=	<i>téléphonique</i> postposed

5 Syntagmatic lexical functions

Redun(<i>ébats</i>)	=	<i>amoureux</i> postposed
Redun(<i>opération</i>)	=	<i>chirurgicale</i> postposed

Russian

Redun(<i>želat'</i>)	=	formal <i>plotski</i>
Redun(<i>zvonok</i>)	=	<i>telefonnyj</i>
Redun(<i>ob''jat'ja</i>)	=	<i>ljubovnye</i>
Redun(<i>operacija</i>)	=	<i>xirurgičeskaja</i>

5.2.0.2.1 Historical remark The distinction between Epit and Redun was introduced in the process of constructing the *DiCo* lexical database (Polguère 2000b) and that of compiling the *Lexique actif du français* = LAF (Mel'čuk & Polguère 2007), where Redun was encoded with the “=” symbol. The LF name Redun was later systematically used in the French Lexical Network [fr-LN] project (Lux-Pogodalla & Polguère 2011). A different convention is used in Mel'čuk (2015b) in order to underscore the functional proximity of both LFs: Epit^{qual} for Epit and Epit^{restr} for Redun.

5.2.0.3 [35] Magn [Lat. *magnus* ‘big, great’]: intensifier

Magn(L) is an adjectival / adverbial lexical unit used by the Speaker as a modifier of L to express a meaning of the type ‘very’, ‘a lot’, ‘intense(ly)’, etc.

Magn: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Magn(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

Magn(<i>sing</i> _(V))	=	‘at the top of N _x ’s lungs / voice’
Magn(<i>drink</i> _(V) [alcohol])	=	‘as a sponge’, heavily, ‘like a fish’
Magn(<i>win</i> _(V))	=	‘with flying colors’
Magn(<i>love</i> _(N))	=	<i>mad</i>
Magn(<i>fear</i> _(N))	=	<i>mortal</i>
Magn(<i>dead</i> _(Adj))	=	‘as a doornail’, <i>stiff</i> anteposed
Magn(<i>pregnant</i> _(Adj))	=	<i>heavily, very</i>
Magn(<i>Thanks!</i>)	=	[.] <i>a lot</i>

French

Magn(<i>chanter</i>)	=	‘à tue-tête’
Magn(<i>boire</i> [alcohol])	=	‘comme un trou’
Magn(<i>gagner</i>)	=	‘haut la main’
Magn(<i>amour</i>)	=	<i>fou</i> postposed
Magn(<i>peur</i>)	=	<i>bleue</i> postposed

Magn(<i>mort</i> _(Adj))	=	<i>raide</i> anteposed
Magn(<i>enceinte</i>)	=	「 <i>jusqu'aux yeux</i> 」
Magn(<i>connu</i>)	=	「 <i>comme le loup blanc</i> 」
Magn(<i>Merci</i> !)	=	<i>bien</i> < <i>beaucoup</i> < <i>infiniment</i> postposed

Russian

Magn(<i>pet'</i>)	=	「 <i>vo ves' golos</i> 」 < 「 <i>vo vsë gorlo</i> 」
Magn(<i>pit'</i> [alcohol])	=	「 <i>kak sapožnik</i> 」
Magn(<i>pobeda</i>)	=	<i>blestjaščaja</i> < <i>polnaja</i>
Magn(<i>ljubov'</i>)	=	<i>bezumnaja, strastnaja</i>
Magn(<i>ranenyyj</i> _(N))	=	☒ <i>tjaželo</i> ...
Magn(<i>beremennaja</i>)	=	colloquial <i>sil'no</i> , ☒ colloquial <i>na snosjax</i>
Magn(<i>izvestnyj</i>)	=	<i>široko</i> anteposed
Magn(<i>spasibo</i>)	=	<i>bol' šoe</i> < <i>ogromnoe</i>

Five remarks concerning this LF follow. Note that remarks 1 to 4 also apply to the two other adjectival / adverbial LFs *Ver* and *Bon*, presented next.

leftmirgin=*,font= The value of Magn(L) often contains fused elements, marked with the ☒ symbol (on fusion, see Ch. 3, Sec. 3.3.3.1, p. li); for instance:

- (1) Magn(*defeat*_(N)) = *crushing, heavy, major, massive, resounding, severe, shattering, stinging, stunning, thumping*
< *complete, thorough, total*, ☒ *rout*_(N)

leftmiirgiin=*,font= A fused Magn of L generally is at the same time its Syn_▷; for instance:¹

- (2) Syn_▷(*defeat*_(N)) = *rout*_(N)

leftmiirgiin=*,font= The LF Magn has an antonymic counterpart AntiMagn:

- (3) AntiMagn(*win*_(V)) = *narrowly*
AntiMagn(*rain*_(N)) = *light*, ☒ *drizzle*_(N)
AntiMagn(*fever*) = *low-grade, slight*

leftmivrgivn=*,font= The meaning of Magn(L) can bear on (= have as target) the meaning of a DSyntA of L rather than on L itself, as is in the prototypical case. This is shown by a DSyntA number in subscript to Magn:

¹See also the comment on example (20), in Ch. 3, Sec. 3.3.3.1, p. li.

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- (4) $\text{Magn}_1(\text{approval}) = \text{general}$ [many Xs approve]
 $\text{Magn}_2(\text{check}_{(N)}) = \text{fat}$ [check's amount Y is significant]
 $\text{Magn}_3(\text{accusation}) = \text{serious} < \text{grave}$ [Z of which X is accused is very bad]

leftmvrgrvn=*,font= The argument L of Magn, as well as that of Ver, must include as LF target in its lexicographic definition a semantic component capable of accepting intensification – for Magn – or evaluation of its “correctness” – for Ver. This remark does not apply to Bon, since any fact or entity is evaluated positively (or negatively) always from a subjective viewpoint. (On LF targets, see Ch. 3, Sec. 3.3.2, p. 1.)

5.2.0.4 fix this

Ver(L) is an adjectival / adverbial lexical unit that the Speaker uses as a modifier of L to express the meaning ‘adequate / as it should be ...’.

Ver: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Ver(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

- Ver(*sing*) = ‘in key’, ‘in tune’
AntiVer(*sing*) = off-key
Ver(*talk_(V)*) = informal turkey
AntiVer(*talk_(V)*) = ‘in riddles’; informal ☒ ‘beat around the bush’
Ver(*aim_(V)*) = straight | postposed
Ver(*truth_(N)*) = naked
Ver(*smile_(N)*) = compelling, frank
AntiVer(*smile_(N)*) = sour; wry
Ver(*success*) = well-deserved
AntiVer(*success*) = underserved
Ver(*choice*) = correct, right
AntiVer(*choice*) = wrong
Ver(*cooked*) = ‘to perfection’

French

- Ver(*chanter*) = juste | postposed
AntiVer(*chanter*) = faux | postposed
Ver(*viser*) = juste | postposed
AntiVer(*viser*) = ‘à côté’
Ver(*sourire_(N)*) = franc
AntiVer(*sourire_(N)*) = forcé

Ver(<i>vérité</i>)	=	「 <i>toute nue</i> 」 postposed, <i>vraie</i> postposed
Ver(<i>succès</i>)	=	<i>mérité</i>
AntiVer(<i>succès</i>)	=	<i>immérité</i>
Ver(<i>choix</i>)	=	<i>approprié, judicieux</i>
AntiVer(<i>choix</i>)	=	<i>erroné</i>
AntiVer(<i>cerveau</i>)	=	colloquial 「 <i>en compote</i> 」
Ver(<i>cuit</i>)	=	「 <i>à point</i> 」
Ver(<i>fait</i> _(Adj))	=	<i>bien</i> anteposed
AntiVer(<i>fait</i> _(Adj))	=	<i>mal</i> anteposed, 「 <i>de travers</i> 」 < 「 <i>en dépit du bon sens</i> 」, ☒ 「 <i>ni fait ni à faire</i> 」

Russian

Ver(<i>govorit'</i> [na ... jazyke])	=	<i>beglo, svobodno, xorošo < prekrasno</i>
AntiVer(<i>govorit'</i> [na ... jazyke])	=	<i>ploxo < edva</i>
Ver(<i>pet'</i>)	=	<i>čisto</i>
AntiVer(<i>pet'</i>)	=	<i>fal' šivo</i>
Ver(<i>ulybka</i>)	=	<i>otkrytaja</i>
AntiVer(<i>ulybka</i>)	=	<i>kislaja</i>
Ver(<i>pravda</i>)	=	<i>čistaja</i>
Ver(<i>vybor</i>)	=	<i>pravil'nyj</i>
AntiVer(<i>vybor</i>)	=	<i>nepravil'nyj, ošibočnyj</i>
Ver(<i>uspex</i>)	=	<i>zaslužennyj</i>
AntiVer(<i>uspex</i>)	=	<i>nezaslužennyj</i>
Ver(<i>sšityj</i>)	=	<i>xorošo < prekrasno</i> , 「 <i>kak na zakaz</i> 」
AntiVer(<i>sšityj</i>)	=	<i>ploxo < bezobrazno</i>
PredAntiVer(<i>mozg</i>)	=	☒ [U X-GEN] 「 <i>šariki za roliki zaexali</i> 」

5.2.0.5 fix me

Bon(L) is an adjectival / adverbial lexical unit that the Speaker uses as modifier of L to express the meaning ‘good’, which constitutes the Speaker’s positive evaluation of L’s referent.

Bon: adjectival / adverbial					
L	V	N	Adj	Adv	Claus
Bon(L)	Adv	Adj	Adv	Adv	Adj / Adv

English

Bon(<i>sing</i>)	=	<i>well < beautifully, delightfully</i> , 「 <i>like an angel</i> 」, 「 <i>like a bird</i> 」, 「 <i>like a nightingale</i> 」
AntiBon(<i>sing</i>)	=	<i>poorly</i>
Bon(<i>choice</i>)	=	<i>fortunate</i>
AntiBon(<i>choice</i>)	=	<i>unfortunate</i>
Bon(<i>weather</i>)	=	<i>good, fine < beautiful</i>
AntiBon(<i>weather</i>)	=	<i>bad < filthy, foul</i>

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French

Bon(<i>chanter</i>)	=	<i>bien</i> < ‘comme un rossignol’, <i>divinement</i>
AntiBon(<i>chanter</i>)	=	<i>mal</i> < ‘comme une casserole’
Magn+AntiBon(<i>chanter</i>)	=	☒ <i>colloquial gueuler</i>
Bon(<i>choix</i>)	=	<i>heureux</i>
AntiBon(<i>choix</i>)	=	<i>malheureux</i>
Bon(<i>temps</i> [weather])	=	<i>beau</i> anteposed < <i>magnifique</i>
AntiBon(<i>temps</i> [weather])	=	<i>mauvais</i> anteposed < ‘à ne pas mettre un chat dehors’, <i>colloquial</i> ‘de chien’

Russian

Bon(<i>pet'</i>)	=	<i>xorošo</i> < <i>prekrasno</i> , ‘kak solovej’ < <i>božestvenno</i>
AntiBon(<i>pet'</i>)	=	<i>ploxo</i> < <i>otvratitel'no</i>
Magn+AntiBon(<i>pet'</i>)	=	☒ <i>colloquial orat'</i>
Bon(<i>vybor</i>)	=	<i>udačnyj</i>
AntiBon(<i>vybor</i>)	=	<i>neudačnyj</i>
Bon(<i>pogoda</i>)	=	<i>xorošaja</i> < <i>prekrasnaja</i> < <i>čudesnaja</i>
AntiBon(<i>pogoda</i>)	=	<i>ploxaja</i> < <i>merzkaia</i> , <i>užasnaja</i>
PredAntiBon(<i>pogoda</i>)	=	☒ ‘V takuju pogodu xorošij xozjain sobaku na ulicu ne vygonit’

5.2.0.5.1 Historical remark A particular case of Bon was previously encoded in the literature as Pos (now abandoned): Pos(L) is an adjectival / adverbial modifier of L that expresses the positive evaluation of L’s Patient by L’s Agent – that is, the evaluator. Since this LF is applicable only to lexical units denoting evaluations, it was decided to replace it with Bon₂ (for the use of actantial subscripts with Magn, Ver and Bon, see Remark iv, p. cxvii):

- (5) Bon₂(*review*_(N)) = *positive* < *glowing*, *rave*
 [The Speaker says that X, the reviewer of Y, finds Y good]
 AntiBon₂(*review*_(N)) = *lukewarm*, *mixed* < *negative* < *devastating*
 [The Speaker says that X, the reviewer of Y, finds Y not very good or outright bad.]

In (5), with L denoting an evaluation, there is a slight semantic shift: contrary to “regular” Bon₂, the evaluation is done here by L’s Agent, and not by the Speaker.

5.2.0.5.2 Magn, Ver and Bon: differences and similarities The adjectival / adverbial LFs Magn, Ver and Bon form a triplet according to their semantic-pragmatic nature, in the sense that quite often two (or even three) of them can be expressed simultaneously by a single lexical unit; for instance:

- (6) a. Ver+Magn(*sleep*_(V)) = *soundly*
 Ver+Magn(*evidence*) = ☒ ‘*smoking gun*’
 [L at a high degree (i.e., Magn) can fairly often be perceived as being as it should (i.e., Ver).]

- b. Bon+Magn(*agree*) = *wholeheartedly*
 Magn+Bon(*smile*_(N)) = *infectious*
 [An inherently positive L at a high degree (Magn) is often perceived as being good (Bon).]
Note. The order of constituent LFs in the example above is inverted with respect to the preceding one because the positive nature of the smile (Bon) dominates its “size” (Magn). For the order of components in configurational LFs, see Ch. 3, Sec. 3.4.2, p. lvi.
- c. Bon+Magn+Ver(*health*) = *good*
 Bon+Magn+Ver(*sleep*_(V)) = *well*, ‘*like a baby*’
 Bon+Magn+Ver(*lover* [sexual performance]) = *good, great < fantastic*

This notwithstanding, the three LFs Magn, Ver and Bon remain distinct and are often contrasted for the same L, as shown in our English illustrations with the verb SING:

- (7) Magn(*sing*) = ‘*at the top of N_X’s voice*’
 Ver(*sing*) = ‘*in key*’, ‘*in tune*’
 Bon(*sing*) = *well < beautifully, delightfully, ‘like an angel’, ‘like a bird’, ‘like a nightingale’*

5.2.0.5.3 A formal shortcut: Degrad [_{Lat.} *degradāre* ‘lower, degrade’] The verbal complex LF IncepPredAnt*i*Bon, which applies to nouns, is very much present in natural language lexicons: that is, it has numerous arguments and numerous possible values. Therefore, we use for this LF a shortcut notation: Degrad ‘undergo degradation’.

Degrad: verbal	
L	N
Degrad(L)	V

English

- Degrad(*milk*_(N)) = ‘*turn / go sour*’_I, *spoil*_(V)
 Degrad(*marriage*) = ‘*turn / go sour*’_{II}
 Degrad(*flower*_(N)) = *wither*
 Degrad(*health*) = *decline*_(V), *deteriorate*
 Degrad(*discipline*) = *crumble*_(V)
 Degrad(*patience*) = ‘*wear thin*’

French

- Degrad(*lait*) = *tourner*
 Degrad(*vin*) = ‘*tourner au vinaigre*’_I
 Degrad(*situation*) = ‘*tourner au vinaigre*’_{II}
 Degrad(*mariage*) = ‘*battre de l’aile*’

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Degrad(<i>fleur</i>)	=	<i>se faner</i>
Degrad(<i>santé</i>)	=	<i>se dégrader, s'étioler</i>
Degrad(<i>discipline</i>)	=	「fouter le camp」
Degrad(<i>patience</i>)	=	<i>s'épuiser</i>

Russian

Degrad(<i>moloko</i>)	=	<i>skisat'</i>
Degrad(<i>vino</i>)	=	<i>prokisat'</i>
Degrad(<i>situacija</i>)	=	<i>uxuđsat' sja</i>
Degrad(<i>cvetok</i>)	=	<i>vjanut'</i>
Degrad(<i>zdorov'e</i>)	=	<i>pošatnut' sja</i>
Degrad(<i>disciplina</i>)	=	<i>rasšat'vat' sja</i>
Degrad(<i>terpenie</i>)	=	<i>istošcat' sja</i>

5.2.0.6 [38] Plus [Lat. *plus* 'more']: positive comparative

The Plus LF means 'more'; it is used non-autonomously – generally in combination with the above three adjectival / adverbial LFs Magn, Ver and Bon – to build the three complex LFs PlusMagn, PlusVer and PlusBon. Plus being non-autonomous, its PoS is irrelevant (inherently, it is an adverbial – a modifier of adjectival LFs). Similar to paradigmatic LFs Sing, Mult, Imper, Perf, Imperf and Result_i (Ch. 4, Sec. 4.8, p. civff.), the LF Plus expresses a meaning of inflectional type: the positive comparative.

PlusMagn, PlusVer and PlusBon are typically found in combination with the complex LF IncepPred 'begin to be' = 'become': IncepPredPlusMagn, IncepPredPlusVer and IncepPredPlusBon. These three LFs take L as DSyntA I, as seen in the illustrations below. As for the LF Plus, it functions as modifier of Magn / Ver / Bon and thus does not impact the syntactic structure of the whole LF formula.

English

IncepPredPlusMagn(<i>rain</i> _(N))	=	「pick up」
IncepPredPlusMagn(<i>battle</i> _(N))	=	「heat up」
IncepPredPlusMagn(<i>tension</i>)	=	<i>escalate</i>
IncepPredPlusMagn(<i>traffic</i> _(N))	=	「build up」, <i>increase</i> _(V)
IncepPredPlusVer(<i>traffic</i> _(N))	=	「ease up」
IncepPredPlusVer(<i>sky</i> _(N))	=	「clear up」
IncepPredPlusBon(<i>situation</i>)	=	<i>improve</i>

French

IncepPredPlusMagn(<i>averse</i>)	=	<i>redoubler</i>
IncepPredPlusMagn(<i>bataille</i>)	=	<i>s'intensifier</i>
IncepPredPlusMagn(<i>tension</i>)	=	<i>monter</i>
IncepPredPlusMagn(<i>circulation</i> [automobile])	=	<i>s'intensifier</i>
IncepPredPlusVer(<i>circulation</i> [automobile])	=	<i>se fluidifier</i>
IncepPredPlusVer(<i>ciel</i>)	=	<i>se dégager</i>

IncepPredPlusVer(<i>vin</i>)	=	<i>se bonifier</i>
CausPredPlusVer(<i>vin</i>)	=	specialized <i>garder</i> [ART ~]
IncepPredPlusBon(<i>situation</i>)	=	<i>s'améliorer</i>

Russian

IncepPredPlusMagn(<i>liven'</i>)	=	<i>usilivat'sja</i>
IncepPredPlusMagn(<i>naprjažennost'</i>)	=	<i>rasti, vozrastat'</i>
IncepPredPlusMagn(<i>dviženie</i> [<i>na ulicax</i>])	=	<i>usilivat'sja</i>
IncepPredPlusVer(<i>nebo</i>)	=	<i>očičat'sja</i>
CausPredPlusVer(<i>vin o</i>)	=	<i>vyderživat'</i> [..o]
IncepPredPlusBon(<i>situacija</i>)	=	<i>ulučsat'sja</i>

Among these LFs, it is IncepPredPlusMagn that is more present in lexicons.

Plus can also be involved in configurational LFs:

English

PlusMagn+IncepOper ₁ (<i>weight</i> _(N))	=	<i>gain</i> _(V) , 'put up' [..]
PlusMagn+IncepOper ₁ (<i>speed</i> _(N))	=	'pick up' [..]
PlusMagn+IncepOper ₁ (<i>popularity</i>)	=	<i>grow</i> [<i>in</i> ..]

French

PlusMagn+IncepOper ₁ (<i>poids</i>)	=	<i>gagner, prendre</i> [<i>du</i> ..]
PlusMagn+IncepOper ₁ (<i>prudence</i>)	=	<i>redoubler</i> [<i>de</i> ..]
PlusBon+Caus ₁ Func ₀ (<i>technique</i>)	=	<i>affiner, améliorer</i> [ART ~]

Russian

PlusMagn+IncepOper ₁ (<i>skorost'</i>)	=	<i>nabirat'</i> [..]
PlusMagn+IncepOper ₁ (<i>usilija</i>)	=	<i>udvaivat', uveličivat'</i> [..]
PlusMagn+IncepOper ₁ (<i>potencial</i>)	=	<i>naraščivat'</i> [..]

Note that between complex and configurational LFs there exist, as a function of L, semantic equivalences of the following type:

- (8) IncepPredPlusMagn(*rain*_(N)) = 'pick up'
 ['rain begins to.be more.heavy']
 ≡
 PlusMagn+IncepFunc₀(*rain*_(N)) = 'pick up'
 ['more.heavy rain begins taking.place']

Equivalence (8) illustrates a particular deep-syntactic paraphrasing rule, namely:

R_{IncepPredPlusMagn}:

$L \leftarrow I - \text{IncepPredPlusMagn}(L) \equiv \text{IncepFunc}_0(L) - I \rightarrow L - \text{ATTR} \rightarrow \text{PlusMagn}(L)$

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5.2.0.6.1 Remark The LF PlusMagn – in isolation or as a part of a complex LF formula – can be abbreviated as Plus for simplicity's sake. This may create a formal ambiguity, which is effectively resolved by the PoS of the entity to which Plus thus applies. Two cases have to be considered.

5.2.0.6.2 1 When Plus applies to an adjective, it is to be interpreted as Plus, such as in (9) below:

- (9) IncepPredPlus(*weak*_(Adj)) = ☐ *weaken*
IncepPredPlus(*solid*_(Adj)) = ☐ *solidify*

5.2.0.6.3 2 When Plus applies not to an adjective, it is to be interpreted as PlusMagn, such as in (10) below:

- (10) IncepPredPlus(*confidence*) = *grow*
IncepPredPlus(*cost*_(N)) = *rise*_(V), 'go up' < *skyrocket*

This remark applies equally to MinusMagn, which can be abbreviated as Minus – see immediately below.

5.2.0.7 [39] Minus [Lat. *minus* 'less']: negative comparative

Minus 'less' is the antonym of Plus: Minus = AntiPlus; as a result, Minus features the same properties as Plus.

English

IncepPredMinusMagn(<i>rain</i> _(N))	=	<i>subside</i>
IncepPredMinusMagn(<i>muscle</i> _(N))	=	'waste away'
IncepPredMinusMagn(<i>fever</i> _(N))	=	'go down'
IncepPredMinusVer(<i>sky</i> _(N))	=	'cloud over'
MinusMagn+IncepOper ₁ (<i>weight</i> _(N))	=	<i>lose</i> [...]

French

IncepPredMinusMagn(<i>muscle</i>)	=	<i>s'atrophier, fondre</i>
IncepPredMinusMagn(<i>averse</i>)	=	<i>se calmer, faiblir</i>
IncepPredMinusMagn(<i>fièvre</i>)	=	<i>tomber</i>
IncepPredMinusVer(<i>ciel</i>)	=	<i>se couvrir</i>
MinusMagn+IncepOper ₁ (<i>poids</i>)	=	<i>perdre</i> [...]

Russian

IncepPredMinusMagn(<i>dožd'</i>)	=	<i>stixat'</i>
IncepPredMinusMagn(<i>muskul</i>)	=	<i>atrofirovat' sja</i>
IncepPredMinusMagn(<i>temperatura</i>)	=	<i>padat', snižat' sja</i>
IncepPredMinusVer(<i>nebo</i>)	=	<i>pokryvat' sja / zavolaktivat' sja oblakami</i>

MinusMagn+IncepOper₁(ves) = *terjat'* [~]

5.3 Prepositional collocates

5.3.0.1 [40] **Loc_{in}** [_{Lat.} *locus* 'place']: position locative preposition

5.3.0.2 [41] **Loc_{ad}**: destination locative preposition

5.3.0.3 [42] **Loc_{ab}**: origin locative preposition

Loc_{in/ad/ab}(L) is a preposition that takes the noun L as complement and means:

Loc_{in}(L) 'in L' = 'having L as location';
 Loc_{ad}(L) 'into L' = 'having L as destination';
 Loc_{ab}(L) 'out_of L' = 'having L as origin'.

In these glosses the notion of location is to be understood in the vaguest, often metaphorical way. Hence, the LFs Loc_{in/ad/ab} describe both spatial and metaphorical (temporal and abstract) localizations.

The three LFs semantically correspond to three grammatical cases, which are found, for instance, in Hungarian and Finnish: locative for Loc_{in}; allative for Loc_{ad}; ablative for Loc_{ab}.

Loc _{in/ad/ab} : adverbial	
L	N
Loc _{in/ad/ab} (L)	Adv

The value of Loc_{in/ad/ab}(L), which is a preposition, is described at the deep-syntactic level as a particular type of Adverb [Adv] (cf. Ch. 3, Sec. 3.5.1, p. lviii).

English

Loc_{in}(*France*) = *in* [~]
 Loc_{ad}(*France*) = *to* [~]
 Loc_{in}(*London*) = *in* [~]
 Loc_{ad}(*London*) = *to* [~]
 Loc_{in}(*face*_(N)) = *on* [ART ~]
 Loc_{ad}(*face*_(N)) = *in* [ART ~]
 Loc_{in/ad}(*imagination*) = *in* [ART ~]

French

Loc_{in/ad}(*France*) = *en* [~]
 Loc_{in/ad}(*Canada*) = *à* [*le* ~] [⇒ *au Canada*]
 Loc_{in/ad}(*Avignon*) = *à, en* [~]

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Loc _{in} (visage)	=	sur [ART ~]
Loc _{ad} (visage)	=	dans [ART ~] < [⌈] en plein [⌋] [~]
Loc _{in/ad} (travail)	=	à [le ~] [⇒ au travail]
Loc _{in/ad} (imagination)	=	dans [ART ~], en [~]
Loc _{in/ad} (rue)	=	Ø _(Prep, loc) 1 [~→N _{street name}] [Ex.: se trouver rue Balzac / rue du Bac]; dans, sur [ART ~]

Note. Ø_(Prep, loc) 1 is a locative zero preposition – see Mel'čuk (2018).

Russian

Loc _{in} (Franci ja)	=	vo 1 [~i]
Loc _{ad} (Franci ja)	=	vo 2 [~ju]
Loc _{in} (London)	=	v 1 [~e]
Loc _{ad} (London)	=	v 2 [~]
Loc _{in} (lic o)	=	na 1 [~e]
Loc _{ad} (lic o)	=	v 2 [~o]
Loc _{in} (ulic a)	=	na 1 [~e]
Loc _{ad} (ulic a)	=	na 2 [~u]
Loc _{in} (vokzal)	=	na 1 [~e]
Loc _{ad} (vokzal)	=	na 2 [~]

Note. The prepositions NA1/NA2 and v(o)1/v(o)2 are different lexemes of the vocables NA and v(o). NA1 and v(o)1 govern the locative case and indicate a location; NA2 and v(o)2 govern the accusative case and indicate a destination.

As one can see from the above illustrations, French systematically does not distinguish values for Loc_{in} and Loc_{ad}. (This is a common feature of Romance languages.) The distinction is, however, quite relevant in other languages – in the first place, in those that have grammatical cases.

Temporal localization, as mentioned earlier, is covered by Loc_{in}, just like physical localization:

English

Loc _{in} (weather _(N))	=	by [ART ~ with a modifier]
Loc _{in} (reign _(N))	=	under [ART ~]
Loc _{in} (antiquity)	=	in [the ~]
Loc _{in} ([⌈] Middle Ages [⌋])	=	in [the ~]
Loc _{in} (epoch)	=	in [ART ~]
Loc _{in} (September)	=	in [~]
Loc _{in} (Monday)	=	Ø _(Prep, temp) , on [~]

French

Loc _{in} (temps)	=	par [ART ~ with a modifier]
Loc _{in} (règne)	=	pendant, sous [ART ~]
Loc _{in} (Antiquité)	=	dans [l'~]
Loc _{in} (Moyen-Âge)	=	à [le ~] [⇒ au Moyen-Âge]
Loc _{in} (époque)	=	à [ART ~]
Loc _{in} (septembre)	=	en [~]

$\text{Loc}_{\text{in}}(\text{lundi}) = \emptyset_{(\text{Prep}, \text{temp})} \mathbf{2} [(\text{ART}) \sim] [\text{Ex.: arriver } (\text{ART}) \text{ lundi}]$

Note. $\emptyset_{(\text{Prep}, \text{temp})} \mathbf{2}$ is the temporal counterpart of the locative zero preposition $\emptyset_{(\text{Prep}, \text{loc})} \mathbf{1}$, which was commented upon in the preceding illustration table.

Russian

$\text{Loc}_{\text{in}}(\text{pogod}|a) = \mathbf{v2} [\sim u \mid \text{with a modifier}]$
 $\text{Loc}_{\text{in}}(\text{carstvovani}|e) = \mathbf{v2} [\sim e]$
 $\text{Loc}_{\text{in}}(\text{antičnost}|\prime) = \mathbf{v1} [\sim i]$
 $\text{Loc}_{\text{in}}(\text{Sredni e Vek}|a^\top) = \mathbf{v2} [\sim e \sim a^\top]$
 $\text{Loc}_{\text{in}}(\text{èpox}|a) = \mathbf{v2} [\sim u]$
 $\text{Loc}_{\text{in}}(\text{sentjabr}|\prime) = \mathbf{v1} [\sim e]$
 $\text{Loc}_{\text{in}}(\text{ponedel'nik}) = \mathbf{v2} [\sim]$

5.3.0.3.1 Historical remark Loc_{in} expressing temporal localization was previously encoded as $\text{Loc}_{\text{in}}^{\text{temp}}$.

5.3.0.4 [43] Instr [_{Lat.} *instrumentum* ‘instrument’]: instrumental preposition

$\text{Instr}(\text{L})$ is a preposition that means ‘[do something] by means of L’ and takes the noun L as DSyntA II .

Instr: adverbial	
L	N
Instr(L)	Adv

English

$\text{Instr}(\text{knife}_{(\text{N})}) = \text{with } [\text{ART } \sim]$
 $\text{Instr}(\text{car}) = \text{by } [\sim], \text{ driving}$
 $\text{Instr}(\text{plane}_{(\text{N})}) = \text{aboard } [\text{ART } \sim], \text{ by } [\sim], \text{ flying}$
 $\text{Instr}(\text{horse}_{(\text{N})}) = \text{on horseback}^\top, \text{ riding}$
 $\text{Instr}(\text{mule}) = \text{on } [\text{ART } \sim]$
 $\text{Instr}(\text{mail}_{(\text{N})}) = \text{by } [\sim]$

French

$\text{Instr}(\text{couteau}) = \grave{\text{a}} [\text{le } \sim] [\Rightarrow \text{au couteau}], \text{ avec } [\text{ART } \sim]$
 $\text{Instr}(\text{voiture}) = \text{en } [\sim]$
 $\text{Instr}(\text{avion}) = \text{en}, \text{ par } [\sim]$
 $\text{Instr}(\text{cheval}) = \text{à dos}^\top [\text{de } \sim] \mid \text{syntactic governor means ‘transport [something]’;}$
 $\text{à } [\sim] \mid \text{syntactic governor means ‘go [somewhere]’}$
 $\text{Instr}(\text{mulet}) = \text{à dos}^\top [\text{de } \sim]$
 $\text{Instr}(\text{courrier}) = \text{par } [\sim]$

Russian

$\text{Instr}(\text{nož}) = \text{na } \sim \text{om}$
 $\text{Instr}(\text{mašin}|a) = \text{na } \mathbf{1} [\sim e], \text{ na } \sim \text{oj}$

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$\text{Instr}(\text{samolët})$	$=$	$\text{na1} [_e], \boxtimes _om$
$\text{Instr}(\text{lošad} ')$	$=$	$(\text{verxom}) \text{na1} [_i], \boxtimes \text{verxom}$
$\text{Instr}(\text{mul})$	$=$	$(\text{verxom}) \text{na1} [_e]$
$\text{Instr}(\text{počt} a)$	$=$	$\text{po} [_e]$

The syntagmatic LF Instr is semantically equivalent to the complex paradigmatic LF $\text{Adv}_1\text{Real}_i$: $\text{Instr}(L) = \text{Adv}_1\text{Real}_i(L)$.² This equality is exemplified in (11), using one of the English illustrations for Instr above: the value of $\text{Instr}(\text{car})$ combined with car is semantically equivalent to $\text{Adv}_1\text{Real}_i(\text{car})$ semantic derivative.

- (11) $\text{Instr}(\text{car}) = \text{by} [_], \boxtimes \text{driving}$
 $\text{Adv}_1\text{Real}_i(\text{car}) = \text{by} \sim, \text{driving}$

5.3.0.5 [44] Propt_i [Lat. *propter* ‘because of’]: causal preposition

$\text{Propt}_i(L)$ is a preposition that means ‘[P takes place] being caused by L’. Its position in the syntactic structure of the sentence is as follows:

- $\text{Propt}_i(L)$ takes the noun L as DSyntA II ;
- it is itself an **ATTR** syntactic dependent of a predicative lexical unit P;
- P takes as DSyntA I L’s DSyntA i .

Figure 5.1 illustrates the participation of Propt_i in the Sem-structure $\Leftrightarrow$ DSynt-structure correspondence with the case of $\text{Propt}_i(\text{boredom})$ used in sentence (12):

- (12) *Rachel dated Joey out_of _{$\text{Propt}_i(\text{boredom})$} boredom.*

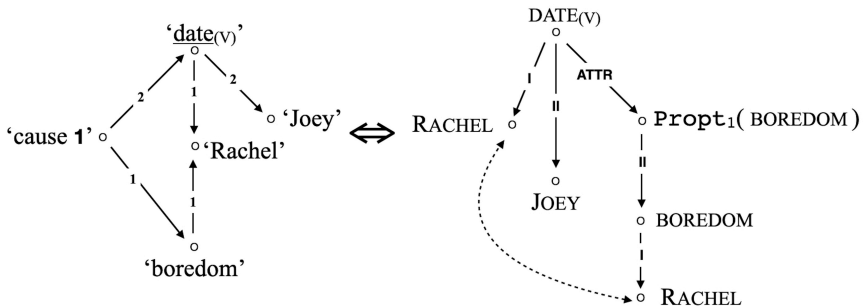


Figure 5.1: Illustration of Propt_i in the Sem $\Leftrightarrow$ DSynt correspondence³

²For Adv_1 , see Ch. 4, [24], p. ci; for Real_i , see [5.4.3.1] below, p. cxxxix.

Because of its meaning, Propt_i is naturally related to the LF Caus (Sec. 5.4.5, [5.4.5.1], p. cxlix), such that it is possible to write the following deep-syntactic paraphrasing rule for Propt_1 :

R_{Propt_1}:
$L_{(V)} - \text{ATTR} \rightarrow \text{Propt}_i(L) - \text{II} \rightarrow L \equiv L \leftarrow \text{I} - \text{CausFunc}_\theta(S_\theta(L_{(V)})) - \text{II} \rightarrow S_\theta(L_{(V)})$
<i>The bar closed for_{$\text{Propt}_1(\text{lack}_{(N)})$} of customers.</i>
$\equiv$
<i>The lack of customers lead_{$\text{CausFunc}_\theta(S_\theta(\text{close}_{(V)}))$} to the closure of the bar.</i>

Propt_i : adverbial	
L	N
$\text{Propt}_i(L)$	Adv

English

$\text{Propt}_1(\text{interest}_{(N)})$	=	<i>out of</i> [ART ~]
$\text{Propt}_1(\text{alcohol})$	=	<i>under the effect</i> [of ~]; $\boxtimes$ <i>under the influence</i> ³
$\text{Propt}_1(\text{combat}_{(N)})$	=	<i>in</i> [~]
$\text{Propt}_1(\text{'heart attack'})$	=	<i>of</i> [(ART) ~]
$\text{Propt}_1(\text{lack}_{(N)})$	=	<i>for</i> [~]
[N_X 's lack of N_Y]		
$\text{Propt}_2(\text{attack})$	=	<i>in</i> [ART ~]

French

$\text{Propt}_1(\text{intérêt})$	=	<i>par</i> [~]
$\text{Propt}_1(\text{alcool})$	=	<i>sous l'emprise</i> [de ART ~]
$\text{Propt}_1(\text{combat})$	=	<i>à</i> [le ~] [$\Rightarrow$ <i>au combat</i>]
$\text{Propt}_1(\text{'crise cardiaque'})$	=	<i>de</i> [(ART) ~]
$\text{Propt}_2(\text{manque})$	=	<i>par</i> [~]
[<i>manque de N_X pour N_Y</i>]		
$\text{Propt}_2(\text{attentat})$	=	<i>dans</i> [ART ~]

Russian

$\text{Propt}_1(\text{interes})$	=	<i>iz</i> [~a]
$\text{Propt}_1(\text{alkogol} ')$	=	<i>pod vozdejstviem</i> [~ja]
$\text{Propt}_1(\text{boj})$	=	<i>v</i> 1 [~u]
$\text{Propt}_1(\text{infarkt})$	=	<i>ot</i> [~a]
$\text{Propt}_2(\text{otsutstvi} e)$	=	<i>iz-za</i> [~ja]
[<i>otsutstvie N_{X-GEN} u N_{Y-GEN}</i>]		
$\text{Propt}_2(\text{pokušeni} e)$	=	<i>v rezul' tate</i> [~ja]

³On the non-agentive 'cause1' vs. the agentive 'cause2', see p. cl.

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5.3.0.5.1 Note The syntagmatic LF Propt_1 , which includes causation, should not be confounded with the non-causative paradigmatic LF Adv_1 ; cf. the following contrast:

- (13) $\text{Adv}_1(\text{affection}) = \text{affectionately, with } \sim$
 vs.
 $\text{Propt}_1(\text{affection}) = \text{out_of } [\sim]$

See also, on Adv_1 , Ch. 4, Sec. 4.7, [24], p. *ciff*. For a detailed study of Propt_1 , see Iordanskaja & Mel'čuk (2007: 503–617).

5.4 Verbal collocates

5.4.1 Copula verb

5.4.1.1 [45] Copul [*Lat. copula*]: verb that means ‘be’

$\text{Copul}(\text{L})$ is a verb that means ‘be_(copula)’ and applies to a nominal or adjectival / adverbial lexical unit L. It takes L as $\text{DSyntA } \Pi$; its $\text{DSyntA } \text{I}$ is the expression of the 1st semantic actant of L.

Copul: verbal			
L	N	Adj	Adv
$\text{Copul}(\text{L})$	V	V	V

English

$\text{Copul}(\text{example})$	=	<i>be</i> [_{ART} $\sim$], <i>serve</i> [<i>as an</i> $\sim$]
$\text{Copul}(\text{challenge}_{(\text{N})})$	=	<i>be, constitute, represent</i> [_{ART} $\sim$]
$\text{Copul}(\text{true})$	=	<i>be, hold</i> [$\sim$]
$\text{IncepCopul}(\text{dead}_{(\text{Adj})})$	=	<i>drop</i> [$\sim$]

French

$\text{Copul}(\text{exemple})$	=	<i>être</i> [_{ART} $\sim$], <i>servir</i> [<i>d'</i> $\sim$]
$\text{Copul}(\text{défi})$	=	<i>être, constituer, représenter</i> [_{ART} $\sim$]
$\text{IncepCopul}(\text{amoureux}_{(\text{Adj})})$	=	<i>tomber</i> [$\sim$]

Russian

$\text{Copul}(\text{primer})$	=	<i>byt', služit'</i> [_{om}]
$\text{IncepCopul}(\text{zamertvo})$	=	<i>upast'</i> [$\sim$]
$\text{IncepCopul}(\text{vljublennyj})$	=	☒ <i>vljubit' sja</i>

Copul is an analytical variant of the paradigmatic LF Pred (Ch. 4, [12], p. xc), so that the following equivalence holds: $\text{Copul}(\text{L}) - \Pi \rightarrow \text{L} \equiv \text{Pred}(\text{L})$. For instance:

- (14) Her declaration was_{Copul(unexpected)} unexpected.
 ≡
 Her declaration 「came as a bolt from the blue」_{Pred(unexpected)}.

Copul does not have numerous values in any given natural language. It is mainly used within complex LFs, as in IncepCopul, etc.

5.4.2 Support [= light] verbs

All LFs *f* introduced in this section are “pure” verbalizers. They are such that:

- *f* applies to a non-verbal lexical unit *L*, and *f*(*L*) takes *L* as DSyntA Ⅰ (= subject) or DSyntA Ⅱ / Ⅲ / ... (= object);
- *f*(*L*) is not selected by the Speaker to express a given meaning, but in order to construct a well-formed verbal syntactic structure with the same meaning as *L*.

Each *f*(*L*) of this type, so to speak, “supports” a non-verbal *L*, by enabling the collocation *f*(*L*)→*L* to function as the syntactic head of the clause; hence the term *support verb* (Gross 1981). Because support verbs are semantically “light” – they do not semantically contribute to the sentence, they are also referred to as *light verbs*, a term generally attributed to Jespersen (1946: Ch. VII, § 7.21ff.).

A support verb is semantically empty in the collocation it forms with *L* in **actual utterances**. But this does not prevent it from having an inherent meaning **in the lexicon**, this meaning becoming redundant with respect to that of *L*. Such is the case, for instance, of the verb FEEL_(V) in the collocation *feel pain*, the verb FACE_(V) in *face an ultimatum*, etc.

There are three types of support verbs according to the position of the collocation base *L* in their DSynt-actantial structure: Oper_i, Func_i and Labor_{ij}. These three LFs have the same PoS table:

Oper _i , Func _i , Labor _{ij} : verbal			
L	N	Adj	Adv
Oper _i /Func _i /Labor _{ij} (L)	V	V	V

A typical support verb collocation is controlled by a nominal base; almost all examples that follow present this type of collocation. However, adjectival / adverbial bases for such collocations are restrictedly possible; a few illustrations are provided at the end of this section.

A support verb LF, being empty by definition, cannot be semantically characterized. We will, thus, supply for each LF described below – Oper_i , Func_i and Labor_{ij} – a prototypical lexical equivalent, so that the reader could have an intuitive feeling of its usage (cf. $\text{DO}_{(V)}$ for Oper_i).

5.4.2.1 [46] **Oper_i** [_{Lat.} *operāri* ‘do, carry out’]: support verb analogous to $\text{DO}_{(V)}$

$\text{Oper}_i(L)$ is a support verb that takes L as DSyntA II and the DSyntA I of L as DSyntA I.

Oper_i participates in the Sem-structure $\Leftrightarrow$ DSynt-structure correspondence shown in Figure 5.2.

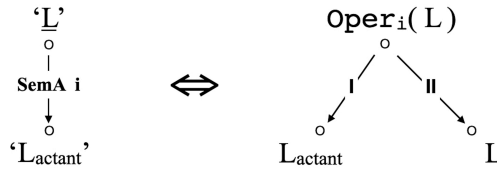


Figure 5.2: Oper_i in the Sem $\Leftrightarrow$ DSynt correspondence

Figure 5.3 illustrates the above correspondence for Oper_1 with $L = \text{FEVER}_{(N)}$:

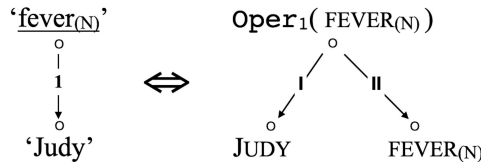


Figure 5.3: Oper_1 in the Sem $\Leftrightarrow$ DSynt correspondence with $L = \text{FEVER}_{(N)}$

Sentences (15a–b) are possible implementations of the DSynt-structure given in Figure 5.3:

- (15) a. Judy **has** a fever.
b. Judy **runs** a fever.

$\text{Oper}_i(L)$ can be an impersonal support verb, which has no DSyntA I. In such case, the notation $\text{Oper}_0(L)$ is used (see French and Russian illustrations below).

English

$\text{Oper}_1(\text{crime})$ = commit, perpetrate [_{ART} ~]

Oper ₁ (supremacy)	=	detain [ART ~]
Oper ₂ (bullying)	=	undergo [~]
Oper ₂ (danger)	=	run [ART ~]
Oper ₂ (applause)	=	draw, get, win [~ with a modifier]
Oper ₃ (order _(N))	=	get, receive [ART ~]
[N _X 's order to V _Y to N _Z]		
Oper ₃ (refusal)	=	meet [with (ART) ~]
[N _X 's refusal to V _Y to N _Z]		

French

Oper ₀ (nuit)	=	[il _(impers)] faire [~]
Oper ₀ (soif)	=	informal [il _(impers)] faire [~]
Oper ₁ (méfait)	=	perpétrer [ART ~]
Oper ₁ (suprémie)	=	détenir [ART ~]
Oper ₂ (calomnie)	=	subir [ART ~]
Oper ₂ (danger)	=	courir [ART ~]
Oper ₂ (applaudissements)	=	recueillir [ART ~]
Oper ₃ (consensus)	=	faire [~]
[concensus entre N _X et N _Y à propos de N _Z]		
Oper ₃ (refus)	=	essuyer [ART ~], se heurter [à ART ~]
[refus de N _X à propos de N _Y à N _Z]		

Russian

Oper ₀ (zapax)	=	tjanut' [..om]
Oper ₀ (proxlada)	=	vejat' [..oj]
Oper ₁ (prestupleniye)	=	soveršit' [..e]
Oper ₁ (prevosxodstvo)	=	imet' [..o]
Oper ₂ (izdevatel'stvo)	=	podvergat' sja [..am]
Oper ₂ (opasnost')	=	podvergat' sja [..i]
Oper ₂ (aplodishment)	=	byt' vstrečen [..ami]
Oper ₃ (prikaz)	=	polučit' [..]
[prikaz N _{X-GEN} N _{Z-DAT} V _{Y-INF}]		
Oper ₃ (otkaz)	=	natolknut' sja [naz ..], polučit' [..]
[otkaz N _{X-GEN} N _{Z-DAT} V _{Y-INF}]		

Oper_i is the most frequent among support verbs. This is explained by the fact that the base of an Oper_i collocation is in most cases the direct object of the verb; and semantic constraints are the strongest between a transitive verb and its direct object – stronger than between a verb and its other actants. It is for this reason that natural languages phraseologize in the first place phrases of the form V—DirO→N.

Understandably, Oper_i has the pride of place in paraphrasing systems of natural languages, as illustrated by the following two cases.

5.4.2.1.1 1 Oper₁(S₀(L_(V)))— $\mapsto$ S₀(L_(V)) is a paraphrase of L_(V):

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(16) Joey joked.

≡

Joey cracked a joke.

On such equivalence, see syntactic paraphrasing rule R_{0per_1} , introduced in Ch. 3, Sec. 3.2.2.3 (p. xlvi).

5.4.2.1.2 2 $0per_2$ is in the same paraphrastic relation – i.e., syntactic conversion – with $0per_1$ as found between the passive and the active voice of a verb, cf. (17a–b):

(17) a. Rachel gave _{$0per_1$} a kiss to Joey.

≡

Rachel kissed_{active voice} Joey.

b. Joey got _{$0per_2$} a kiss from Rachel.

≡

Joey was kissed_{passive voice} by Rachel.

5.4.2.1.3 **Transfer of L's actants to support verbs** In collocations $0per_i(L) - \text{II} \rightarrow L$, the DSyntAs of L can migrate – i.e., be transferred under paraphrasing – to $0per_i(L)$. In other words, $0per_i(L)$ can host, besides its own two obligatory DSyntAs, additional DSyntAs coming from L. This transfer of actants is known as (*syntactic*) *raising* – see Ch. 2, Sec. 2.3, p. xxxi. It is illustrated in (18) with $0per_1(\textit{crime})$.⁴

(18) Joey committed _{$0per_1$} a terrible crime – $\text{II} \rightarrow$ **against decency**.

≡

a terrible crime that Joey committed – $\text{III} \rightarrow$ **against decency**

The transfer of actants from the collocation base to the verbal collocate may be performed for all types of support verbs LFs – $0per_i$, $Func_i$ and $Labor_{ij}$, as well as for corresponding complex LFs (for instance, $CausFunc_0$).

FURTHER READING. On support / light verbs, see Gross (1981), Mel'čuk (2022) and Fleischhauer & Riccio (2025); on transfer of actants from a collocation base to the collocative verb, see Alonso Ramos (2004: 253–272, 2007).

⁴The two utterances in (i) are semantically quasi-equivalent: they express the same propositional meaning, but have different communicative organizations.

5.4.2.2 [47] **Func_i** [_{Lat.} **functionāre* '[to] function']: support verb analogous to 'TAKE PLACE'

Func_i(L) is a support verb that takes L as DSyntA _i and the DSyntA _i of L as DSyntA _{II}. However, it can also have only one actantial dependent: L as DSynt _i. In such a case – an “absolutely intransitive” construction – the LF is denoted as Func₀.

Func_i participates in the Sem-structure $\Leftrightarrow$ DSynt-structure correspondence shown in Figure 5.4; it is to be compared to the Oper_i(L) correspondence in Figure 5.2.

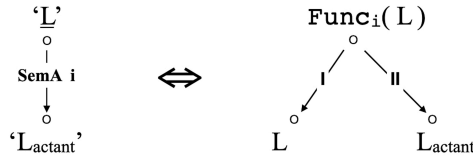


Figure 5.4: Func_i in the Sem $\Leftrightarrow$ DSynt correspondence

Figure 5.5 illustrates the above correspondence for Func₁ with L = RESPONSIBILITY:

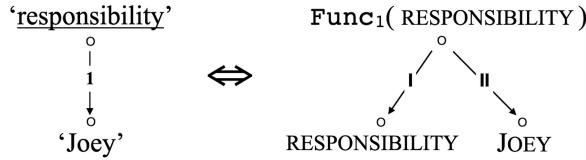


Figure 5.5: Func₁ in the Sem $\Leftrightarrow$ DSynt correspondence with L = RESPONSIBILITY

Sentences (19a–b) are possible implementations of the DSynt-structure given in Figure 5.5:

- (19) a. The responsibility **lies** with Joey.
b. The responsibility **rests** with Joey.

English

Func ₀ (<i>thunder</i> _(N))	=	<i>clap</i> _(V) , <i>crash</i> _(V) , <i>roll</i> _(V) , <i>rumble</i> _(V)
Magn+Func ₀ (<i>fire</i> _(N) [disaster])	=	<i>rage</i> _(V)
Magn+Func ₁ (<i>fire</i> _(N) [disaster])	=	<i>rip</i> _(V) , <i>sweep</i> _(V) [through N _X]
Func ₁ (<i>responsibility</i> [of N _X])	=	<i>lie</i> _(V) , <i>rest</i> _(V) [with N _X]
Func ₁ (<i>smell</i> _(N) [of N _X])	=	<i>come</i> _(V) [from N _X]

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Func ₂ (<i>danger</i> [for N _Y])	=	<i>threaten</i> [N _Y]
Func ₂ (<i>accusation</i> [against N _Y])	=	<i>hang</i> _(V) [over N _Y]
Func ₃ (<i>conversation</i>)	=	<i>be</i> [about N _Z]
[conversation between N _X and N _Y about N _Z]		

French

Func ₀ (<i>tonnerre</i>)	=	<i>gronder, retentir</i>
Magn+Func ₀ (<i>incendie</i>)	=	' <i>faire rage</i> '
Func ₁ (<i>responsabilité</i> [de N _X])	=	<i>incomber</i> [à N _X]
Func ₁ (<i>odeur</i> [de N _X])	=	<i>se dégager, émaner</i> [de N _X]
Func ₂ (<i>danger</i> [pour N _Y])	=	<i>menacer</i> [N _Y]
Func ₂ (<i>accusation</i> [contre N _Y])	=	<i>frapper</i> [N _Y]
Func ₃ (<i>conversation</i>)	=	<i>porter</i> [sur N _Z]
[conversation entre N _X et N _Y sur N _Z]		

Russian

Func ₀ (<i>grom</i>)	=	<i>gremet', gromyxat'</i>
Magn+Func ₀ (<i>požar</i>)	=	<i>buševat'</i>
Func ₁ (<i>otvetstvennost'</i> [N _X -GEN])	=	<i>ležat'</i> [na1 N _X -PREPOS]
Func ₁ (<i>zapax</i> [N _X -GEN])	=	<i>idti, isxodit'</i> [ot N _X -GEN]
Func ₂ (<i>opasnost'</i> [dlja N _Y -GEN])	=	<i>grozit', ugrožat'</i> [N _Y -DAT]
Func ₂ (<i>obvinenie</i> [protiv N _Y -GEN])	=	<i>navisat'</i> [nad N _Y -INSTR]
Func ₃ (<i>razgovor</i>)	=	<i>idti</i> [o N _Z -PREPOS], <i>vertet'sja</i> [vokrug N _Z -GEN]
[razgovor meždu N _X -INSTR i N _Y -INSTR o N _Z -PREPOS]		

5.4.2.3 [48] Labor_{ij} [Lat. *laborāre* '[to] work']: support verb analogous to SUBMIT [N₁ to N₂]

Labor_{ij}(L) is a support verb that takes L as its most oblique (= most peripheral) DSyntA. Its DSyntA *i* corresponds to L's DSyntA *i* and its DSyntA *II* corresponds to L's DSyntA *j*.

Labor_{ij} participates in the Sem-structure ⇔ DSynt-structure correspondence shown in Figure 5.6.

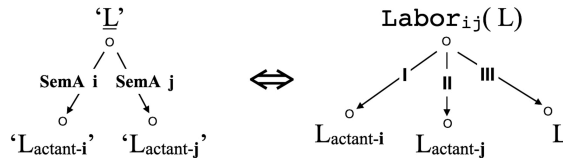


Figure 5.6: Labor_{ij} in the Sem ⇔ DSynt correspondence

Figure 5.7 illustrates the above correspondence for Labor₁₂ with L = APPLAUSE:

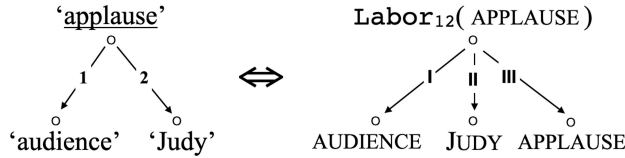


Figure 5.7: Labor_{12} in the $\text{Sem} \Leftrightarrow \text{DSynt}$ correspondence with $L = \text{APPLAUSE}$

The DSynt-structure given in Figure 5.7 is expressed by sentence (20):

(20) The audience **meets** Judy with applause.

English

$\text{Labor}_{12}(\text{esteem}_{(N)} [\text{of } N_X \text{ for } N_Y])$	=	<i>hold</i> [N_Y in $\sim$]
$\text{Labor}_{12}(\text{use}_{(N)} [\text{by } N_X \text{ of } N_Y])$	=	<i>put</i> [N_Y to $\sim$]
$\text{Labor}_{21}([N_X,] \text{result}_{(N)} [\text{of } N_Y])$	=	<i>have</i> [N_X as a $\sim$]

French

$\text{Labor}_{12}(\text{estime} [\text{de } N_X \text{ pour } N_Y])$	=	<i>avoir, tenir</i> [N_Y en $\sim$]
$\text{Labor}_{12}(\text{soins} [\text{de } N_X \text{ destinés à } N_Y])$	=	<i>entourer</i> [N_Y de $\sim$]
$\text{Labor}_{21}([N_X,] \text{résultat} [\text{de } N_Y])$	=	<i>avoir</i> [N_X comme $\sim$]

Russian

$\text{Labor}_{12}(\text{vlást}' [N_{X-\text{GEN}} \text{ nad } N_{Y-\text{INSTR}}])$	=	<i>deržat'</i> [$N_{Y-\text{ACC}}$ pod (svoej) $\sim$ 'ju]
$\text{Labor}_{12}(\text{zabot} \text{a} [N_{X-\text{GEN}} \text{ o } N_{Y-\text{PREPOS}}])$	=	<i>okružít'</i> [$N_{Y-\text{ACC}}$ $\sim$ oj]
$\text{Labor}_{21}([N_{X-\text{NOM}},] \text{rezul'tat} [N_{Y-\text{GEN}}])$	=	<i>imet'</i> [$N_{X-\text{ACC}}$ $\sim$ om $\langle$ v kačestve $\sim$ a $\rangle$]

Figure 5.6 and the above illustrations describe the prototypical case, where the argument L of Labor_{ij} is bi-actantial. In such a case, $\text{Labor}_{ij}(L)$ governs three DSyntAs, L filling the most oblique DSynt-slot. However, there are L s with more than two DSyntAs. As a result of the transfer (that is, syntactic raising) of L 's DSyntAs to the support verb – see above, p. cxxxiv, one obtains Labor verbs with more than three DSyntAs. Take, for instance, the noun INHERITANCE 'belongings Y that X inherits from Z ', and its French and Russian counterparts: they control three four-actantial Labor collocates, as illustrated below.

English

$\text{Labor}_{123}(\text{inheritance})$	=	<i>receive</i> [$N_{Y \Leftrightarrow II}$ from $N_{Z \Leftrightarrow III}$ as $\sim_{L \Leftrightarrow IV}$]
$\text{Labor}_{321}(\text{inheritance})$	=	<i>leave</i> _(V) [$N_{Y \Leftrightarrow II}$ to $N_{X \Leftrightarrow III}$ as $\sim_{L \Leftrightarrow IV}$]
$\text{Labor}_{213}(\text{inheritance})$	=	<i>come</i> _(V) [$\text{to } N_{X \Leftrightarrow II}$ from $N_{Z \Leftrightarrow III}$ as $\sim_{L \Leftrightarrow IV}$]

French

$\text{Labor}_{123}(\text{héritage})$	=	<i>obtenir, recevoir</i> [$N_{Y \Leftrightarrow II}$ de $N_{Z \Leftrightarrow III}$ en $\sim_{L \Leftrightarrow IV}$]
$\text{Labor}_{321}(\text{héritage})$	=	<i>laisser, léguer</i> [$N_{Y \Leftrightarrow II}$ à $N_{X \Leftrightarrow III}$ en $\sim_{L \Leftrightarrow IV}$]

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Labor₂₁₃(*héritage*) = specialized *échoir*, *venir* [*à* N_{X↔II} *de* N_{Z↔III} *en* ~_{L↔IV}]

Russian

Labor₁₂₃(*nasledstv|o*) = *polučit'* [N-ACC_{Y↔II} *ot* N-GEN_{Z↔III} *v* 2 ~_{0L↔IV}]

Labor₃₂₁(*nasledstv|o*) = *ostavit'* [N-ACC_{Y↔II} N-DAT_{X↔III} *v* 2 ~_{0L↔IV}]

Labor₂₁₃(*nasledstv|o*) = *dostat' sja* [N-DAT_{X↔II} *ot* N-GEN_{Z↔III} *v* 2 ~_{0L↔IV}]

As we have seen, a collocational support verb Labor_{ij} relegates the collocation's base L to the most oblique syntactic role: thus, L occupies the least privileged syntactic position, and this is anti-iconic. As a consequence, cases of Labor_{ij} are the least frequent ones among support verbs.

It is important to emphasize that the three support verb LFs are syntactically converse to each other, so that Func₁(L) = Conv₂₁Oper₁(L), etc.:

- (21) a. Oper₁(*analysis*) = *carry out*, *conduct*_(V), *make*_(V) [ART ~]
 Oper₂(*analysis*) = *undergo* [ART ~]
 b. Func₁(*analysis*) = *come*_(V) [from N_X]
 Func₂(*analysis*) = *concern*_(V) [N_Y]
 c. Labor₁₂(*analysis*) = *subject*_(V) [N_Y to ART ~]

To round off the present section, we will consider three cases of support verb collocations having a non-nominal base.

5.4.2.3.1 1 In SAE languages, all adjectives used as attributive complement of a copula verb have the same “joker” Oper₁ support verb; for English, French and Russian: BE_(copula), ÊTRE_(copula) and BYT'_(copula).

Oper₁(L_(Adj)) is equivalent to Copul(L_(Adj)) – Sec. 5.4.1, [5.4.1.1], p. cxxx. Since the corresponding phrases are free (non-phraseologized), they are not to be considered here.

5.4.2.3.2 2 However, adjectives / adverbs denoting meteorological phenomena behave differently. They have a specific Oper_i: an impersonal Oper₀.

English

Oper₀(*hot / misty*) = [*it*_(impers)] *be*_(meteor) [~]
It was hot / misty in Montreal.

French

Oper₀(*chaud*) = [*il*_(impers)] *faire*_(meteor) [~]
Il faisait chaud à Montréal.

Oper₀(*brumeux*) = [*il*_(impers)] *faire*_(meteor), [*ce*_(impers)] *être*_(meteor) [~]
Il faisait (C'était) brumeux à Montréal.

Russian

$\text{Oper}_0(\text{žarko} / \text{tumanno}) = [\emptyset_{(\text{impers})}] \text{byt}'_{(\text{meteor})} [-]$
V Monreale bylo žarko / tumanno.

5.4.2.3.3 3 ‘Prep $\rightarrow$ N’ adverbial idioms that control several Oper_1 collocates are not uncommon, as illustrated by the English idiom ‘ON CLOUD NINE’ ‘extremely happy’:⁵

(22) $\text{Oper}_1(\text{‘on cloud nine’}) = \text{be}, \text{float}_{(V)}, \text{fly}_{(V)} [-]$

5.4.3 Realization verbs

This section introduces three types of verbal collocates that are similar in their syntactic behavior to the three support verbs described in the previous subsection. Semantically, however, *realization verbs* are very different. They are not empty, but express a specific meaning: ‘fulfill an inherent requirement of L’ – in the most general sense (more specifically, ‘realize’, ‘use’, ‘accomplish’, etc.). This semantic load justifies the name *realization verbs*. Just as the support verb LFs, the three realization verb LFs are mutual conversives.

A fourth LF that is semantically related to realization verb LFs is also introduced at the end of this section: *Prepar*. It is, so to speak, a realization verb by extension.

All four verbal LFs presented in this section have the same PoS table:

Real _i , Fact _i , Labreal _{ij} , Prepar: verbal		
L	V	N
Real _i / Fact _i / Labreal _{ij} / Prepar(L)	V	V

This table shows that the realization verbs prototypically have a nominal base. The shading of the column for verbal bases indicates that these cases present some complications – see Remark 3 below, p. cxli.

5.4.3.1 [49] Real_i [Lat. *realis* ‘real’] : realization verb that means ‘fulfill [L]’

Real_i(L) is a realization verb that takes L as DSyntA Π and the DSyntA ι of L as DSyntA ι . It controls the same syntactic structure as support verbs of the type Oper_i .

⁵See also Mel’čuk & Polguère (2021: 136) for a detailed analysis of the French idiom ‘DANS LA SEMOULE’ (lit. ‘in the semolina’) $\approx$ ‘not knowing what to do in a difficult situation’ and its Oper_1 collocates *être* ‘be’, *nager* ‘swim’, *patauger* ‘wade’, *pédaler* ‘pedal’.

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English

$\text{Real}_1(\text{promise}_{(N)})$	=	<i>keep</i> [ART ~]
$\text{Real}_2(\text{expectations})$	=	‘ <i>come up</i> ’ [to ART ~], <i>fulfill</i> [ART ~]; ‘ <i>live up</i> ’ [to ART ~]
$\text{Real}_3(\text{order}_{(N)})$	=	‘ <i>carry out</i> ’, <i>execute</i> , <i>fulfill</i> [ART ~]
[N_X ’s order to V_Y to N_Z]		
$\text{Real}_1(\text{plane}_{(N)})$	=	<i>pilot</i> _(V) [ART ~]

French

$\text{Real}_1(\text{promesse})$	=	<i>tenir</i> [ART ~]
$\text{Real}_2(\text{attentes})$	=	<i>répondre</i> [à ART ~]
$\text{Real}_3(\text{ordre})$	=	<i>exécuter</i> , <i>suivre</i> [ART ~]
[ordre de N_X de V_Y à N_Z]		
$\text{Real}_1(\text{avion})$	=	<i>piloter</i> [ART ~]
$\text{Real}_1(\text{‘chasse d’eau’})$	=	<i>actionner</i> , <i>activer</i> , <i>tirer</i> [ART ~]

Russian

$\text{Real}_1(\text{obeščani} e)$	=	<i>sderžat’</i> , <i>vypolnit’</i> [~e]
$\text{Real}_2(\text{ožidani} ja)$	=	<i>opravdat’</i> [~ja]
$\text{Real}_3(\text{prikaz})$	=	<i>vypolnit’</i> [~]
[prikaz N_X -GEN V_Y -INF N_Z -DAT]		
$\text{Real}_1(\text{samolët})$	=	<i>pilotirovat’</i> , <i>vesti</i> [~]

The argument L of Real_i – just as of the two other realization verbs – must contain in its lexicographic definition at least one particular semantic component which expresses a “requirement” of L and thus is compatible with the meaning of realization. It is this component that is the target for Real_i .⁶ Consider, for instance, the definition of the noun DEBT:

X’s debt of Y to Z : sum of money Y that X got from Z
 • that X is **expected to pay back** to Z

Figure 5.8: Lexicographic definition of DEBT

The boldfaced component corresponds to the inherent requirement of DEBT, which is targeted in the Real_1 collocations: ‘*pay back*’, ‘*pay off*’, *repay a debt*.

The following four remarks concerning Real_i are in order.

5.4.3.1.1 1 The elements of the value of $\text{Real}_i(L)$ may differ by the degree of realization. Three degrees are distinguished, using the superscripts ^I, ^{II} and ^{III}:

- (23) a. $\text{Real}_1^I(\text{suitcase})$ = *fill*_(V), *pack*_(V) [ART ~]
 $\text{Real}_1^{II}(\text{suitcase})$ = *carry*_(V) [ART ~]; *drag*_(V) [ART ~]; *wheel*_(V) [ART ~]
 $\text{Real}_1^{III}(\text{suitcase})$ = *empty*_(V), *unpack* [ART ~]

⁶See the presentation of targets of LFs in Ch. 3, Sec. 3.3.2, p. 1.

- b. $\text{Real}_2^{\text{I}}(\text{task}_{(\text{N})}) = \text{tackle}_{(\text{V})} [\text{ART } \sim]$
 $\text{Real}_2^{\text{II}}(\text{task}_{(\text{N})}) = \text{fulfill} [\text{ART } \sim]$

Some details on the distinction between Real_1^{I} and the LF Prepar are given below ([5.4.3.4], p. cxliii).

5.4.3.1.2 2 The “realizer” of $\text{Real}_1(\text{L})$ can be an external participant of the situation denoted by L (cf. Ch. 2, Sec. 2.2, p. xxvi). In such cases, a special actantial subscript Ω is used: Real_Ω . The symbol Ω indicates that the DSyntA of $\text{Real}_\Omega(\text{L})$ does not correspond to one of L’s DSyntAs . Here are a few illustrations:

- (24) $\text{Real}_\Omega(\text{mountain}) = \text{climb}_{(\text{V})}, \text{scale}_{(\text{V})} [\text{ART } \sim]$
 $\text{Real}_\Omega(\text{costs}_{(\text{N})}) = \text{cover}_{(\text{V})} [\text{ART } \sim]$
 $\text{Real}_\Omega(\text{baby}_{(\text{N})}) = \text{‘take care’} [\text{of ART } \sim], \boxtimes \text{babysit}$
 $\text{Real}_\Omega(\text{pulse}_{(\text{N})}) = \text{feel}_{(\text{V})}, \text{take}_{(\text{V})} [\text{ART } \sim]$

5.4.3.1.3 **Historical remark** In previous lexicographic projects – in particular, in the development of the *DiCo* lexical database (Polguère 2000b) –, subscripts “ $\diamond$ ” or “@” have been used in place of “ Ω .”

5.4.3.1.4 3 Real_1 can also apply to a verb. In this case, $\text{Real}_1(\text{L})$ normally gives fused values:

- (25) $\text{Real}_1(\text{approach}_{(\text{V})} [\text{N}_Y]) = \boxtimes \text{reach}_{(\text{V})} [\text{N}_Y]$
 $\text{Real}_1(\text{look}_{(\text{V})} [\text{for N}_Y]) = \boxtimes \text{find}_{(\text{V})} [\text{N}_Y]$
 $\text{Real}_1(\text{pursue} [\text{N}_Y]) = \boxtimes \text{catch}_{(\text{V})} [\text{N}_Y]$
 $\text{Real}_1(\text{stagger}_{(\text{V})}) = \boxtimes \text{fall}_{(\text{V})}$
 $\text{Real}_2(\text{rock}_{(\text{V})} [\text{N}_Y]) = \boxtimes \text{fall asleep}$

These fused $\text{Real}_1(\text{L})$ should not be mistaken for cases of $\text{Result}_1(\text{L})$, which are in a **paradigmatic** relation with L (Ch. 4, [30], p. cviii). The situation denoted by $\text{Real}_1(\text{L})$, contrary to $\text{Result}_1(\text{L})$, is not necessarily a result of the situation denoted by L. For instance, one can fall asleep without being rocked – cf. last illustration in (25).

5.4.3.1.5 4 Because it is not semantically empty, Real_1 – and the two other LFs of realization as well – frequently combines with *Anti*. As a result, $\text{AntiReal}_1(\text{L})$ expresses the meaning ‘do the opposite of fulfilling an inherent requirement of L’.

- (26) a. $\text{Real}_1(\text{impulse}_{(\text{N})}) = \text{yield}_{(\text{V})} [\text{to ART } \sim]$
 $\text{AntiReal}_1(\text{impulse}_{(\text{N})}) = \text{control}_{(\text{V})}, \text{resist}_{(\text{V})} [\text{ART } \sim]$

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- b. $\text{Real}_1(\text{'red light'}) = \text{stop}_{(V)} [\text{at ART } \sim]$
 $\text{AntiReal}_1(\text{'red light'}) = \text{jump}_{(V)}, \text{run}_{(V)} [\text{ART } \sim]$
- c. $\text{Real}_2(\text{exam}) = \text{pass}_{(V)} [\text{ART } \sim]$
 $\text{AntiReal}_2(\text{exam}) = \text{fail}_{(V)} [\text{ART } \sim]$

5.4.3.2 [50] **Fact_i** [*Lat. factum* ‘action, fact’]: realization verb that means ‘[L] be fulfilled’

$\text{Fact}_i(L)$ is a realization verb that takes L as DSyntA *i* and the expression of the DSyntA *i* of L as DSyntA *ii*.

This verb of realization is a converse of Real_i ; it controls the same syntactic configuration as the support verb Func_i .

Fact_0 (i.e., with *i* = 0) means that the verbal collocate does not govern an object expressing a DSyntA of L.

English

- $\text{Fact}_0(\text{dream}_{(N)}) = \text{'come true'}$
 $\text{Fact}_1(\text{fright}_{(N)}) = \text{paralyze } [N_X]$
 $\text{Fact}_2(\text{insolence}) = \text{offend, shock}_{(V)} [N_Y]$

French

- $\text{Fact}_0(\text{rêve}) = \text{se réaliser}$
 $\text{Fact}_1(\text{effroi}) = \text{paralyser } [N_X]$
 $\text{Fact}_2(\text{insolence}) = \text{choquer, offenser, offusquer } [N_Y]$

Russian

- $\text{Fact}_0(\text{mečta}) = \text{sbyt'sja}$
 $\text{Fact}_1(\text{užas}) = \text{paralizovat'} [N_X\text{-ACC}]$
 $\text{Fact}_2(\text{naglost'}) = \text{potrjasti, vozmutit'} [N_Y\text{-ACC}]$

5.4.3.3 [51] **Labreal_{ij}** [hybrid of **Labor** and **Real**]: realization verb that means ‘submit to the action [of L], thereby fulfilling [L]’

$\text{Labreal}_{ij}(L)$ is a realization verb that takes L as its most oblique (= most peripheral) DSyntA. Its DSyntA *i* corresponds to the DSyntA *i* of L and its DSyntA *ii* corresponds to the DSyntA *j* of L. Labreal_{ij} controls the same syntactic configuration as support verbs of the Labor_{ij} type.

The communicative effect of Labreal_{ij} is to give a peripheral position to the expression of L in the sentence, while communicatively favoring the expression of the actants *i* and *j*.

English

- $\text{Labreal}_{12}(\text{microwave}_{(N)}) = \text{heat}_{(V)} [N_Y \text{ in ART } \sim], \boxtimes \text{microwave}_{(V)} [N_Y]$
 $\text{Labreal}_{12}(\text{trap}_{(N)}) = \text{catch}_{(V)} [N_Y \text{ in ART } \sim], \boxtimes \text{trap}_{(V)} [N_Y]$

Labreal ₁₂ (<i>imagination</i>)	=	<i>see</i> _(V) [N _Y in (ART) ~], ☒ <i>imagine</i> [N _Y]
Labreal ₂₃ (<i>mailbox</i>)	=	<i>drop</i> _(V) , <i>put</i> [N _Z in / into ART ~]
[<i>mailbox of N_X into which N_Y puts N_Z</i>]		

French

Labreal ₁₂ (<i>micro-ondes</i>)	=	<i>passer</i> [N _Y au ~]
Labreal ₁₂ (<i>piège</i>)	=	<i>prendre</i> [N _Y dans ART ~], ☒ <i>piéger</i> [N _Y]
Labreal ₁₂ (<i>imagination</i>)	=	<i>voir</i> [N _Y en ~], ☒ <i>imaginer</i> [N _Y]
Labreal ₂₃ (<i>boîte aux lettres</i> ⁷)	=	<i>déposer, glisser, mettre</i> [N _Z dans ART ~]
[<i>boîte aux lettres de N_X dans laquelle N_Y met N_Z</i>]		

Russian

Labreal ₁₂ (<i>mikrovolnovk a</i>)	=	<i>razogret'</i> [N _Y -ACC v1 ~e]
Labreal ₁₂ (<i>lovušk a</i>)	=	<i>pojmat'</i> [N _Y -ACC v2 ~u]
Labreal ₁₂ (<i>voobraženi e</i>)	=	<i>videt'</i> [N _Y -ACC v1 svoëm ~i], ☒ <i>voobrazit'</i> [N _Y -ACC]
Labreal ₂₃ (<i>počtovyj jaščik</i> ⁷)	=	<i>brosit', opustit'</i> [N _Z -ACC v2 ~]
[<i>počtovyj jaščik</i> ⁷ N _X -GEN, <i>kuda</i> N _Y -NOM <i>pomeščæet</i> N _Z -ACC]		

5.4.3.4 [52] **Prepar** [Lat. *praeparāre* 'prepare']: verb that means 'prepare [L] for functioning'

Prepar takes as argument L only nouns that are semantic quasi-predicates.⁷ It means 'cause that L is ready to function / to be used'.

Prepar: verbal	
L	N
Prepar(L)	V

Prepar is equivalent to CausPredAble₁Fact₀: it denotes an action that makes it possible for the realization of L to take place. It is the reason why it is introduced in the section dedicated to realization verbs.

English

Prepar(<i>knife</i> _(N))	=	<i>sharpen</i> [ART ~]
Prepar(<i>shoes</i>)	=	<i>shine</i> _(V) [ART ~]
Prepar(<i>needle</i> _(N))	=	<i>thread</i> [ART ~]
Prepar(<i>oven</i> _(N))	=	<i>preheat</i> [ART ~]
Prepar(<i>vocal cords</i> ⁷)	=	☒ <i>clear</i> [N _X 's] <i>throat</i> ⁷

French

Prepar(<i>couteau</i>)	=	<i>aiguiser</i> [ART ~]
Prepar(<i>chaussures</i>)	=	<i>cirer</i> [ART ~]
Prepar(<i>aiguille</i>)	=	<i>enfiler</i> [ART ~]

⁷On quasi-predicates, see Ch. 2, Sec. 2.2, p. xxv.

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Prepar(*four* [pour la cuisine]) = *préchauffer* [ART ~]
 Prepar(^Γ*cordes vocales*^Γ) = ^Γ*s'éclaircir la gorge*^Γ; ^Γ*specialized vocaliser*

Russian

Prepar(*nož*) = *točit'* [~]
 Prepar(*botink*|*i*) = *čistit'*, *načistit'* [~i]
 Prepar(*igolk*|*a*) = *vdet' nitku* [v2 ~u]
 Prepar(*duxovk*|*a*) = *razogret'* [~u]
 Prepar(^Γ*golosovye svjazki*^Γ) = ^Γ*kašljanut'*, ^Γ*otkašljat' sja*, ^Γ*pročistit' glotku*^Γ,
^Γ*pročistit' gorlo*^Γ, ^Γ*prokašljat' sja*

Prepar is semantically related to two other syntagmatic LFs, for which it should not be mistaken: **A)** Real_1^{I} and **B)** CausPredAble_1 .

A) Examples in (27a–b) illustrate the difference between Prepar and Real_1^{I} (= the first degree of Real_1 – see [5.4.3.1], Remark 1, p. cxl):

- (27) a. $\text{Prepar}(\text{rifle}_{(\text{N})}) = \text{load}_{(\text{V})}$ [ART ~]
 vs.
 $\text{Real}_1^{\text{I}}(\text{rifle}_{(\text{N})}) = \text{cock}_{(\text{V})}$ [ART ~]; *shoulder*_(V) [ART ~]
 b. $\text{Prepar}(\text{horse}_{(\text{N})}) = \text{saddle}_{(\text{V})}$ [ART ~]
 vs.
 $\text{Real}_1^{\text{I}}(\text{horse}_{(\text{N})}) = \text{mount}_{(\text{V})}$ [ART ~], ^Γ*mount up*^Γ

The examples in (27) show that the user of L's referent does not necessarily correspond to the DSyntA₁ of Prepar, while it is, by definition, the DSyntA₁ of Real_1^{I} : the person who loads the rifle is not necessarily the shooter, whereas it is necessarily the shooter who shoulders it; the person who saddles the horse is not necessarily the rider, whereas it is necessarily the rider who gets into the saddle.

B) As stated earlier, Prepar is semantically equivalent to $\text{CausPredAble}_1\text{Fact}_0$, and therefore it can be easily confused with the complex LF CausPredAble_1 , illustrated in (28).

- (28) $\text{Caus}_1\text{PredAble}_1(\text{sleep}_{(\text{V})}) = \supset^{\Gamma} \text{lie down}^{\Gamma}$
 $\text{Caus}_1\text{PredAble}_1(\text{exercise}_{(\text{V})}) = \supset^{\Gamma} \text{warm up}^{\Gamma}$

The difference between Prepar and CausPredAble_1 is that the latter applies to semantic predicates, while Prepar applies only to semantic quasi-predicates.

5.4.3.4.1 Historical remark In previous publications, Prepar was not presented as intrinsically related to realization verbs and was occupying a different position in lists of LFs. Prepar was also conceptualized as being applicable to verbs, with

the use of an actantial subscript indicating which DSyntA of L is the DSyntA_I of the verbal collocate – see, e.g., Mel’čuk (2015b: 231).

5.4.4 Phasic verbs

LFs presented in this section – Incep, Fin and Cont – are *phasic verbs*, which apply only to verbal lexical units L: they take L as DSyntA_{II} and the expression of the DSyntA_I of L as DSyntA_I. As a result, a phasic LF hooks up to L without altering L’s actantial structure, which implies that it does not carry an actantial subscript. The three phasic LF’s Incep, Fin and Cont respectively mean ‘begin’, ‘cease’ and ‘continue’, i.e., they denote the three phases of the dynamic unfolding of L: beginning, cessation and continuation. In the extralinguistic reality, these three phases temporally follow each other in a different order: beginning, continuation and cessation. From a semantic viewpoint, however, the meaning of continuation is the most complex one: ‘X continues’ = ‘X does not cease’; it has, therefore, to be considered last.

The following two equations express semantic relations holding between phasic LF’s:

$$\begin{aligned}\text{Fin}(L) &= \text{IncepNon}(L) \\ \text{Cont}(L) &= \text{NonIncepNon}(L)\end{aligned}$$

A phasic LF, because of its meaning, applies only to Ls denoting non-punctual facts. Consequently, by saying *begin to explode* one considers an explosion from a special perspective (e.g., an explosion considered as a physical process that is technologically analyzable or describing explosions of multiple entities).

All three phasic LF’s have the following PoS table:

Incep, Fin, Cont: verbal	
L	V
Incep / Fin / Cont(L)	V

As Incep, Fin and Cont apply to verbs only, they require the introduction of a support verb – or of Copul – in case of a non-verbal argument. They frequently appear, therefore, in complex LF’s. Note that in English, and in many other languages as well, cases of application of “pure” phasic LF’s (simple LF’s with a verbal L argument) systematically return fused values (see illustrations below).

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5.4.4.1 [53] Incep [Lat. *incipĕre*]: verb that means ‘begin’

English

Incep(<i>sleep</i> _(V))	=	☒ <i>fall asleep</i>
Incep(<i>drive</i> _(V))	=	☒ [↑] <i>take the wheel</i> [↑]
IncepOper ₁ ([↑] <i>in love</i> [↑])	=	<i>fall</i> [-]
IncepOper ₁ (<i>cold</i> _(N))	=	<i>catch</i> [ART -]
IncepFunc ₀ (<i>day</i>)	=	<i>break</i> _(V) , <i>dawn</i> _(V)
IncepFunc ₀ (<i>night</i>)	=	literary <i>descend</i> , <i>fall</i> _(V)

French

Incep(<i>dormir</i>)	=	☒ <i>s’endormir</i>
Incep(<i>conduire</i>)	=	☒ [↑] <i>prendre le volant</i> [↑]
IncepOper ₁ (<i>amoureux</i> _(Adj))	=	<i>tomber</i> [-]
IncepOper ₁ (<i>rhume</i>)	=	<i>attraper</i> , colloquial <i>choper</i> [ART -], ☒ <i>s’enrhumer</i>
IncepFunc ₀ (<i>jour</i>)	=	<i>poindre</i> < <i>se lever</i>
IncepFunc ₀ (<i>nuit</i>)	=	<i>tomber</i>

Russian

Incep(<i>spat</i> [↑])	=	☒ <i>zasnut</i> [↑]
Incep(<i>vesti</i> [↑] [mašinu])	=	☒ [↑] <i>sest</i> [↑] <i>za rul</i> [↑] [↑]
IncepOper ₁ (<i>vljublennyj</i> _(Adj))	=	☒ <i>vljubit</i> [↑] <i>sja</i>
IncepOper ₁ (<i>nasmork</i>)	=	<i>podxvatit</i> [↑] , <i>pojmat</i> [↑] [-]
IncepFunc ₀ (<i>den</i> [↑])	=	<i>zabrezžit</i> [↑] < <i>zanjat</i> [↑] <i>sja</i> ; ☒ <i>nastat</i> [↑] <i>utro</i>
IncepFunc ₀ (<i>noč</i> [↑])	=	<i>nastat</i> [↑] , literary <i>opustit</i> [↑] <i>sja</i> , <i>prijti</i>

5.4.4.2 [54] Fin [Lat. *finĭre*]: verb that means ‘cease’

English

Fin(<i>sleep</i> _(V))	=	☒ <i>awake</i> _(V) , ☒ [↑] <i>wake up</i> [↑]
FinOper ₁ (<i>sleep</i> _(N))	=	<i>awake</i> _(V) [from ART -]
FinOper ₁ (<i>illness</i>)	=	<i>recover</i> [from ART -]
FinOper ₁ (<i>efforts</i>)	=	<i>abandon</i> _(V) [-]
FinOper ₁ ([↑] <i>in love</i> [↑])	=	☒ [↑] <i>fall out of love</i> [↑]
FinFunc ₀ (<i>fog</i> _(N))	=	<i>lift</i> _(V)

French

Fin(<i>dormir</i>)	=	☒ <i>s’éveiller</i> , ☒ <i>se réveiller</i>
FinOper ₁ (<i>sommeil</i>)	=	<i>émerger</i> , <i>sortir</i> [de ART -]
FinOper ₁ (<i>maladie</i>)	=	<i>guérir</i> [de ART -]
FinFunc ₀ (<i>brouillard</i>)	=	<i>se dissiper</i>

Russian

Fin(<i>spat</i> [↑])	=	☒ <i>prosnut</i> [↑] <i>sja</i>
FinOper ₁ (<i>bolezni</i> [↑])	=	<i>izlečit</i> [↑] <i>sja</i> < <i>opravit</i> [↑] <i>sja</i> [ot -i]
FinOper ₁ (<i>usilija</i> _(N))	=	<i>prekratit</i> [↑] [-ia]
FinOper ₁ (<i>vljublennyj</i> _(Adj))	=	☒ <i>razljubit</i> [↑]

FinFunc₀(*tuman*) = *rassejat' sja*

5.4.4.3 [55] Cont [Lat. *continuāre*]: verb that means ‘continue’

English

ContOper₁(*impassive*) = *remain* [.]
 ContOper₁(*silence*_(N)) = *maintain* [.]
 ContFunc₀(*doubt*_(N)) = *persist*

French

ContOper₁(*impassible*) = ☐^r *rester de marbre*^r
 ContOper₁(*silence*) = *garder* [ART ~]
 ContFunc₀(*doute*) = *persister*

Russian

ContOper₁(*besstrastn|yj*) = *ostavat' sja* [~ym]
 ContOper₁(*molčani|e*) = *xranit'* [~e]
 ContFunc₁(*somnenie*) = *ne pokidat'* [N_X-ACC]

For semantic reasons, we introduce in the present section also the non-phasic verbal LF Prox, as it is directly linked to the expression of the beginning or cessation phase of a fact L.

5.4.4.4 [56] Prox [Lat. *proximāre* ‘[to] approach’]: verb that means ‘be on the verge’ [of L]’

Prox(L) is a verbal lexical unit that denotes the proximity of the initial or final phase of L. It takes the verbal lexical unit L as DSyntA II and the expression of the DSyntA I of L as DSyntA I.

Prox: verbal	
L	V
Prox(L)	V

Prox often combines with two other phasic LFs: Incep and Fin. It is clearly not compatible with Cont, as it would be odd to say that someone is on the verge of continuing to do something. ProxIncep is abridged as Prox.

English

Prox(*die*_(V)) = ☐^r *be fighting for life*^r, ☐^r *be hovering between life and death*^r
 A₁Prox(*die*_(V)) = ☐^r *at the point of death*^r, ☐^r *at the death's door*^r
 ProxOper₁(*collapse*_(N)) = ☐^r *be on the brink*^r [of ~]
 ProxFunc₀(*revolt*_(N)) = *brew*_(V), *rumble*_(V)

5 Syntagmatic lexical functions

ProxFin(「take place」)	=	「near _(V) [A _{(poss)X}] end」
ProxFinFact ₀ (train _(N))	=	pull to the station

French

Prox(mourir)	=	agoniser
A ₁ Prox(mourir)	=	「à l'article de la mort」
ProxOper ₁ (victoire)	=	courir [à ART ...]
ProxFunc ₀ (révolte)	=	gronder
ProxFin(「avoir lieu」)	=	「toucher à sa fin」
ProxFinFact ₀ (train)	=	entrer en gare

Russian

Prox(umeret')	=	「borot'sja za svoju žizn'」
A ₁ Prox(umeret')	=	「na poroge smerti」; 「odnoj nogoj v mogile」
A ₁ Prox(krax)	=	「na grani」 [-a]
ProxFunc ₀ (mjatež)	=	gotovit'sja, podgotovljat'sja
ProxFin(「imet' mesto」)	=	「blizit'sja k (svoemu) koncu」
ProxFinFact ₀ (poezd)	=	podxodit' k vokzalu (stancii) < podxodit' k peronu

5.4.5 Verbs of causing

The three LFs presented in this section – Caus ‘cause’, Liqu ‘cause to cease’ and Perm ‘not cause to cease’ (= ‘let continue’) – are *verbs of causing*, which take as DSyntA II a verbal lexical unit L and as DSyntA I the Cause / Causer.

Most of the time, the argument L of a verb of causing is another verbal LF (CausOper₁, CausReal₁, CausCont, etc.). L being a genuine verb of the language is, of course, possible, but much rarer in such languages as English and French, which have a regular means of expressing causation (see the remark on “Joker values” below).

This Cause / Causer is, generally speaking, an additional semantic and deep-syntactic actant – additional with respect to the situation denoted by L. For this reason the LFs of causing do not carry actantial subscripts. However, it is possible that the Cause / Causer actant coincides with one of L's actants; then, the LF of causing receives an actantial subscript corresponding to this actant of L (Caus₁, Caus₂, ..., Liqu₁, Liqu₂, ..., Perm₁, Perm₂, ...) – see also Ch. 3, Sec. 3.2.3, p. xlvii.

5.4.5.0.1 Terminological remark By *verb* or *LF of causing*, we mean a verb or a verbal LF that denotes causation as such. A verb that denotes a fact that is not a causation proper, but includes a causation, is called *causative verb*. For instance, *make* [something happens] is a verb of causing, whereas *kill* ‘do something that causes someone to die’ is a causative verb.

The three LFs of causing are intimately related to the three phasic LFs (Sec. 5.4.4) and both sets of LFs have the same PoS table.

Caus, Liqu, Perm: verbal	
L	V
Caus / Liqu / Perm(L)	V

5.4.5.1 [57] **Caus** [_{Lat.} *causāre*]: verb that means ‘cause [to begin]’

Cause that L carries one of the following two meanings:

- ‘cause that L begins’, if L is non-punctual – with the indication of the beginning phase;
- ‘cause that L takes place’, if L is punctual – without the indication of a phase.

In theory, one should use the complex LF CausIncep in the first case and the simple LF Caus in the second case. To simplify, however, CausIncep is abridged as Caus, and it is the punctual or non-punctual semantic nature of L that allows for the disambiguation of the formula.

5.4.5.1.1 Joker values In English, the joker Caus for verbs is MAKE_(V) (*make* [someone] *fall / read ...*); in French, it is FAIRE (*faire tomber / lire ...*); Russian does not have such a joker Caus.

English

Caus(<i>sleep</i> _(V)) [= CausIncep(<i>sleep</i> _(V))]	=	<i>put</i> [N _X to ~]
Caus(<i>colloquial</i> ‘ <i>shut up</i> ’)	=	☒ <i>hush</i> _(V) [N _X]; ☒ <i>mute</i> _(V) [N _X]
Caus ₁ Oper ₁ (<i>prisoner</i>)	=	☒ ‘ <i>give oneself up</i> ’
CausFunc _θ (<i>wound</i> _(N))	=	<i>inflict</i> [ART ~]
Caus ₁ Func _θ ([<i>sports</i>] <i>record</i> _(N))	=	<i>set</i> [ART ~]
Caus ₁ Fact _θ ([<i>electric</i>] <i>light</i> _(N))	=	‘ <i>switch on</i> ’, ‘ <i>turn on</i> ’ [ART ~]

French

Caus(<i>dormir</i>) [= CausIncep(<i>dormir</i>)]	=	☒ <i>endormir</i> [N _X]
Caus(<i>se taire</i>)	=	☒ <i>colloquial</i> ‘ <i>clouer le bec</i> ’ [à N _X]
Caus ₁ Oper ₁ (<i>prisonnier</i>)	=	<i>se constituer</i> [~]
CausFunc _θ (<i>lésion</i>)	=	<i>entraîner, provoquer</i> [ART ~]
Caus ₁ Func _θ (<i>record</i>)	=	<i>établir</i> [ART ~]
Caus ₁ Fact _θ (<i>lumière</i> [<i>électrique</i>])	=	<i>allumer</i> [ART ~]

Russian

Caus(<i>spat</i> ‘) [= CausIncep(<i>spat</i> ‘)]	=	☒ <i>usypit</i> ‘ [N _X -ACC]; ☒ <i>ubajukat</i> ‘ [N _X -ACC] [by rocking]
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5 Syntagmatic lexical functions

Caus(zamolč at')	= zastavit' [N _X -ACC .at'], ☒offensive ↑ zatknut' [N _X -DAT] rot ⁷
Caus ₁ Oper ₁ (plennyj _(N))	= ☒ ↑ sdat' sja (v plen) ⁷
CausFunc ₀ (povreždeni _(e))	= pričinit' [..e]
Caus ₁ Func ₀ (rekord)	= postavit', ustanovit' [..]
Caus ₁ Fact ₀ (svet [èlektričestvo])	= vključit', zažeč' [..]

As shown in the above illustrations, for nominal arguments, Caus is used in combination with a support verb LF or a realization verb LF. This behavior is similar to that of phasic verbs and concerns all LF's of causing.

5.4.5.1.2 Non-agentive vs. agentive causation The LF Caus and the two other LF's of causation encompass both types of causation (Kahane & Mel'čuk 2006, Mel'čuk 2012: Ch. 5), that is, the non-agentive 'cause 1' (≈ 'X is the cause of Y') and the agentive 'cause 2' (≈ 'X is the Causer of Y').

5.4.5.2 [58] Liqu [Lat. *liquidāre]: verb that means 'cause to cease'

The meaning of the LF Liqu is built on that of Fin ([5.4.4.2], p. cxlvi):

$$\text{Liqu}(L) = \text{CausFin}(L).$$

English

Liqu(sleep _(V))	= ☒ awake [N _X], ☒ wake [N _X]
Liqu ₍₁₎ Func ₁ (smile _(N))	= wipe [ART ~ off N _X 's face]
LiquFunc ₀ (ambiguity)	= resolve [ART ~]
Liqu ₁ Fact ₀ (engine)	= ↑ shut off ⁷ [ART ~]

French

Liqu(dormir)	= ☒ éveiller [N _X]; ☒ réveiller [N _X]
Liqu ₁ Oper ₁ (sourire)	= se départir [de ART ~]
LiquFunc ₀ (ambiguïté)	= lever [ART ~]
Liqu ₁ Fact ₀ (moteur)	= couper [ART ~]

Russian

Liqu(spat')	= ☒ razbudit' [N _X -ACC]
Liqu ₁ Oper ₁ (ulybk a)	= ↑ steret' s lica ⁷ [..u]
LiquFunc ₀ (neodnoznačnosť')	= razrešit' [..']; snjat' [..']
Liqu ₁ Fact ₀ (motor)	= vyključit' [..]

5.4.5.3 fixme

fix me

5.4.5.3.1 Joker values In English, the joker Perm for verbs is LET_(V) (as in *Let me go*); in French, it is LAISSER; just as for Caus (see above, p. cxlix), there is no equivalent Perm in Russian.

English

Perm ₂ Oper ₁ (<i>member</i>)	=	<i>accept</i> [N _X as ~]
Perm ₁ IncepPredMinusMagn(<i>vigilance</i>)	=	<i>decrease, diminish</i> [ART ~]
NonPerm ₁ Func ₀ (<i>smile</i> _(N))	=	<i>suppress</i> [ART ~]

French

PermOper ₁ (<i>abandon</i>)	=	<i>laisser</i> [N _Y à l'~]
Perm ₁ IncepPredMinusMagn(<i>vigilance</i>)	=	<i>relâcher</i> [ART ~]
NonPerm ₁ Func ₀ (<i>sourire</i> _(N))	=	<i>retenir</i> [ART ~]

Russian

Perm ₂ Oper ₁ (<i>člen</i>)	=	<i>prinjat'</i> [N _X -ACC v2 ~y]
Perm ₁ IncepPredMinusMagn(<i>bditel'nost'</i>)	=	<i>oslabit'</i> [~']
NonPerm ₁ Func ₀ (<i>ulybk a</i>)	=	<i>prjatat', starat'sja skryt'</i> [~u]

The three LFs of causing are connected by the following equations – to be compared to the equations connecting phasic LFs (cf. p. cxlv):

$$\text{Perm}(L) = \text{NonLiqu}(L) = \text{NonCausFin}(L).$$

5.4.6 Verbs of mode of functioning

The three LFs presented in this section – Obstr, Stop and Excess – are verbs that take a nominal lexical unit L as DSyntA₁ and denote three abnormal (approximately, AntiVer) modes of functioning of L's referent: hindered functioning, interrupted functioning and excessive functioning.

By virtue of their semantic content (and syntactic behavior), these LFs are linked to two previously introduced LFs:

- the realization verb LF Fact₀ ([5.4.3.2], p. cxlii), which denotes L's functioning;
- the support verb LF Func₀ ([5.4.2.2], p. cxxxv), in case L itself denotes a functioning.

The individual presentations of Obstr, Stop and Excess below explain these relations.

LFs of mode of functioning are mainly used to form a configurational LF with Symp_{t_{ijk}} ([5.4.7.4], p. clvii). They all have the following PoS table:

5 Syntagmatic lexical functions

Obstr, Stop, Excess: verbal	
L	N
Obstr / Stop / Excess(L)	V

5.4.6.1 [60] **Obstr** [Lat. *obstruere* 'obstruct']: verb that means 'not function as it should'

Obstr is equivalent to either the complex LF AntiVerFact₀ or the configurational LF AntiVer+Func₀, depending on L's meaning.

English

Obstr(<i>memory</i>) [= AntiVerFact ₁ (<i>memory</i>)]	=	'play tricks' [on N _X]
Obstr(<i>negociations</i>) [= AntiVerFact ₀ (<i>negociations</i>)]	=	<i>stall</i> _(V) ; 'drag on'
Conv ₂₁ Obstr(<i>breath</i>) [= AntiVer+Oper ₁ (<i>breath</i>)]	=	<i>be short</i> [of ~]; <i>cannot catch</i> [A _(poss) X ~]

French

Obstr(<i>mémoire</i>) [= AntiVerFact ₀ (<i>mémoire</i>)]	=	<i>flancher</i>
Obstr(<i>négociations</i>) [= AntiVerFact ₀ (<i>négociations</i>)]	=	<i>patiner</i> < 'être au point mort'
Obstr(<i>souffle</i>) [= AntiVer+Func ₀ (<i>souffle</i>)]	=	<i>manquer</i> [à N _X]

Russian

Obstr(<i>pamjat'</i>) [= AntiVerFact ₁ (<i>pamjat'</i>)]	=	<i>podvodit'</i> [N _X -ACC]
Obstr(<i>peregovory</i>) [= AntiVerFact ₀ (<i>peregovory</i>)]	=	<i>buksovat'</i> , <i>probuksovyvat'</i> ; <i>zastřjat'</i> , <i>zastoporit' sja</i>
Obstr(<i>dyxani e</i>) [= AntiVer+Func ₁ (<i>dyxani e</i>)]	=	<i>ne xvatat'</i> [N _X -DAT ~ja]
Conv ₂₁ Obstr(<i>dyxani e</i>) [= AntiVer+Oper ₁ (<i>dyxani e</i>)]	=	☒ <i>zadyxat' sja</i>

5.4.6.2 **fixme**

Stop is equivalent to either AntiVerFinFunc₀ or AntiVerFinFact₀, depending on L's meaning.

English

Stop(<i>match</i> _(N)) [= AntiVerFinFunc ₀ (<i>match</i> _(N))]	=	<i>be interrupted</i>
Stop(<i>heart</i> _(N)) [= AntiVerFinFact ₀ (<i>heart</i> _(N))]	=	<i>fail</i> _(V) , 'give out'
Stop(<i>engine</i>) [= AntiVerFinFact ₀ (<i>engine</i>)]	=	<i>stall</i> _(V)

French

Stop(<i>match</i>) [= AntiVerFinFunc ₀ (<i>match</i>)]	=	<i>s'interrompre</i>
Stop(<i>cœur</i>) [= AntiVerFinFact ₀ (<i>cœur</i>)]	=	<i>s'arrêter</i> , <i>flancher</i>
Stop(<i>moteur</i>) [= AntiVerFinFact ₀ (<i>moteur</i>)]	=	<i>caler</i>

Russian

Stop(<i>matč</i>) [= AntiVerFinFunc ₀ (<i>matč</i>)]	=	<i>prervat' sja</i>
Stop(<i>serdce</i>) [= AntiVerFinFact ₀ (<i>serdce</i>)]	=	<i>ostanovit' sja</i>
Stop(<i>motor</i>) [= AntiVerFinFact ₀ (<i>motor</i>)]	=	<i>zagloxnut'</i>

5.4.6.3 [62] **Excess** [_{Lat.} **excessus*]: verb that means ‘function with excess’

Excess is equivalent to either the complex LF $\text{Magn}^{\text{excess}}\text{Fact}_\theta$ or the configurational LF $\text{Magn}^{\text{excess}}+\text{Func}_\theta$, depending on L’s meaning (the notation $\text{Magn}^{\text{excess}}$ means ‘too much’).

English

$\text{Excess}(\text{heart}_{(N)}) [= \text{Magn}^{\text{excess}}\text{Fact}_\theta(\text{heart}_{(N)})]$	=	$\text{race}_{(V)}$
$\text{Excess}(\text{thoughts}) [= \text{Magn}^{\text{excess-quant}}+\text{Func}_\theta(\text{thoughts})]$	=	$\text{run}_{(V)} < \text{race}_{(V)}$ (through $A_{(\text{poss})X}$ head ⟨mind⟩)

French

$\text{Excess}(\text{cœur}) [= \text{Magn}^{\text{excess}}\text{Fact}_\theta(\text{cœur})]$	=	‘ <i>battre la chamade</i> ’
$\text{Excess}(\text{pensées}) [= \text{Magn}^{\text{excess-quant}}+\text{Func}_\theta(\text{pensées})]$	=	<i>se bousculer</i> (dans $A_{(\text{poss})X}$ tête ⟨esprit⟩)

Russian

$\text{Excess}(\text{serdce}) [= \text{Magn}^{\text{excess}}\text{Fact}_\theta(\text{serdce})]$	=	‘ <i>byt’ gotovo vyskočit’ iz grudi’</i> , <i>kolotit’ sja</i>
$\text{Excess}(\text{mysli}) [= \text{Magn}^{\text{excess-quant}}+\text{Func}_\theta(\text{mysli})]$	=	<i>kružit’ sja</i> ⟨ <i>roit’ sja</i> , <i>tesnit’ sja</i> ⟩ v_1 $A_{(\text{poss})X}$ <i>golove</i> ⟨ <i>mozgu</i> , <i>ume</i> ⟩

5.4.7 Other verbal collocates

The four LFs presented in this section – Son, Manif, Invol_v and Sympt_{ijk} – are verbs that take a nominal lexical unit L as DSyntA₁ and denote situations that are, in a sense, external with respect to L. These external situations come in four types:

‘Son(L)’	=	‘L produces its typical sound’ (29a);
‘Manif(L)’	=	‘L manifests itself in something’ (29b);
‘Invol _v (L)’	=	‘L impacts something’ (29c);
‘Sympt _{ijk} (L)’	=	‘Something happens, being a (metaphorical) symptom of psychological / physiological state of L’ (29d).

- (29) a. The floorboards **creak**_{Son(floorboard)}.
 b. Desire **shines**_{Manif(desire_(N))} in his eyes.
 c. The wind **caresses**_{Invol_v(wind_(N))} his face.
 d. His lips are **quivering**_{Excess^{trem}(lips)—Sympt₂₃(desire_(N))} with desire.

In the case of Manif, Invol_v and Sympt_{ijk}, the verbal collocate puts L into relation with an additional participant, not foreseen by L’s meaning. This participant is denoted in the above examples by *eyes* (29b), *face* (29c) and *lips* (29d).

All four verbal LFs presented in this section have the same PoS table, namely:

5 Syntagmatic lexical functions

Son, Manif, Invol _v , Sympt _{ijk} : verbal	
L	N
Son/Manif/Invol _v /Sympt _{ijk} (L)	V

5.4.7.1 [63] Son [Lat. *sonāre* '[to] sound']: verb that means 'produce a typical sound'

Son is a verb that takes the nominal lexical unit L as DSyntA 1.

English

Son(<i>wind</i> _(N))	=	<i>murmur</i> _(V) , <i>whisper</i> _(V) < <i>howl</i> _(V) , <i>roar</i> _(V)
Son(<i>stream</i> _(N))	=	<i>murmur</i> _(V) < <i>bubble</i> _(V)
Son(<i>fire</i> _(N))	=	<i>crackle</i> _(V)
Son(<i>dog</i> _(N))	=	<i>yap</i> _(V) < <i>bark</i> _(V) < <i>howl</i> _(V) ; <i>growl</i> _(V)
Son(<i>fly</i> _(N))	=	<i>buzz</i> _(V)
Son(<i>steps</i> _(N))	=	<i>sound</i> _(V)
Son(<i>whip</i> _(N))	=	<i>crack</i> _(V)
Son(<i>S₁</i> (<i>sleep</i> _(V)))	=	<i>snore</i> _(V)

Note. The last example is a case of composition of LFs (Ch. 3, Sec. 3.4.1, p. liv). *X snores* means 'X produces the typical sound_{Son} of a sleeper_{S₁}(*sleep*_(V))'. From the viewpoint of LF combination, *S₁* applies to *sleep*_(V) and returns *sleeper*, after which Son applies to *sleeper* [= *S₁*(*sleep*_(V))] to produce *snore*_(V). Such a composition of LFs is characteristic of Son, since it describes a sound produced by an entity *S₁* involved in a particular situation L – e.g., Son(*S₁*([physical] *pain*_(N))) = *scream*_(V).

French

Son(<i>vent</i>)	=	<i>gémir</i> , <i>murmurer</i> < <i>siffler</i> < <i>hurler</i> , <i>mugir</i> , <i>rugir</i>
Son(<i>ruisseau</i>)	=	<i>chanter</i> ; <i>gazouiller</i> ; <i>murmurer</i>
Son(<i>flamme</i>)	=	<i>crépiter</i>
Son(<i>chien</i>)	=	<i>japper</i> < <i>aboyer</i> < <i>hurler</i>
Son(<i>mouche</i>)	=	<i>bourdonner</i>
Son(<i>pas</i> _(N))	=	<i>résonner</i>
Son(<i>fouet</i>)	=	<i>claquer</i>
Son(<i>S₁</i> (<i>dormir</i>))	=	<i>ronfler</i>

Russian

Son(<i>veter</i>)	=	<i>svistet'</i> < <i>revet'</i> ; <i>vyt'</i> , <i>zavyvat'</i>
Son(<i>ručej</i>)	=	<i>žurčat'</i>
Son(<i>ogon'</i>)	=	<i>potreskivat'</i> < <i>treščat'</i>
Son(<i>sobaka</i>)	=	<i>gavkat'</i> , <i>tjavkat'</i> < <i>lajat'</i> < <i>vyt'</i> ; <i>ryčat'</i>
Son(<i>muxa</i>)	=	<i>žužžat'</i>
Son(<i>šagi</i>)	=	<i>poslyšat' sja</i> < <i>razdat' sja</i>
Son(<i>xlyst</i>)	=	<i>ščelknut'</i>
Son(<i>S₁</i> (<i>spat'</i>))	=	<i>xrapet'</i>

The LF Son may be semantically related to two other syntagmatic LFs, depending on the meaning of their argument L.

- L denotes a sound. In this case, $\text{Son}(L) = \text{Func}_\theta(L)$.

$$(30) \quad \begin{aligned} \text{Son}(\text{thunder}_{(N)}) &= \text{Func}_\theta(\text{thunder}_{(N)}) = \text{rumble}_{(V)} \\ \text{Son}(\text{gunshot}) &= \text{Func}_\theta(\text{gunshot}) = \text{‘ring out’} \end{aligned}$$

- L denotes an artefact designed to produce a sound. In this case, $\text{Son}(L) = \text{Fact}_\theta(L)$.

$$(31) \quad \begin{aligned} \text{Son}(\text{bell}_{(N)}) &= \text{Fact}_\theta(\text{bell}_{(N)}) = \text{ring}_{(V)} \\ \text{Son}(\text{gun}_{(N)}) &= \text{Fact}_\theta(\text{gun}_{(N)}) = \text{‘bang out’} \end{aligned}$$

The meaning of Son is often expressed in natural languages by onomatopoeic interjections, i.e., clausative lexical units. This implies that the LF Son naturally combines with the structural derivative LF Claus_θ (see Ch. 4, [11], p. lxxxix), as illustrated below.

English

Claus _θ Son(<i>dog</i> _(N))	=	<i>Bow-wow!, Woof-woof!</i>
Claus _θ Son(<i>cat</i> _(N))	=	<i>Meow!; Purr-purr!</i>
Claus _θ Son(<i>doorbell</i>)	=	<i>Ding-dong!, Ring-ring!</i>
Claus _θ Son(<i>collision</i>)	=	<i>Bam!</i>

French

Claus _θ Son(<i>chien</i>)	=	<i>Ouah, ouah !</i>
Claus _θ Son(<i>chat</i>)	=	<i>Miaou !; Ronron !</i>
Claus _θ Son(<i>sonnette</i>)	=	<i>Ding-dong !, Drelin !, Dring !</i>
Claus _θ Son(<i>goutte</i> [<i>d'eau</i>])	=	<i>Plic !, Ploc !</i>

Russian

Claus _θ Son(<i>sobaka</i>)	=	<i>Gav, gav!</i>
Claus _θ Son(<i>koška</i>)	=	<i>Mjau!</i>
Claus _θ Son(<i>zvonok</i>)	=	<i>Dzin', dzin'!</i>
Claus _θ Son(<i>kapli</i>)	=	<i>Kap, kap!</i>

5.4.7.2 [64] **Manif** [_{Lat.} *manifestāre* ‘manifest’]: verb that means ‘manifest itself in something’

Manif is a verb that takes as DSyntA₁ the nominal lexical unit L and as DSyntA_{II} the expression of an additional participant – i.e., the “something” in which L is manifested.

In the value’s government pattern of the illustrations below, “N” stands for the nominal expression of the additional participant.

5 Syntagmatic lexical functions

English

Manif(<i>tiredness</i>)	=	<i>show</i> _(V) [Loc _{in} N] [<i>Tiredness</i> was showing in his eyes (< on his face).]
Manif(<i>joy</i> _(N))	=	<i>beam</i> _(V) [Loc _{in} N] < <i>explode</i> [Loc _{in} N]
A ₂ Manif(<i>joy</i> _(N))	=	<i>joyful</i> [joyful look]
A ₂ Manif(<i>admiration</i>)	=	<i>admiring</i> < full of ~

French

Manif(<i>fatigue</i>)	=	<i>se lire</i> [Loc _{in} N] [<i>La fatigue</i> se lisait dans ses yeux (sur son visage)]
Manif(<i>joie</i>)	=	<i>briller, luire, rayonner</i> [Loc _{in} N]
A ₂ Manif(<i>joie</i>)	=	<i>joyeux</i> [regard joyeux]
A ₂ Manif(<i>admiration</i>)	=	<i>admiratif</i> < plein de (ART) ~, rempli de (ART) ~

Russian

Manif(<i>ustalost'</i>)	=	<i>čitat' sja</i> [Loc _{in} N] [<i>Ustalost'</i> čitalas' v ego glazax (< na ego lice).]
Manif(<i>radost'</i>)	=	<i>svetit' sja</i> [Loc _{in} N] < <i>sverkat</i> [Loc _{in} N]
A ₂ Manif(<i>radost'</i>)	=	<i>radostnyj</i> [radostnyj vzgljad]
A ₂ Manif(<i>vosxiščeni e</i>)	=	<i>vosxiščennyj</i> < <i>polnyj</i> ~ja < <i>ispolnennyj</i> ~ja

5.4.7.3 [65] Invol_v [Lat. *involvĕre* 'drag along']: verb that means 'impact something'

Invol_v controls the same deep-syntactic configuration as Manif: it is a verb that takes as DSyntA I the nominal lexical unit L and as DSyntA II the designation of an additional participant, that is, of the entity / fact impacted by L.

L's impact on something can be good or bad for the referent of L – DSyntA I of Invol_v(L) – or for the thing impacted – DSyntA II of Invol_v(L). Therefore, Invol_v is often combined with Bon_{1/2} and AntiBon_{1/2}, as shown in illustrations below.

English

Invol _v (<i>crowd</i> _(N))	=	<i>fill</i> _(V) [N]; <i>rush</i> _(V) [Loc _{ad} N]
Invol _v (<i>voice</i> _(N))	=	<i>fill</i> _(V) [N]
Bon ₂ Invol _v (<i>river</i>)	=	<i>irrigate</i> [N]
AntiBon ₂ Invol _v (<i>river</i>)	=	<i>flood</i> _(V) [N]
AntiBon ₁ Invol _v (<i>car</i>)	=	<i>crash</i> _(V) , <i>smash</i> _(V) [into N]
AntiBon ₂ Invol _v (<i>car</i>)	=	<i>hit</i> _(V) , <i>mow</i> _(V) , 'run over' [N N is a living being]

Note. The LF (Anti)Bon₁ in the last four complex LFs characterizes the effect produced by the fact denoted by Invol_v on the DSyntA I of Invol_v. Thus, *The car crashed*_{AntiBon₁Invol_v(car)} into N implies a negative effect_{AntiBon} for the car [= DSyntA I of Invol_v], while *The car ran over*_{AntiBon₂Invol_v(car)} N implies a negative effect_{AntiBon} for N [= DSyntA II of Invol_v].

French

Invol _v (<i>foule</i>)	=	<i>emplir</i> [N]; <i>déferler</i> [Loc _{ad} N]
Invol _v (<i>voix</i>)	=	<i>emplir</i> [N]
Bon ₂ Invol _v (<i>fleuve</i>)	=	<i>irriguer</i> [N]
AntiBon ₂ Invol _v (<i>fleuve</i>)	=	<i>inonder</i> [N]

AntiBon ₁ Involv(<i>voiture</i>)	=	<i>s'écraser</i> [contre N]
AntiBon ₂ Involv(<i>voiture</i>)	=	<i>faucher, renverser</i> [N N is a living being]
Russian		
Involv(<i>tolpa</i>)	=	<i>zapolnit'</i> [N-ACC]; <i>xlynut'</i> [Loc _{ad} N]
Involv(<i>golos</i>)	=	<i>napolnit'</i> [N-ACC] < <i>zapolnit'</i> [N-ACC]
Bon ₂ Involv(<i>reka</i>)	=	<i>orošat'</i> [N-ACC]
AntiBon ₂ Involv(<i>reka</i>)	=	<i>navodnit'</i> < <i>zalit'</i> , <i>zatopit'</i> [N-ACC]
AntiBon ₁ Involv(<i>avtomobil'</i>)	=	<i>naletet'</i> [na2 N-ACC], <i>vrezat' sja</i> [v2 N-ACC]
AntiBon ₂ Involv(<i>avtomobil'</i>)	=	<i>pereexat'</i> , <i>sbit'</i> [N-ACC] < <i>zadavit'</i> [N-ACC N is a living being]

5.4.7.4 [66] Sympt_{ijk} [Lat. **symptoma*]: verb that means 'show symptoms [of L] in a part of one's body [literally or metaphorically]'

The LF Sympt_{ijk} is meant to describe *somatic phrasemes* denoting psychological states – e.g., *shake with fear* – or physiological states – e.g., *choke with thirst*. It is transgressive as regards the very notion of LF, for at least two reasons.

5.4.7.4.1 1 From the viewpoint of its domain of application – i.e., the range of Ls that Sympt_{ijk} can take as argument, it is very constrained. Sympt_{ijk} can take as argument only nouns of feeling – more precisely, nominal lexical units that denote psychological or physiological states of human beings (marginally, of animals). Its inclusion within the system of standard LFs, however, is justified by the fact that Sympt_{ijk} reflects the anthropocentric organization of natural languages.

5.4.7.4.2 2 From the viewpoint of its structure, Sympt_{ijk} always functions in combination with one of the three LFs of mode of functioning (Sec. 5.4.6, p. cliff): Obstr, Stop or Excess ([5.4.6.1], [??], [5.4.6.3]), these LFs applying to their own argument L', different from L. More precisely, Sympt_{ijk}(L) is encoded using a very peculiar template shown in (32), where "X" refers to the DSyntA₁ of L, that is, to the human being:

(32) Obstr / Stop / Excess(L'('body element of X'))—Sympt_{ijk}(L).

For the LFs Obstr and Excess, the following superscripts may be used to specify the aspect of obstruction or excessive functioning:

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Table 5.1: Superscripts for *Obstr* and *Excess* in the context of *Sympt_{ijk}*

<i>color</i> : ‘with respect to color’	<i>motor</i> : ‘with respect to movements’
<i>dim</i> : ‘with respect to dimension / size’	<i>stat</i> : ‘with respect to the vertical position’
<i>express</i> : ‘with respect to expression’	<i>trem</i> : ‘with respect to trembling’
<i>fulg</i> : ‘with respect to brightness’	<i>t°</i> : ‘with respect to temperature’

Syntactically, *Sympt_{ijk}* is a verb that controls three DSyntAs:

- One corresponds to X, i.e., the DSyntA I of L; it is designated in *Sympt_{ijk}* formulas by the subscript “₁”.
- One corresponds to X’s body element that reacts to L; it is designated by the subscript “₂”.
- One corresponds to L; it is designated by the subscript “₃”.

Finally, in the formula *Sympt_{ijk}*, the **linear order** of actantial subscripts is to be interpreted as follows:

- i (first position) is the DSyntA I of *Sympt_{ijk}*;
- j (second position) is the DSyntA II of *Sympt_{ijk}*;
- k (third position) is the DSyntA III of *Sympt_{ijk}*.

English

<i>Obstr^{color}</i> (<i>face</i>)— <i>Sympt₁₃</i> (<i>fear_(N)</i>)	= [<i>N_X</i>] <i>turned pale</i> ⟨ <i>livid</i> ⟩ [with <i>...</i>]
<i>Stop</i> (<i>body</i>)— <i>Sympt₁₃</i> (<i>fear_(N)</i>)	= [<i>N_X</i>] <i>be paralyzed, freeze_(V)</i> [with <i>...</i>]
<i>Excess^{trem}</i> (<i>body</i>)— <i>Sympt₁₃</i> (<i>fear_(N)</i>)	= [<i>N_X</i>] <i>shake_(V), tremble_(V)</i> [with <i>...</i>]
<i>Excess^{trem}</i> (<i>lips</i>)— <i>Sympt₂₃</i> (<i>fear_(N)</i>)	= [<i>N_X</i> ’s <i>lips</i>] <i>quiver_(V)</i> [with <i>...</i>]
<i>Excess^{motor}</i> (<i>excrements</i>)— <i>Sympt₁₃</i> (<i>fear_(N)</i>)	= vulgar [<i>N_X</i>] <i>shit one’s pants</i> [with <i>...</i>] literal; vulgar [<i>N_X</i>] ‘ <i>shit one’s pants</i> ’ [with <i>...</i>] metaph.

Note. This case is chosen for its clear-cut character to illustrate the fact that *Sympt_{ijk}*(L) can be literal or metaphorical. We do not indicate this difference in other illustrations.

<i>Excess^{dim}</i> (<i>eyes</i>)— <i>Sympt₂₃</i> (<i>fright_(N)</i>)	= [<i>N_X</i> ’s <i>eyes</i>] ‘ <i>pop out</i> ’ of their sockets [from <i>...</i>]
<i>Excess^{express}</i> (<i>eyes</i>)— <i>Sympt₂₃</i> (<i>passion</i>)	= [<i>N_X</i> ’s <i>eyes</i>] <i>be ablaze</i> [with <i>...</i>]
<i>Excess^{fulg}</i> (<i>eyes</i>)— <i>Sympt₂₃</i> (<i>joy_(N)</i>)	= [<i>N_X</i> ’s <i>eyes</i>] <i>be shining</i> ⟨ <i>radiant</i> ⟩ [with <i>...</i>]
<i>Excess^{fulg}</i> (<i>eyes</i>)— <i>Sympt₃₂</i> (<i>joy_(N)</i>)	= [<i>...</i>] <i>be shining</i> [in <i>N_X</i> ’s eyes]
<i>Obstr^{stat}</i> (<i>legs</i>)— <i>Sympt₂₃</i> (<i>fear_(N)</i>)	= [<i>N_X</i> ’s <i>legs</i>] ‘ <i>give way</i> ’ [from <i>...</i>]
<i>Excess^{t°}</i> (<i>cheeks</i>)— <i>Sympt₂₃</i> (<i>shame_(N)</i>)	= [<i>N_X</i> ’s <i>cheeks</i>] <i>burn</i> [from <i>...</i>]

French

<i>Obstr^{color}</i> (<i>visage</i>)— <i>Sympt₁₃</i> (<i>peur</i>)	= [<i>N_X</i>] <i>blêmir</i> [de <i>...</i>]
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Stop(<i>corps</i>)—Sympt ₁₃ (<i>peur</i>)	=	[N _X] <i>être paralysé</i> [de .]
Excess ^{trem} (<i>corps</i>)—Sympt ₁₃ (<i>peur</i>)	=	[N _X] <i>trembler</i> [de .]
Excess ^{trem} (<i>lèvres</i>)—Sympt ₂₃ (<i>peur</i>)	=	[les lèvres de N _X] <i>trembler</i> [de .]
Excess ^{motor} (<i>excréments</i>)—Sympt ₁₃ (<i>peur</i>)	=	vulgar [N _X] <i>chier dans son froc</i> [de .]
Excess ^{dim} (<i>yeux</i>)—Sympt ₁₂₃ (<i>effroi</i>)	=	[N _X] <i>écarquiller</i> [les yeux d'.]
Excess ^{express} (<i>visage</i>)—Sympt ₂₃ (<i>joie</i>)	=	[le visage de N _X] <i>rayonner</i> [de .]
Excess ^{fulg} (<i>yeux</i>)—Sympt ₂₃ (<i>désir</i>)	=	[les yeux de N _X] <i>briller</i> [de .]
Obstr ^{stat} (<i>jambes</i>)—Sympt ₂₁₃ (<i>effroi</i>)	=	∅ [les jambes de N _X] <i>se dérober</i> [sous N _X]
Excess ^t (<i>joues</i>)—Sympt ₂₃ (<i>honte</i>)	=	[les joues de N _X] <i>brûler</i> [de .]
Russian		
Obstr ^{color} (<i>lico</i>)—Sympt ₁₃ (<i>strax</i>)	=	[N _X -NOM] <i>poblednet' < stat' ves' belym</i> [ot .a]
Stop(<i>telo</i>)—Sympt ₁₃ (<i>strax</i>)	=	[N _X -NOM] <i>ocepenet', okamenet'</i> [ot .a]
Excess ^{trem} (<i>telo</i>)—Sympt ₁₃ (<i>strax</i>)	=	[N _X -NOM] <i>drožat', trjastis'</i> (vsem telom) [ot .a]
Excess ^{trem} (<i>guby</i>)—Sympt ₂₃ (<i>strax</i>)	=	[guby u N _X -GEN] <i>drožat', trjastis'</i> [ot .a]
Excess ^{motor} (<i>èskrementy</i>)—Sympt ₁₃ (<i>strax</i>)	=	colloquial [N _X -NOM] <i>naklast' < naložit'> v</i> <i>štany</i> [ot .a], colloquial <i>naklast' < naložit'></i> <i>polnye štany</i> [ot .a], colloquial <i>obdelat'sja</i> [ot ..a]
Excess ^{dim} (<i>glaza</i>)—Sympt ₂₃ (<i>užas</i>)	=	[glaza u N _X -GEN] <i>vylezti iz orbit</i> [ot .a]
Excess ^{express} (<i>glaza</i>)—Sympt ₂₃ (<i>strast' '</i>)	=	[glaza N _X -GEN] <i>goret'</i> [ognëm ..i]
Excess ^{fulg} (<i>glaza</i>)—Sympt ₂₃ (<i>radost' '</i>)	=	[glaza N _X -GEN] <i>lučit'sja</i> [..ju], <i>sverkat'</i> [ot ..i]
Excess ^{fulg} (<i>glaza</i>)—Sympt ₃₂ (<i>radost' '</i>)	=	[v glazax N _X -GEN] <i>sverkat'</i> [..']
Excess ^{fulg} (<i>glaza</i>)—Sympt ₃₂ (<i>otčajanije</i>)	=	[v glazax N _X -GEN] <i>čitat'sja</i> [..e]
Obstr ^{stat} (<i>nogi</i>)—Sympt ₂₃ (<i>strax</i>)	=	[nogi N _X -GEN] <i>sdelat'sja < stat'> kak vatnye</i> [ot ..a]
Excess ^t (<i>ščėki</i>)—Sympt ₂₃ (<i>styd</i>)	=	[ščėki u N _X -GEN] <i>goret'</i> [ot ..a]

Note. More illustrations for Sympt_{ijk} can be found in the lexical entry of ANGER_(N), given in the Appendix (p. cxcvii ff.).

5.4.7.4.3 Alternative approach to Sympt_{ijk} The above presentation of the LF Sympt_{ijk} corresponds to the “classical” approach, where Sympt_{ijk} is described as a double-shot LF with two arguments (Iordanskaja 1972, 1986).

An alternative approach – introduced in Mel'čuk & Polguère (2021) – considers that the Speaker does not necessarily have in mind a given lexical unit denoting a body element when expressing Sympt_{ijk}(L). Only the actual body element could matter, regardless of which lexical unit expresses it. As a matter of fact, all above illustrations where the subscript “₂” does not appear next to Sympt correspond to phrasemes where the body element is not even lexicalized. For instance, when saying *Joey_X turns pale with fear* [Sympt₁₃(*fear*_(N))], the Speaker may think of Joey's face without having any expressive goal regarding specific lexical units such as FACE_(N) or **colloquial** MUG_(N) or **literary** VISAGE, etc. By contrast, Sympt₁₃(*fear*_(N)) is indeed a collocate of the lexical unit FEAR_(N), the base of the collocation.

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For this, in the alternative approach to Sympt_{ijk} , the *Obstr*, *Stop* and *Excess* components of Sympt_{ijk} formulas are not considered as actual **lexical** functions, their arguments being **meanings** rather than lexical units. Thus, pattern (32) given above for Sympt_{ijk} can be rephrased as (33):

(33) *Obstr* / *Stop* / *Excess* ('body element of X')— $\text{Sympt}_{ijk}(L)$.

and the above illustrations should follow pattern (33) with arguments of *Obstr*, *Stop* and *Excess* written in the form of meanings (i.e., between '...') rather than lexical units (in *italics*) – for instance:

(34) $\text{Obstr}^{\text{color}}(\text{'face'})\text{—Sympt}_{13}(\textit{fear}_{(N)}) = [N_X] \textit{turned pale} \langle \textit{livid} \rangle [\textit{with -}]$
 $\text{Stop}(\text{'body'})\text{—Sympt}_{13}(\textit{fear}_{(N)}) = [N_X] \textit{be paralyzed, freeze}_{(V)} [\textit{with -}]$

5.5 Non-LF approaches to the description of collocations

Though paradigmatic LFs can be considered primary from a theoretical point of view,⁸ the originality and relevance of LFs are better understood from the perspective of the description of collocations – i.e., from the perspective of syntagmatic LFs. For instance, the first glimpse at the notion of LF that we offer in the Introduction of the book starts with the analysis of the two English collocations *pay tribute* and *give recognition* (Ch. 1, Sec.1.1, p. xiff). As a matter of fact, LFs were invented in the early 1960s as a solution to modeling the problem of phraseologically constrained lexical choice in Machine Translation, with the conceptualization of the two iconic syntagmatic LFs *Magn* and *Oper*₁ (Mel'čuk 2015b: Ch. 14, Sec. 8, p. 269). It is thus reasonable to conclude this chapter on syntagmatic LFs – often perceived as LFs par excellence – with an overview of alternative means of description of collocations found in current lexical resources: dictionaries (5.5.1) and lexical databases (5.5.2).

5.5.1 Collocations in dictionaries

General-purpose dictionaries, as a rule, do not systematically describe collocations. One common strategy used by lexicographers is to simply mention some collocations in the entries for lexical units that are collocation bases, in particular, in lexicographic examples. For instance, the entry for the lexeme **DISTINCTION**_{1a} of the online *American Heritage Dictionary of the English Language* (American Heritage 2022) “smuggles” an *Oper*₁ for this noun in a lexicographic example:

⁸Cf. Chapter 4, where the paradigmatic LF *Syn* is presented as central in the system of LFs (Sec. 4.1, p. lxxi).

“economists making[= 0per₁] an ongoing distinction between domestic and foreign markets.”

The *Longman Dictionary* (LDOCE 2025) provides more collocations in its entry for the corresponding lexeme – numbered **DISTINCTION 1**, but with minimal information. It simply lists phrases (35a–c) together with illustrative examples, under the lexicographic definition:

- (35) a. *distinction between*
[i.e., part of **DISTINCTION 1**’s government pattern – see below in footnote 10,
the so-called “grammatical collocations” of the *BBJ Dictionary*]
- b. *clear / sharp distinction*
[i.e., Magn(*distinction 1*)]
- c. *make / draw a distinction*
[i.e., 0per₁(*distinction 1*)]

In a dictionary where collocations are classified, this classification is mainly based on the PoS of the collocate or on the syntactic structure of the collocations. Such is prototypically the case in *dictionaries of collocations* (also called *collocation(s) dictionaries*). For instance, the *Oxford collocations dictionary for students of English* (OCD 2022) lists four groups of collocations in its entry for **ANGER_(N)**, presented and commented in Table 5.2.

Table 5.2: Classified collocations of **ANGER_(N)** in the OCD (2022)

OCD’s description	Comments
<u>adj.</u> bitter, black [...] terrible genuine, real [...]	A significant proportion of collocates listed in this section of the OCD entry correspond to the two modifying syntagmatic LFs Magn and Ver. Note that the symbol “ ” is used in the OCD to separate different subclasses of collocates that correspond to different (families of) LFs.
<u>quant.</u> burst, fit, outburst <i>He slammed the door in a fit of anger.</i>	The collocations these lexical units form with their base ANGER_(N) correspond to values of the paradigmatic LF Sing – cf. the lexicographic entry for ANGER_(N) in the Appendix (p. cxcviii). On analytical values of paradigmatic LFs, see Ch. 3, Sec. 3.3.3.1 (p. li).
verb + anger be filled with, feel, shake with, tremble with [...] express, give vent to, show [...]	This section of the OCD entry lists a heterogeneous set of collocates, from the viewpoint of syntagmatic LFs: in the case of collocates listed here, 0per ₁ , Sympt ₁₃ and Perm ₁ Fact ₀ .
anger + verb boil over/up, bubble up [...] abate, drain [...]	This section contains two families of collocations corresponding to IncepPredPlusMagn vs. IncepPredMinusMagn.

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Note that the online *Longman Dictionary* (LDOCE 2025) – mentioned above to illustrate the strategies used in general-purpose dictionaries – encapsulates a full-fledged dictionary of collocations. In the case of the collocations (35b–c), that are embedded in the entry for **DISTINCTION 1a**, they are further described among other collocations in a table positioned at the end of the vocable’s entry (instead of being inserted in the entry for **DISTINCTION 1a**, where it belongs). As for most dictionaries of collocations, a syntactic classification is used in this table: two sections “verbs” and “adjectives” under which collocations are grouped according to the PoS of collocates.

The *BBi Combinatory Dictionary of English* (Benson et al. 2010)⁹ deserves a special attention: while not explicitly using LFs to describe collocations, it has borrowed from the LF approach the idea of classifying lexical collocations¹⁰ both syntactically and semantically (see the dictionary’s preface, p. ix). The classification used by this dictionary is based on explicitly defined *types*. For instance, the L4 type is defined as follows (Benson et al. 2010: xxxiii):

L4 collocations consist of a noun and verb; the verb names an action characteristic of the person or thing designated by the noun: *adjectives modify*, *alarms go off* (ring, sound), *bees buzz* (sting, swarm), *blizzards rage*, *blood circulates* (clots, congeals, flows, runs), *bombs explode* (go off), etc. The *Dictionary* does not include predictable combinations such as *bakers bake*, *boxers box*, *cooks cook*, *dancers dance*, *fencers fence*, etc.

The L4 type encompasses at least the two LFs Fact_i and Son , together with their phasic and causal variants – cf. $\text{IncepSon}(\text{alarm}_{(N)}) = \text{‘go off’}$. As a rule, the BBI classification system is significantly coarser than that of LFs, which is justified by the fact that this dictionary targets the general public. For the same reason, type names (L1, L2, ...) are not mentioned in the entries. Types are used to group and list collocations in a canonical order throughout the dictionary,¹¹ and illustrative phrases are used to help the users grasp the semantic-syntactic value of collocations.

⁹This dictionary is known to be the first general-public dictionary of English collocations; its first edition appeared in 1986 under the title *The BBI Combinatory Dictionary of English: A guide to word combinations*.

¹⁰The BBI dictionary distinguishes between *lexical* collocations – i.e., constrained combinations of lexical units that we model by means of syntagmatic LFs – and *grammatical* collocations – i.e., prepositional/ conjunctive constructions controlled by verbs, nouns, etc., to express their actants (*cope with*, *rely on / upon*, ...). These phrases are actually not collocations as they pertain to the active syntactic valence of predicative lexical units, modelled by government patterns in Explanatory Combinatorial Dictionaries (see Ch. 2, Sec. 2.3, p. xxxii).

¹¹In the same vein, LF relations are always listed in a canonical order in Explanatory Combinatorial Dictionaries – cf. Table 3.5 (Ch. 3, Sec. 3.8, p. lxvii).

Presenting an extensive review of the description of collocations in dictionaries is beyond the scope of the present section and we have to limit ourselves to a selection of representative and well-known lexicographic approaches. This also applies to our next topic: the description of collocations in lexical databases that are not making use of LFs.

FURTHER READING. On techniques used by lexicographers for embedding collocational information in general-purpose dictionaries, see Gouws (2015); on dictionaries of collocations, see Hausmann (1989) and Corpas Pastor (2016); on the use of popularization formulas to paraphrase LF relations, see Polguère (2000a), Mel'čuk & Polguère (2007) and L'Homme (2008); on the automatic extraction and syntactic classification of collocations for the *Antidote* software suite, see Charest et al. (2010).¹²

5.5.2 Collocations in lexical databases

By *lexical databases*, we mean descriptions of lexical entities that have the following properties:

- the information they contain is structured, stored and made available as computational databases (simple spreadsheets, relational SQL tables, etc.);
- though they can eventually be accessed directly by human users, they are meant to be compatible with computational exploitations, such as in Natural Language Processing;
- for this reason, they abide to well-defined formalization principles, though the degree for formalization can fluctuate significantly among lexical databases.

As it is the case with dictionaries, we have to operate a drastic choice among all existing lexical resources. We focus on one of them, the set of *lexicon-grammar tables*, which offers a broad-scale description of the French lexicon and has given rise to significant exploitation in linguistic research and Natural Language Processing.

The *lexicon-grammar* (Gross 1988b, 1994) originated from the *distributional analysis* approach put forward by Zellig Harris (Harris 1954, Sahlgren 2008). It was oriented toward the lexicographic construction of lexical resources deemed to be exploitable in Natural Language Processing, with a focus on the French language.¹³

¹²The current bilingual distribution of *Antidote* (Antidote 2025) contains a range of linguistic resources: dictionaries, grammars and text correctors. Its dictionaries of collocations – named *Dictionary of combinations*, for English, and *Dictionnaire des cooccurrences*, for French – list collocations according to the syntactic dependencies that connect bases and their collocates.

¹³For an application to English, see Garcia-Vega & Machonis (2014).

5 Syntagmatic lexical functions

The study of phraseology is at the center of the lexicon-grammar, which is well-known for its description of support verbs (Sec. 5.4.2, p. cxxxi^{ff}), seen as tools for verbalizing nominal predicates and, therefore, as sentence-builders.¹⁴ As for formalization, support verb collocations are described in tables where rows correspond to predicative nouns and columns correspond to potential verbal collocates. The acceptability vs. non-acceptability of a given collocation is indicated in the corresponding table cell by “+” vs. “–” signs, as illustrated in Figure 5.9.¹⁵

<ID>	<ENT>N	avoir = concevoir	avoir = éprouver	avoir = ressentir	Det =: un	Det =: un-Modif	Det =: du	Prép1 =: à l'égard de
1	admiration	+	+	+	-	+	+	-
2	affection	+	+	+	-	+	+	-
3	amitié	+	+	+	-	+	+	-
4	amour	+	+	+	-	+	+	-
5	animosité	+	+	+	-	+	+	-
6	antipathie	+	+	+	-	+	+	+
7	appréhension	-	-	+	-	+	+	-
8	attachement	-	+	+	-	+	+	+

Figure 5.9: Extract from the N_aa.lgt lexicon-grammar table¹⁶

Explanations. The second column lists French nouns of feelings: *admiration*, *affection*, *amitié* ‘friendship’, *amour* ‘love_(N)’, *animosité* ‘animosity’, *antipathie* ‘antipathy’, *appréhension* ‘apprehension’ and *attachement* ‘attachment’. Columns 3 to 5 indicate the (in)compatibility with French support verbs *concevoir* lit. ‘conceive’, *éprouver* ‘experience_(V)’ and *ressentir* ‘feel’. The specific class of support verbs treated here is identified with the formula “avoir =” in reference to the joker Oper_1 for French nouns of feeling: *avoir* ‘have’. The remaining columns are used to specify the other syntactic elements of the collocation (determiners and noun complements) – cf. government patterns of value elements in LF applications (Ch. 3, Sec. 3.1.2, p. xl).

Lexicon-grammar tables describe all three families of support verbs identified in terms of LFs as Oper_i , Func_i and Labor_{ij} ; they also encompass verbal collocates that incorporate phasic meanings. Interestingly, working on support verbs for nouns of feelings lead lexicon-grammar research to take into consideration phraseological expressions corresponding to the Sympt_{ijk} LF – cf. the construc-

¹⁴Cf. Maurice Gross’s semantic-syntactic credo: “The unit of meaning is the elementary sentence, and not the word” (Gross 1988b: 182).

¹⁵This is a simplification; in fact, lexicon-grammar tables make use of a richer set of cell values.

¹⁶Lexicon-grammar tables are archived on the website of the LIGM laboratory: <https://infolingua.univ-eiffel.fr> (last consulted 15/12/2025).

tion of a “local grammar”¹⁷ of French nouns of feelings in Gross (1988a: Sec. 4.2) and Gross (1995: Sec. 5).

To conclude, let us mention that there exist numerous lexical databases that are using LFs to encode collocations (and paradigmatic lexical relations); to name a few: for French, the French Lexical Network [fr-LN] (Lux-Pogodalla & Polguère 2011),¹⁸ for Spanish, DiCE (Alonso Ramos et al. 2010) and Diretes (Barrios Rodríguez & Boguslavsky 2019); for English and Russian, the multilingual dictionaries of the ETAP-3 linguistic processor (Boguslavsky et al. 2004).

FURTHER READING. On the automatic enrichment of the Princeton WordNet with collocational information, see Espinosa Anke et al. (2016); on collocational information in the FrameNet lexical database, see Ruppenhofer et al. (2002); for connections between syntagmatic LFs and collocational data in FrameNet, see Langer (2005: 3–5) and Bouveret & Fillmore (2008).

¹⁷Local grammars are formal graph-based representations of the structure of utterances used in Natural Language Processing (in particular, for sentence and phrase extraction from corpora).

¹⁸The fr-LN is presented in detail in Chapter 6.

6 Lexical networks

This concluding chapter examines the role played by lexical functions [LFs] in the structure of natural language lexicons and, consequently, their relevance in the structuring of lexicon models. Because lexicons are large informational systems involved in complex mental processes (speech production / understanding, language acquisition, reasoning with elaborate concepts, etc.), they can be expected to have a structure that supports their involvement in such processes, and it is where LFs come into play.

The preceding chapters deal with the relevance of LFs from a dual perspective:

- the writing of lexical entries – cf. the LF zone of corresponding lexical entries, as illustrated with the entry for $\text{ANGER}_{(N)}$ (Appendix, p. cxcvii^{ff.});¹
- the modeling of lexical selection in speech production (Ch. 3, Sec. 3.2.2.2, p. xliv) – cf. the role of LFs in deep-syntactic structures and in deep-syntactic paraphrasing rules described throughout the book.

We now take a broader view on the system of standard LFs and explain the key role they play in organizing the Speaker's *lexical knowledge*. The implication of what will be presented now is twofold. Firstly, the system of lexical relations induced by LF applications can be used to determine the shape of new *models of the lexicon* – for short, *lexicon models* – and, therefore, can be exploited to develop a new type of lexicography that goes beyond traditional dictionary writing (as demonstrated in Sec. 6.3, p. cxcii^{ff.}). Secondly, in terms of applicability, lexicon models primarily structured by LF relations are expected to be better *data structures* than ordinary – textual – dictionaries, in the sense that they are better tools for the study and description of essential lexicon-related phenomena, such as lexical acquisition and lexical access / selection by the Speaker.

The present chapter contains the following sections:

- What is the “shape” of natural language lexicons? (6.1)

¹LF applications that systematically illustrate the presentation of each individual LFs in Chapters 4 and 5 are in reality part of lexicographic information: an expression of the form “ $f(L) = L'$ ” belongs to the LF zone of L 's lexical entry.

- Lexical Systems as network lexicon models (6.2)
- Proof of concept: the French Lexical Network [fr-LN] (6.3)

6.1 What is the “shape” of natural language lexicons?

The question of the structure of lexical knowledge – and, consequently, the formal structure of models that describe it – has been largely disregarded by lexicologists and lexicographers in favor of a dictionary-based approach. Under this approach, the *macrostructure* (Wiegand 1996) of a lexicon model is an alphabetically sorted list of lexical entries; this is the common way to structure most of the known dictionaries. However, there have been notable exceptions, for instance:

- For dictionaries, the *production dictionaries*, inaugurated for English by the *Longman Language Activator* (Summers 1993). Its thesaurus-like macrostructure gives the users access to lexical entries via a set of conceptual keywords, thus breaking the linear alphabetical barrier and allowing for a semantic navigation through lexical knowledge.²
- For lexical databases, the *-Net models* (Fontenelle 2012), such as *WordNets* (Princeton WordNet, EuroWordNet, etc.), which adopt a *taxonomic* perspective on the structuring of lexical knowledge (Fellbaum 1998), or *FrameNet*, which is based on lexical relations induced by Frame Semantics (Fillmore et al. 2003).

Given this state of affairs, we have to answer the following question: What is the “shape” of a natural language lexicon? Of course, a real lexicon of natural language is a purely informational (abstract) entity and does not have actual physical shape. We use the figurative expression *shape of the lexicon* in order to stress the fact that, in advanced formal lexicology and lexicography, we need adequate structuring and visualizations of lexicon models (see Figure 6.1, p. clxx) that could support reasoning on such knowledge representations.

To develop the notion of lexicon model, we will introduce two postulates: one, immediately below, and the second, in Section 6.2 (p. clxxv).

Postulate 1: structure of lexicons. Natural language lexicons are, first and foremost, sets of lexical entities connected by lexical relations. Formally, they are *lexical networks*; therefore, lexicon models should be structured as such.

²Cf. the illustration given in Rundell & Ham (1994: 176): the ANGRY concept, which gives access to (dictionary entries for) “lexical realizations” such as: *angry, cross, enrage, fly off the handle, furious, incensed, mad, maddening, mollify, rage, stormy, wind someone up*.

6.1 What is the “shape” of natural language lexicons?

This postulate is taken for granted – by definition, a postulate does not call for a proof. However, its **adequacy** can be evaluated in the context of descriptive lexicology – i.e., lexicography – and in that of its applications – language teaching, Natural Language Processing, etc. Postulate 1 is also in line with a whole array of observations made in disciplines as varied as psycholinguistics, speech disorder research or neurolinguistics.³

Lexical networks are often associated in contemporary research with the notion of *Mental Lexicon* – roughly, lexical knowledge as stored in the brain (Aitchison 2012, De Deyne et al. 2017). We believe that network models of the lexicon – in particular, Lexical Systems presented in Sec. 6.2, p. clxxiii^{ff}. – are indeed plausible models of the Mental Lexicon (Polguère 2016).

However, the sole compliance to the principle of representing lexical knowledge as lexical network is insufficient to properly determine the formal nature of resulting lexicon models. The lexicon is a set of lexical entities, but which ones? They are connected by relations, but which ones? The nature of these entities and these relations determines the “true” formal structure of lexicon models. To illustrate this, let us contrast two approaches to structuring lexical networks: taxonomic networks, exemplified with the Princeton WordNet, vs. small-world networks, exemplified with the French Lexical Network [fr-LN] presented in Sec. 6.3 (p. cxcii^{ff}).

1) The Princeton WordNet, for English,⁴ is a computational lexicon model – a lexical database – that follows taxonomic principles postulated in the context of psycholinguistic research (Miller et al. 1990).⁵ It is probably the most well-known and most widely used lexical database in Natural Language Processing. In the Princeton WordNet, lexical units called *senses* are grouped into small sets of (quasi-)synonyms called *synsets*, and it is those synsets that are linked within the original WordNet’s lexical network. They constitute lexical classes primarily connected by the relation of hyperonymy / hyponymy. This entails a global taxonomic organization of the lexicon model, characterized by its creators as being an *inheritance system* (Miller et al. 1990: 242).

³This is no place for a detailed review of the huge literature on the relevance / application of lexical networks in these areas. We content ourselves with the mention of particularly significant texts.

⁴We focus here on the Princeton WordNet for the English language. The WordNet approach has been applied to a broad range of natural languages and has also been adapted to the construction of multilingual lexical resources (Fellbaum & Vossen 2012).

⁵A taxonomy – in the most general sense – is a model of a vast natural domain organized as a hierarchy of classes and subclasses of entities. Linnaeus’s classification of species is generally cited as prototypical taxonomic model in science. For a presentation and critique of this taxonomy, see Ereshefsky (1994).

2) The fr-LN is a computational lexicon model built according to the principles of Explanatory Combinatorial Lexicology (Lux-Pogodalla & Polguère 2011). Its structure has the characteristic properties of small-world networks (see details in Secs. 6.2 and 6.3); it can be loosely characterized as being a “social network” of lexical entities (mainly, lexical units) connected by paradigmatic and syntagmatic lexical relations, that are, in most part, LF relations. Though not taxonomic, its small-world organization puts order in the apparent mess of the lexicon: this organization allows for the computation of meaningful lexical clusters and, more generally, of topological proximity⁶ – cf. the geometric distribution of lexical nodes in Figure 6.1.

Figure 6.1 visualizes the formal gap between the two approaches, with a sample of WordNet’s taxonomy centred around the lexical unit WATERCOURSE 2 – a member of the “{ STREAM 1, WATERCOURSE 2 }” *synset*, next to a sample of the fr-LN centred around the corresponding French lexical unit ‘COURS D’EAU’.

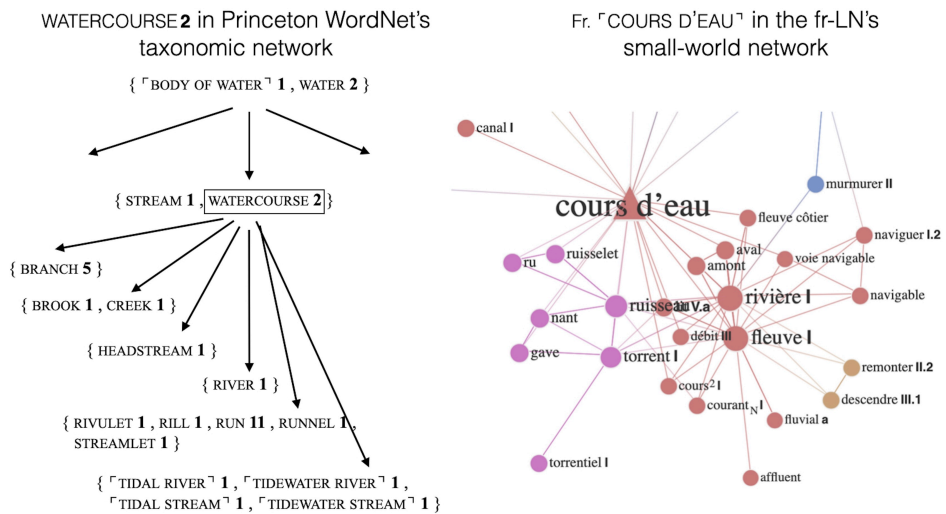


Figure 6.1: Taxonomic vs. small-world lexical networks

Though both the Princeton WordNet and the fr-LN ambition to reflect the organization of the Mental Lexicon, the above figure clearly shows how different structuring principles adopted for lexical network models induce different characteristic “shapes” of these models.⁷

⁶On the topological proximity of lexical nodes in networks / graphs, see Sec. 6.2.2.2, p. cxc.

⁷For arguments in favor of choosing small-world lexical networks over taxonomic ones, see Pol-

6.1 What is the “shape” of natural language lexicons?

We present in Section 6.2 (p. clxxiii^{ff.}) lexicon models called Lexical Systems, of which the fr-LN is a particular instance, and highlight the fundamental role played by LF relations in such network lexicon models. Before we do that, let us briefly examine paradigmatic lexical relations encoded in WordNet from the perspective of LFs.

6.1.0.0.1 WordNet’s paradigmatic relations from the LF perspective The taxonomic organization of WordNet – illustrated in Figure 6.1 above – operates only in the nominal “sector” of this lexical database (Miller 1990), which is subdivided according to part of speech classification (nouns, verbs, adjectives and adverbs) and follows different structuring principles for each part of speech.

Furthermore, WordNet has a hybrid network structure where *senses* themselves (and not only *syntsets*) can be connected. Such is the case for WordNet’s so-called *morphosemantic links* that were grafted onto the original version of the database (Miller & Fellbaum 2003): they indicate lexical relations that would be encoded by us with paradigmatic LFs. In WordNet, these links are not typed, but identified under the same label: *derivation*, described as being a “derivational morphology pointer.” This is illustrated in Figure 6.2, which shows the WordNet subnetwork of derivation links extracted by taking the sense MARRY_(V) 1 as point of origin and recursively following these links until all available derivation links are exhausted.⁸

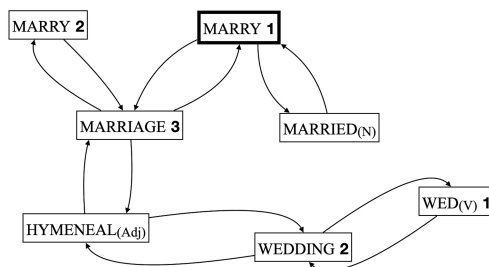


Figure 6.2: Subnetwork of derivation links in WordNet originating from MARRY_(V) 1

In order to clarify the nature of WordNet senses appearing in the above figure, we list in Table 6.1 their definitions in WordNet – called *glosses* (Miller et al. 1990: 240) – together with the relevant neighboring polysemy.

guère (2014) and Gotkova et al. (2024). The computable nature of small-world lexical networks as data structures is presented in Sec. 6.2.2 (p. clxxiii^{ff.}).

⁸The present discussion is based on data extracted from version 3.1 of the Princeton WordNet.

Table 6.1: Extract of WordNet’s polysemy around the vocable MARRY

WordNet senses	WordNet glosses
MARRY 1	<i>take in marriage</i>
MARRY 2	<i>perform a marriage ceremony</i>
MARRIAGE 1	<i>the state of being a married couple voluntarily joined for life (or until divorce)</i>
MARRIAGE 2	<i>two people who are married to each other</i>
MARRIAGE 3	<i>the act of marrying; the nuptial ceremony</i>
MARRIED _(N)	<i>a person who is married</i>
HYMENEAL _(Adj)	<i>of or relating to a wedding or marriage</i>
WEDDING 1	<i>the social event at which the ceremony of marriage is performed</i>
WEDDING 2	same gloss as its synonym MARRIAGE 3: it belongs to the same synset ⁹
WED _(V) 1	same gloss as its synonym MARRY 1
WED _(V) 2	same gloss as its synonym MARRY 2

Let us make four comments about Figure 6.2, taking the perspective of the encoding of paradigmatic relations by means of LFs.

1) The derivation links encode a very heterogeneous set of paradigmatic relations that can be discriminated with LFs; e.g., in Figure 6.2:

- (1) MARRY 1 $\vdash S_0 \rightarrow$ MARRIAGE 3
 MARRIAGE 3 $\vdash V_0 \rightarrow$ MARRY 1
 MARRY 1 $\vdash S_{1/2} \text{Perf} \rightarrow$ MARRIED_(N)

This is a known issue with derivation links in WordNet. For a proposal to improve the description, see Fellbaum et al. (2009).

2) A derivation link does not necessarily encode a morphological derivation, in spite of being termed *morphosemantic* in the WordNet literature. Such is the case of the link between MARRIAGE 3 and HYMENEAL_(Adj) in Figure 6.2: a case of A_0 , in terms of LFs. Because morphology is not necessarily involved, these links can be considered as encoding **semantic** derivations, just like paradigmatic LFs.

3) The encoding of semantic derivations in WordNet can be implemented in surprising ways. For instance, the absence of SPOUSE in Figure 6.2 is somehow strange, given the presence of the very marginal term MARRIED_(N). The noun SPOUSE happens to be connected to MARRIAGE 2 – not MARRIAGE 3 – by the WordNet relation of meronymy. Because MARRIAGE 2 is glossed as denoting a set of two persons rather than a social relation between two individuals (cf. Table 6.1),¹⁰

⁹WordNet’s glosses are associated to synsets, not to individual senses.

¹⁰WordNet illustrates this sense with the following example: *his second marriage was happier than the first*.

SPOUSE is modeled as denoting a “unit” of a marriage² and therefore does not surface in Figure 6.2.

This last observation illustrates the importance of organizing the modeling of paradigmatic – and syntagmatic! – relations by means of a coherent, well-defined **system** of such relations that guides the process of exploring – not just encoding – lexical relations.

6.2 Lexical Systems as network lexicon models

A *Lexical System* (Polguère 2009, 2014) – with capitalized initials – is a lexicon model structured as a network of lexical entities connected, primarily, by paradigmatic and syntagmatic LF relations. Though built according to the principles of Explanatory Combinatorial Lexicology, Lexical Systems have a formal structure radically different from that of Explanatory Combinatorial Dictionaries (Mel’čuk & Žolkovskij 1984, Mel’čuk & Žolkovskij 2016, Mel’čuk et al. 1984, 1988, 1992, 1999), these latter being textual (therefore, linear) representations of lexical knowledge.

This section contains two parts: main characteristics of Lexical Systems (6.2.1) and formal properties of Lexical Systems (6.2.2).

The Lexical System approach has been applied to the construction of several lexical networks [*LN*s], in particular, for French [*fr-LN*], English [*en-LN*] and Russian [*ru-LN*]. Because at the present time only the *fr-LN* (Sec. 6.3) can be considered as being a full-fledged Lexical System, our illustrations are mostly borrowed from this particular lexical database.¹¹

6.2.1 Main characteristics of Lexical Systems

6.2.1.1 LF relations: the backbone of Lexical Systems

Going back to Postulate 1 (the network structure of lexicons: Sec. 6.1, p. clxviii), we can characterize Lexical Systems as informational structures made up of two types of elements:

1. lexical entities, as defined in Chapter 2 (Sec. 2.1, p. xxiv) – primarily, lexical units, but also, compositional phrasemes¹² and word-formation means analogous to lexemes;

¹¹For information on exploratory lexicographic work conducted on the *en-LN*, see Gader et al. (2014); on the *ru-LN*, see Krylosova (2017) and Mikhel (2022).

¹²I.e., collocations and (linguistic) clichés. An illustration of a lexical entity of the *fr-LN* that is

2. lexical relations between these entities, as defined in Chapter 3 (Sec. 3.2.1, p. xli) – primarily, LF relations (other relations that play an important structuring role in Lexical Systems are presented in Sec. 6.2.1.2, p. clxxviii^{ff}).

Elements of the first type correspond to the **nodes** of the lexical network and elements of the second type correspond to the **arcs** of the network. Figure 6.3 illustrates these two notions; it is a blow-up of a segment of the fr-LN’s subnetwork given in Figure 6.1 (p. clxx) that features:

- two lexical nodes – Fr. CANAL I ‘[water] canal’ and ‘COURS D’EAU’ ‘watercourse’,
- connected by an arc corresponding to the LF relation Gener; i.e., in terms of LF application, it encodes $\text{Gener}(\text{canal I}) = \text{‘cours d’eau’}$.

As a rule, an LF application $f(L)$ translates within a Lexical System as a set of arcs labeled with the LF f , that originate from the lexical node L and lead to each individual value elements of the LF application. They are called *outgoing arcs*, relative to L , and *incoming arcs*, relative to these elements.

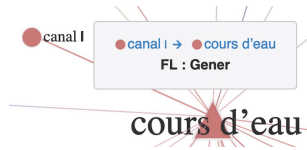


Figure 6.3: The LF application $\text{Gener}(\text{canal I } \text{‘[water] canal’}) = \text{‘cours d’eau’}$ ‘watercourse’ as represented in the fr-LN

Visualization of Lexical Systems. In the present chapter, we make use of a dedicated *network browser*, called *Spiderlex*, for the visualization of the content of Lexical Systems (Ollinger et al. 2020, Ollinger & Polguère 2020). Spiderlex exploits the mathematical properties of the small-world structure of Lexical Systems (Sec. 6.2.2, p. clxxxiii^{ff}) and plays an important role for the utilization of lexical data both by lexicographers and users of the fr-LN – see “New lexicography of lexical networks,” Sec. 6.3, p. cxciv. Note that the orientation of arcs and their labels are not displayed by default, as in Figure 6.1, in order to keep the visualization legible; this information can be shown by pointing at individual arcs, as in Figure 6.3.¹³

a linguistic cliché can be found in Figure 6.10 (Sec. 6.2.2.2, p. cxcii): Fr. *Fumer tue* ‘Smoking kills’ (bottom left corner of the figure). This cliché is a pragmateme (Mel’čuk 2020) used as a warning to smokers, written on packs of cigarettes.

¹³Data from the fr-LN used in this chapter reflect the state of this constantly evolving model at time of writing. In the pop-up of Figure 6.3, “FL” stands for Fr. *fonction lexicale*.

The notion of Lexical System, first introduced in Polguère (2009), gradually took shape out of the practice of computational Explanatory Combinatorial Lexicography (Steinlin et al. 2005, Polguère & Mel'čuk 2006). It was based on the observation that the system of simple standard LFs – in particular LFs identified in this book as being central (Ch. 3, Sec. 3.8, p. lxvi) – was a good candidate as main contributor to the structuring of lexical knowledge, better by all means than mainstream taxonomic principles. This has led to a second postulate, this time concerning the structure of Lexical Systems.

Postulate 2: structure of Lexical Systems. Lexical Systems, as network lexicon models, are primarily structured by standard LF relations.

Postulate 2 implies that LF relations form the backbone of Lexical Systems. From a methodological viewpoint, the lexicographic process of building a Lexical System for a given natural language is driven by the “weaving” of standard LF relations within the lexical network (Lux-Pogodalla & Polguère 2011, Polguère 2012b).

Two properties of standard LFs can be put forward in support of the adoption of Postulate 2:

- 1) Standard LFs express constrained meanings (Ch. 3, Sec. 3.1.1, p. xxxvii), present in all natural languages; these meanings determine fundamental paradigmatic and syntagmatic relations between lexical units.
- 2) A standard LF is essentially related to the meaning of each individual lexical unit *L* it takes as argument, in the sense that it is anchored in *L*'s definition – cf. the notion of LF target (Ch. 3, Sec. 3.3.2, p. l). Conversely, a standard LF can serve as a revealer of the presence of a particular semantic chunk in *L*'s meaning (Iordanskaja & Polguère 2005).

6.2.1.1.1 Relational profile of a node / lexical unit Because of properties 1) and 2) above, it is justified to consider that each node *n* of a Lexical System – and the corresponding lexical unit *L* – has a *relational profile* consisting of two sets: 1) the set of *n*'s outgoing arcs and 2) the set of its incoming arcs.

1) The set of *n*'s outgoing arcs – i.e., *L*'s *outgoing profile* – is the main constituent of *L*'s relational profile as it is a reflection of the essential semantic and combinatorial properties of *L*. It corresponds to the information listed in the LF zone of *L*'s lexical entry in an Explanatory Combinatorial Dictionary – see Appendix, p. cxcviii^{ff}.

2) The set of *n*'s incoming arcs – i.e., *L*'s *incoming profile* – is a relational information that is **not** present in the LF zone of *L*'s lexical entry in an Explanatory Combinatorial Dictionary and, more importantly, that is simply not encoded in this type of model. In a Lexical System, by contrast, incoming profiles are embedded in the model's network structure and can be accessed directly. This is illustrated in Table 6.2, with an adaptation of the Spiderlex visualization of both the current outgoing and incoming profiles of *Fr. SENTIMENT 1a* 'feeling' in the fr-LN. The complementarity of the two profiles is well illustrated by the contrast between the unique outgoing Gener relation (towards *Fr. état psychique* 'psychological state') and the large set of incoming Gener relations, which characterizes *Fr. SENTIMENT 1a* as being a generic term (i.e., classifying lexical unit) in the French lexicon.

Table 6.2: Outgoing vs. incoming profiles of *Fr. SENTIMENT 1a* 'feeling' in the fr-LN

Outgoing LF relations →	Incoming LF relations ←
Syn _⊃ : <i>émotion 2</i>	Syn _⊃ for <i>émotion 2</i>
Syn _⊂ : <i>sensation 1, spec affect a; spec état émotionnel</i>	Syn _⊂ for <i>affect a – état émotionnel – sensation 1</i>
Gener : <i>(spec) état psychique</i>	Gener for <i>admiration – allégresse – amour 1.1a – amour 1.1b – animosité 1 – appitoiement a – appitoiement b – aversion – bonheur 1 – cafard – chagrin_(N) 1 – colère 1 – compassion – contentement – curiosité 1.2 – déception – dégoût – déprime – désespoir 1 – désir 1 – désir 2 – dévastation 1.2 – doute – effroi – émoi – émotion 1 – empathie – étonnement – fierté 1.1 – frayeur – "froid dans le dos" – frousse – frustration – fureur 1 – gratitude – haine 1 – haine 2 – honte 1.1 – horreur 1 – horreur – humiliation 2 – jalousie 1 – "mal au cœur" – "mal du pays" – malaise 3 – mécontentement – mélancolie 1.1 – mépris – nausée – nostalgie 1 – orgueil 2a – peur 1 – plaisir 1.1 – rage – ravissement – reconnaissance – répugnance 1 – répulsion 1 – ressentiment – satisfaction – sentiment 1b – souci 1 – spleen – tristesse 1 – trousse</i>
S _{loc} : <i>cœur 11.1; âme 1, esprit 1.1</i>	S ₂ for <i>éprouver a</i>
Magn : <i>vif 11.2, profond_(Adj) 11, fort_(Adj) 11.2</i>	S ₂ ^{prototyp} for <i>emoji</i>
Oper ₁ : <i>éprouver a [ART -], ressentir [ART -]; nourrir 11.1 [ART -], avoir 11.2 [ART -]</i>	S ₃ ^{prototyp} for <i>chantage 11</i>
Func ₁ : <i>habiter 11 [N_X] < étreindre 11 [N_X] < submerger 11 [N_X]</i>	
IncepFunc ₁ : <i>naître 11 [en N_X], envahir 11 [N_X], saisir 11 [N_X], prendre 11 [N_X] < s'emparer 11.2 [de N_X]</i>	
Caus ₁ Func ₂ : <i>transférer 11 [ART - sur N_Y]</i>	
Fact ₁ : <i>guider 11 [N_X], animer 11 [N_X]</i>	
IncepManif : <i>se peindre [sur N]</i>	

Table 6.2 shows that Lexical Systems' data need not be visualized graphically. Textual, dictionary-like visualizations similar to lexical entries of a traditional dictionary can also be generated from the same data. For this reason, the construction of Lexical Systems is presented in Polguère (2012a) as being a *lexicography*

of virtual dictionaries.

Incoming profiles could play an important role in applications of lexical network models, for instance, in language teaching. Thus, the meaning of the generic term PRECIPITATION 1 ‘type of weather condition’¹⁴ is better taught to language learners by relying on this term’s incoming profile, i.e., by listing some of the most typical lexical units for which PRECIPITATION 1 is a Gener – RAIN_(N), SNOW_(N), HAIL_(N), etc. – as most dictionaries do.¹⁵ This is probably a better pedagogical strategy than trying to first provide learners with a lexicographic definition that can only be non-specific and will be felt as more confusing than explanatory.

This observation does not necessarily apply to any generic terms, in particular to generic terms with more specific meanings, for which a lexicographic definition can be a good starting point in vocabulary teaching – e.g., generic terms such as BIRD, TREE_(N), etc. Collocative lexical units – lexical units that are essentially selected by the Speaker as collocates in collocations – are another illustration of the usefulness of incoming profiles. Thus, the intensifier HEAVY_(Adj) 2 [*heavy rain*]¹⁶ controls very few outgoing LF relations and its relational profile resides essentially in all its incoming Magn relations – the set of all lexical units for which it is a Magn (ACCENT_(N), ACCUSATION, APPLAUSE, BATTLE_(N), BLOW_(N), EMPHASIS, FOG_(N), etc.). The meaning of HEAVY_(Adj) 2 is simply that of Magn – the same constrained meaning as for dozens of adjectival intensifiers in English; consequently, its incoming profile is as important as its definition in the acquisition of this lexical unit by language learners.¹⁷

6.2.1.1.2 Semantic weight of LF relations All LF relations found in the relational profile of a given lexical unit L do not carry the same *semantic weight*.¹⁸ Indeed, LFs are very diverse with respect to the semantic proximity they **inherently** imply – or not – between their argument and their value. For instance,

¹⁴The lexicographic number of PRECIPITATION 1 comes from the LDOCE.

¹⁵See, for instance, the largely “enumerative” definition of LDOCE: ‘any form of water, such as rain, snow, sleet, or hail, that falls to the earth’s surface’. On defining so-called *functional collective superordinates* within the Natural Semantic Metalanguage approach, see Goddard (2017).

¹⁶The lexicographic number of HEAVY_(Adj) 2 comes from the LDOCE.

¹⁷It is interesting to contrast two types of dictionaries of collocations in the light of the notion of relational profile: (i) those whose entries are collocation bases and that provide the list of collocates they select – see Benson et al. (2010) for English and Bosque (2006) for Spanish; (ii) those whose entries are collocates and that provide the list of corresponding bases in collocations – see Bosque (2010) for Spanish. The former describe the outgoing relational profile of bases, while the latter describe the incoming relational profile of collocates.

¹⁸The notion of semantic weight is further developed in Sec. 6.2.2, p. clxxxiii ff.

$\text{Syn}(L) = L'$, $\text{V}_0(L) = L'$, $\text{A}_1(L) = L'$, etc., obviously imply a strong *meaning relation* between L and L' , as L' is a semantic derivative of L . By contrast, $\text{Magn}(L) = L'$, $\text{Loc}_{in}(L) = L'$, $\text{Oper}_1(L) = L'$, etc., do not inherently imply any meaning relation between L and L' ; such relation may exist “incidentally,” for a given case of $f(L) = L'$, but not by virtue of the meaning ‘ σ^f ’ of f – e.g., $\text{Magn}(\text{appetite}) = \text{ravenous}$ – where *appetite* means approximately ‘desire to eat’ and *ravenous*, when used “freely” (i.e., not as an intensifier) means ‘who has a strong desire / need to eat’.

In a Lexical System, each arc α of the network that corresponds to an LF relation carries a semantic weight which is determined by the meaning ‘ σ^f ’ of the corresponding LF f . Prototypically, arcs corresponding to paradigmatic LFs (i.e., leading to semantic derivatives) tend to carry a stronger semantic weight than those corresponding to syntagmatic LFs. We will see shortly (Sec. 6.2.1.2) that the notion of semantic weight also applies to non-LF relations.

Before proceeding with the examination of lexical relations in Lexical Systems that are not based on LFs, it is important to note that a given node n of a Lexical System is a complex informational entity that encapsulates all the properties of the corresponding lexical unit L not accounted for by LF relations: grammatical characteristics, lexicographic definition, etc. That is, the lexical properties specified in n together with n ’s relational profile are informationally equivalent to a lexical entry for L . Of course, in the context of the lexicography of lexical networks (Polguère 2014), special lexicographic tools are required to construct individual nodes with their individual properties and to weave the relations between nodes (see “New lexicography of lexical networks,” Sec. 6.3, p. cxciv).

6.2.1.2 Non-LF relations between lexical units

Three non-LF relations play an important role in Lexical Systems:

- 1) copolysemy relation, holding between two lexical units of a polysemous vocable;
- 2) idiom-component relation, holding between an idiom and a lexeme that participates in its surface-syntactic structure;
- 3) meaning-inclusion relation, holding between a lexical unit and another lexical unit that is an element of its lexicographic definition.

To make the presentation of these relations more straightforward and concrete, we base the following explanation mostly on data borrowed from the fr-LN.¹⁹

6.2.1.2.1 1) Copolysemy relation The fr-LN – as should be the case of any full-fledged Lexical System – contains the explicit modeling of the *polysemy structure* of vocables through the encoding of so-called *copolysemy relations* between lexical units of a vocable. A copolysemy relation – that is, the corresponding arc of the lexical network – is oriented from the lexical unit that is the source of the “copolysemy derivation” toward the “derived” lexical unit. In other words, $L_1 \vdash_{\text{copolysemy}} L_2$ reads as ‘lexical unit L_1 has lexical unit L_2 as its copolysemy extension / metonymy / metaphor / ...’; this relation participates in the polysemy structure of a given vocable.

Some copolysemy relations have a strong semantic weight in the network: they imply that the two lexical units they connect belong to identical semantic fields. This is the case of the metonymy relation that links **Fr. CIGARETTE I** ‘cigarette’ to **Fr. CIGARETTE II** ‘habit of smoking *cigarettes*’²⁰ – see Figure 6.4. Other copolysemy relations have weak (or, even, null) semantic weight – for instance, the metaphor relation (based on shape) that links **Fr. CIGARETTE I** to the appositive noun **Fr. CIGARETTE III** ‘[piece of clothing X] long and tight against the body – ‘as if’ it had the shape of a *cigarette*’ (*une robe cigarette* ‘a cigarette dress’). Such relations connect lexical units that are not expected to belong to the same semantic fields.



Figure 6.4: Polysemy of **Fr. CIGARETTE** as represented in the fr-LN

¹⁹The order of presentation of the three types of non-LF relations is determined by the expertise that has been gained from lexicographic work on the fr-LN: from the most extensively described relations to those whose encoding is still at an experimental stage. Finally, note that the relational profile of a node / lexical unit, while primarily based on LF relations, also includes the three relations that are presented in this subsection.

²⁰Cf. *Julie veut arrêter la cigarette* ‘Julie wants to stop smoking cigarettes’. There is no corresponding lexical unit in the English vocable **CIGARETTE**. The polysemy of **Fr. CIGARETTE** is chosen as illustration for two reasons: (i) it is not identical to that of **Eng. CIGARETTE**, which highlights the relevance of explicitly encoding polysemy structures, in particular, for the study of regular polysemy (Apresjan 1974, Peters & Peters 2000, Polguère 2022); (ii) though not too rich, it displays both metonymy and metaphor copolysemy relations, which we use later to illustrate the modeling of lexical proximity in lexical networks (Sec. 6.2.2.2, p. cxci).

A polysemous vocable is formally encoded in a Lexical System as a subnetwork consisting of nodes connected by copolysemy arcs, as shown in Figure 6.4. The *basic lexical unit* of a given vocable – i.e., the lexical unit that controls the polysemy structure of this vocable – corresponds to the unique node in the subnetwork that does not have an incoming copolysemy arc in its relational profile.

6.2.1.2.2 2) Idiom-component relation In a Lexical System, an *idiom-component relation* encodes the presence of a lexeme in the surface-syntactic structure [SSyntS] of an idiom (Pausé 2017a,b). This is illustrated in Figure 6.5, with the SSyntS of the French nominal idiom ‘FRAS KASSÉ’ lit. ‘arm broken’ – which means approximately ‘bungler’ – and the corresponding idiom-component relations encoded in the fr-LN.

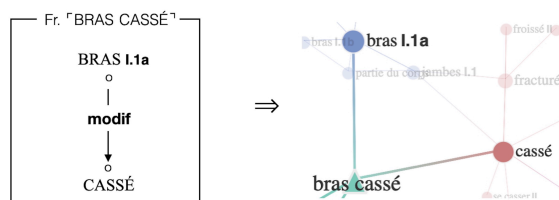


Figure 6.5: SSyntS of the idiom Fr. ‘FRAS KASSÉ’ ‘bungler’ together with the two corresponding idiom-component relations as encoded in the fr-LN²¹

Idiom-component relations belong to the outgoing relational profile of idioms and, conversely, to the incoming relational profile of the lexemes that compose the idiom. The list of all idioms in which a given lexeme participates is thus directly available in the network structure of a Lexical System and can be shown in the lexical entry of this lexeme, as it is the common practice in Explanatory Combinatorial Dictionaries; see the mention of the idiom ‘ANGER MANAGEMENT’ at the end of the lexical entry for ANGER_(N) (the Appendix, p. cc).²² We give in 6.3 the list of all idioms for which the lexeme BRAS 1.1a ‘arm’ (cf. Figure 6.5) is

²¹For now, the SSyntS of an idiom is not encoded as such in the fr-LN. Each idiom is associated with a structural pattern that is simply a string of parts of speech slots – e.g., N+Adj, for ‘FRAS KASSÉ’. These slots are filled with actual lexical units in the description of each individual idiom by encoding idiom-component relations – for details, see Pausé (2017a).

²²In the lexicography of Lexical Systems, however, this information is encoded while building the description of the idiom, whereas it is usually done while writing the lexical entry of the embedded lexemes in the case of Explanatory Combinatorial Dictionaries, and commercial dictionaries as well.

a component in the fr-LN, at the time of writing. This list is automatically extracted from this lexeme's incoming relational profile and displayed in textual visualizations of its lexical entry.

Table 6.3: List of idioms with BRAS_{1.1a} 'arm' as lexemic component in the fr-LN

「À BRAS-LE-CORPS」 ¹	lit. 'with arm-the-body'	'with both arms around someone's body'
「BAISSER LES BRAS」 ²	lit. 'lower the arms'	'give up'
「BRAS CASSÉ」 ³	lit. 'broken arm'	'bungler'
「BRAS DE FER」 ⁴	lit. 'arm of iron'	'arm-wrestling'
「BRAS DE MORPHÉE」 ⁵	lit. 'arms of Morpheus'	'state of sleeping / dreaming'
「BRAS DROIT」 ⁶	lit. 'right arm'	'right-hand person'
「BRAS LONG」 ⁷	lit. 'long arm'	'the long arm'
「COUPER BRAS ET JAMBES」 ⁸	[à N _Y] lit. 'cut.off arms and legs [to Y]'	'take the steam out of [person Y]'
「LES BRAS EN TOMBENT」 ⁹	[à N _X] lit. 'The arms because.of.this fall [to X]'	'X is completely flabbergasted'

An idiom-component relation is **inherently** not a meaning relation: it has a null semantic weight. However, it is not necessarily devoid of **semantic relevance**: it can participate in establishing a semantically relevant path between two lexical nodes of the lexical network – on the important notion of path in a graph/network, see Sec. 6.2.2.1, p. clxxxvi. Such is the case, obviously, for weak idioms (e.g., 「FRYING PAN」), whose definition contains the semantemes of all their lexical components: here, idiom-component relations are somehow doubling meaning-inclusion relations – described in 3) below. More unexpectedly, strong idioms (e.g., 「PUSH [someone's] BUTTONS」) can participate in establishing semantically relevant paths in a lexical network, especially when they are based on metaphors that are transparent / active in the current state of the language. To illustrate this, let us consider the idiom 「AS SNOW」, that functions as Magn collocate (i.e., intensifier) for adjectives such as COLD_(Adj), PURE OR WHITE_(Adj). One of the two idiom-component relations this idiom controls is part of a two-step path connecting the adjective WHITE_(Adj) and the noun SNOW_(N), where each step of the path has a null semantic weight due to the nature of the lexical relation it encodes, as shown in Table 6.4. This unsemantic two-step path is doubling in the lexical network a direct meaning-inclusion relation: WHITE_(Adj) is semantically included in SNOW_(N).²³

²³See 'white flakes' in the definition of SNOW_(N) in Ch. 4, Sec. 4.2, p. lxxxii.

Table 6.4: Two-step path from $\text{WHITE}_{(\text{Adj})}$ to $\text{SNOW}_{(\text{N})}$ based on inherently non-meaning relations²⁴

Steps	Relations	Comments
1.	$\text{WHITE}_{(\text{Adj})} \vdash_{\text{Magn}} \rightarrow \ulcorner \text{AS SNOW} \urcorner$	LF relation – i.e., $\text{Magn}(\text{white}_{(\text{Adj})}) = \ulcorner \text{as snow} \urcorner$
2.	$\ulcorner \text{AS SNOW} \urcorner \vdash_{\text{idiom-comp.} \rightarrow} \text{SNOW}_{(\text{N})}$	$\text{SNOW}_{(\text{N})}$ participates in the SSyntS of the idiom $\ulcorner \text{AS SNOW} \urcorner$

The explicit encoding of the above network path within a lexicon model helps account for the great proximity that exists in English between the semantic field of ‘snow’ and the semantic field of ‘whiteness [things that are white]’. This proximity is not a purely conceptual fact: it is linguistically determined. It results from existence in the English language of lexemes and phrasemes – such as the idiom $\ulcorner \text{AS SNOW} \urcorner$ – that are based on the inclusion of the semanteme ‘white’ in the definition of $\text{SNOW}_{(\text{N})}$.

The analysis we just performed illustrates the important role that idiom-component relations can play in the structuring of the Mental Lexicon. By encoding these relations in Lexical Systems, we account better for the multiplicity of ways the Speaker can navigate from one lexical unit to another during the activation of lexical knowledge; we also account for “associations of ideas” that take place in the Speaker’s mind and can influence lexical selection.

6.2.1.2.3 3) Meaning-inclusion relation A lexicographic definition, as specified in Mel’čuk & Polguère (2018), is a paraphrase of the defined lexical unit in terms of semantically simpler lexical units.²⁵ Consequently, each defining element in the definition of $\text{ANGER}_{(\text{N})}$ given in Table 6.5 (see also the Appendix, p. cxcvii) is a semanteme corresponding to a specific lexical unit of English.

From a lexical network viewpoint, this inclusion of lexical units (more precisely, the corresponding semantemes) in the definition translates into a set of *meaning-inclusion relations* originating from $\text{ANGER}_{(\text{N})}$ and leading to included lexical units.

²⁴Remember that $\text{Magn}(\text{L}) = \text{L}'$ entails an inherently non-meaning relation between L and L' – Sec. 6.2.1.1, “Semantic weight of LF relations,” p. clxxvii.

²⁵Lexical unit L_1 is semantically simpler than lexical unit L_2 if and only if L_1 appears in the semantic decomposition of L_2 ’s meaning, but not vice versa.

Table 6.5: Lexicographic definition of $ANGER_{(N)}$

X 's anger at Y for Z	: X 's unpleasant unfavorable feeling directed at person Y and caused by Z – such that
	Stimulus
	• X believes that Y is responsible for Z • Z is undesirable for X
	Effect
	• X wants to do something hostile to Y (in order to stop Z) • X enters in a mental state such that X can easily lose self-control

Meaning-inclusion relations induced by lexicographic definitions are not all equally relevant as regards semantic weight. For instance, the three relations below, induced by the definition of $ANGER_{(N)}$, are listed by order of decreasing semantic weight:

- $ANGER_{(N)} \vdash \text{meaning inclusion} \rightarrow FEELING_{(N)}$ – strong semantic weight;
- $ANGER_{(N)} \vdash \text{meaning inclusion} \rightarrow HOSTILE_{(Adj)}$ – average semantic weight;
- $ANGER_{(N)} \vdash \text{meaning inclusion} \rightarrow RESPONSIBLE$ – weak semantic weight.

In many instances, meaning-inclusion relations establish lexical connections that play a significant role in the lexicon and cannot be accounted for by LFs. An LF relation $L \vdash f \rightarrow L'$ (i.e., $f(L) = L'$) is, first of all, a **combinatorial** relation, that may or may not also be semantic: the meaning ' σ^f ' of f is expressed in combination with L by the lexical entity L' . By contrast, an $L \vdash \text{meaning inclusion} \rightarrow L'$ relation is necessarily a purely meaning relation: the fact that the semanteme ' L' ' is included in the semanteme ' L '.

6.2.2 Formal properties of Lexical Systems

Lexical Systems have a formal structure known in graph theory as small-world network. This characteristic has strong implications for Lexical Systems' computability, so that it is essential to give some consideration to this topic. To present formal properties of Lexical Systems, we start with the introduction of some basic notions of graph theory the knowledge of which is presupposed for the construction and study of lexical networks (Sec. 6.2.2.1). Next, we use these notions to explain how the informational proximity between lexical units in the Mental Lexicon can be modeled by exploiting the small-world structure of Lexical Systems (Sec. 6.2.2.2).

We apologize in advance both (i) to readers who are unfamiliar with mathematical notions as they may find what follows too technical and (ii) to readers with background in graph theory for the simplifications / approximations we operate.

6.2.2.1 Some basics of graph theory

Until now, we have used the term *network* in a rather broad sense, which can be glossed as ‘system of individual entities connected by one-to-one relations’. Actual networks can be **artificial** (designed) systems: they have been devised as such, to serve specific purposes – for instance, a city metro system, where metro stations are individual entities and tracks / tunnels connecting stations are relations between these entities. Networks can also be **natural** systems: they happen to exist without having been consciously designed; we are specially interested here in networks constituted of a huge number of entities – for instance, the system formed in the brain by neurones and their synaptic connections. These very large networks that are natural systems found in our surrounding or remote environment are called *real-world networks*. Note that, in our use of the term, a real-world network can be the product of human activity – for instance, the network formed by hyperlinked webpages of the Internet. What matters is that this structure has emerged naturally, unlike the structure of a city metro system, which is relatively small and has been consciously designed.

Following Postulate 1 (p. clxviii), Mental Lexicons are natural real-world networks and, following Postulate 2 (p. clxxv), Lexical Systems are *human-made* networks whose structure is expected to be analogous to the *natural* structure of Mental Lexicons. Therefore, Lexical Systems are considered here as real-world networks as well.

Networks are studied as formal objects by *graph theory*, a branch of mathematics customarily traced back to Euler (1741): it deals with the formal representation and calculus of network systems. Let us provide some terminological clarification:

- we use the term *graph* to designate a mathematical object that underlies a network stripped off of all (or most of) the attributes of its nodes and arcs;
- for the sake of simplicity, our presentation of graphs and their properties uses the terms *node* and *arc*, more habitual in linguistic works, even though two other terms are generally favored in graph theory literature, respectively: *vertex* (pl. *vertices*) and *edge*.

6.2.2.1.1 Graphs and their representation Graphs are mathematical – therefore, informational – objects. They should not be confused with *graph representations*, which are used throughout the present chapter. Figure 6.6 gives three common modes of representation for a given *undirected graph* G , i.e., a graph whose arcs are not oriented: *geometric representation*, *set-theoretic representation* and *adjacency matrix*.

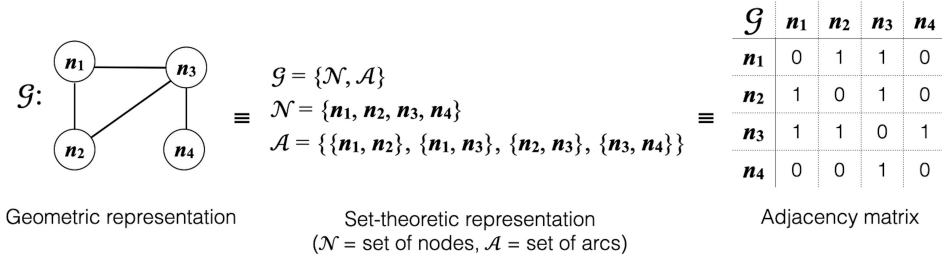


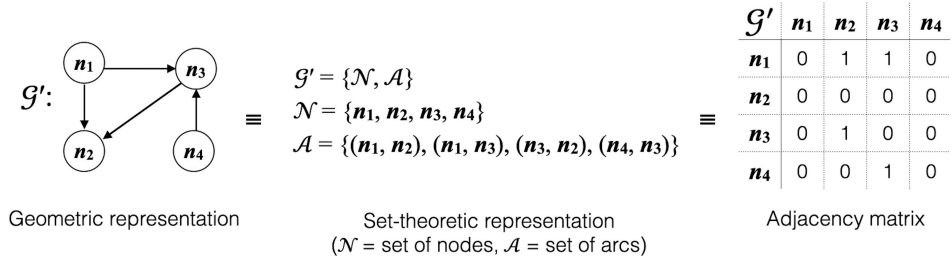
Figure 6.6: Three modes of representing a given undirected graph $\mathcal{G}$

- 1) The geometric representation is almost self-explanatory. In our figure, we put node names inside the circles representing nodes in order to highlight the fact that such names are only node IDs, and not attributes of the nodes.²⁶ A graph is entirely defined by its *topology*: its system of nodes and connecting arcs. Consequently, in a graph geometric representation, the spacial distribution of nodes and arcs is irrelevant from a formal point of view and only serves legibility purposes.²⁷ The relative positioning of nodes in the geometric representation of Figure 6.6 can be totally upset (n_4 at the very top, n_1 below n_3 , n_2 positioned much closer to n_3 , etc.); it will remain the representation of the same graph $\mathcal{G}$ as long as connections established by the arcs remain identical.
- 2) The set-theoretic representation defines a graph as a mathematical set comprised of two subsets: (i) a set of nodes and (ii) a set of arcs; this latter set contains itself unordered two-element sets encoding non-oriented binary relations between nodes.
- 3) The adjacency matrix is a table where each row and each column correspond to a node of the graph; each cell of the table indicates whether an arc connects the corresponding nodes: 1 means that the nodes are connected and 0 means that they are not. The adjacency matrix for graph $\mathcal{G}$ (Figure 6.6) is symmetrical relative to the table's top-left to bottom-right diagonal; this comes from the fact that, the arcs being non-oriented, any n - n' arc is equally an n' - n arc.

We exemplified the three modes of representation above with an undirected graph for pedagogical purposes. However, lexical relations examined in Sec. 6.2.1 are all oriented and, therefore, the graphs underlying Lexical Systems are *directed graphs*. Figure 6.7 gives the three representations of a directed graph $\mathcal{G}'$.

²⁶This is in reference to what was said earlier about the distinction we operate between the terms *network* and *graph* (= network stripped off of non-structural information).

²⁷We will show (Sec. 6.2.2.2, p. cxcff.) that geometric distribution of nodes can nonetheless be used to encode additional information about the graph's overall topology.

Figure 6.7: Three modes of representing a given directed graph $\mathcal{G}'$

Differences with the representations of Figure 6.6 (an undirected graph) are as follows:

- arcs of the geometric representations are **arrows** – they indicate the *starting node* and the *end node* for each relation;
- the set of arcs $\mathcal{A}$ in the set-theoretic representation contains (ordered) **couples** of nodes – instead of (unordered) two-element sets of nodes;
- 1s and 0s in the adjacency matrix take into account, for each cell, which node is the starting node and which node is the end node of the corresponding relation; note that the adjacency matrix of graph $\mathcal{G}'$ is non-symmetrical.

One of the advantages of the geometric representation of graphs over the two other representations is that it offers a clear visualization of possible paths between nodes. A *path* in a graph is a sequence of arcs that connects a sequence of nodes where neither arcs nor nodes are repeated. In an oriented graph, one can consider *directed paths* – the sequence of arcs follows the orientation of the arcs – or *undirected paths* – the orientation of arcs is not relevant. For instance, the geometric representation of Figure 6.7 clearly shows:

- two directed paths from n_1 to n_2 , namely: n_1-n_2 and $n_1-n_3-n_2$;
- two undirected paths from n_1 to n_4 , namely: $n_1-n_3-n_4$ and $n_1-n_2-n_3-n_4$.

The notion of path is central in the definition of small-world networks (paragraph “Lexical Systems as small-world networks,” p. clxxxviii*ff.*).

6.2.2.1.2 Structural properties of Lexical System graphs Graphs are characterized by structural properties that pertain, primarily, (i) to the nature of their arcs – for instance, we have already established that we are concerned here with directed graphs only – and (ii) to the types of node-arc configurations they contain. We concentrate on *Lexical System graphs* – referred to as $\mathcal{G}_{LS}$ – and identify four properties (A–D) that characterize them as a specific type of directed graph. These properties are listed below and summarized / exemplified in Figure 6.8 (p. clxxxviii).

A) Labeled graph. Each arc of a $\mathcal{G}_{LS}$ is defined by (i) a starting node, (ii) an end node **and** (iii) a *label* that identifies the lexical relation encoded by the arc: LF relation, copolysemy relation, etc. This characterizes a $\mathcal{G}_{LS}$ as an oriented *labeled graph*.

B) Weighed graph. When presenting LF and non-LF relations encoded in Lexical Systems, we have indicated that these relations can be distinguished according to their semantic weight (cf. Sec. 6.2.1.1, “Semantic weight of LF relations,” p. clxxvii). This information is encoded in a $\mathcal{G}_{LS}$ for each arc of the graph, concurrently with the arc label, which characterizes a $\mathcal{G}_{LS}$ as a *weighed graph*.

C) Multigraph. Two nodes of a $\mathcal{G}_{LS}$ can be connected by more than one arc – for instance, when two lexical units are connected simultaneously by an LF relation and a meaning-inclusion relation, as in (2). This results in *multiple arcs* in a $\mathcal{G}_{LS}$, which characterizes it as a *multigraph*.

(2) $\text{Gener}(gas_{(N)}) = \textit{substance}$
 $gas_{(N)} \mapsto \text{meaning inclusion} \rightarrow \text{SUBSTANCE}$ [= central component in the definition of $gas_{(N)}$]

D) Cyclic graph. A $\mathcal{G}_{LS}$ contains *cycles*, i.e., directed paths that originate and end at the same given node. This characterizes a $\mathcal{G}_{LS}$ as a *cyclic graphs*. For instance, the set of LF relations in (3) is encoded as a two-step cycle in the corresponding $\mathcal{G}_{LS}$.

(3) $S_{\theta}(\textit{present}_{(V)}) = \textit{presentation}$
 $V_{\theta}(\textit{presentation}) = \textit{present}_{(V)}$

6.2.2.1.3 Note The cycle in (3) also corresponds to a multigraph configuration – see property C) above.

The above four properties are recapitulated in Figure 6.8, with illustrations that apply to several properties wherever it is convenient to do so.

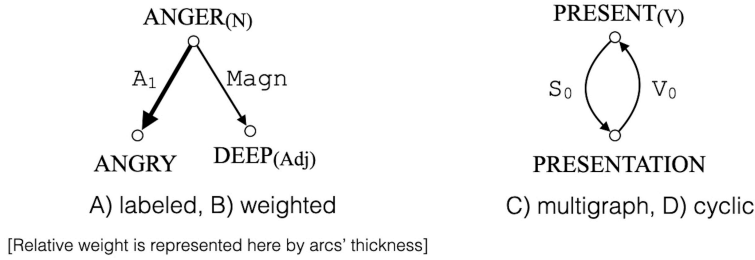


Figure 6.8: Four structural properties of a Lexical System graph $\mathcal{G}_{LS}$

6.2.2.1.4 Lexical Systems as small-world networks Properties examined above characterize a $\mathcal{G}_{LS}$ at a *microscopic* level, i.e., at the level of individual nodes and relations they participate in. We are now taking a bird's-eye view of such graphs in order to contrast their microscopic and *macroscopic* structures. For this, we focus on a natural law that tends to apply to all real-world networks (as characterized in Sec. 6.2.2.1, p. clxxxiv).

Graph theory has determined that graphs of real-world networks have characteristic topological properties that identify them as *small-world networks* (Watts & Strogatz 1998). This term is a reference to the well-known “small-world phenomenon” in social networks,²⁸ which can be presented as follows: there is a high probability for any two randomly selected individuals i_1 and i_2 to be connected through a relatively short chain of acquaintances (Kleinberg 2000: 163) – hence the expression *It's a small world!* Though this phenomenon was originally conceptualized from the perspective of social and psychological studies (Milgram 1967), it is a characteristic feature of real-world networks of any kind. The topological structure of a small-world network has (at least) four properties described below; in order to anchor what follows more directly to our topic of interest (i.e., Lexical Systems), these properties are introduced as properties of a given $\mathcal{G}_{LS}$.²⁹

1) Mainly connected graph. A $\mathcal{G}_{LS}$ may contain some isolated tiny subgraphs (possibly, isolated nodes); however, the bulk of its structure forms one huge *connected graph*: a graph in which there exists at least one path connecting any two nodes.

²⁸By *social network*, we mean here social structures made up of individuals having social interactions. We are not referring to social media platforms on the Internet.

²⁹We try to keep what follows as less “mathematical” as possible, at the expense of accuracy. See Gaume et al. (2010) for a purely mathematical characterization of lexical graphs as small-world networks; see Gader et al. (2014: Sec. 3) for the measurement of the *pedigree* of Lexical Systems as small-world networks.

2) Sparse graph. The arc-to-node ratio in a $\mathcal{G}_{LS}$ is very low. Consequently, there is an extremely low probability for any two nodes in a $\mathcal{G}_{LS}$ to be **directly** connected. This characterizes a $\mathcal{G}_{LS}$ as a *sparse graph*.³⁰

3) It's a small world, nonetheless! In spite of its sparsity, a $\mathcal{G}_{LS}$ displays the small-world phenomenon in the sense that any two nodes of $\mathcal{G}_{LS}$, no matter how distant they may be in the topological structure of the graph, are connected by at least one relatively short path. This phenomenon is mainly made possible by two factors:

1. Though most nodes of $\mathcal{G}_{LS}$ are weakly connected, there are nodes that function as lexical hubs from a relational viewpoint; for instance:

- generic terms that identify large lexical classes (ANIMAL, INSTRUMENT, FEELING, ...) and are starting nodes of many Gener arcs;
- collocative lexical units forming collocations with a large spectrum of potential bases – for instance, the Magn value HEAVY_(Adj) 2 mentioned in Sec. 6.2.1.1, p. clxxvii.

2. Many copolysemy and idiom-component relations in a $\mathcal{G}_{LS}$ establish direct paths between semantic areas that are otherwise very far apart; for instance:

- CRANE_(N) I [animals] $\vdash$ copolysem. metaphor $\rightarrow$ CRANE_(N) II [machines];
- $\ulcorner$ GO BANANAS $\urcorner$ [feelings / behaviors] $\vdash$ idiom-comp. $\rightarrow$ BANANA [fruits].

4) High clustering rate. A $\mathcal{G}_{LS}$ is sparse at a macroscopic level but dense at a microscopic level, where a profusion of small node clusters are found. In mathematical terms, a $\mathcal{G}_{LS}$ has a high *clustering rate*: there is a much higher probability for two nodes n_1 and n_2 to be directly connected if both of them are directly connected to a node n_3 than if they are randomly selected. This is easily illustrated with LF relations, where various clustering *inference rules* such as (4) and (5) can be established – Figure 6.9 visualizes linguistic illustrations given in (4)–(5).

- (4) **If** $f(L_1) = L_2, L_3$
then there exists $\text{Syn}/\text{Syn}_\neg/\text{Syn}_\subset/\text{Syn}_\cap(L_2) = L_3$,³¹ for instance:
- $S_2(\text{married}) = \text{spouse}_{(N)}, \ulcorner \text{better half} \urcorner$
 - $\text{Syn}_\cap(\text{spouse}_{(N)}) = \ulcorner \text{better half} \urcorner$

³⁰The mathematical measurement used to evaluate this sparsity is the *average node degree* of the graph (the average number of arcs connected to each node) – for lexical networks, see Gaume et al. (2010: Sec. 2).

³¹This is true because all the elements of an LF value must be (quasi-)synonymous.

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- (5) **If** $f(L_1) = L_2$ **and** $f'(L_2) = L_3$
then there is an above-average probability that there exists $f''(L_1) = L_3$; for instance:
- $\text{Conv}_{21}(\text{defeat}_{(N)}) = \text{victory}$ **and** $A_1(\text{victory}) = \text{victorious}$
 - $A_2(\text{defeat}_{(N)}) = \text{victorious}$

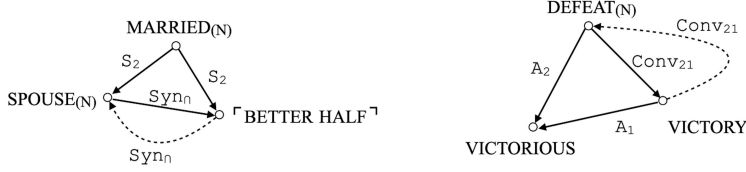


Figure 6.9: Two LF-based lexical clusters – solid arrows correspond to illustrations given in (4)–(5) and dashed arrows are additional clustering LF relations

6.2.2.2 Lexical vs. topological proximity

Formal properties presented above show that $\mathcal{G}_{LS}$ are not *random graphs*, i.e., graphs in which nodes and connecting arcs are randomly distributed.³² They are structured as small-world networks, whose topological properties can be exploited to access, study and reason on the information they encode. The main advantage of small-world lexical networks over the better-known lexical taxonomies (Sec. 6.1, p. clxviii ff.) is that they do not adhere to one primary classifying / hierarchization principle; consequently, they allow for very flexible and varied types of computation.

To illustrate the computability of small-world networks, we focus here on *lexical proximity* (whether semantic or combinatorial) that can be inferred from the topology of $\mathcal{G}_{LS}$. This is well illustrated in visualizations performed by the Spiderlex graph browser (cf. Sec. 6.2.1.1, “Visualization of Lexical Systems,” p. clxxiv), where clusters of lexical entities are identified by a stochastic algorithm that computes *topological proximity* between lexical nodes (Gaume 2004, 2008). This proximity is visually represented by two means:

- the geometric distribution of nodes in the visualization – the closer the nodes from the viewpoint of the graph topology, the closer they are positioned;

³²In mathematical / statistical terms, a random graph is a graph in which the distribution of node degrees follows a Poisson law; by contrast, the degree distribution of a small-world graph follows a power law (Gaume 2008: 849).

- the color coding of node clusters identified by the topological analysis – these clusters correspond to semantic-combinatorial classes of lexical entities.

6.2.2.2.1 Lexical spaces We have established earlier that $\mathcal{G}_{LS}$ s possess a high clustering rate (Sec. 6.2.2.1, p. clxxxix): at the microscopic level, their nodes tend to form small clusters of lexical entities that are mutually interrelated. Such clusters are of course identified by the topological analysis of a $\mathcal{G}_{LS}$. However, at the macroscopic level, broader clusters are also identified; they bring together lexical entities that are more loosely related but form a distinct semantic-combinatorial *lexical space* in the graph and, hopefully, in the Mental Lexicon this graph encodes.

Figure 6.10 below serves to illustrate the notion of lexical space and, more broadly, the outcome of a $\mathcal{G}_{LS}$'s topological analysis. It shows:

1. lexical proximity identified by means of a topological analysis of the fr-LN;
2. visualization means used by Spiderlex to represent lexical spaces.

In this illustration, lexical proximity is computed on the polysemy structure of the French vocable CIGARETTE, which has already been explained in Sec. 6.2.1.2 (p. clxxxix).³³

Let us make two comments on Figure 6.10 in order to illustrate the linguistic relevance of the topological analysis and geometrization algorithms that have lead to this visualization.

1) The basic lexical unit CIGARETTE I and the metonymic sense CIGARETTE II 'habit of smoking *cigarettes*' form together a single lexical space, while the metaphoric sense CIGARETTE III '[piece of clothing X] long and tight against the body – as if it had the shape of a *cigarette*' is singled out, at the center of its own lexical space. These two spaces are located far apart, their connection holding by only one thread: the metaphor copolysemy relation between CIGARETTE I and CIGARETTE III. This reflects the fact that a metonymy belongs to the same broad conceptual domain as its source, while a metaphor can totally escape the gravitation field, so to speak, of the copolyseme it originates from.

³³To work on the complete polysemy of CIGARETTE, three topological explorations are performed as so-called *random walks* (Gaume 2008: Sec. 3): one for each sense of the vocable taken as point of origine of the random walk. The three lexical nodes that are origines of random walks appear in Figure 6.10 as triangles, rather than discs.

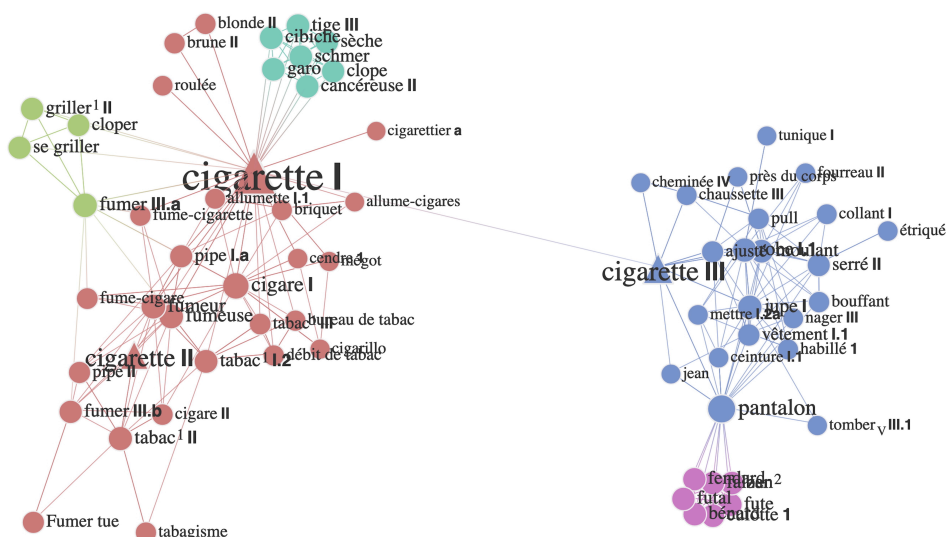


Figure 6.10: Lexical proximity computed on the polysemy of Fr. CIGARETTE in the fr-LN

2) The two lexical spaces identified above are themselves constituted of lexical clusters that display their own lexical cohesion. Such is the case, in CIGARETTE I + II lexical space, of the small cluster constituted of FUMER III ‘smoke_(V)’ – that is $\text{Real}_1(\text{cigarette I})$ – and three of its colloquial synonyms: CLOPER, GRILLER¹ II [ART .] and SE GRILLER [ART .] (see top left corner of Figure 6.10). This is an extremely tight cluster – and it is geometrically displayed as such – for two linguistic reasons: (i) the lexical units it contains are all values of Syn for each other³⁴ and (ii) they are additionally all values of the same LF application $\text{Real}_1(\text{cigarette I})$.

FURTHER READING. On the Lexical System approach to the modeling of polysemy, see Polguère (2018); on small-world lexical networks in lexicology / lexicography, see Loiseau et al. (2010), Polguère (2014, 2016); in psycholinguistics, see Gaume et al. (2002, 2008), De Deyne et al. (2017); in Natural Language Processing, see Biemann (2012).

6.3 Proof of concept: the French Lexical Network [fr-LN]

In lieu of conclusion, we present briefly the lexicographic enterprise that produced the first large-scale Lexical System: the *French Lexical Network* [fr-LN]. Biblio-

³⁴In graph terminology, this cluster is called a *clique*: a (sub)graph where every two nodes are connected.

graphic references on important aspects of the fr-LN project are provided at the end of the section.

The construction of the fr-LN is an ongoing long-term endeavor that started at the ATILF CNRS laboratory (Nancy, France) in 2011. An initial four-year funded project (called *RELIEF*, see Lux-Pogodalla & Polguère 2011) achieved the following tasks:

- 1a** training of a team of lexicographers in the theory and practice of Explanatory Combinatorial Lexicology / Lexicography;
- 1b** implementation (in parallel with **1a**) of a lexicographic editor tailor-made for the lexicography of Lexical Systems, whose capabilities were gradually incremented as the project proceeded with lexicographic description (Gader et al. 2012);
- 2** construction of a nucleus fr-LN for a core French vocabulary of around 3,700 vocables;
- 3** standard lexicographic work proceeding by important semantic fields (feelings, geographical elements, animals, ...).

This initial stage of the project resulted in a Lexical System that modeled a significant sample of the French lexicon and that later expanded into a full-fledged small-world network (as defined in Sec. 6.2.2, p. clxxxiii*ff*).

The fr-LN is in constant evolution both in terms of the number of lexical entities it describes and in terms of depth / accuracy of the description. At the time of writing, its coverage can be quantified as in Table 6.6.

Table 6.6: Current coverage of the fr-LN

Lexical entities, i.e., network nodes	: 30,115
Vocables, i.e., lexicographic entries	: 19,042
Polysemy rate (= Lexical entities / Vocables)	: 1.58
Lexical relations, i.e., network arcs	: 85,554
LF relations	: 67,205
Copolysemy relations	: 9,934
Idiom-component relations	: 8,113
Meaning-inclusion relations	: 302

The above table shows that the fr-LN, in its current state, is a lexical network of significant proportions: approximately 30,000 lexical nodes for 85,500 lexical

relations. In accordance with the properties of Lexical Systems (cf. Sec. 6.2.1.1, p. clxxiii*ff.*), LF relations form the bulk of the relational structure of the model. Meaning-inclusion relations are underrepresented as the work on the writing of lexicographic definitions is still at an embryonic stage.

In the absence of a dedicated lexicographic team, the fr-LN now evolves mainly “on demand,” according to the needs of researchers, language teachers, etc. who are collaborators on the project or are making use of the resource. The fr-LN’s lexicographic data is available in open access, to be used for Natural Language Processing or computational linguistic work – see ATILF (2024a), for the network itself, and ATILF (2024b), for the distribution of lexicographic examples embedded in the fr-LN.

6.3.0.0.1 New lexicography of lexical networks The fr-LN project is mature enough – in terms of coverage and applications it has generated – to be seen as a proof of concept for Lexical Systems as plausible models of the Mental Lexicon. It demonstrates that network lexical models primarily structured by standard LF relations can offer new insights into the organization of lexical knowledge and, very importantly, that such models can indeed be lexicographically built.

As regards this last point, it is essential to stress the importance of the technological aspects of this new lexicography of lexical networks. Modern physics or astronomy could not exist without advanced technological means (cyclotrons, orbiting telescopes, etc.); similarly, though at a much more modest level, the lexicography of lexical networks has been made possible by the development of a sophisticated editor that allows lexicographers to literally jump from one lexical entity to another, encoding lexical relations while simultaneously creating new lexical nodes in the process, if the targeted lexical entity is not already present in the network.³⁵ This lexicographic “gesture” ensures that lexical entries always point to (the lexical entry of) actual lexical entities and forces lexicographers to encode a lexical relation as a link between two lexical entities that are (at least, minimally) modeled in the lexical network. Anyone who has extensively used standard dictionaries (for language teaching, translation, etc.) knows that this is far from being the case in such models. The system of lexical relations present in a standard dictionary – that has been termed the dictionary’s *mediostructure* (Wiegand 1996, Gouws & Prinsloo 1998) – is only modeled by cross-references in the

³⁵We do not illustrate the use of the Lexical System lexicographic editor (named *Dicet*, see Gader et al. 2012) as a next generation editor (*ItsyBitsy*) is under development at the time of writing and will soon replace it.

text of the *microstructure*. At the scale of the whole lexicon, however, this cross-referencing is necessarily approximate for logistic reasons. Only by building a lexical model that is, so to speak, a *mediostructure* (i.e., a lexical network) can one ensure that the modeling of lexical relations – the essence of the lexicon’s structure (cf. Postulate 1, Sec. 6.1, p. clxviii) – is truly valid and unambiguous.

Finally, essential technological means include visualization and navigation tools such as Spiderlex, that has been extensively used for illustration purposes in the present chapter. Their purpose is threefold: improve access to the content of Lexical Systems, visualize complex topological information and, finally, help lexicographers diagnose missing or wrongly encoded information.

FURTHER READING. On the strategy adopted to organize the fr-LN’s initial growth based on the encoding of standard paradigmatic LF relations that are central in the system of LFs,³⁶ see Lux-Pogodalla & Polguère (2011) and Polguère & Sikora (2013); on the small-world structure of the fr-LN, see Gader et al. (2014); on the implementation of analogical reasoning in the fr-LN, see Ollinger (2014); on lexicographic edition of Lexical Systems, see Gader et al. (2012) and Polguère (2012b); on the modeling of terminologies within general language Lexical Systems, see Ingrosso & Polguère (2015) and Gotkova et al. (2024).

³⁶Roughly, those paradigmatic LFs whose ID is boxed in Table 3.5 (Ch. 3, p. lxxvii).

Appendix: A sample lexical entry

The lexical entry of the noun $ANGER_{(N)}$ given below is intended to illustrate the use of lexical functions in lexicographic description. This is a reworked version of the entry presented in Iordanskaja & Mel'čuk (2022: Sec. 5, 42–46).

$ANGER_{(N)}$, noun, uncountable

Definition

X's anger at Y for Z : X's unpleasant unfavorable feeling directed at person Y and caused by Z – such that

Stimulus

- X believes that Y is responsible for Z
- Z is undesirable for X

Effect

- X wants to do something hostile to Y (in order to stop Z)
- X enters in a mental state such that X can easily lose self-control

Government pattern

'X' $\Leftrightarrow$ I	'Y' $\Leftrightarrow$ II	'Z' $\Leftrightarrow$ III
1. –possess $\rightarrow$ N's	1. –obl-obj1 $\rightarrow$ <i>against</i> N	1. –obl-obj2 $\rightarrow$ <i>for</i> N
2. –subj-adnom $\rightarrow$ <i>of</i> N	2. –obl-obj1 $\rightarrow$ <i>at</i> N	2. –obl-obj2 $\rightarrow$ <i>for</i> V _{GER}
3. –modif $\rightarrow$ A _θ (N)	3. –obl-obj1 $\rightarrow$ <i>toward</i> N	3. –obl-obj2 $\rightarrow$ <i>over</i> N
		4. –obl-obj2 $\rightarrow$ <i>over</i> V _{GER}

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Illustrations for the government pattern

- I.1 + II.1 *Judy's <Her> anger against co-workers is not helpful.*
 I.1 + II.2 + III.2 *They wanted to show their anger at the Government for signing the pact.*
 I.2 + II.2 + III.3 *The anger of seniors at the Government over the health care reform will impact the elections.*
 II.3 + III.2 *She felt anger toward her parents for leaving her there.*
 I.3 *Popular [= A₀(people)] anger is rising.*

Lexical functions [LFs]

Paradigmatic LFs

Syn _⊃	: <i>fury; indignation; outrage; formal wrath; archaic ire</i>
Syn _⊂	: <i>rage</i>
Anti _⊂	: <i>gratefulness; gratitude</i>
Gener	: <i>feeling [of ~] / [angry] feeling; emotion [of ~] / [angry] emotion</i>
Magn+Figur	: <i>fire of ~, flame of ~ < firestorm of ~</i>
S ₂	: <i>object, subject, target [of ART ~]</i>
S ₃	: <i>cause, object, subject [of ART ~], reason [for ART ~]</i>
A ₁	: <i>angry, in ~</i>
Magn+A ₁	: <i>filled with ~, full of ~ < consumed by ~, colloquial 'blue in the face', fuming, colloquial 'hot under the collar'</i>
AntiVer+Magn+A ₁	: <i>steamed-up [I don't see why you got steamed-up about this slip the tongue.]</i>
Able ₁	: <i>irascible; hot-tempered, short-tempered; short-fused</i>
PredAble ₁	: <i>'be on a short fuse', 'have a short fuse'</i>
Adv ₁	: <i>angrily, in ~, with ~</i>
Sing	: <i>~ attack; burst of ~; fit of ~; flash of ~; gust of ~</i>

Syntagmatic LFs

Germ	: <i>seeds [of ~]</i>
Magn	: <i>deep, great < fierce, terrible < extreme; red < black; towering; unbridled, uncontrollable; a lot [of ~]; 'over the top' anteposed; ☒ fury; ☒ rage</i>
Magn ₁ ^{quant}	: <i>widespread</i>
AntiMagn	: <i>slight; 'a little bit' [of ~]; ☒ annoyance</i>
IncepPredPlusMagn	: <i>'builds up', grows, mounts, rises, 'swells up', 'wells [inside N_X]</i>

6.3 Proof of concept: the French Lexical Network [fr-LN]

CausPredPlusMagn	: <i>fuel, stir</i> [(ART) ~]
IncepPredMinusMagn	: <i>cools, fades, 'goes away', subsides</i>
CausPredMinusMagn	: <i>calm, soothe</i> [ART ~]
Ver	: <i>righteous</i>
Bon+Ver	: <i>noble</i>
Oper ₁	: <i>feel</i> [(ART) ~]
IncepOper ₁	: ☒ <i>slang</i> 'have a cow', ☒ <i>slang</i> 'have kittens'
Magn+Oper ₁	: <i>blaze</i> [with (ART) ~]; <i>boil, burst, seethe</i> [with (ART) ~]
Magn+IncepOper ₁	: <i>erupt</i> [in ~], <i>fly</i> [into (an) ~], ☒ 'blow up' [at N _Y], ☒ <i>colloquial</i> 'blow a fuse', ☒ <i>colloquial</i> 'blow a gasket', ☒ <i>colloquial</i> 'blow [N _X 's] stack', ☒ <i>colloquial</i> 'blow [N _X 's] top', ☒ <i>explode</i> , ☒ <i>colloquial</i> 'fly off the handle', ☒ <i>colloquial</i> 'go ballistic', ☒ <i>colloquial</i> 'go through the roof', ☒ <i>colloquial</i> 'hit the ceiling', ☒ <i>colloquial</i> 'see red'
CausOper ₁	: ☒ <i>anger</i> _(V) [N _X]
IncepOper ₂ or Caus ₂ Oper ₂	: <i>attract, draw</i> [(ART) ~]
Caus ₍₂₎ Func ₀	: <i>provoke</i> [(ART) ~]; <i>arouse, spark, trigger</i> [(ART) ~]
Caus ₍₂₎ ContFunc ₀	: <i>feed</i> [ART ~]
Magn+Func ₁	: <i>boils, bubbles, seethes</i> [in N _X / in N _X 's gut / in N _X 's soul]; <i>fills, overcomes, seizes</i> [N _X]; <i>devours</i> [N _X] (from inside); ☒ 'Smoke <Steam> is coming <pouring> out' [of N _X 's ears]
Magn+IncepFunc ₁	: <i>explodes</i> [in N _X]
IncepFunc ₂	: <i>falls</i> [on N _Y]
Real ₁ , 'hostile action'	: <i>vent</i> [A _(poss) X ~on N _Y]
Note. The semanteme configuration in subscript to the name of an LF identifies its target, i.e., the semantic component in the definition of the headword (in this case, ANGER _(N)) on which bears the meaning of the LF (see Ch. 3, Sec. 3.3.2, p. 1).	
AntiVerReal ₁ , 'hostile action'	: 'take out' [A _(poss) X ~on N _Ω], ☒ 'take it out' [on N _Ω] Ω is not responsible for Z
Adv ₁ Real ₁ , 'hostile action'	: 'out of' ~
SingS ₀ Real ₁ , 'hostile action'	: <i>outburst</i> [of ~], [angry] <i>outburst</i>
Real ₁ , 'loss of self-control'	: 'be beside oneself' [with ~]
Real ₂ , 'hostile action'	: <i>endure</i> [N _X 's ~]
Fact ₀ , 'loss of self-control'	: <i>explodes</i>
Perm ₁ Fact ₀	: 'give vent' [to (ART) ~], 'let out', <i>unleash, vent</i> _(V) [(ART) ~]
NonPerm ₁ Fact ₀	: <i>contain, control, stifle, suppress</i> [A _(poss) X ~]
A ₂ NonPerm ₁ Fact ₀	: <i>pent-up; suppressed</i>
Fact ₁ , 'loss of self-control'	: <i>blinds</i> [N _X]
Excess ^{motor} (<i>body</i>)—Sympt ₁₃	: [N _X] <i>is shaking <trembling></i> [with ~]
Obstr(<i>breath</i>)—Sympt ₁₃	: [N _X] <i>is choking</i> [of ~]
Excess ^{motor} (<i>brows</i>)—Sympt ₁₂₃	: [N _X] <i>furrows</i> [A _(poss) X <i>brows</i> ([in ~])]
Excess ^{express} (<i>eyes</i>)—Sympt ₂₃	: [N _X 's eyes] <i>shine</i> [with ~]

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Excess ^{express} (eyes)—Sympt ₁	: ∅[N _X] 'looks daggers' [at N _Y]
Excess ^{express} (eyes)—Sympt ₃₁	: seethes [in N _X 's eyes]
Excess ^{motor} (face)—Sympt ₁	: ∅[N _X] snarles [at N _Y]
Excess ^{color} (face)—Sympt ₂₃	: [N _X 's face] turns red ([with -])
Excess ^{color} (face)—Sympt ₁₃	: [N _X] is flushed <is red> ([with -])
Excess ^{motor} (teeth)—Sympt ₁₂₃	: [N _X] grinds [A _{(poss)X}] teeth ([in -])
Obstr ^{color} (face)—Sympt ₂₃	: [N _X 's face] blanches [with -]; [N _X 's face] 'clouds over' ([with -])
Obstr ^{color} (face)—Sympt ₁₃	: [N _X] is livid [with -]
Excess ^{motor} (feet)—Sympt ₁₂₃	: [N _X] stomps <stomps> [A _{(poss)X}] feet [in -], ∅ thumps [A _{(poss)X}] feet
Excess(voice)—Sympt ₁₃	: [N _X] shouts <yells> [in -]

Examples

Examples come from the Corpus of Contemporary American English = COCA (Davies 2008).

- (6) Our anger generally tells us a lot more about ourselves than about the object of our anger.
- (7) Well, you're both anti-social misfits, filled with anger and rage.
- (8) Henrik looks at him with anger and, in an act of aggressive sublimation, knocks off the lamp and books from his father's table.
- (9) She came out publicly in a fit of anger when a fan harassed her online.
- (10) Their stories are outrageous and heartbreaking and fuel my anger.
- (11) This righteous anger was a relief; it masked his sense of shame.
- (12) Suddenly my suppressed anger explodes. How could he ever do this to me?
- (13) Even if protesters do not vent their anger at the troops, there are some who argue that any demonstrations will cause psychic wounds.
- (14) She had to endure the anger of friends and neighbors who saw her as a threat to their jobs in the mining industry.
- (15) Sam jumps up from the table, trembling with anger.
- (16) He was almost always on the verge of outright shouting in anger during the show.

Idiom

◇ 「ANGER MANAGEMENT」

Lexical Functions: The Naked Truth

Set blurb on back with \BackBody{my blurb}